

AWS Certification Questions

[!NOTE] This document contains practice questions and detailed explanations for AWS Certification.

Question Q1

[!TIP]

Correct Answer:

B. Amazon Inspector

EXPLANATION

Here's how each option stacks up: -

A.

AWS Trusted Advisor Focuses on best practices and cost optimization. While it provides guidance on improving security and performance, it doesn't perform deep vulnerability assessments of EC2 instances. -

B.

Amazon Inspector Designed for automated security assessments. It scans EC2 instances for vulnerabilities and deviations from security best practices, making it the ideal choice for identifying application-level risks and infrastructure misconfigurations. -

C.

AWS Config  Tracks configuration changes and compliance. Useful for auditing and compliance, but it doesn't actively scan for vulnerabilities. -

D.

Amazon Guard Duty  Threat detection service. It monitors for malicious activity and anomalies, but doesn't assess application vulnerabilities or infrastructure best practices.

Amazon Inspector is purpose-built for this scenario—it performs automated security assessments to help you identify and remediate vulnerabilities in your EC2 workloads. Here's why this is the most operationally efficient solution: - AWS Storage Gateway (File Gateway) 

Bridges on-premises environments with cloud storage. It allows users to store files locally while asynchronously backing them up to Amazon S3. This setup retains local performance and provides scalable cloud storage, making it ideal for centralized users with large file needs. Let's break down the other options: -

A.

S3 bucket per user with mounting utility Not operationally efficient. S3 is object storage, not a native file system. Mounting S3 buckets requires third-party tools and lacks the performance and compatibility of traditional file systems. -

C.

Amazon Work Spaces + Work Docs Overkill and misaligned. This shifts the entire user environment to virtual desktops, which is unnecessary if only file storage is the issue. Work Docs is more for collaboration than large-scale file storage. -

D.

EC2 with EBS shared over network Complex and not scalable. Sharing EBS volumes across multiple users requires custom configuration and doesn't scale well. It also introduces single points of failure and network bottlenecks. Storage Gateway's file gateway mode is purpose-built for this scenario—offering seamless integration, local caching, and cloud scalability. This aligns with AWS security best practices for granting permissions to services: - IAM Roles for EC2 
Secure and scalable. Assigning an IAM role to an EC2 instance allows it to obtain temporary credentials via the instance metadata service. This avoids hardcoding secrets and simplifies permission management. Let's break down the other options: -

A.

Hard code IAM credentials in the application Highly insecure. This exposes sensitive credentials in code, which can be leaked or misused. -

B.

Store credentials in a text file on the instance Still insecure. Even if not hardcoded, storing secrets on disk is risky and violates best practices. -

D.

Modify the S3 bucket policy to allow any service to upload. Overly permissive and dangerous. This could expose your bucket to unintended access and potential abuse. Using IAM roles ensures that credentials are rotated automatically and scoped appropriately. It's the gold standard for secure access between AWS services. Under the AWS Shared Responsibility Model, responsibilities are divided between AWS and the customer:

- AWS is responsible for:
 - Physical security of infrastructure (including DynamoDB servers) → ❌ Not the customer's responsibility
 - Patching and maintenance of managed services like DynamoDB → ❌ AWS handles this
 - Encryption of data at rest (if using AWS-managed keys) → ❌ AWS can manage this unless customer opts for their own keys
- Customer is responsible for:
 - ✅ Access control and permissions

Customers must define who can access their DynamoDB tables using IAM policies and roles. This includes setting up fine-grained access, managing users, and ensuring least privilege. This distinction ensures that while AWS secures the underlying infrastructure, customers maintain control over how their data is accessed and used.

Question 6

**Which option is a perspective that includes foundational capabilities of the AWS Cloud Adoption Framework (AWS CAF)?

**

OPTIONS

A. Sustainability

B. Performance efficiency

C. Governance

D. Reliability

[!TIP]

Correct Answer:

C. Governance

EXPLANATION

Explanation: -

C.

Governance Correct. Governance is one of the six AWS CAF perspectives and includes foundational capabilities such as compliance, risk management, and policy enforcement. It helps organizations maintain control as they adopt cloud technologies. -

A.

Sustainability Incorrect. While sustainability is a growing concern in cloud operations, it is not one of the six formal AWS CAF perspectives. -

B.

Performance efficiency Incorrect. This is a pillar of the AWS Well-Architected Framework, not a CAF perspective. It focuses on using cloud resources efficiently, but doesn't address foundational governance capabilities. -

D.

Reliability Incorrect. Like performance efficiency, reliability is part of the Well-Architected Framework, not the CAF. It deals with system availability and fault tolerance.

Question 7

******A company is running and managing its own Docker environment on Amazon EC2 instances. The company wants an alternative to help manage cluster size, scheduling, and environment maintenance. Which AWS service meets these requirements?

OPTIONS

A. AWS Lambda

B. Amazon RDS

C. AWS Fargate

D. Amazon Athena

[!TIP]

Correct Answer:

C. AWS Fargate

EXPLANATION

Explanation: -

C.

AWS Fargate Correct. Fargate is a serverless compute engine for containers that eliminates the need to manage EC2 instances. It handles cluster scaling, scheduling, and infrastructure maintenance automatically, making it ideal for replacing manual Docker management on EC2. -

A.

AWS Lambda Incorrect. Lambda is designed for running short-lived functions, not managing containerized applications or clusters. It doesn't support long-running container workloads or complex scheduling. -

B.

Amazon RDS Incorrect. RDS is a managed database service. It has nothing to do with container orchestration or Docker environments. -

D.

Amazon Athena Incorrect. Athena is a query service for analyzing data in S3 using SQL. It's unrelated to container management or infrastructure scheduling.

Question 8

**A company wants to run a NoSQL database on Amazon EC2 instances. Which task is the responsibility of AWS in this scenario?

**

OPTIONS

- A. Update the guest operating system of the EC2 instances
- B. Maintain high availability at the database layer
- C. Patch the physical infrastructure that hosts the EC2 instances
- D. Configure the security group firewall

[!TIP]

Correct Answer:

- C. Patch the physical infrastructure that hosts the EC2 instances

EXPLANATION

Explanation: -

C.

Patch the physical infrastructure that hosts the EC2 instances Correct. Under the AWS Shared Responsibility Model, AWS is responsible for the security and maintenance of the underlying physical infrastructure, including hardware patching and data center operations. -

A.

Update the guest operating system of the EC2 instances Incorrect. The guest OS is part of the customer's responsibility. Since the company is managing its own EC2 instances, it must handle OS updates and patches. -

B.

Maintain high availability at the database layer Incorrect. AWS does not manage application-level availability unless using managed services like Dynamo DB or RDS. In this case, the company is running its own No SQL database, so it must architect for high availability. -

D.

Configure the security group firewall Incorrect. Security group configuration is a customer responsibility. AWS provides the tools, but the customer must define and manage the rules.

Question 9

**Which AWS services or tools can identify rightsizing opportunities for Amazon EC2 instances? (Choose two.)

**

OPTIONS

A. AWS Cost Explorer

B. AWS Billing Conductor

C. Amazon CodeGuru

D. Amazon SageMaker

E. AWS Compute Optimizer

[!TIP]

Correct Answer:

A. AWS Cost Explorer and E. AWS Compute Optimizer

EXPLANATION

Explanation: -

A.

AWS Cost Explorer Correct. Cost Explorer includes a rightsizing recommendations feature that analyzes EC2 usage patterns and highlights underutilized instances. It helps identify cost-saving opportunities by suggesting downsizing or termination. -

E.

AWS Compute Optimizer Correct. Compute Optimizer uses Cloud Watch metrics to assess EC2 resource utilization and recommends optimal instance types. It's tailored for performance and cost optimization. -

B.

AWS Billing Conductor Incorrect. This tool is used for customizing billing and cost allocation across accounts, not for analyzing EC2 usage or rightsizing. -

C.

Amazon Code Guru Incorrect. Code Guru is focused on code quality and performance profiling, not infrastructure optimization. -

D.

Amazon Sage Maker Incorrect. Sage Maker is a machine learning platform and does not provide EC2 rightsizing insights.

REFERENCES

<https://docs.aws.amazon.com/cost-management/latest/userguide/ce-rightsizing.html>

Question 10

**Which of the following are benefits of using AWS Trusted Advisor? (Choose two.)

**

OPTIONS

- A. Providing high-performance container orchestration
- B. Creating and rotating encryption keys
- C. Detecting underutilized resources to save costs
- D. Improving security by proactively monitoring the AWS environment
- E. Implementing enforced tagging across AWS resources

[!TIP]

Correct Answer:

C. Detecting underutilized resources to save costs and D. Improving security by proactively monitoring the AWS environment

EXPLANATION

Explanation: -

C.

Detecting underutilized resources to save costs Correct. Trusted Advisor analyzes your AWS environment and flags underutilized resources like idle EC2 instances or unattached EBS volumes, helping reduce unnecessary spend. -

D.

Improving security by proactively monitoring the AWS environment Correct. Trusted Advisor provides security checks such as open ports, overly permissive IAM roles, and MFA usage, helping you strengthen your cloud security posture. -

A.

Providing high-performance container orchestration Incorrect. This is the domain of services like Amazon ECS or EKS, not Trusted Advisor. -

B.

Creating and rotating encryption keys Incorrect. AWS Key Management Service (KMS) handles key creation and rotation, not Trusted Advisor. -

E.

Implementing enforced tagging across AWS resources Incorrect. Tag enforcement requires custom automation or AWS Config rules. Trusted Advisor may flag missing tags but doesn't enforce them.

REFERENCES

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Question 11

**Which of the following is an advantage that users experience when they move on-premises workloads to the AWS Cloud?

**

OPTIONS

- A. Elimination of expenses for running and maintaining data centers
- B. Price discounts that are identical to discounts from hardware providers
- C. Distribution of all operational controls to AWS
- D. Elimination of operational expenses

[!TIP]

Correct Answer:

- A. Elimination of expenses for running and maintaining data centers

EXPLANATION

Explanation: -

A.

Elimination of expenses for running and maintaining data centers Correct. Migrating to AWS removes the need for physical infrastructure, including power, cooling, hardware maintenance, and real estate — significantly reducing capital and operational costs. -

B.

Price discounts that are identical to discounts from hardware providers Incorrect. AWS pricing models differ from traditional hardware vendor discounts. AWS offers savings through Reserved Instances, Savings Plans, and spot pricing — not hardware-style discounts. -

C.

Distribution of all operational controls to AWS Incorrect. AWS handles infrastructure-level operations, but customers retain responsibility for application-level controls, configurations, and security under the Shared Responsibility Model. -

D.

Elimination of operational expenses Incorrect. While some operational costs are reduced, others remain — such as cloud service usage, monitoring, and support. Operational expenses are not eliminated, just shifted and optimized.

REFERENCES

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

Question 12

**A company wants to manage deployed IT services and govern its infrastructure as code (IaC) templates. Which AWS service will meet this requirement?

**

OPTIONS

A. AWS Resource Explorer

B. AWS Service Catalog

C. AWS Organizations

D. AWS Systems Manager

[!TIP]

Correct Answer:

B. AWS Service Catalog

EXPLANATION

Explanation: -

B.

AWS Service Catalog Correct. AWS Service Catalog allows organizations to centrally manage approved IT services, including infrastructure as code (IaC) templates. It helps enforce governance, standardize deployments, and control access to provisioned resources. -

A.

AWS Resource Explorer Incorrect. This tool helps users search and discover AWS resources across accounts and regions, but it doesn't manage or govern IaC templates. -

C.

AWS Organizations Incorrect. AWS Organizations helps manage multiple AWS accounts and apply service control policies, but it doesn't handle service deployment or IaC governance directly. -

D.

AWS Systems Manager Incorrect. Systems Manager provides operational insights and automation for AWS resources, but it's not designed for cataloging and governing IaC templates.

REFERENCES

<https://aws.amazon.com/servicecatalog/>

Question 13

**Which AWS service or tool helps users visualize, understand, and manage spending and usage over time?

**

OPTIONS

A. AWS Organizations

B. AWS Pricing Calculator

C. AWS Cost Explorer

D. AWS Service Catalog

[!TIP]

Correct Answer:

C. AWS Cost Explorer

EXPLANATION

Explanation: -

C.

AWS Cost Explorer Correct. Cost Explorer provides interactive graphs and reports to help users analyze AWS spending and usage patterns over time. It supports filtering by service, account, and tags, and includes forecasting and rightsizing recommendations. -

A.

AWS Organizations Incorrect. AWS Organizations helps manage multiple AWS accounts and apply consolidated billing, but it doesn't offer detailed visualization or usage analysis. -

B.

AWS Pricing Calculator Incorrect. This tool estimates costs for planned AWS usage, but it doesn't track or visualize actual usage over time. -

D.

AWS Service Catalog Incorrect. Service Catalog manages approved products and templates for deployment, not cost tracking or visualization.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

Question 14

**A company is using a central data platform to manage multiple types of data for its customers. The company wants to use AWS services to discover, transform, and visualize the data. Which combination of AWS services should the company use to meet these requirements? (Choose two.)

**

OPTIONS

A. AWS Glue

B. Amazon Elastic File System (Amazon EFS)

C. Amazon Redshift

D. Amazon QuickSight

E. Amazon Quantum Ledger Database (Amazon QLDB)

[!TIP]

Correct Answer:

A. AWS Glue and D. Amazon QuickSight

EXPLANATION

Explanation: -

A.

AWS Glue Correct. AWS Glue is a serverless data integration service that helps discover, catalog, and transform data. It's ideal for ETL (extract, transform, load) workflows and integrates well with other AWS analytics services. -

D.

Amazon Quick Sight Correct. Quick Sight is a scalable business intelligence tool that enables users to visualize data through dashboards and reports. It supports data sources like S3, Redshift, Athena, and more. -

B.

Amazon EFS Incorrect. EFS is a scalable file storage service for EC2 instances. It doesn't provide data discovery, transformation, or visualization capabilities. -

C.

Amazon Redshift Incorrect. Redshift is a data warehouse optimized for analytics, but it doesn't handle data discovery or transformation directly. It can be used in conjunction with Glue and Quick Sight, but isn't the best standalone answer for this question. -

E.

Amazon QLDB Incorrect. QLDB is a ledger database designed for immutable, cryptographically verifiable records. It's not intended for general-purpose data transformation or visualization.

<https://aws.amazon.com/glue> and <https://aws.amazon.com/quicksight>

REFERENCES

<https://aws.amazon.com/glue>

<https://aws.amazon.com/quicksight>

Question 15

**A global company wants to migrate its third-party applications to the AWS Cloud. The company wants help from a global team of experts to complete the migration faster and more reliably in accordance with AWS internal best practices. Which AWS service or resource will meet these requirements?

**

OPTIONS

A. AWS Support

B. AWS Professional Services

C. AWS Launch Wizard

D. AWS Managed Services (AMS)

[!TIP]

Correct Answer:

B. AWS Professional Services

EXPLANATION

Explanation: -

B.

AWS Professional Services Correct. This team consists of AWS experts who work directly with customers to accelerate cloud adoption and migration. They provide hands-on guidance, architectural best practices, and project management tailored to complex workloads and enterprise needs. -

A.

AWS Support Incorrect. AWS Support helps troubleshoot issues and provides guidance, but it doesn't offer full-scale migration planning or execution. -

C.

AWS Launch Wizard Incorrect. Launch Wizard simplifies deployment of specific workloads like SAP or SQL Server, but it doesn't provide expert consulting or migration strategy. -

D.

AWS Managed Services (AMS) Incorrect. AMS operates and manages AWS infrastructure post-deployment. It's not designed for initial migration or third-party application onboarding.

REFERENCES

https://d1.awsstatic.com/partner-network/APN_Consulting-Benefits_Brochure-Digital.pdf

Question 16

**An e-learning platform needs to run an application for 2 months each year. The application will be deployed on Amazon EC2 instances. Any application downtime during those 2 months must be avoided. Which EC2 purchasing option will meet these requirements MOST cost-effectively?

**

OPTIONS

A. Reserved Instances

B. Dedicated Hosts

C. Spot Instances

D. On-Demand Instances

[!TIP]

Correct Answer:

D. On-Demand Instances

EXPLANATION

Explanation: -

D.

On-Demand Instances Correct. On-Demand is the most cost-effective and flexible option for short-term, predictable workloads like a 2-month annual application. It ensures availability without long-term commitment or upfront costs, and avoids the risk of interruption. -

A.

Reserved Instances Incorrect. Reserved Instances offer cost savings for workloads that run continuously over a 1- or 3-year term. They're not ideal for workloads that only run 2 months per year. -

B.

Dedicated Hosts Incorrect. These are used for licensing compliance or physical isolation needs, and are significantly more expensive. Not suitable for short-term, cost-sensitive workloads. -

C.

Spot Instances Incorrect. Spot Instances are the cheapest option but can be interrupted with little notice. Since the application must avoid downtime, Spot is too risky.

REFERENCES

<https://aws.amazon.com/ec2/pricing/>

Question 17

**A developer wants to deploy an application quickly on AWS without manually creating the required resources. Which AWS service will meet these requirements?

**

OPTIONS

A. Amazon EC2

B. AWS Elastic Beanstalk

C. AWS CodeBuild

D. Amazon Personalize

[!TIP]

Correct Answer:

B. AWS Elastic Beanstalk

EXPLANATION

Explanation: -

B.

AWS Elastic Beanstalk Correct. Elastic Beanstalk is a Platform as a Service (PaaS) that automatically handles resource provisioning, load balancing, scaling, and monitoring.

Developers simply upload their code, and Beanstalk deploys the application with minimal manual setup. -

A.

Amazon EC2 Incorrect. EC2 provides raw virtual machines. Developers must manually configure networking, storage, and scaling — which contradicts the goal of quick, automated deployment. -

C.

AWS Code Build Incorrect. Code Build is a CI/CD tool for compiling source code and running tests. It doesn't deploy applications or manage infrastructure. -

D.

Amazon Personalize Incorrect. Personalize is a machine learning service for building recommendation systems. It's unrelated to application deployment.

Question 8

******A company is storing sensitive customer data in an Amazon S3 bucket. The company wants to protect the data from accidental deletion or overwriting. Which S3 feature should the company use to meet these requirements?

OPTIONS

A. S3 Lifecycle rules

B. S3 Versioning

C. S3 bucket policies

D. S3 server-side encryption

[!TIP]

Correct Answer:

EXPLANATION

Explanation: -

B.

S3 Versioning Correct. Versioning preserves, retrieves, and restores every version of an object stored in an S3 bucket. It protects against accidental overwrites and deletions by maintaining historical versions of files. -

A.

S3 Lifecycle rules Incorrect. Lifecycle rules automate transitions between storage classes and object expiration, but they don't prevent deletion or overwriting. -

C.

S3 bucket policies Incorrect. Bucket policies control access permissions, but they don't inherently protect against accidental deletion or overwriting unless explicitly configured with deny rules. -

D.

S3 server-side encryption Incorrect. Encryption protects data confidentiality, not integrity or version control. It doesn't prevent overwrites or deletions.

REFERENCES

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Versioning.html>

Question 19

****Which AWS service provides the ability to manage infrastructure as code?**

OPTIONS

A. AWS CodePipeline

B. AWS CodeDeploy

C. AWS Direct Connect

D. AWS CloudFormation

[!TIP]

Correct Answer:

D. AWS CloudFormation

EXPLANATION

Explanation: -

D.

AWS Cloud Formation Correct. Cloud Formation enables users to define and provision AWS infrastructure using declarative templates written in JSON or YAML. It automates resource creation, updates, and deletion — making it the core service for managing infrastructure as code (IaC). -

A.

AWS Code Pipeline Incorrect. Code Pipeline automates CI/CD workflows, but it doesn't define or manage infrastructure resources directly. -

B.

AWS Code Deploy Incorrect. Code Deploy handles application deployment to EC2, Lambda, or on-prem servers, but it doesn't manage infrastructure provisioning. -

C.

AWS Direct Connect Incorrect. Direct Connect establishes dedicated network connections between on-premises environments and AWS. It's unrelated to infrastructure as code.

REFERENCES

Question 20

**An online gaming company needs to choose a purchasing option to run its Amazon EC2 instances for 1 year. The web traffic is consistent, and any increases in traffic are predictable. The EC2 instances must be online and available without any disruption. Which EC2 instance purchasing option will meet these requirements MOST cost-effectively?

**

OPTIONS

A. On-Demand Instances

B. Reserved Instances

C. Spot Instances

D. Spot Fleet

[!TIP]

Correct Answer:

B. Reserved Instances

EXPLANATION

Explanation: -

B.

Reserved Instances Correct. Reserved Instances offer significant cost savings (up to 72%) compared to On-Demand pricing for predictable, steady workloads over a 1- or 3-year term. Since the traffic is consistent and uptime is critical, Reserved Instances are the most cost-effective and reliable choice. -

A.

On-Demand Instances Incorrect. While flexible, On-Demand pricing is the most expensive option for long-term workloads. It's better suited for short-term or unpredictable usage. -

C.

Spot Instances Incorrect. Spot Instances are the cheapest option but can be interrupted with little notice. They're unsuitable for workloads that require guaranteed availability. -

D.

Spot Fleet Incorrect. Spot Fleets manage a collection of Spot Instances and can include On-Demand fallback, but they still carry interruption risk and aren't ideal for guaranteed uptime over a year.

Question 21

******Which AWS service or feature allows a user to establish a dedicated network connection between a company's on-premises data center and the AWS Cloud?

OPTIONS

A. AWS Direct Connect

B. VPC peering

C. AWS VPN

D. Amazon Route 53

[!TIP]

Correct Answer:

A. AWS Direct Connect

EXPLANATION

Explanation: -

A.

AWS Direct Connect Correct. AWS Direct Connect establishes a dedicated, private network connection between an on-premises data center and AWS. It offers consistent performance, lower latency, and higher bandwidth compared to internet-based connections. -

B.

VPC peering Incorrect. VPC peering connects two VPCs, not an on-premises environment to AWS. It's used for internal AWS network routing. -

C.

AWS VPN Incorrect. VPN provides secure connectivity over the public internet. While it connects on-prem to AWS, it's not a dedicated connection and may suffer from latency or bandwidth variability. -

D.

Amazon Route 53 Incorrect. Route 53 is a DNS service used for domain name resolution and routing traffic, not for establishing network connections.

REFERENCES

<https://aws.amazon.com/directconnect>

Question 22

****Which option is a physical location of the AWS global infrastructure?**

OPTIONS

A. AWS DataSync

B. AWS Region

C. Amazon Connect

D. AWS Organizations

[!TIP]

Correct Answer:

B. AWS Region

EXPLANATION

Explanation: -

B.

AWS Region Correct. An AWS Region is a physical location around the world where AWS clusters data centers. Each Region consists of multiple Availability Zones and represents a core component of AWS's global infrastructure. -

A.

AWS Data Sync Incorrect. Data Sync is a data transfer service used to move data between on-premises storage and AWS. It's not a physical location. -

C.

Amazon Connect Incorrect. Amazon Connect is a cloud-based contact center service. It's an application, not a physical infrastructure component. -

D.

AWS Organizations Incorrect. AWS Organizations helps manage multiple AWS accounts centrally. It's a governance tool, not a physical location.

REFERENCES

https://aws.amazon.com/about-aws/global-infrastructure/regions_az

Question 23

**A company wants to protect its AWS Cloud information, systems, and assets while performing risk assessment and mitigation tasks. Which pillar of the AWS Well-Architected Framework is supported by these goals?

**

OPTIONS

A. Reliability

B. Security

C. Operational excellence

D. Performance efficiency

[!TIP]

Correct Answer:

B. Security

EXPLANATION

Explanation: -

B.

Security Correct. The Security pillar focuses on protecting data, systems, and assets. It includes identity and access management, detective controls, infrastructure protection, data protection, and incident response — all aligned with risk assessment and mitigation. -

A.

Reliability Incorrect. Reliability ensures workloads perform as intended and can recover from failures, but it doesn't directly address data protection or risk mitigation. -

C.

Operational excellence Incorrect. This pillar emphasizes monitoring, incident response, and continuous improvement of operations — not security or risk management. -

D.

Performance efficiency Incorrect. Performance efficiency is about using computing resources effectively and adapting to changing requirements, not securing assets or assessing risk.

REFERENCES

<https://docs.aws.amazon.com/wellarchitecture/latest/security-pillar/conclusion.html>

Question 24

**What is the purpose of having an internet gateway within a VPC?

**

OPTIONS

A. To create a VPN connection to the VPC

B. To allow communication between the VPC and the internet

C. To impose bandwidth constraints on internet traffic

D. To load balance traffic from the internet across Amazon EC2 instances

[!TIP]

Correct Answer:

B. To allow communication between the VPC and the internet

EXPLANATION

Explanation: -

B.

To allow communication between the VPC and the internet Correct. An internet gateway enables resources within a VPC (like EC2 instances) to send and receive traffic from the internet. It's a horizontally scaled, redundant, and highly available component. -

A.

To create a VPN connection to the VPC Incorrect. VPN connections use virtual private gateways or AWS Client VPN, not internet gateways. -

C.

To impose bandwidth constraints on internet traffic Incorrect. Internet gateways do not control bandwidth. Bandwidth management requires other tools like traffic shaping or EC2 instance types. -

D.

To load balance traffic from the internet across Amazon EC2 instances Incorrect. Load balancing is handled by services like Elastic Load Balancing (ELB), not internet gateways.

Question 25

**A company is running a monolithic on-premises application that does not scale and is difficult to maintain. The company has a plan to migrate the application to AWS and divide the

application into microservices. Which best practice of the AWS Well-Architected Framework is the company following with this plan?

**

OPTIONS

A. Integrate functional testing as part of AWS deployment

B. Use automation to deploy changes

C. Deploy the application to multiple locations

D. Implement loosely coupled dependencies

[!TIP]

Correct Answer:

D. Implement loosely coupled dependencies

EXPLANATION

Explanation: -

D.

Implement loosely coupled dependencies Correct. Migrating from a monolithic architecture to microservices aligns with the Well-Architected best practice of designing systems with loosely coupled components. This improves scalability, maintainability, and fault isolation. -

A.

Integrate functional testing as part of AWS deployment Incorrect. While important, this relates to operational excellence and deployment hygiene, not architectural decoupling. -

B.

Use automation to deploy changes Incorrect. This supports agility and reliability, but doesn't directly address the architectural shift from monolith to microservices. -

C.

Deploy the application to multiple locations Incorrect. This relates to the reliability pillar and high availability, not modular design.

REFERENCES

<https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html>

Question 26

**A company has an AWS account. The company wants to audit its password and access key rotation details for compliance purposes. Which AWS service or tool will meet this requirement?

**

OPTIONS

A. IAM Access Analyzer

B. AWS Artifact

C. IAM credential report

D. AWS Audit Manager

[!TIP]

Correct Answer:

C. IAM credential report

EXPLANATION

Explanation: -

C.

IAM credential report Correct. This tool generates a downloadable CSV file listing all IAM users and their credential details — including password last used, access key age, and whether MFA is enabled. It's purpose-built for auditing identity hygiene and credential rotation. -

A.

IAM Access Analyzer Incorrect. This tool identifies resources shared externally (e.g., public buckets or cross-account access), not credential rotation. -

B.

AWS Artifact Incorrect. Artifact provides access to AWS's own compliance reports (like SOC or ISO), not your account's credential data. -

D.

AWS Audit Manager Incorrect. Audit Manager helps automate evidence collection for broader audits, but it doesn't directly report on IAM credential rotation.

Question 27

******A company wants to receive a notification when a specific AWS cost threshold is reached. Which AWS services or tools can the company use to meet this requirement? (Choose two.)

OPTIONS

A. Amazon Simple Queue Service (Amazon SQS)

B. AWS Budgets

C. Cost Explorer

D. Amazon CloudWatch

E. AWS Cost and Usage Report

[!TIP]

Correct Answer:

B. AWS Budgets and D. Amazon CloudWatch

EXPLANATION

Explanation: -

B.

AWS Budgets Correct. AWS Budgets lets you set custom cost and usage thresholds. You can configure alerts via email or SNS when spending exceeds your defined budget. It's purpose-built for proactive cost monitoring. -

D.

Amazon Cloud Watch Correct. Cloud Watch can be integrated with AWS Budgets to trigger alarms and notifications based on budget thresholds. It's also used to monitor usage metrics and automate responses. -

A.

Amazon SQS Incorrect. SQS is a messaging service for decoupling applications. It doesn't monitor or notify based on cost thresholds. -

C.

Cost Explorer Incorrect. Cost Explorer helps visualize and analyze historical spending, but it doesn't support threshold-based notifications. -

E.

AWS Cost and Usage Report Incorrect. This provides detailed billing data for analysis, but it's passive and doesn't trigger alerts.

Question 28

****Which AWS service or resource provides answers to the most frequently asked security-related questions that AWS receives from its users?**

OPTIONS

A. AWS Artifact

B. Amazon Connect

C. AWS Chatbot

D. AWS Knowledge Center

[!TIP]

Correct Answer:

D. AWS Knowledge Center

EXPLANATION

Explanation: -

D.

AWS Knowledge Center Correct. The AWS Knowledge Center is a public resource that provides answers to frequently asked questions, especially around security, billing, and service usage. It's designed to help users quickly find trusted guidance from AWS. -

A.

AWS Artifact Incorrect. AWS Artifact provides access to AWS's compliance reports and agreements (like SOC, ISO), but it doesn't answer general security FA Qs. -

B.

Amazon Connect Incorrect. Amazon Connect is a cloud contact center service. It's used for customer service operations, not for answering AWS security questions. -

C.

AWS Chatbot Incorrect. AWS Chatbot enables Dev Ops teams to receive alerts and run commands via Slack or Microsoft Teams. It's not a knowledge base or FAQ resource.

Question 29

**Which tasks are customer responsibilities, according to the AWS shared responsibility model? (Choose two.)

**

OPTIONS

A. Configure the AWS provided security group firewall

B. Classify company assets in the AWS Cloud

C. Determine which Availability Zones to use for Amazon S3 buckets

D. Patch or upgrade Amazon DynamoDB

E. Select Amazon EC2 instances to run AWS Lambda on

[!TIP]

Correct Answer:

A. Configure the AWS provided security group firewall and B. Classify company assets in the AWS Cloud

EXPLANATION

Explanation: -

A.

Configure the AWS provided security group firewall Correct. Security groups are customer-managed. AWS provides the infrastructure, but customers must configure rules to control inbound/outbound traffic. -

B.

Classify company assets in the AWS Cloud Correct. Customers are responsible for identifying and classifying their data (e.g., public, confidential, regulated) and applying appropriate controls. -

C.

Determine which Availability Zones to use for Amazon S3 buckets Incorrect. S3 is a regional service. AWS automatically manages Availability Zone placement for durability and availability. -

D.

Patch or upgrade Amazon Dynamo DB Incorrect. Dynamo DB is a fully managed service. AWS handles patching and maintenance. -

E.

Select Amazon EC2 instances to run AWS Lambda on Incorrect. Lambda abstracts the infrastructure. Customers don't choose EC2 instances — AWS manages the runtime environment.

Question 30

**Which of the following are pillars of the AWS Well-Architected Framework? (Choose two.)

**

OPTIONS

A. Availability

B. Reliability

C. Scalability

D. Responsive design

[!TIP]

Correct Answer:

B. Reliability and E. Operational excellence

EXPLANATION

Explanation: -

B.

Reliability Correct. This pillar focuses on ensuring a workload performs its intended function correctly and consistently. It includes strategies for fault tolerance, recovery, and distributed system design. -

E.

Operational excellence Correct. This pillar emphasizes monitoring, incident response, and continuous improvement of operations. It's about running workloads effectively and evolving procedures over time. -

A.

Availability Incorrect. Availability is a key concept within the Reliability pillar, but it's not a standalone pillar itself. -

C.

Scalability Incorrect. Scalability is addressed within the Performance Efficiency pillar, but it's not one of the five named pillars. -

D.

Responsive design Incorrect. This is a web development concept, not part of the AWS Well-Architected Framework. (6 pillars: Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization and sustainability)

Question 31

**Which AWS service or feature is used to send both text and email messages from distributed applications?

**

OPTIONS

A. Amazon Simple Notification Service (Amazon SNS)

B. Amazon Simple Email Service (Amazon SES)

C. Amazon CloudWatch alerts

D. Amazon Simple Queue Service (Amazon SQS)

[!TIP]

Correct Answer:

A. Amazon Simple Notification Service (Amazon SNS)

EXPLANATION

Explanation:

A.

Amazon SNS  Correct. SNS is a fully managed pub/sub messaging service that can deliver notifications via multiple protocols — including email, SMS (text messages), and HTTP endpoints. It's ideal for distributed applications that need to push alerts or updates to users or systems.

B.

Amazon SES  Incorrect. SES is used specifically for sending email, not SMS. It's great for transactional or marketing emails but doesn't support text messaging.

C.

Amazon CloudWatch alerts ❌ Incorrect. CloudWatch can trigger alarms and send notifications, but it relies on SNS to deliver those messages. It's not a standalone messaging service.

D.

Amazon SQS ❌ Incorrect. SQS is a message queuing service for decoupling application components. It doesn't send notifications to users — it's for internal message processing.

REFERENCES

<https://aws.amazon.com/sns/>

Question 32

**A user needs programmatic access to AWS resources through the AWS CLI or the AWS API. Which option will provide the user with the appropriate access?

**

OPTIONS

A. Amazon Inspector

B. Access keys

C. SSH public keys

D. AWS Key Management Service (AWS KMS) keys

[!TIP]

Correct Answer:

B. Access keys

EXPLANATION

Explanation:

B.

Access keys  Correct. Access keys (consisting of an access key ID and secret access key) are used to authenticate programmatic requests to AWS services via the CLI or API. They're tied to IAM users or roles and are essential for non-console access.

A.

Amazon Inspector  Incorrect. Inspector is a security assessment service that scans for vulnerabilities. It doesn't provide access credentials.

C.

SSH public keys  Incorrect. SSH keys are used to connect to EC2 instances via SSH, not for accessing AWS services through the CLI or API.

D.

AWS KMS keys  Incorrect. KMS keys are used for encryption and decryption of data. They don't grant access to AWS resources.

Question 34

**What does the concept of agility mean in AWS Cloud computing? (Choose two.)

**

OPTIONS

A. The speed at which AWS resources are implemented

B. The speed at which AWS creates new AWS Regions

C. The ability to experiment quickly

D. The elimination of wasted capacity

E. The low cost of entry into cloud computing

[!TIP]

Correct Answer:

A. The speed at which AWS resources are implemented and C. The ability to experiment quickly

EXPLANATION

Explanation:

A.

The speed at which AWS resources are implemented  Correct. Agility in AWS means you can provision infrastructure in minutes, enabling rapid iteration and deployment.

C.

The ability to experiment quickly  Correct. AWS empowers teams to test ideas without long procurement cycles. You can spin up environments, run tests, and shut them down — all with minimal friction.

B.

The speed at which AWS creates new AWS Regions  Incorrect. While AWS expands globally, this is a provider-side capability, not a user-facing agility feature.

D.

The elimination of wasted capacity  Incorrect. That's more aligned with cost optimization, not agility.

E.

The low cost of entry into cloud computing  Incorrect. This supports accessibility and affordability, but doesn't directly reflect agility.

Question 35

**A company needs to block SQL injection attacks. Which AWS service or feature can meet this requirement?

**

OPTIONS

A. AWS WAF

B. AWS Shield

C. Network ACLs

D. Security groups

[!TIP]

Correct Answer:

A. AWS WAF

EXPLANATION

Explanation:

A.

AWS WAF (Web Application Firewall)  Correct. AWS WAF protects web applications from common threats like SQL injection and cross-site scripting (XSS). It lets you define custom rules or use managed rule groups to block malicious traffic at the application layer.

B.

AWS Shield  Incorrect. AWS Shield protects against DDoS attacks, not application-layer threats like SQL injection.

C.

Network ACLs ❌ Incorrect. Network ACLs operate at the subnet level and filter traffic based on IP and port — they don't inspect payloads for SQL injection.

D.

Security groups ❌ Incorrect. Security groups control inbound/outbound traffic at the instance level, but they don't analyze request content for threats.

Question 36

**Which AWS service or feature identifies whether an Amazon S3 bucket or an IAM role has been shared with an external entity?

**

OPTIONS

A. AWS Service Catalog

B. AWS Systems Manager

C. AWS IAM Access Analyzer

D. AWS Organizations

[!TIP]

Correct Answer:

C. AWS IAM Access Analyzer

EXPLANATION

Explanation:

C.

AWS IAM Access Analyzer  Correct. This tool helps you identify resources (like S3 buckets, IAM roles, KMS keys) that are shared with external entities. It analyzes policies and flags unintended exposure, making it essential for security audits and access reviews.

A.

AWS Service Catalog  Incorrect. Service Catalog helps manage approved products and services for deployment — not access visibility.

B.

AWS Systems Manager  Incorrect. Systems Manager is used for operational tasks like patching, automation, and inventory — not access analysis.

D.

AWS Organizations  Incorrect. Organizations helps manage multiple AWS accounts centrally, but doesn't inspect resource-level sharing.

Question 37

******A cloud practitioner needs to obtain AWS compliance reports before migrating an environment to the AWS Cloud. How can these reports be generated?

OPTIONS

A. Contact the AWS Compliance team

B. Download the reports from AWS Artifact

C. Open a case with AWS Support

D. Generate the reports with Amazon Macie

[!TIP]

Correct Answer:

B. Download the reports from AWS Artifact

EXPLANATION

Explanation:

B.

AWS Artifact  Correct. AWS Artifact is the central resource for accessing AWS's compliance documentation, including SOC reports, ISO certifications, and other audit artifacts. It's self-service and available directly from the AWS Management Console.

A.

Contact the AWS Compliance team  Incorrect. You don't need to contact AWS directly — Artifact provides instant access to the reports.

C.

Open a case with AWS Support  Incorrect. Support cases are for troubleshooting or account issues, not for retrieving standard compliance documentation.

D.

Generate the reports with Amazon Macie  Incorrect. Macie is a data security service that identifies sensitive data (like PII) — it doesn't generate compliance reports.

Question 38

**An ecommerce company has migrated its IT infrastructure from an on-premises data center to the AWS Cloud. Which cost is the company's direct responsibility?

**

OPTIONS

A. Cost of application software licenses

B. Cost of the hardware infrastructure on AWS

C. Cost of power for the AWS servers

D. Cost of physical security for the AWS data center

[!TIP]

Correct Answer:

A. Cost of application software licenses

EXPLANATION

Explanation:

A.

Cost of application software licenses  Correct. Under the AWS shared responsibility model, customers are responsible for managing and paying for their own software — including licensing, configuration, and updates.

B.

Cost of the hardware infrastructure on AWS  Incorrect. AWS owns and operates the hardware. Customers pay for usage (e.g., EC2 hours), but not the infrastructure itself.

C.

Cost of power for the AWS servers  Incorrect. AWS handles power, cooling, and physical infrastructure costs as part of its cloud service.

D.

Cost of physical security for the AWS data center  Incorrect. AWS is responsible for securing its physical facilities. Customers focus on logical security (e.g., IAM, encryption).

Question 39

******A company is setting up AWS Identity and Access Management (IAM) on an AWS account. Which recommendation complies with IAM security best practices?

OPTIONS

- A. Use the account root user access keys for administrative tasks
- B. Grant broad permissions so that all company employees can access the resources they need
- C. Turn on multi-factor authentication (MFA) for added security during the login process
- D. Avoid rotating credentials to prevent issues in production applications

[!TIP]

Correct Answer:

- C. Turn on multi-factor authentication (MFA) for added security during the login process

EXPLANATION

Explanation:

C.

Turn on multi-factor authentication (MFA)  Correct. MFA adds an extra layer of protection by requiring a second form of verification. It's a core IAM best practice, especially for privileged users and the root account.

A.

Use the account root user access keys for administrative tasks  Incorrect. AWS strongly advises against using the root user for daily operations. Instead, create IAM users with scoped permissions.

B.

Grant broad permissions to all employees ❌ Incorrect. This violates the principle of least privilege. Permissions should be tightly scoped to match job roles.

D.

Avoid rotating credentials ❌ Incorrect. Regular credential rotation reduces the risk of compromise and is a recommended security hygiene practice.

Question 40

**Elasticity in the AWS Cloud refers to which of the following? (Choose two.)

**

OPTIONS

- A. How quickly an Amazon EC2 instance can be restarted
- B. The ability to rightsize resources as demand shifts
- C. The maximum amount of RAM an Amazon EC2 instance can use
- D. The pay-as-you-go billing model
- E. How easily resources can be procured when they are needed

[!TIP]

Correct Answer:

B. The ability to rightsize resources as demand shifts and E. How easily resources can be procured when they are needed

EXPLANATION

Explanation:

B.

The ability to rightsize resources as demand shifts  Correct. Elasticity means you can scale resources up or down based on demand — automatically or manually — to avoid overprovisioning or underperformance.

E.

How easily resources can be procured when they are needed  Correct. Elasticity also refers to the ability to quickly provision infrastructure without long lead times, enabling rapid response to changing workloads.

A.

How quickly an Amazon EC2 instance can be restarted  Incorrect. That's more about availability or recovery time, not elasticity.

C.

The maximum amount of RAM an Amazon EC2 instance can use  Incorrect. This relates to instance sizing, not elasticity.

D.

The pay-as-you-go billing model  Incorrect. That's a feature of cloud economics, not elasticity. It supports cost optimization, but doesn't define scaling behavior.

Question 41

**Which service enables customers to audit API calls in their AWS accounts?

**

OPTIONS

A. AWS CloudTrail

B. AWS Trusted Advisor

C. Amazon Inspector

[!TIP]

Correct Answer:

A. AWS CloudTrail

EXPLANATION

Explanation:

A.

AWS CloudTrail  Correct. CloudTrail records AWS API calls and events across services. It logs who made the call, when, from where, and what actions were taken — making it essential for auditing, security analysis, and compliance.

B.

AWS Trusted Advisor  Incorrect. Trusted Advisor provides best practice recommendations for cost, performance, and security — but doesn't log API activity.

C.

Amazon Inspector  Incorrect. Inspector scans for vulnerabilities in EC2 instances and container workloads — it doesn't track API calls.

D.

AWS X-Ray  Incorrect. X-Ray helps developers trace requests through distributed applications — it's focused on performance debugging, not auditing API usage.

Question 42

**What is a customer responsibility when using AWS Lambda according to the AWS shared responsibility model?

**

OPTIONS

- A. Managing the code within the Lambda function
- B. Confirming that the hardware is working in the data center
- C. Patching the operating system
- D. Shutting down Lambda functions when they are no longer in use

[!TIP]

Correct Answer:

- A. Managing the code within the Lambda function

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of the AWS shared responsibility model, especially in the context of serverless computing. With AWS Lambda, AWS abstracts away infrastructure, OS, and runtime management — but you still own the logic.

WHY A IS CORRECT

Lambda is serverless, but the function code — its logic, dependencies, and behavior — is your responsibility. You must: Write and maintain the code Handle input validation, error handling, and security logic Ensure the code doesn't expose secrets or violate compliance This aligns with the "security in the cloud" portion of the shared model: AWS secures the platform, you secure your workloads.

DISTRACTOR BREAKDOWN

****B. Confirming that the hardware is working in the data center **** Incorrect. AWS owns and operates the physical infrastructure. You never touch or inspect hardware in serverless.

****C. Patching the operating system **** Incorrect. AWS handles OS patching for Lambda. You don't manage the runtime environment.

****D. Shutting down Lambda functions when they are no longer in use **** Incorrect. Lambda is event-driven and auto-scales. You don't manually shut it down — it runs only when triggered.

REAL-WORLD TIE-IN

Imagine you deploy a Lambda function that processes user uploads. If it fails due to bad input or leaks data, AWS won't fix your code — that's your domain. But if the underlying server crashes, AWS handles recovery.

Question 43

****A company has 5 TB of data stored in Amazon S3. The company plans to occasionally run queries on the data for analysis. Which AWS service should the company use to run these queries in the MOST cost-effective manner?**

OPTIONS

A. Amazon Redshift

B. Amazon Athena

C. Amazon Kinesis

D. Amazon RDS

[!TIP]

Correct Answer:

B. Amazon Athena

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match data access patterns with cost-efficient analytics services. The key clues are: Data is already in S3 Queries are occasional Cost optimization is critical

WHY B IS CORRECT

Amazon Athena is a serverless, pay-per-query service that lets you run SQL directly on S3 data. No infrastructure to manage. You pay only for the amount of data scanned — ideal for occasional, ad hoc analysis. Supports formats like CSV, JSON, Parquet, and integrates with Glue Data Catalog for schema management. This is the canonical choice for low-frequency, high-volume S3 analytics.

DISTRACTOR BREAKDOWN

****A. Amazon Redshift **** Incorrect. Redshift is a powerful data warehouse, but it requires loading data into its cluster and incurs ongoing compute/storage costs. Overkill for occasional queries.

****C. Amazon Kinesis **** Incorrect. Kinesis is for real-time streaming data, not querying static S3 datasets. Doesn't fit the batch analysis use case.

****D. Amazon RDS **** Incorrect. RDS is a managed relational database. You'd need to import the data and pay for instance uptime — inefficient for sporadic querying.

REAL-WORLD TIE-IN

Imagine a data science team wants to run monthly reports on archived logs stored in S3. With Athena, they can: Write SQL directly against the bucket Avoid ETL pipelines Pay only for what they scan This is a classic “query-in-place” architecture — fast, cheap, and scalable.

REFERENCES

<https://aws.amazon.com/athena>

Question 44

**Which AWS service can be used at no additional cost?

**

OPTIONS

A. Amazon SageMaker

B. AWS Config

C. AWS Organizations

D. Amazon CloudWatch

[!TIP]

Correct Answer:

C. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cost boundaries within AWS services, especially which ones are free to use by default versus those that incur charges based on usage or configuration.

WHY C IS CORRECT

AWS Organizations is a free service that lets you centrally manage and govern multiple AWS accounts. You can: Create organizational units (OUs) Apply Service Control Policies (SCPs) Consolidate billing There's no charge for using Organizations itself — you only pay for resources used within member accounts. This is a foundational governance tool, especially for enterprises managing multiple environments (e.g., dev, prod, audit).

DISTRACTOR BREAKDOWN

****A. Amazon SageMaker **** Incorrect. Sage Maker is a powerful ML platform, but it incurs charges for notebook instances, training jobs, and model hosting.

****B. AWS Config **** Incorrect. AWS Config tracks resource configurations and changes — but it charges per configuration item recorded and per rule evaluation.

****D. Amazon CloudWatch **** Incorrect. Cloud Watch has a free tier, but metrics, logs, dashboards, and alarms beyond the free limits are billed. It's not “free” in the absolute sense.

REAL-WORLD TIE-IN

If you're setting up a multi-account structure for a startup or enterprise, AWS Organizations lets you enforce security boundaries and billing controls without adding cost overhead. It's a strategic tool for scaling governance.

REFERENCES

Question 45

**Which AWS Cloud Adoption Framework (AWS CAF) capability belongs to the people perspective?

**

OPTIONS

A. Data architecture

B. Event management

C. Cloud fluency

D. Strategic partnership

[!TIP]

Correct Answer:

C. Cloud fluency

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Cloud Adoption Framework (CAF) — specifically its six perspectives: Business, People, Governance, Platform, Security, and Operations. Each perspective contains capabilities that guide cloud transformation. The People

perspective focuses on organizational culture, skills, and change management — not infrastructure or tooling.

WHY C IS CORRECT

Cloud fluency refers to the ability of individuals and teams to understand cloud concepts, tools, and practices. It's a capability under the People perspective, which aims to: Upskill teams Foster cloud-native thinking Align roles with cloud responsibilities This is essential for successful adoption — tech alone doesn't transform an organization; people do.

DISTRACTOR BREAKDOWN

****A. Data architecture **** Incorrect. This belongs to the Platform perspective, which covers infrastructure, compute, storage, and data design.

****B. Event management **** Incorrect. This is part of the Operations perspective, focused on monitoring, incident response, and reliability.

****D. Strategic partnership **** Incorrect. This aligns with the Business perspective, which covers vendor relationships, financial modeling, and value realization.

REAL-WORLD TIE-IN

In a cloud migration, you might have: Engineers learning Terraform Finance teams understanding cost models Executives grasping cloud ROI All of that falls under cloud fluency — a people-first capability that drives adoption beyond tooling.

Question 46

****A** a company wants to make an upfront commitment for continued use of its production Amazon EC2 instances in exchange for a reduced overall cost. Which pricing options meet these

requirements with the LOWEST cost? (Choose two.)

**

OPTIONS

A. Spot Instances

B. On-Demand Instances

C. Reserved Instances

D. Savings Plans

E. Dedicated Hosts

[!TIP]

Correct Answer:

C. Reserved Instances and D. Savings Plans

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to align long-term workload stability with cost optimization strategies in EC2. The key phrase is “upfront commitment” — which rules out flexible or interruptible models. 🔍 Why C and D Are Correct

C.

Reserved Instances (RIs) ☒ Correct. RIs offer up to 75% savings compared to On-Demand pricing when you commit to a 1- or 3-year term. You choose instance type, region, and payment plan (All Upfront, Partial Upfront, or No Upfront).

D.

Savings Plans  Correct. Savings Plans provide similar discounts to RIs but with more flexibility. You commit to a specific dollar amount per hour over 1 or 3 years, and AWS applies the discount across eligible compute usage — including EC2, Fargate, and Lambda. These are the two most cost-effective options for predictable, long-term EC2 workloads.

SOURCES

DISTRACTOR BREAKDOWN

****A. Spot Instances **** Incorrect. Spot is the cheapest per hour, but it's interruptible and not suitable for production workloads requiring guaranteed uptime.

****B. On-Demand Instances **** Incorrect. On-Demand is the most flexible but most expensive. No commitment = no discount.

****E. Dedicated Hosts **** Incorrect. These are for licensing or compliance needs. They're more expensive and not designed for cost savings through commitment.

REAL-WORLD TIE-IN

If you're running a stable production workload (e.g., web servers, backend APIs), committing via RIs or Savings Plans can cut EC2 costs by half or more. Use Cost Explorer or Compute Optimizer to model savings before committing.

Question 47

****A** A company wants to migrate its on-premises relational databases to the AWS Cloud. The company wants to use infrastructure as close to its current geographical location as possible.

Which AWS service or resource should the company use to select its Amazon RDS deployment area?

**

OPTIONS

A. Amazon Connect

B. AWS Wavelength

C. AWS Regions

D. AWS Direct Connect

[!TIP]

Correct Answer:

C. AWS Regions

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS global infrastructure and how to strategically place workloads based on latency, compliance, and proximity. The key phrase is “as close to its current geographical location as possible.”

WHY C IS CORRECT

AWS Regions are the primary geographic boundaries for deploying services like Amazon RDS. Each Region contains multiple Availability Zones and is designed to be isolated and resilient. When migrating databases, choosing the right Region ensures: Low latency to users or on-prem systems Data residency compliance Cost optimization (data transfer, inter-region charges) You select the Region during RDS provisioning — it’s the first decision in cloud placement strategy.

DISTRACTOR BREAKDOWN

****A. Amazon Connect **** Incorrect. This is a contact center service — unrelated to infrastructure placement or database migration.

****B. AWS Wavelength **** Incorrect. Wavelength extends AWS services to telecom edge locations for ultra-low latency — used in mobile apps, not database hosting.

****D. AWS Direct Connect **** Incorrect. Direct Connect provides private network links between on-prem and AWS, but doesn't determine where RDS is deployed. It's a connectivity layer, not a placement tool.

REAL-WORLD TIE-IN

If your company is based in Nairobi, you'd likely choose the Africa (Cape Town) Region to minimize latency. You'd then use Direct Connect or VPN for hybrid access — but the Region choice drives the RDS location.

Question 48

****A company is exploring the use of the AWS Cloud, and needs to create a cost estimate for a project before the infrastructure is provisioned. Which AWS service or feature can be used to estimate costs before deployment?**

OPTIONS

A. AWS Free Tier

B. AWS Pricing Calculator

C. AWS Billing and Cost Management

D. AWS Cost and Usage Report

[!TIP]

Correct Answer:

B. AWS Pricing Calculator

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to distinguish between pre-deployment planning tools and post-deployment billing tools. The key phrase is “before the infrastructure is provisioned.”

WHY B IS CORRECT

The AWS Pricing Calculator is a pre-deployment tool that lets you: Model costs for services like EC2, S3, RDS, Lambda, etc. Choose instance types, storage, data transfer, and usage patterns. Generate sharable estimates for budgeting and approval. It's ideal for forecasting cloud spend before committing resources.

DISTRACTOR BREAKDOWN

****A. AWS Free Tier **** Incorrect. The Free Tier offers limited usage at no cost, but it's not a cost estimation tool. It's a pricing model, not a calculator.

****C. AWS Billing and Cost Management **** Incorrect. This dashboard shows actual usage and charges after deployment. It's reactive, not predictive.

****D. AWS Cost and Usage Report (CUR) **** Incorrect. CUR provides granular billing data for existing usage — useful for audits and optimization, but not for estimating future costs.

REAL-WORLD TIE-IN

Before launching a new analytics pipeline on EC2 and S3, you'd use the Pricing Calculator to:
Estimate monthly compute and storage costs
Compare Reserved vs On-Demand pricing
Present a budget to stakeholders
This is foundational for cloud financial operations (FinOps).

REFERENCES

<https://docs.aws.amazon.com/pricing-calculator/latest/userguide/what-is-pricing-calculator.html>

Question 49

**A company is building an application that needs to deliver images and videos globally with minimal latency. Which approach can the company use to accomplish this in a cost effective manner?

**

OPTIONS

- A. Deliver the content through Amazon CloudFront
- B. Store the content on Amazon S3 and enable S3 cross-region replication
- C. Implement a VPN across multiple AWS Regions
- D. Deliver the content through AWS PrivateLink

[!TIP]

Correct Answer:

A. Deliver the content through Amazon CloudFront

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match global content delivery needs with the right AWS service architecture. The key goals are: Global reach Minimal latency Cost efficiency

WHY A IS CORRECT

Amazon CloudFront is AWS's Content Delivery Network (CDN). It caches content (images, videos, etc.) at edge locations worldwide, reducing latency by serving users from the nearest point. It integrates seamlessly with S3, EC2, and custom origins. You pay based on data transfer and request volume — far cheaper than replicating content across Regions or maintaining VPNs. This is the canonical solution for media-heavy, latency-sensitive global apps.

DISTRACTOR BREAKDOWN

****B. S3 cross-region replication **** Incorrect. While it improves availability and durability, it doesn't reduce latency or optimize delivery. It also incurs extra storage and replication costs.

****C. VPN across multiple AWS Regions **** Incorrect. VPNs are for secure private connectivity, not content delivery. They're expensive and introduce latency — the opposite of what's needed here.

****D. AWS PrivateLink **** Incorrect. Private Link is for secure access to VPC endpoints — useful for internal services, not public media delivery.

REAL-WORLD TIE-IN

If you're building a photo-sharing app or video platform, CloudFront lets you: Cache thumbnails and media at edge Serve users in Nairobi, Tokyo, or São Paulo with sub-second latency Avoid duplicating storage or compute across Regions It's the backbone of scalable, performant media delivery.

REFERENCES

<https://aws.amazon.com/cloudfront/>

Question 50

**Which option is a benefit of the economies of scale based on the advantages of cloud computing?

**

OPTIONS

A. The ability to trade variable expense for fixed expense

B. Increased speed and agility

C. Lower variable costs over fixed costs

D. Increased operational costs across data centers

[!TIP]

Correct Answer:

C. Lower variable costs over fixed costs

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of economies of scale — a foundational economic principle that cloud providers leverage to reduce costs for customers. The key idea is that as usage increases, providers can spread fixed costs across more customers, resulting in lower per-unit costs.

WHY C IS CORRECT

Cloud providers like AWS operate massive infrastructure at global scale. They aggregate demand from millions of customers, allowing them to: Purchase hardware and bandwidth in bulk Optimize data center operations Reduce per-user costs These savings are passed on to customers via lower variable pricing — especially in pay-as-you-go models. As AWS puts it: “By using cloud computing, you can achieve a lower variable cost than you can get on your own.”

DISTRACTOR BREAKDOWN

****A. Trade variable expense for fixed expense **** Incorrect. This reverses the actual cloud benefit. Cloud computing lets you avoid fixed expenses (like data center builds) and pay only for what you use.

****B. Increased speed and agility **** Incorrect. While true for cloud benefits, this relates to agility, not economies of scale.

****D. Increased operational costs across data centers **** Incorrect. This contradicts the principle — economies of scale reduce operational costs, not increase them.

REAL-WORLD TIE-IN

If you run a startup in Nairobi, you don’t need to build a data center. You tap into AWS’s global infrastructure and pay only for compute, storage, and bandwidth you use — benefiting from the scale of AWS’s operations without upfront investment.

SOURCES

REFERENCES

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

Question 52

**A company is developing an application that uses multiple AWS services. The application needs to use temporary, limited-privilege credentials for authentication with other AWS APIs. Which AWS service or feature should the company use to meet these authentication requirements?

**

OPTIONS

A. Amazon API Gateway

B. IAM users

C. AWS Security Token Service (AWS STS)

D. IAM instance profiles

[!TIP]

Correct Answer:

C. AWS Security Token Service (AWS STS)

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of temporary credential management in multi-service AWS applications. The key phrases are: Temporary credentials Limited privileges Cross-service API access This is classic STS territory — the backbone of secure, ephemeral access in AWS.

WHY C IS CORRECT

AWS STS issues temporary security credentials (access key ID, secret key, session token) that:
Expire after a defined duration
Are scoped by IAM policies
Can be assumed by users, roles, or federated identities
Ideal for: Cross-account access Federated identity providers (e.g., SAML, OIDC)
Applications needing short-lived access to AWS APIs
This aligns with least privilege and credential rotation best practices.

DISTRACTOR BREAKDOWN

****A. Amazon API Gateway **** Incorrect. API Gateway manages HTTP AP Is — it doesn't issue AWS credentials. It can validate tokens, but not generate them.

****B. IAM users **** Incorrect. IAM users have long-term credentials, which are risky for apps. They're static and harder to rotate securely.

****D. IAM instance profiles **** Incorrect. Instance profiles attach IAM roles to EC2 instances — useful for EC2-based apps, but not for general-purpose credential issuance across services.

REAL-WORLD TIE-IN

Imagine a Lambda function that needs to access S3 and DynamoDB. Instead of hardcoding credentials, it can assume a role via STS, get temporary credentials, and operate securely — all without manual rotation.

SOURCES

REFERENCES

<https://docs.aws.amazon.com/STS/latest/APIReference/Welcome.html>

<https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html>

Question 54

**Which AWS service is always provided at no charge?

**

OPTIONS

A. Amazon S3

B. AWS Identity and Access Management (IAM)

C. Elastic Load Balancers

D. AWS WAF

[!TIP]

Correct Answer:

B. AWS Identity and Access Management (IAM)

EXPLANATION

CONCEPTUAL FRAMING

This question tests your knowledge of AWS pricing models, especially which services are foundational and free for all customers. The key phrase is “always provided at no charge” — not just under the Free Tier, but permanently free.

WHY B IS CORRECT

IAM is AWS’s core service for managing users, roles, and permissions. It is always free, regardless of usage volume or account type. You can: Create and manage IAM users and groups Define granular permissions via policies Enable MFA and enforce security best practices IAM is essential for secure access control, and AWS makes it universally available at no cost.

SOURCES

DISTRACTOR BREAKDOWN

****A. Amazon S3 **** Incorrect. S3 has a Free Tier (5 GB), but beyond that, storage and requests are billed.

****C. Elastic Load Balancers **** Incorrect. EL Bs incur charges based on hours active and data processed — not free.

****D. AWS WAF **** Incorrect. AWS WAF is a paid service for web application firewall protection — billed per rule and request.

REAL-WORLD TIE-IN

When setting up a new AWS account, you can immediately: Create IAM roles for EC2, Lambda, and S3 Enforce least privilege access Audit actions via CloudTrail All without incurring any IAM-related charges — making it the foundation of secure cloud operations.

REFERENCES

<https://aws.amazon.com/free/>

Question 55

**To reduce costs, a company is planning to migrate a NoSQL database to AWS. Which AWS service is fully managed and can automatically scale throughput capacity to meet database workload demands?

**

OPTIONS

A. Amazon Redshift

B. Amazon Aurora

C. Amazon DynamoDB

D. Amazon RDS

[!TIP]

Correct Answer:

C. Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match database type (NoSQL) with the right AWS-native service, while factoring in cost efficiency and scalability. The key phrases are: NoSQL Fully managed Auto-scaling throughput

WHY C IS CORRECT

Amazon DynamoDB is AWS's flagship NoSQL database service. It's: Fully managed — no servers, patching, or provisioning Auto-scaling — adjusts read/write throughput based on demand Highly available and durable — with multi-AZ replication Cost-efficient — supports on-demand and provisioned capacity modes It's ideal for key-value and document workloads — think user profiles, session data, IoT telemetry.

SOURCES

DISTRACTOR BREAKDOWN

****A. Amazon Redshift **** Incorrect. Redshift is a data warehouse — optimized for OLAP, not No SQL. It's columnar and doesn't auto-scale throughput.

****B. Amazon Aurora **** Incorrect. Aurora is a relational database (MySQL/PostgreSQL compatible). It's not No SQL and doesn't match the workload type.

****D. Amazon RDS **** Incorrect. RDS supports relational engines (MySQL, PostgreSQL, Oracle, SQL Server). It's not designed for No SQL workloads or auto-scaling throughput.

REAL-WORLD TIE-IN

If you're migrating a MongoDB-like workload with unpredictable traffic spikes, DynamoDB's on-demand mode lets you scale seamlessly — no provisioning, no throttling, no wasted capacity.

REFERENCES

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/AutoScaling.html>

<https://aws.amazon.com/blogs/database/migrating-nosql-workloads-to-amazon-dynamodb/>

Question 57

**A company has a test AWS environment. The application testing can be interrupted and does not need to run continuously. Which Amazon EC2 purchasing option will meet these requirements MOST cost-effectively?

**

OPTIONS

A. On-Demand Instances

B. Dedicated Instances

C. Spot Instances

D. Reserved Instances

[!TIP]

Correct Answer:

C. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match workload tolerance (interruptible, non-continuous) with the most cost-effective EC2 pricing model. The key clues are: Test environment Interruptible Cost-sensitive

WHY C IS CORRECT

Spot Instances let you bid on unused EC2 capacity at up to 90% discount compared to On-Demand. Ideal for: Batch jobs CI/CD pipelines Fault-tolerant test environments AWS can reclaim the instance with 2-minute warning, so it's not suitable for critical workloads — but perfect for non-continuous testing.

SOURCES

DISTRACTOR BREAKDOWN

****A. On-Demand Instances **** Incorrect. Flexible but expensive — you pay per hour/second with no discount. Better for unpredictable workloads, not cost-efficient for testing.

****B. Dedicated Instances **** Incorrect. These run on hardware dedicated to a single customer — useful for compliance, but very costly and unnecessary for testing.

****D. Reserved Instances **** Incorrect. Reserved Instances require long-term commitment (1–3 years) — not suitable for short-term, interruptible testing.

REAL-WORLD TIE-IN

If you're testing a new image-processing pipeline in Nairobi, Spot Instances let you: Run large-scale tests cheaply Accept interruptions without data loss Scale up/down based on availability This is the backbone of cost-efficient experimentation in the cloud.

REFERENCES

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-spot-instances.html>

<https://docs.aws.amazon.com/wellarchitected/latest/cost-optimization-pillar/select-the-best-pricing-model.html>

Question 58

**Which AWS service gives users the ability to discover and protect sensitive data that is stored in Amazon S3 buckets?

**

OPTIONS

A. Amazon Macie

B. Amazon Detective

C. Amazon GuardDuty

D. AWS IAM Access Analyzer

[!TIP]

Correct Answer:

A. Amazon Macie

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of data security and classification in AWS — specifically how to detect and protect sensitive data in S3. The key phrases are: Discover sensitive data Protect S3 buckets This is the domain of Amazon Macie, AWS's data privacy and security service.

WHY A IS CORRECT

Amazon Macie uses machine learning and pattern matching to: Discover sensitive data (e.g., PII, credentials, financial info) Classify and label S3 objects Alert on public access or misconfigured buckets It integrates with Security Hub and EventBridge for automated remediation. Macie is purpose-built for data loss prevention (DLP) in S3.

SOURCES

DISTRACTOR BREAKDOWN

****B. Amazon Detective **** Incorrect. Detective helps investigate security incidents using logs and relationships — it doesn't scan S3 for sensitive data.

****C. Amazon GuardDuty **** Incorrect. Guard Duty detects threats like compromised IAM roles or EC2 instances — not data classification.

****D. AWS IAM Access Analyzer **** Incorrect. Access Analyzer identifies resource policies that allow external access, but doesn't scan or classify S3 content.

REAL-WORLD TIE-IN

If you store customer data in S3, Macie can: Flag files with unencrypted PII Alert if a bucket becomes public Help you meet GDPR, HIPAA, or PCI compliance It's essential for data governance and breach prevention.

REFERENCES

<https://docs.aws.amazon.com/macie/latest/userguide/what-is-macie.html>

<https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/security-services.html>

Question 59

**Which of the following services can be used to block network traffic to an instance? (Choose two.)

**

OPTIONS

A. Security groups

B. Amazon Virtual Private Cloud (Amazon VPC) flow logs

C. Network ACLs

D. Amazon CloudWatch

E. AWS CloudTrail

[!TIP]

Correct Answer:

A. Security groups and C. Network ACLs

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of network-level access control in AWS — specifically which services can actively block traffic, versus those that observe or log it. The key phrase is “block network traffic.” 🔍 Why A and C Are Correct

A.

Security groups ☒ Correct. Security groups are stateful firewalls attached to EC2 instances. You can: Allow or deny inbound/outbound traffic by protocol, port, and source Block traffic by not allowing it — anything not explicitly allowed is denied

C.

Network ACLs (NACLs) ☒ Correct. NACLs are stateless firewalls at the subnet level. You can: Explicitly deny traffic by rule number Apply rules to all resources in a subnet Use them for coarse-grained blocking (e.g., IP ranges) Together, SGs and NACLs form the dual perimeter defense in AWS VPCs.

SOURCES

DISTRACTOR BREAKDOWN

****B. VPC flow logs **** Incorrect. Flow logs record traffic metadata (source, destination, protocol, action) — but don’t block anything.

****D. Amazon CloudWatch **** Incorrect. Cloud Watch monitors metrics and logs — it’s an observability tool, not a firewall.

****E. AWS CloudTrail **** Incorrect. Cloud Trail logs API calls — useful for auditing, but not for blocking traffic.

REAL-WORLD TIE-IN

If you're running a public-facing EC2 instance and want to block traffic from a suspicious IP range: Use a NACL deny rule at the subnet level Tighten SG rules to allow only trusted ports and sources This is the backbone of defense-in-depth in AWS networking.

REFERENCES

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Security.html

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

Question 60

**Which AWS service can identify when an Amazon EC2 instance was terminated?

**

OPTIONS

A. AWS Identity and Access Management (IAM)

B. AWS CloudTrail

C. AWS Compute Optimizer

D. Amazon EventBridge

[!TIP]

Correct Answer:

B. AWS CloudTrail

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of event-level auditing in AWS — specifically how to track resource lifecycle actions like EC2 termination. The key phrase is “identify when... terminated” — which implies historical API visibility.

WHY B IS CORRECT

AWS CloudTrail records API calls made in your account, including: `TerminateInstances` `StopInstances` `RunInstances` Each event includes: Timestamp Caller identity Source IP Request parameters This makes CloudTrail the canonical source of truth for EC2 lifecycle events.

SOURCES

DISTRACTOR BREAKDOWN

****A. IAM **** Incorrect. IAM defines who can perform actions, but doesn't log when those actions occur.

****C. AWS Compute Optimizer **** Incorrect. Compute Optimizer analyzes resource usage for recommendations — it doesn't track instance termination events.

****D. Amazon EventBridge **** Incorrect. Event Bridge can react to events, but it doesn't retain historical logs. It's an event router, not an audit trail.

REAL-WORLD TIE-IN

If an EC2 instance disappears unexpectedly, CloudTrail lets you: Confirm who terminated it Review the exact time and method Trigger alerts or remediation via EventBridge This is essential for security forensics and operational accountability.

REFERENCES

https://docs.aws.amazon.com/AWSEC2/latest/APIReference/API_TerminateInstances.html

<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-event-reference.html>

Question 62

**Which AWS service supports a hybrid architecture that gives users the ability to extend AWS infrastructure, AWS services, APIs, and tools to data centers, co-location environments, or on-premises facilities?

**

OPTIONS

A. AWS Snowmobile

B. AWS Local Zones

C. AWS Outposts

D. AWS Fargate

[!TIP]

Correct Answer:

C. AWS Outposts

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of hybrid cloud architecture — where workloads span both on-premises and AWS environments. The key phrase is “extend AWS infrastructure... to on-premises.”

WHY C IS CORRECT

AWS Outposts delivers AWS infrastructure and services directly to your data center or co-location facility. You get: Native AWS APIs and tools Managed hardware installed on-prem Seamless integration with AWS Regions Ideal for: Low-latency workloads Data residency requirements Hybrid modernization Outposts is the flagship hybrid solution for consistent cloud/on-prem experience.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS Snowmobile **** Incorrect. Snowmobile is for mass data transfer — not hybrid infrastructure. It’s a truck-sized storage device.

****B. AWS Local Zones **** Incorrect. Local Zones bring compute/storage closer to users — but they’re still AWS-managed infrastructure, not on-prem extensions.

****D. AWS Fargate **** Incorrect. Fargate is a serverless container engine — it runs in AWS, not on-prem, and doesn’t support hybrid deployment.

REAL-WORLD TIE-IN

If you’re running latency-sensitive workloads in Nairobi but need AWS APIs locally, Outposts lets you: Deploy EC2, EBS, RDS on-prem Use CloudFormation and IAM as usual Sync with

AWS Region for hybrid orchestration It's the backbone of cloud-native infrastructure in non-cloud environments.

REFERENCES

<https://aws.amazon.com/outposts/>

<https://docs.aws.amazon.com/whitepapers/latest/hybrid-cloud-architecture/aws-hybrid-cloud-architecture.html>

Question 62

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**

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C. AWS Outposts

D. AWS Fargate

[!TIP]

Correct Answer:

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EXPLANATION

CONCEPTUAL FRAMING

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REFERENCES

<https://aws.amazon.com/outposts/>

<https://docs.aws.amazon.com/whitepapers/latest/hybrid-cloud-architecture/aws-hybrid-cloud-architecture.html>

Question 63

**Which AWS service can run a managed PostgreSQL database that provides online transaction processing (OLTP)?

**

OPTIONS

A. Amazon DynamoDB

B. Amazon Athena

C. Amazon RDS

D. Amazon EMR

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match relational database engines with OLTP workloads in AWS. The key phrases are: Managed PostgreSQL Online Transaction Processing (OLTP)

WHY C IS CORRECT

Amazon RDS (Relational Database Service) supports: PostgreSQL, MySQL, Oracle, SQL Server, and MariaDB Fully managed provisioning, patching, backups, and scaling OLTP workloads — transactional apps like e-commerce, banking, and inventory systems You can deploy PostgreSQL with: Multi-AZ replication Read replicas Encryption and IAM integration RDS is the go-to for managed relational OLTP databases.

SOURCES

DISTRACTOR BREAKDOWN

****A. Amazon DynamoDB **** Incorrect. Dynamo DB is No SQL — not relational, and not PostgreSQL-compatible.

****B. Amazon Athena **** Incorrect. Athena is a serverless query engine for S3 — it doesn't run databases or support OLTP.

****D. Amazon EMR **** Incorrect. EMR is for big data processing — not transactional databases. It runs Hadoop, Spark, etc.

REAL-WORLD TIE-IN

If you're migrating a legacy PostgreSQL app to AWS, RDS lets you: Preserve schema and ACID properties Offload operational overhead Scale reads with replicas It's the backbone of cloud-native OLTP modernization.

REFERENCES

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_PostgreSQL.htm

<https://aws.amazon.com/rds/>

Question 64

**A company wants to provide managed Windows virtual desktops and applications to its remote employees over secure network connections. Which AWS services can the company use to meet these requirements? (Choose two.)

**

OPTIONS

A. Amazon Connect

B. Amazon AppStream 2.0

C. Amazon WorkSpaces

D. AWS Site-to-Site VPN

E. Amazon Elastic Container Service (Amazon ECS)

[!TIP]

Correct Answer:

B. Amazon AppStream 2.0 and C. Amazon WorkSpaces

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match remote workforce needs with desktop and application streaming services in AWS. The key goals are: Managed Windows environments Secure remote access Scalable delivery 🔍 Why B and C Are Correct

B.

Amazon AppStream 2.0 ✅ Correct. AppStream 2.0 streams individual applications (e.g., Excel, CAD, IDEs) to users via browser — no local install needed. It's ideal for: App-specific delivery Secure, scalable access BYOD environments

C.

Amazon WorkSpaces ✅ Correct. WorkSpaces provides full virtual desktops — Windows or Linux — with persistent storage and secure access. It's ideal for: Remote employees Managed desktop environments Integration with AD and MFA Together, they offer flexible remote access — full desktop or app-only.

SOURCES

DISTRACTOR BREAKDOWN

****A. Amazon Connect **** Incorrect. Connect is a contact center service — not for desktop or app delivery.

****D. AWS Site-to-Site VPN **** Incorrect. VPN provides network connectivity, not desktop or app streaming. It's a supporting layer, not a delivery service.

****E. Amazon ECS **** Incorrect. ECS runs containerized workloads — not suitable for desktop or GUI app delivery.

REAL-WORLD TIE-IN

If your Nairobi-based team needs secure access to Windows desktops and Excel apps: Use WorkSpaces for full desktop environments Use AppStream 2.0 for lightweight app delivery Both integrate with IAM, AD, and CloudWatch for governance This is the backbone of secure, scalable remote work infrastructure.

REFERENCES

<https://aws.amazon.com/workspaces/>

<https://aws.amazon.com/appstream2/>

Question 65

****A company wants to monitor for misconfigured security groups that are allowing unrestricted access to specific ports. Which AWS service will meet this requirement?**

OPTIONS

A. AWS Trusted Advisor

B. Amazon CloudWatch

C. Amazon GuardDuty

D. AWS Health Dashboard

[!TIP]

Correct Answer:

A. AWS Trusted Advisor

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to identify configuration-level security checks — specifically for security groups that expose ports like SSH (22) or RDP (3389) to the world. The key phrase is “monitor for misconfigured security groups.”

WHY A IS CORRECT

AWS Trusted Advisor provides real-time guidance on: Security best practices Cost optimization Fault tolerance Performance It includes a security check for: “Security Groups – Unrestricted Access” → Flags open ports with 0.0.0.0/0 access It’s ideal for configuration hygiene and audit readiness.

SOURCES

DISTRACTOR BREAKDOWN

****B. Amazon CloudWatch **** Incorrect. Cloud Watch monitors metrics and logs — it doesn’t inspect security group configurations.

****C. Amazon GuardDuty **** Incorrect. Guard Duty detects threats and anomalies — not misconfigurations. It reacts to behavior, not setup.

****D. AWS Health Dashboard **** Incorrect. Health Dashboard shows service status and account-level issues — not security group misconfigurations.

REAL-WORLD TIE-IN

If your EC2 instance has port 22 open to the world, Trusted Advisor will: Flag it as a security risk
Recommend tightening the rule Help you align with CIS benchmarks It's the backbone of proactive security posture management.

REFERENCES

<https://docs.aws.amazon.com/awssupport/latest/user/trusted-advisor-security-checks.html>

<https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/security-best-practices.html>

Question 66

****Which AWS service is a key-value database that provides sub-millisecond latency on a large scale?**

OPTIONS

A. Amazon DynamoDB

B. Amazon Aurora

C. Amazon DocumentDB (with MongoDB compatibility)

D. Amazon Neptune

[!TIP]

Correct Answer:

A. Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match database type (key-value) with performance characteristics at scale. The key phrases are: Key-value database Sub-millisecond latency Large scale

WHY A IS CORRECT

Amazon DynamoDB is AWS's fully managed NoSQL key-value and document database. It offers: Single-digit millisecond or sub-millisecond latency Automatic scaling to millions of requests per second Global tables for multi-region replication On-demand or provisioned capacity modes Ideal for: Session stores User profiles IoT telemetry Gaming leaderboards

SOURCES

DISTRACTOR BREAKDOWN

****B. Amazon Aurora **** Incorrect. Aurora is a relational database — not key-value, and not optimized for sub-millisecond latency.

****C. Amazon DocumentDB **** Incorrect. Document DB is a document database (Mongo DB-compatible) — not key-value, and latency is higher.

****D. Amazon Neptune **** Incorrect. Neptune is a graph database — designed for relationship queries, not key-value access.

REAL-WORLD TIE-IN

If you're building a real-time leaderboard or session store for a mobile app in Nairobi, DynamoDB lets you: Serve requests with sub-millisecond latency Scale globally without manual provisioning Avoid schema rigidity It's the backbone of high-performance NoSQL architecture.

REFERENCES

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.html>

<https://aws.amazon.com/products/databases/>

Question 67

****Which AWS services or features provide disaster recovery solutions for Amazon EC2 instances? (Choose two.)**

OPTIONS

A. EC2 Reserved Instances

B. EC2 Amazon Machine Images (AMIs)

C. Amazon Elastic Block Store (Amazon EBS) snapshots

D. AWS Shield

[!TIP]

Correct Answer:

B. EC2 Amazon Machine Images (AMIs) and C. Amazon EBS snapshots

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of disaster recovery (DR) strategies for EC2 — specifically how to preserve and restore compute state. The key goals are: Recover EC2 instances Preserve configuration and data Enable fast redeployment 🔍 Why B and C Are Correct

B.

EC2 AMIs ✅ Correct. AMIs capture: OS configuration Installed packages Application state Attached volumes (if bundled) You can launch new EC2 instances from AMIs in any Region — ideal for rapid recovery.

C.

Amazon EBS snapshots ✅ Correct. Snapshots preserve: Block-level data from EBS volumes Can be used to restore volumes or create new AMIs Stored in S3 for durability Ideal for data recovery and backup rotation Together, AMIs and snapshots form the core of EC2 disaster recovery.

SOURCES

DISTRACTOR BREAKDOWN

****A. EC2 Reserved Instances **** Incorrect. Reserved Instances are a billing construct — they don't preserve or recover EC2 state.

****D. AWS Shield **** Incorrect. Shield protects against DDoS attacks, not instance recovery.

****E. Amazon GuardDuty **** Incorrect. Guard Duty detects threats and anomalies, not backup or recovery.

REAL-WORLD TIE-IN

If your EC2 instance in Nairobi fails due to corruption or deletion: Restore the EBS volume from a snapshot Launch a new instance from the AMI Resume operations with minimal downtime

This is the backbone of resilient EC2 architecture.

REFERENCES

<https://docs.aws.amazon.com/whitepapers/latest/disaster-recovery-workshop/disaster-recovery-workshop.html>

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>

Question 68

****Which AWS service provides command line access to AWS tools and resources directly from a web browser?**

OPTIONS

A. AWS CloudHSM

B. AWS CloudShell

C. Amazon WorkSpaces

D. AWS Cloud Map

[!TIP]

Correct Answer:

B. AWS CloudShell

EXPLANATION

CONCEPTUAL FRAMING

This question tests your familiarity with developer tooling in AWS, especially browser-based environments for CLI access. The key phrase is “command line access... from a web browser.”

WHY B IS CORRECT

AWS CloudShell is a browser-based shell environment that provides: Pre-authenticated CLI access to your AWS account Pre-installed tools (AWS CLI, Python, Node.js, etc.) Persistent home directory (1 GB) No local setup required Ideal for: Quick scripting CLI experimentation Lightweight automation It's the fastest way to get shell access without provisioning EC2 or installing anything.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS CloudHSM **** Incorrect. Cloud HSM provides hardware security modules — not CLI access.

****C. Amazon WorkSpaces **** Incorrect. Work Spaces delivers virtual desktops, not shell environments.

****D. AWS Cloud Map **** Incorrect. Cloud Map is a service discovery tool — not for CLI access.

REAL-WORLD TIE-IN

If you're troubleshooting IAM policies or testing CLI commands in Nairobi: Open CloudShell in your browser Run `aws iam list-users` or `aws s3 ls` No setup, no EC2, no credentials needed It's the backbone of frictionless cloud CLI access.

REFERENCES

<https://docs.aws.amazon.com/cloudshell/latest/userguide/what-is-cloudshell.html>

<https://aws.amazon.com/developer/tools/>

Question 69

****A network engineer needs to build a hybrid cloud architecture connecting on-premises networks to the AWS Cloud using AWS Direct Connect. The company has a few VPCs in a single AWS Region and expects to increase the number of VPCs to hundreds over time. Which AWS service or feature should the engineer use to simplify and scale this connectivity as the VPCs increase in number?**

OPTIONS

A. VPC endpoints

B. AWS Transit Gateway

C. Amazon Route 53

D. AWS Secrets Manager

[!TIP]

Correct Answer:

B. AWS Transit Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to architect scalable hybrid connectivity — especially when dealing with many VPCs and Direct Connect. The key phrases are: Hybrid cloud Direct Connect Hundreds of VPCs Simplify and scale connectivity

WHY B IS CORRECT

AWS Transit Gateway acts as a central hub for: VPC-to-VPC routing On-premises connectivity via Direct Connect or VPN Scalable, simplified network management Benefits: One attachment per VPC — no need for peering mesh Route tables for segmentation Integration with Direct Connect Gateway for hybrid routing It's the backbone of hub-and-spoke network architecture in AWS.

SOURCES

DISTRACTOR BREAKDOWN

****A. VPC endpoints **** Incorrect. Endpoints allow private access to AWS services, not VPC-to-VPC or hybrid routing.

****C. Amazon Route 53 **** Incorrect. Route 53 is a DNS service — not for network routing or connectivity.

****D. AWS Secrets Manager **** Incorrect. Secrets Manager stores credentials — unrelated to network architecture.

REAL-WORLD TIE-IN

If your Nairobi-based enterprise plans to scale from 5 to 500 VPCs: Use Transit Gateway to attach each VPC Connect on-prem via Direct Connect Gateway Manage routing centrally with route tables This is the backbone of scalable hybrid cloud networking.

REFERENCES

<https://docs.aws.amazon.com/vpc/latest/tgw/what-is-transit-gateway.html>

<https://docs.aws.amazon.com/directconnect/latest/UserGuide/direct-connect-gateways.html>

Question 70

****A company wants to establish a schedule for rotating database user credentials. Which AWS service will support this requirement with the LEAST amount of operational overhead?**

OPTIONS

A. AWS Systems Manager

B. AWS Secrets Manager

C. AWS License Manager

D. AWS Managed Services

[!TIP]

Correct Answer:

B. AWS Secrets Manager

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to identify automated secrets management in AWS — especially for database credentials. The key phrases are: Rotate credentials Least operational overhead

WHY B IS CORRECT

AWS Secrets Manager provides: Automatic rotation of secrets (e.g., database passwords, API keys) Native integration with RDS, Redshift, and DocumentDB Rotation via Lambda functions or built-in templates Secure storage with encryption and access control It's purpose-built for low-touch secrets lifecycle management.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS Systems Manager **** Incorrect. SSM can store parameters, but doesn't natively rotate database credentials.

****C. AWS License Manager **** Incorrect. License Manager tracks software licenses — unrelated to credentials.

****D. AWS Managed Services **** Incorrect. AMS is a third-party operations service — not a native credential rotation tool.

REAL-WORLD TIE-IN

If your Nairobi-based team runs RDS PostgreSQL, Secrets Manager lets you: Rotate credentials every 30 days Avoid manual updates in apps Enforce least-privilege access via IAM It's the backbone of secure, automated secrets hygiene. Which AWS service is used to provide encryption for Amazon EBS?

A.

AWS Certificate Manager B. AWS Systems Manager C. AWS KMS D. AWS Config

CONCEPTUAL FRAMING

This question tests your understanding of data-at-rest encryption in AWS — specifically how EBS volumes are encrypted using customer-managed or AWS-managed keys. The key phrase is “provide encryption for Amazon EBS.”

WHY C IS CORRECT

AWS Key Management Service (KMS) provides: Encryption keys for EBS, S3, RDS, Redshift, and more Automatic integration with EBS for volume encryption Customer-managed keys (CMKs) and AWS-managed keys Audit logging via CloudTrail EBS volumes can be encrypted at creation using KMS keys — no need for manual encryption logic.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS Certificate Manager **** Incorrect. ACM manages SSL/TLS certificates — not encryption keys for EBS.

****B. AWS Systems Manager **** Incorrect. SSM helps with automation and patching — not encryption.

****D. AWS Config **** Incorrect. Config tracks resource configuration changes — it doesn't encrypt anything.

REAL-WORLD TIE-IN

If your Nairobi-based team provisions EC2 instances with sensitive data: Use KMS to encrypt EBS volumes Control access via IAM and key policies Monitor usage via CloudTrail It's the backbone of secure, compliant block storage encryption.

REFERENCES

<https://docs.aws.amazon.com/secretsmanager/latest/userguide/rotating-secrets.html>

<https://aws.amazon.com/secrets-manager/>

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EBSEncryption.html>

<https://docs.aws.amazon.com/kms/latest/developerguide/overview.html>

Question 72

****A company wants to manage its AWS Cloud resources through a web interface. Which AWS service will meet this requirement?**

OPTIONS

A. AWS Management Console

B. AWS CLI

C. AWS SDK

D. AWS Cloud9

[!TIP]

Correct Answer:

A. AWS Management Console

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to match user interface modality with resource management needs. The key phrase is “manage... through a web interface.”

WHY A IS CORRECT

AWS Management Console is the primary web-based UI for: Launching and managing AWS services Viewing dashboards, metrics, and billing Configuring IAM, EC2, S3, RDS, and more It's ideal for: Visual workflows Non-CLI users Quick access to service settings It's the backbone of browser-native AWS orchestration.

SOURCES

DISTRACTOR BREAKDOWN

****B. AWS CLI **** Incorrect. CLI is command-line based — not a web interface.

****C. AWS SDK **** Incorrect. SDKs are for programmatic access via code — not a UI.

****D. AWS Cloud9 **** Incorrect. Cloud9 is a browser-based IDE — useful for coding, not for managing AWS resources broadly.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to: Launch EC2 instances Configure IAM roles Monitor billing and usage The Management Console provides a centralized, visual interface — no terminal or code required.

REFERENCES

<https://aws.amazon.com/console/>

<https://docs.aws.amazon.com/gettingstarted/latest/console/getting-started-console.html>

Question 73

****Which of the following are advantages of the AWS Cloud? (Choose two.)**

OPTIONS

A. Trade variable expenses for capital expenses

B. High economies of scale

C. Launch globally in minutes

D. Focus on managing hardware infrastructure

E. Overprovision to ensure capacity

[!TIP]

Correct Answer:

B. High economies of scale and C. Launch globally in minutes

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of cloud value propositions — especially around cost efficiency and deployment agility. The key themes are: Economics of scale Global reach Operational abstraction 🔍 Why B and C Are Correct

B.

High economies of scale ✅ Correct. AWS aggregates massive infrastructure demand across millions of customers, enabling: Lower per-unit costs Efficient resource utilization Competitive pricing

C.

Launch globally in minutes ✅ Correct. With AWS Regions and Availability Zones, you can: Deploy workloads worldwide Reduce latency Meet data residency requirements Do it all via console, CLI, or SDK — no shipping hardware

SOURCES

DISTRACTOR BREAKDOWN

****A. Trade variable expenses for capital expenses **** Incorrect. AWS lets you trade capital expenses for variable expenses — not the other way around.

****D. Focus on managing hardware infrastructure **** Incorrect. AWS abstracts hardware management — you focus on code, not cables.

****E. Overprovision to ensure capacity **** Incorrect. AWS encourages right-sizing and auto-scaling — overprovisioning is wasteful and discouraged.

REAL-WORLD TIE-IN

If your Nairobi-based startup wants to launch a global app: Use AWS to deploy in multiple Regions instantly Pay only for what you use Avoid hardware procurement delays It's the backbone of agile, cost-efficient cloud innovation.

REFERENCES

<https://aws.amazon.com/benefits/>

<https://docs.aws.amazon.com/wellarchitected/latest/framework/wellarchitected-framework.html>

Question 74

****Which AWS Cloud benefit is shown by an architecture's ability to withstand failures with minimal downtime?**

OPTIONS

A. Agility

B. Elasticity

C. Scalability

D. High availability

[!TIP]

Correct Answer:

D. High availability

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of resilience engineering in cloud architecture — specifically how systems maintain uptime despite failures. The key phrase is “withstand failures with minimal downtime.”

WHY D IS CORRECT

High availability (HA) refers to: Architectures designed to minimize downtime Use of multi-AZ deployments, load balancing, and failover mechanisms Continuous operation even during hardware, software, or network failures AWS supports HA through: Elastic Load Balancing Multi-AZ RDS Auto Scaling groups Route 53 health checks It’s the backbone of resilient, fault-tolerant cloud systems.

SOURCES

DISTRACTOR BREAKDOWN

****A. Agility **** Incorrect. Agility refers to speed of innovation and iteration, not fault tolerance.

****B. Elasticity **** Incorrect. Elasticity is about scaling resources up/down based on demand — not availability.

****C. Scalability **** Incorrect. Scalability ensures performance under load — not resilience during failure.

REAL-WORLD TIE-IN

If your Nairobi-based app must stay online during an AZ outage: Deploy EC2 instances across multiple AZs Use ELB to route traffic Enable RDS Multi-AZ for database failover That's how AWS delivers high availability by design.

REFERENCES

<https://docs.aws.amazon.com/wellarchitected/latest/reliability-pillar/reliability-pillar.html>

<https://aws.amazon.com/architecture/high-availability/>

Question 75

****A developer needs to maintain a development environment infrastructure and a production environment infrastructure in a repeatable fashion. Which AWS service should the developer use to meet these requirements?**

OPTIONS

A. AWS Ground Station

B. AWS Shield

C. AWS IoT Device Defender

D. AWS CloudFormation

[!TIP]

Correct Answer:

D. AWS CloudFormation

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of infrastructure automation — specifically how to define and replicate environments using code. The key phrases are: Development and production environments Repeatable fashion

WHY D IS CORRECT

AWS CloudFormation enables: Infrastructure as Code (IaC) using JSON or YAML templates Automated provisioning of resources like EC2, RDS, IAM, VPC, etc. Version-controlled, repeatable deployments Stack updates, rollbacks, and drift detection It's the backbone of environment parity and deployment hygiene.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS Ground Station **** Incorrect. Ground Station is for satellite data ingestion — unrelated to infrastructure management.

****B. AWS Shield **** Incorrect. Shield provides DDoS protection, not environment provisioning.

****C. AWS IoT Device Defender **** Incorrect. Device Defender secures IoT fleets — not general infrastructure.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to: Spin up dev and prod stacks with identical VPCs, EC2s, and IAM roles Track changes via Git Automate deployments via CI/CD CloudFormation delivers repeatable, auditable infrastructure workflows.

REFERENCES

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/Welcome.html>

<https://docs.aws.amazon.com/wellarchitected/latest/operational-excellence-pillar/infrastructure-as-code.html>

Question 76

****Which task is the customer's responsibility, according to the AWS shared responsibility model?**

OPTIONS

A. Maintain the security of the AWS Cloud

B. Configure firewalls and networks

C. Patch the operating system of Amazon RDS instances

D. Implement physical and environmental controls

[!TIP]

Correct Answer:

B. Configure firewalls and networks

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Shared Responsibility Model — which divides security and compliance duties between AWS and the customer. The key is knowing where AWS stops and the customer begins.

WHY B IS CORRECT

Customers are responsible for: Security “in” the cloud — including firewall rules, VPC configurations, IAM policies, and OS-level controls Managing network access, security group rules, and NACLs AWS handles: Security “of” the cloud — physical infrastructure, hypervisors, and global network backbone

SOURCES

DISTRACTOR BREAKDOWN

****A. Maintain the security of the AWS Cloud **** Incorrect. AWS handles the cloud's physical and foundational security.

****C. Patch the operating system of Amazon RDS instances **** Incorrect. RDS is a managed service — AWS patches the OS.

****D. Implement physical and environmental controls **** Incorrect. AWS owns and secures the data centers — customers never touch the hardware.

REAL-WORLD TIE-IN

If your Nairobi-based team deploys EC2 instances: You must configure security groups and NACLs You must manage IAM roles and access policies AWS ensures the underlying infrastructure is secure It's the backbone of cloud-native security accountability.

REFERENCES

<https://aws.amazon.com/compliance/shared-responsibility-model/>

<https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/security-best-practices.html>

Question 77

****Which AWS service helps deliver highly available applications with fast failover for multi-Region and Multi-AZ architectures?**

OPTIONS

A. AWS WAF

B. AWS Global Accelerator

C. AWS Shield

D. AWS Direct Connect

[!TIP]

Correct Answer:

B. AWS Global Accelerator

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of global availability and routing resilience — especially for applications spanning multiple Regions and Availability Zones. The key phrases are: Highly available applications Fast failover Multi-Region and Multi-AZ

WHY B IS CORRECT

AWS Global Accelerator provides: Static IPs that front your application globally Intelligent routing to the nearest healthy endpoint Automatic failover across Regions and AZs Performance boost via AWS global network backbone Ideal for: Latency-sensitive apps Disaster recovery Multi-Region deployments It's the backbone of global resilience and performance routing.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS WAF **** Incorrect. WAF protects against web exploits — it doesn't route or failover.

****C. AWS Shield **** Incorrect. Shield defends against DDoS — not routing or availability.

****D. AWS Direct Connect **** Incorrect. Direct Connect provides private network links — not global failover or routing.

REAL-WORLD TIE-IN

If your Nairobi-based app serves users in Frankfurt and Singapore: Global Accelerator routes traffic to the nearest healthy endpoint. If Frankfurt fails, traffic shifts to Singapore instantly. No DNS propagation delays. That's how AWS delivers fast, fault-tolerant global access.

REFERENCES

<https://docs.aws.amazon.com/global-accelerator/latest/dg/what-is-global-accelerator.html>

<https://aws.amazon.com/architecture/high-availability/>

Question 78

****A company has a set of ecommerce applications. The applications need to be able to send messages to each other. Which AWS service meets this requirement?**

OPTIONS

A. AWS Auto Scaling

B. Elastic Load Balancing

C. Amazon Simple Queue Service (Amazon SQS)

D. Amazon Kinesis Data Streams

[!TIP]

Correct Answer:

C. Amazon Simple Queue Service (Amazon SQS)

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of application integration patterns — especially how to decouple services using message queues. The key phrase is “send messages to each other.”

WHY C IS CORRECT

Amazon SQS is a fully managed message queuing service that enables: Asynchronous communication between distributed components Decoupling of microservices, ecommerce modules, and backend systems Reliable delivery with retry logic and dead-letter queues Ideal for: Order processing Inventory updates Payment notifications It's the backbone of resilient, loosely coupled ecommerce architecture.

SOURCES

DISTRACTOR BREAKDOWN

****A. AWS Auto Scaling **** Incorrect. Auto Scaling adjusts compute resources — it doesn't handle messaging.

****B. Elastic Load Balancing **** Incorrect. ELB distributes traffic — not messages between applications.

****D. Amazon Kinesis Data Streams **** Incorrect. Kinesis is for real-time streaming data — not general-purpose messaging.

REAL-WORLD TIE-IN

If your Nairobi-based ecommerce platform has separate services for checkout, inventory, and shipping: Use SQS to queue order events Each service processes messages independently Scale consumers based on demand That's how AWS enables modular, scalable ecommerce workflows.

REFERENCES

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/welcome.html>

<https://aws.amazon.com/messaging/>

Question 79

****What are the benefits of consolidated billing for AWS Cloud services? (Choose two.)**

OPTIONS

A. Volume discounts

B. A minimal additional fee for use

C. One bill for multiple accounts

D. Installment payment options

E. Custom cost and usage budget creation

[!TIP]

Correct Answer:

A. Volume discounts and C. One bill for multiple accounts

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS Organizations and consolidated billing — especially how it helps centralize payments and optimize costs across multiple accounts. The key themes are: Cost efficiency Billing simplicity Multi-account management 🔍 Why A and C Are Correct

A.

Volume discounts ☒ Correct. AWS applies tiered pricing across all accounts in an organization — enabling: Shared usage aggregation Lower per-unit costs for services like S3, EC2, and data transfer

C.

One bill for multiple accounts ☒ Correct. Consolidated billing lets you: Receive a single invoice Track charges per account Simplify accounting and budgeting

SOURCES

DISTRACTOR BREAKDOWN

****B.** A minimal additional fee for use ****** Incorrect. Consolidated billing is free — no extra fee.

****D.** Installment payment options ****** Incorrect. AWS does not offer installment plans — billing is monthly.

****E.** Custom cost and usage budget creation ****** Incorrect. Budgeting is handled via AWS Budgets, not consolidated billing itself.

REAL-WORLD TIE-IN

If your Nairobi-based company has 12 AWS accounts: Use consolidated billing to unify invoices
Share volume discounts across accounts
Track usage with Cost Explorer and Budgets
It's the backbone of cost-aware multi-account cloud governance.

REFERENCES

<https://docs.aws.amazon.com/organizations/latest/userguide/orgs-consolidated-billing.html>

<https://aws.amazon.com/pricing/>

Question 80

****A** user wants to review all Amazon S3 buckets with ACLs and S3 bucket policies in the S3 console. Which AWS service or resource will meet this requirement?

OPTIONS

A. S3 Multi-Region Access Points

B. S3 Storage Lens

C. AWS IAM Identity Center (AWS Single Sign-On)

D. Access Analyzer for S3

[!TIP]

Correct Answer:

D. Access Analyzer for S3

EXPLANATION

CONCEPTUAL FRAMING

This question tests your ability to identify visibility tools for S3 access control — especially around ACLs and bucket policies. The key phrase is “review... in the S3 console.”

WHY D IS CORRECT

Access Analyzer for S3 provides: A dashboard in the S3 console showing buckets with public access Visibility into ACLs and bucket policies Alerts for unintended exposure Integration with IAM Access Analyzer for deeper insights It’s the backbone of S3 access visibility and policy hygiene.

SOURCES

DISTRACTOR BREAKDOWN

****A. S3 Multi-Region Access Points **** Incorrect. These simplify cross-region access — not policy review.

****B. S3 Storage Lens **** Incorrect. Storage Lens provides usage and activity metrics, not access policy visibility.

****C. AWS IAM Identity Center **** Incorrect. Identity Center manages user access across accounts — not bucket-level AC Ls.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to audit S3 buckets for public exposure: Use Access Analyzer for S3 in the console Identify buckets with 0.0.0.0/0 or AllUsers access Remediate with tighter policies or bucket-level blocks That's how AWS delivers policy-aware S3 governance.

REFERENCES

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/access-analyzer.html>

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/security-best-practices.html>

Question 81

****What is the best resource for a user to find compliance-related information and reports about AWS?**

OPTIONS

A. AWS Artifact

B. AWS Marketplace

C. Amazon Inspector

D. AWS Support

[!TIP]

Correct Answer:

A. AWS Artifact

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of compliance documentation access — especially how AWS provides audit-ready reports for certifications, attestations, and regulatory frameworks. The key phrase is “compliance-related information and reports.”

WHY A IS CORRECT

AWS Artifact is the central portal for: Accessing compliance reports (SOC, ISO, PCI, HIPAA, etc.) Viewing agreements like Business Associate Addendums (BAAs) Downloading audit artifacts for internal or external review It’s designed for: Security teams Auditors Risk and compliance officers It’s the backbone of compliance transparency and documentation access.

SOURCES

DISTRACTOR BREAKDOWN

****B. AWS Marketplace **** Incorrect. Marketplace is for buying third-party software — not compliance reports.

****C. Amazon Inspector **** Incorrect. Inspector scans for vulnerabilities — it doesn't provide compliance documentation.

****D. AWS Support **** Incorrect. Support helps with troubleshooting — not direct access to compliance reports.

REAL-WORLD TIE-IN

If your Nairobi-based fintech startup needs PCI DSS documentation for auditors: Log into AWS Artifact Download the latest PCI report Share with your compliance team That's how AWS delivers audit-ready compliance access on demand.

REFERENCES

<https://aws.amazon.com/artifact/>

<https://aws.amazon.com/compliance/>

Question 82

****Which AWS service enables companies to deploy an application close to end users?**

OPTIONS

A. Amazon CloudFront

B. AWS Auto Scaling

C. AWS AppSync

D. Amazon Route 53

[!TIP]

Correct Answer:

A. Amazon CloudFront

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of content delivery architecture — especially how AWS reduces latency by serving content from edge locations. The key phrase is “deploy... close to end users.”

WHY A IS CORRECT

Amazon CloudFront is AWS’s Content Delivery Network (CDN) that: Caches content at global edge locations Delivers static and dynamic content with low latency Integrates with S3, EC2, Lambda@Edge, and custom origins Supports HTTPS, geo restrictions, and signed URLs It’s the backbone of proximity-aware application performance.

SOURCES

DISTRACTOR BREAKDOWN

****B. AWS Auto Scaling **** Incorrect. Auto Scaling adjusts compute resources — it doesn’t deploy content geographically.

****C. AWS AppSync **** Incorrect. App Sync manages Graph QL AP Is — not edge delivery.

****D. Amazon Route 53 **** Incorrect. Route 53 handles DNS routing — not content caching or proximity optimization.

REAL-WORLD TIE-IN

If your Nairobi-based app serves users in Frankfurt, Mumbai, and São Paulo: CloudFront caches content at edge locations near each user Reduces latency and origin load Improves global UX without provisioning infrastructure in every Region That's how AWS delivers fast, scalable content delivery across continents.

REFERENCES

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>

<https://aws.amazon.com/edge/>

Question 83

****Which AWS service or feature improves network performance by sending traffic through the AWS worldwide network infrastructure?**

OPTIONS

A. Route table

B. AWS Transit Gateway

C. AWS Global Accelerator

[!TIP]

Correct Answer:

C. AWS Global Accelerator

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of network performance optimization — especially how AWS routes traffic across its global backbone. The key phrase is “sending traffic through the AWS worldwide network infrastructure.”

WHY C IS CORRECT

AWS Global Accelerator provides: Static IPs for global entry points Intelligent routing to the nearest healthy endpoint Performance boost via AWS’s private backbone (not public internet) Automatic failover across Regions and Availability Zones It’s the backbone of latency-aware, fault-tolerant global routing.

SOURCES

DISTRACTOR BREAKDOWN

****A. Route table **** Incorrect. Route tables define VPC-level routing — they don’t optimize global traffic.

****B. AWS Transit Gateway **** Incorrect. Transit Gateway connects VP Cs and on-prem networks — not global acceleration.

****D. Amazon VPC **** Incorrect. VPC is a network container — it doesn't route traffic across AWS's backbone.

REAL-WORLD TIE-IN

If your Nairobi-based app serves users in Tokyo and São Paulo: Global Accelerator routes traffic through AWS's backbone Reduces latency and jitter Ensures fast failover if a Region goes down That's how AWS delivers high-performance global access with resilience baked in.

REFERENCES

<https://docs.aws.amazon.com/global-accelerator/latest/dg/what-is-global-accelerator.html>

<https://aws.amazon.com/blogs/networking-and-content-delivery/>

Question 84

****Which AWS service provides highly durable object storage?**

OPTIONS

A. Amazon S3

B. Amazon Elastic File System (Amazon EFS)

C. Amazon Elastic Block Store (Amazon EBS)

[!TIP]

Correct Answer:

A. Amazon S3

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of storage durability guarantees — especially for object storage designed to survive hardware failures and regional disruptions. The key phrase is “highly durable object storage.”

WHY A IS CORRECT

Amazon S3 (Simple Storage Service) offers: 99.999999999% (11 nines) durability Multi-AZ replication within a Region Optional cross-Region replication Versioning, lifecycle policies, and encryption Ideal for: Backups Media assets Data lakes Static website hosting It’s the backbone of resilient, scalable object storage.

SOURCES

DISTRACTOR BREAKDOWN

****B. Amazon EFS **** Incorrect. EFS is a file system — not object storage, and durability is lower than S3.

****C. Amazon EBS **** Incorrect. EBS is block storage for EC2 — not designed for object-level durability.

****D. Amazon FSx **** Incorrect. FSx provides managed file systems (e.g., Windows File Server, Lustre) — not object storage.

REAL-WORLD TIE-IN

If your Nairobi-based team needs to store product images, logs, and backups: Use S3 for object-level access Enable versioning and cross-Region replication Rely on 11 nines durability for long-term retention That's how AWS delivers durable, scalable storage with zero infrastructure overhead.

REFERENCES

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>

<https://aws.amazon.com/products/storage/>

Question 85

****Which responsibility belongs to AWS when a company hosts its databases on Amazon EC2 instances?**

OPTIONS

A. Database backups

B. Database software patches

C. Operating system patches

D. Operating system installations

[!TIP]

Correct Answer:

D. Operating system installations

EXPLANATION

🧠 **Clarification and Framing** You're absolutely right to challenge the previous answer — and here's the refined breakdown: When a company launches an EC2 instance, AWS provides the base operating system image (e.g., Amazon Linux, Ubuntu, Windows Server) as part of the AMI (Amazon Machine Image). That initial installation is AWS's responsibility — it's baked into the provisioning process. Once the instance is running: AWS installs the OS via the AMI you select. You (the customer) are responsible for patching, configuring, and maintaining that OS and any software (like databases) you install on top of it.

WHY D IS CORRECT

Operating system installations are part of the EC2 provisioning process handled by AWS. You choose the AMI, and AWS spins up the instance with the OS pre-installed. This is distinct from patching, which is your responsibility unless using managed services like RDS.

SOURCES

REAL-WORLD TIE-IN

If your Nairobi-based team launches EC2 instances with Amazon Linux: AWS installs the OS from the AMI. You configure the OS, install PostgreSQL, and manage backups. AWS handles the

infrastructure and initial OS provisioning That's the subtle but critical distinction between installation and maintenance.

REFERENCES

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Question 6

**Which of the following are advantages of moving to the AWS Cloud? (Choose two.)

**

OPTIONS

A. The ability to turn over the responsibility for all security to AWS

B. The ability to use the pay-as-you-go model

C. The ability to have full control over the physical infrastructure

D. No longer having to guess what capacity will be required

E. No longer worrying about users access controls

[!TIP]

Correct Answer:

B. The ability to use the pay-as-you-go model and D. No longer having to guess what capacity will be required

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of cloud cost models and scalability benefits — especially how AWS reduces upfront investment and enables dynamic resource provisioning. 🔍 Why B and D

Are Correct

B.

Pay-as-you-go model ☒ Correct. AWS charges only for what you use — no upfront hardware costs, no long-term commitments. This: Improves cash flow Enables experimentation Aligns cost with actual usage

D.

No longer guessing capacity ☒ Correct. AWS supports: Auto Scaling and Elastic Load Balancing On-demand provisioning Avoidance of overprovisioning or underutilization

SOURCES

AWS Cloud Benefits AWS Pricing Models

DISTRACTOR BREAKDOWN

****A. Turn over all security to AWS **** Incorrect. AWS handles security of the cloud, but customers must manage security in the cloud — including IAM, encryption, and access controls.

****C. Full control over physical infrastructure **** Incorrect. AWS abstracts physical infrastructure — customers control virtual resources, not hardware.

****E. No longer worrying about user access controls **** Incorrect. IAM and access management are customer responsibilities under the shared responsibility model.

REAL-WORLD TIE-IN

If your Nairobi-based startup launches a new app: You avoid upfront server purchases You scale EC2 and RDS based on traffic You pay only for what you use That's how AWS delivers cost-efficient, scalable innovation.

Question 87

**Which AWS service is a hybrid cloud storage service that provides on-premises users access to virtually unlimited cloud storage?

**

OPTIONS

A. AWS DataSync

B. Amazon S3 Glacier

C. AWS Storage Gateway

D. Amazon Elastic Block Store (Amazon EBS)

[!TIP]

Correct Answer:

C. AWS Storage Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of hybrid cloud architecture — especially how AWS bridges on-premises environments with cloud-based storage. The key phrase is “on-premises users access to virtually unlimited cloud storage.”

WHY C IS CORRECT

AWS Storage Gateway enables: Seamless integration between on-premises applications and Amazon S3, Glacier, and EBS Local caching for low-latency access File, volume, and tape gateway modes Virtually unlimited cloud storage via S3 backend It's the backbone of hybrid cloud storage extension and archival workflows.

SOURCES

AWS Storage Gateway Overview Hybrid Cloud Architectures

DISTRACTOR BREAKDOWN

****A. AWS DataSync **** Incorrect. Data Sync moves data between on-prem and AWS — it doesn't provide continuous access.

****B. Amazon S3 Glacier **** Incorrect. Glacier is for archival storage — not hybrid access.

****D. Amazon EBS **** Incorrect. EBS is block storage for EC2 — not hybrid or on-prem accessible.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to: Archive backups from on-prem servers Access files locally while storing them in S3 Avoid managing physical tape libraries Storage Gateway delivers hybrid access with cloud-scale durability.

Question 88

******A company plans to migrate to AWS and wants to create cost estimates for its AWS use cases. Which AWS service or tool can the company use to meet these requirements?

OPTIONS

A. AWS Pricing Calculator

B. Amazon CloudWatch

C. AWS Cost Explorer

D. AWS Budgets

[!TIP]

Correct Answer:

A. AWS Pricing Calculator

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of pre-migration cost planning — especially how AWS helps forecast expenses before deploying workloads. The key phrase is “create cost estimates.”

WHY A IS CORRECT

AWS Pricing Calculator enables: Pre-deployment cost modeling for EC2, S3, RDS, Lambda, and more Custom configurations (e.g., instance types, storage tiers, usage hours) Exportable estimates for budgeting and stakeholder review Ideal for: Migration planning Business case

development Comparing architectures It's the backbone of forecast-driven cloud financial planning.

SOURCES

AWS Pricing Calculator AWS Cost Management Tools

DISTRACTOR BREAKDOWN

****B. Amazon CloudWatch **** Incorrect. Cloud Watch monitors performance and usage — it doesn't estimate costs.

****C. AWS Cost Explorer **** Incorrect. Cost Explorer analyzes past usage and spend — not future estimates.

****D. AWS Budgets **** Incorrect. Budgets track actual spend against thresholds — not initial cost modeling.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to migrate a monolith to microservices: Use Pricing Calculator to model EC2, Lambda, and S3 costs Share estimates with finance and leadership Refine architecture based on budget constraints That's how AWS enables cost-aware migration planning with precision.

Question 9

****Which tool should a developer use to integrate AWS service features directly into an application?**

OPTIONS

A. AWS Software Development Kit

B. AWS CodeDeploy

C. AWS Lambda

D. AWS Batch

[!TIP]

Correct Answer:

A. AWS Software Development Kit (SDK)

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of application-level AWS integration — especially how developers embed AWS functionality into their codebase. The key phrase is “integrate AWS service features directly into an application.”

WHY A IS CORRECT

AWS SDKs provide: Language-specific libraries (e.g., Python, JavaScript, Java, Go, .NET) Direct access to AWS services like S3, DynamoDB, EC2, and IAM Built-in authentication, retries, and error handling Support for both synchronous and asynchronous calls It's the backbone of programmatic AWS service access from within your application logic.

SOURCES

DISTRACTOR BREAKDOWN

****B. AWS CodeDeploy **** Incorrect. Code Deploy automates deployment — it doesn't embed AWS features into your app.

****C. AWS Lambda **** Incorrect. Lambda runs code in response to events — it's a compute service, not an integration tool.

****D. AWS Batch **** Incorrect. Batch runs large-scale jobs — not for embedding AWS features in app code.

REAL-WORLD TIE-IN

If your Nairobi-based app needs to: Upload files to S3 Query DynamoDB Assume IAM roles for secure access Use the AWS SDK to integrate those features directly into your application logic.

Question 92

****A company wants to operate a data warehouse to analyze data without managing the data warehouse infrastructure. Which AWS service will meet this requirement?**

OPTIONS

A. Amazon Aurora

B. Amazon Redshift Serverless

C. AWS Lambda

D. Amazon RDS

[!TIP]

Correct Answer:

B. Amazon Redshift Serverless

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of serverless analytics architecture — especially how AWS enables data warehousing without infrastructure management. The key phrase is “analyze data without managing the data warehouse infrastructure.”

WHY B IS CORRECT

Amazon Redshift Serverless provides: Fully managed data warehouse capabilities Automatic scaling and on-demand compute No need to provision or manage clusters Seamless integration with S3, Glue, and BI tools It’s the backbone of serverless, scalable analytics workflows.

SOURCES

Amazon Redshift Serverless Overview AWS Analytics Services

DISTRACTOR BREAKDOWN

****A. Amazon Aurora **** Incorrect. Aurora is a relational database, not a data warehouse.

****C. AWS Lambda **** Incorrect. Lambda runs code — it's not designed for analytical querying or warehousing.

****D. Amazon RDS **** Incorrect. RDS is for transactional databases, not optimized for analytical workloads.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to: Run SQL queries on large datasets Avoid provisioning clusters Pay only for query time Redshift Serverless delivers analytics at scale with zero infrastructure overhead.

REFERENCES

<https://aws.amazon.com/redshift/>

Question 93

****How does AWS Cloud computing help businesses reduce costs? (Choose two.)**

OPTIONS

A. AWS charges the same prices for services in every AWS Region

B. AWS enables capacity to be adjusted on demand

C. AWS offers discounts for Amazon EC2 instances that remain idle for more than 1 week

D. AWS does not charge for data sent from the AWS Cloud to the internet

E. AWS eliminates many of the costs of building and maintaining on-premises data centers

[!TIP]

Correct Answer:

B. AWS enables capacity to be adjusted on demand and E. AWS eliminates many of the costs of building and maintaining on-premises data centers

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cost optimization principles in cloud computing — especially how AWS reduces capital expenditure and enables elastic scaling. 🔍 Why B and E Are Correct

B.

Capacity adjusted on demand ☒ Correct. AWS supports: Auto Scaling, on-demand provisioning, and serverless architectures Avoids overprovisioning and underutilization Aligns spend with actual usage

E.

Eliminates on-premises infrastructure costs ☒ Correct. AWS removes the need for: Physical servers, cooling, power, and real estate Hardware refresh cycles and maintenance Large upfront investments

SOURCES

AWS Cloud Benefits AWS Cost Optimization

DISTRACTOR BREAKDOWN

****A. Same prices in every Region **** Incorrect. AWS pricing varies by Region due to infrastructure and operational costs.

****C. Discounts for idle EC2 **** Incorrect. AWS charges for running instances — idle time still incurs cost unless stopped.

****D. No charge for outbound data **** Incorrect. AWS does charge for data transferred out to the internet.

REAL-WORLD TIE-IN

If your Nairobi-based startup migrates from on-prem to AWS: You avoid buying servers and networking gear You scale EC2 and Lambda based on traffic You pay only for what you use — no idle waste That's how AWS delivers elastic, infrastructure-free cost efficiency.

Question 94

****A company wants to grant users in one AWS account access to resources in another AWS account. The users do not currently have permission to access the resources. Which AWS service will meet this requirement?**

OPTIONS

A. IAM group

B. IAM role

C. IAM tag

D. IAM Access Analyzer

[!TIP]

Correct Answer:

B. IAM role

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cross-account access patterns — especially how AWS enables trusted delegation between accounts. The key phrase is “grant users in one AWS account access to resources in another.”

WHY B IS CORRECT

IAM roles support: Cross-account access via trust policies Temporary credentials using sts:AssumeRole Fine-grained permissions scoped to specific resources You define: A role in Account B with a trust policy allowing Account A users to assume it A permission policy attached to the role for resource access It's the backbone of secure, auditable cross-account delegation.

SOURCES

IAM Roles for Cross-Account Access AWS STS AssumeRole

DISTRACTOR BREAKDOWN

****A. IAM group **** Incorrect. Groups manage users within a single account — not cross-account access.

****C. IAM tag **** Incorrect. Tags help organize resources — they don't grant access.

****D. IAM Access Analyzer **** Incorrect. Analyzer helps review access policies — it doesn't grant permissions.

REAL-WORLD TIE-IN

If your Nairobi-based team in Account A needs access to S3 buckets in Account B: Create a role in Account B with s3:ListBucket permissions Add a trust policy allowing Account A's IAM users to assume the role Use STS to generate temporary credentials That's how AWS enables secure, scalable cross-account collaboration.

Question 5

****Which task is the responsibility of AWS when using AWS services?**

OPTIONS

A. Management of IAM user permissions

B. Creation of security group rules for outbound access

C. Maintenance of physical and environmental controls

D. Application of Amazon EC2 operating system patches

[!TIP]

Correct Answer:

C. Maintenance of physical and environmental controls

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model — especially the boundary between cloud provider duties and customer responsibilities. The key phrase is “responsibility of AWS.”

WHY C IS CORRECT

AWS is responsible for: Security of the cloud, which includes: Physical data center security
Environmental controls (power, cooling, fire suppression) Hardware maintenance and access restrictions
These controls are part of AWS’s compliance with standards like: ISO 27001 SOC 1/2/3 PCI DSS

SOURCES

AWS Shared Responsibility Model AWS Physical Security Whitepaper

DISTRACTOR BREAKDOWN

****A. IAM user permissions **** Incorrect. Customers manage IAM users, groups, roles, and policies.

****B. Security group rules **** Incorrect. Customers define inbound/outbound rules for EC2 and other services.

****D. EC2 OS patches **** Incorrect. Customers are responsible for patching the OS on EC2 unless using managed services like RDS or ECS.

REAL-WORLD TIE-IN

If your Nairobi-based team deploys EC2 instances: You configure IAM, security groups, and OS patches AWS secures the physical servers, racks, and data center access You focus on security in the cloud, while AWS handles security of the cloud That's how AWS delivers infrastructure-level trust and compliance.

Question 96

**A company wants to automate infrastructure deployment by using infrastructure as code (IaC). The company wants to scale production stacks so the stacks can be deployed in multiple AWS Regions. Which AWS service will meet these requirements?

**

OPTIONS

A. Amazon CloudWatch

B. AWS Config

C. AWS Trusted Advisor

D. AWS CloudFormation

[!TIP]

Correct Answer:

D. AWS CloudFormation

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of IaC automation and multi-Region scalability — especially how AWS enables consistent infrastructure provisioning across environments. The key phrase is “automate infrastructure deployment... in multiple AWS Regions.”

WHY D IS CORRECT

AWS CloudFormation provides: Declarative templates for provisioning AWS resources Support for stack sets to deploy across multiple accounts and Regions Version-controlled infrastructure definitions Integration with CI/CD pipelines and change sets It’s the backbone of scalable, repeatable infrastructure automation.

SOURCES

CloudFormation StackSets AWS IaC Best Practices

DISTRACTOR BREAKDOWN

****A. Amazon CloudWatch **** Incorrect. Cloud Watch monitors resources — it doesn’t deploy them.

****B. AWS Config **** Incorrect. Config tracks resource configurations — it doesn’t provision infrastructure.

****C. AWS Trusted Advisor **** Incorrect. Trusted Advisor provides recommendations — it doesn’t automate deployments.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to deploy EC2, RDS, and S3 stacks in Frankfurt, Tokyo, and São Paulo: Use CloudFormation StackSets Define resources in a single template Deploy consistently across Regions with audit tagging That’s how AWS delivers IaC-driven global infrastructure orchestration.

Question 97

**Which option is an AWS Cloud Adoption Framework (AWS CAF) platform perspective capability?

**

OPTIONS

A. Data architecture

B. Data protection

C. Data governance

D. Data science

[!TIP]

Correct Answer:

A. Data architecture

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Cloud Adoption Framework (CAF) — especially how the Platform perspective supports technical readiness for cloud adoption. The key phrase is “platform perspective capability.”

WHY A IS CORRECT

The Platform perspective focuses on: Designing, deploying, and optimizing cloud workloads
Building scalable, secure, and resilient architectures It includes capabilities such as: Data architecture Application architecture Infrastructure Integration Deployment Monitoring and logging

SOURCES

AWS CAF Platform Perspective AWS CAF Overview

DISTRACTOR BREAKDOWN

****B. Data protection **** Incorrect. Part of the Security perspective, which covers confidentiality, integrity, and availability.

****C. Data governance **** Incorrect. Falls under the Governance perspective, focused on compliance, policies, and risk.

****D. Data science **** Incorrect. Not a formal CAF capability — it's a discipline, not a framework component.

REAL-WORLD TIE-IN

If your Nairobi-based team is designing a cloud-native analytics platform: Use the Platform perspective to define data architecture Align with best practices for scalability, performance, and resilience Coordinate with Governance and Security perspectives for compliance and protection That's how AWS CAF scaffolds holistic, capability-driven cloud adoption.

Question 98

****A company is running a workload in the AWS Cloud. Which AWS best practice ensures the MOST cost-effective architecture for the workload?**

OPTIONS

A. Loose coupling

B. Rightsizing

C. Caching

D. Redundancy

[!TIP]

Correct Answer:

B. Rightsizing

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cost optimization within the AWS Well-Architected Framework — especially how to align resource usage with actual demand. The key phrase is “MOST cost-effective architecture.”

WHY B IS CORRECT

Rightsizing means: Selecting the optimal instance type, size, and configuration for each workload. Avoiding overprovisioning (which wastes money) and underprovisioning (which hurts performance). Using tools like AWS Compute Optimizer, Cost Explorer, and CloudWatch metrics to guide decisions. It's the cornerstone of cost-aware cloud architecture.

SOURCES

AWS Cost Optimization Pillar Best Practices for Cost Optimization Practical Guide to Cost-Effective AWS Architecture

DISTRACTOR BREAKDOWN

****A. Loose coupling **** Incorrect. Improves scalability and resilience — not directly tied to cost savings.

****C. Caching **** Incorrect. Enhances performance — may reduce cost indirectly, but not the most cost-effective strategy.

****D. Redundancy **** Incorrect. Boosts availability — often increases cost due to resource duplication.

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 workloads: Use Compute Optimizer to identify oversized instances Switch from m5.4xlarge to m5.2xlarge if utilization is low Save thousands monthly while maintaining performance That's how AWS enables precision-tuned infrastructure for cost efficiency.

Question 9

****A company is using a third-party service to back up 10 TB of data to a tape library. The on-premises backup server is running out of space. The company wants to use AWS services for the backups without changing its existing backup workflows. Which AWS service should the company use to meet these requirements?**

OPTIONS

- A. Amazon Elastic Block Store (Amazon EBS)
- B. AWS Storage Gateway
- C. Amazon Elastic Container Service (Amazon ECS)
- D. AWS Lambda

[!TIP]

Correct Answer:

B. AWS Storage Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of hybrid cloud backup strategies — especially how AWS enables seamless integration with existing on-prem workflows. The key phrase is “without changing its existing backup workflows.”

WHY B IS CORRECT

AWS Storage Gateway offers: Tape Gateway mode, which emulates physical tape libraries
Integration with existing backup software (e.g., Veeam, Veritas, NetBackup) Automatic migration of virtual tapes to Amazon S3 Glacier or Deep Archive Local caching for low-latency access It's the backbone of non-disruptive cloud-based backup extension.

SOURCES

AWS Storage Gateway – Tape Gateway Hybrid Backup Architecture

DISTRACTOR BREAKDOWN

****A. Amazon EBS **** Incorrect. EBS is block storage for EC2 — not designed for backup workflows or tape emulation.

****C. Amazon ECS **** Incorrect. ECS runs containerized applications — irrelevant to backup infrastructure.

****D. AWS Lambda **** Incorrect. Lambda executes code — not suitable for emulating tape libraries or integrating with backup software.

REAL-WORLD TIE-IN

If your Nairobi-based team uses a legacy backup tool: Deploy Tape Gateway on-prem Point your backup software to the gateway Let AWS handle tape lifecycle and archival to Glacier That's how AWS delivers cloud-scale backup without workflow disruption.

Question 100

****Which tasks are the customer's responsibility, according to the AWS shared responsibility model? (Choose two.)**

OPTIONS

A. Establish the global infrastructure

B. Perform client-side data encryption

C. Configure IAM credentials

D. Secure edge locations

E. Patch Amazon RDS DB instances

[!TIP]

Correct Answer:

B. Perform client-side data encryption and C. Configure IAM credentials

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model — especially the boundary between AWS-managed infrastructure and customer-managed configurations and data security. 🔍 Why B and C Are Correct

B.

Client-side data encryption ☒ Correct. Customers are responsible for: Encrypting data before sending it to AWS Managing encryption keys if not using AWS KMS Ensuring confidentiality and integrity of sensitive data

C.

IAM credential configuration ☒ Correct. Customers must: Create and manage IAM users, roles, and policies Rotate credentials and enforce MFA Apply least privilege principles

SOURCES

AWS Shared Responsibility Model IAM Best Practices

DISTRACTOR BREAKDOWN

****A. Establish global infrastructure **** Incorrect. AWS handles all physical infrastructure — data centers, Regions, and networking.

****D. Secure edge locations **** Incorrect. AWS secures edge locations like Cloud Front PO Ps and Global Accelerator endpoints.

****E. Patch Amazon RDS DB instances **** Incorrect. AWS patches the OS and database engine for managed services like RDS — unless using EC2-hosted databases.

REAL-WORLD TIE-IN

If your Nairobi-based team stores sensitive customer data: Encrypt it client-side before uploading to S3 Use IAM roles with scoped permissions Let AWS handle the physical security and managed service patching That's how AWS enables secure, scalable cloud operations with clear boundaries.

Question 101

****A company is using the AWS Free Tier for several AWS services for an application. What will happen if the Free Tier usage period expires or if the application use exceeds the Free Tier usage limits?**

OPTIONS

A. The company will be charged the standard pay-as-you-go service rates for the usage that exceeds the Free Tier usage.

B. AWS Support will contact the company to set up standard service charges.

C. The company will be charged for the services it consumed during the Free Tier period, plus additional charges for service consumption after the Free Tier period.

D. The company's AWS account will be frozen and can be restarted after a payment plan is established.

[!TIP]

Correct Answer:

A. The company will be charged the standard pay-as-you-go service rates for the usage that exceeds the Free Tier usage.

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS Free Tier boundaries — especially how billing transitions once usage exceeds limits or the promotional period ends. The key phrase is “exceeds the Free Tier usage limits.”

WHY A IS CORRECT

AWS Free Tier includes: 12-month introductory offers (e.g., 750 hours of EC2 per month) Always-free services (e.g., Lambda, DynamoDB with limited usage) Once limits are exceeded or the period ends: AWS automatically bills at standard pay-as-you-go rates No manual intervention or support setup required

SOURCES

AWS Free Tier FAQ AWS Pricing Overview

DISTRACTOR BREAKDOWN

****B. AWS Support contacts the company **** Incorrect. Billing is automated — support does not initiate contact for standard charges.

****C. Charged for services during Free Tier **** Incorrect. Free Tier usage is free — charges apply only after limits are exceeded.

****D. Account frozen until payment plan **** Incorrect. AWS does not freeze accounts unless there's a billing issue or overdue payment, not simply for exceeding Free Tier.

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 and S3 workloads: Monitor usage with AWS Budgets and Cost Explorer Set alerts for approaching Free Tier thresholds Avoid surprise charges by rightsizing and scheduling workloads That's how AWS enables transparent, usage-based billing with proactive cost controls.

Question 102

****Which AWS service uses machine learning to help discover, monitor, and protect sensitive data that is stored in Amazon S3 buckets?**

OPTIONS

A. AWS Shield

B. Amazon Macie

C. AWS Network Firewall

D. Amazon Cognito

[!TIP]

Correct Answer:

B. Amazon Macie

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of data security automation — especially how AWS uses machine learning to detect sensitive information in S3. The key phrase is “discover, monitor, and protect sensitive data.”

WHY B IS CORRECT

Amazon Macie provides: ML-powered detection of PII, financial data, credentials, and secrets
Continuous monitoring of S3 buckets for public access, encryption status, and anomalies
Automated alerts and integration with Security Hub and EventBridge It’s the backbone of intelligent, scalable data protection in AWS.

SOURCES

Amazon Macie Overview Macie Use Cases

DISTRACTOR BREAKDOWN

****A. AWS Shield **** Incorrect. Protects against DDoS attacks — not for data discovery or classification.

****C. AWS Network Firewall **** Incorrect. Secures VPC traffic — not S3 data inspection.

****D. Amazon Cognito **** Incorrect. Manages user authentication and identity — not data protection.

REAL-WORLD TIE-IN

If your Nairobi-based team stores customer records in S3: Use Macie to scan for PII and access risks Automate alerts for public buckets or unencrypted objects Integrate findings into your security workflow That's how AWS delivers ML-driven data sensitivity awareness at scale.

Question 03

****According to the AWS shared responsibility model, which of the following is exclusively the responsibility of AWS?**

OPTIONS

- A. Patching of the guest operating system
- B. Security awareness and training
- C. Physical and environmental controls
- D. Development of an IAM password policy

[!TIP]

Correct Answer:

C. Physical and environmental controls

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of the AWS shared responsibility model, which divides cloud security responsibilities between AWS and the customer. The key phrase is “exclusively the responsibility of AWS.”

WHY C IS CORRECT

AWS is responsible for security of the cloud, which includes: Physical security of data centers
Environmental controls like power, cooling, and fire suppression
Hardware maintenance and access restrictions
These responsibilities are part of AWS’s compliance with standards like ISO 27001, SOC 1/2/3, and PCI DSS.

DISTRACTOR BREAKDOWN

****A. Patching of the guest operating system **** Incorrect. Customers are responsible for patching the OS on EC2 instances and other unmanaged services.

****B. Security awareness and training **** Incorrect. Customers must train their teams on cloud security best practices.

****D. IAM password policy development **** Incorrect. Customers configure IAM policies, including password strength and rotation rules.

REAL-WORLD TIE-IN

If your Nairobi-based team deploys EC2 and S3: You manage IAM, OS patches, and user training
AWS secures the physical servers, racks, and facilities
You focus on security in the cloud, while AWS handles security of the cloud
That’s how AWS enables compliance-ready infrastructure with clear operational boundaries.

Question 104

**What can users do using AWS Marketplace? (Select two.)

**

OPTIONS

- A. Sell unused Amazon EC2 Spot Instances
- B. Sell solutions to other AWS users
- C. Buy third-party software that runs on AWS
- D. Purchase AWS security and compliance documents
- E. Order AWS Snowball

[!TIP]

Correct Answer:

B. Sell solutions to other AWS users and C. Buy third-party software that runs on AWS

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS Marketplace capabilities — especially how it enables software discovery, procurement, and distribution. The key phrase is “what can users do.” 🔍 Why B and C Are Correct

B.

Sell solutions to other AWS users ☒ Correct. ISVs and developers can: List SaaS, AMI, container, and data products Set pricing, licensing, and deployment models Reach global AWS customers

C.

Buy third-party software that runs on AWS  Correct. Customers can: Discover and deploy vetted software Use 1-click deployment for EC2, ECS, EKS Integrate billing with AWS usage

SOURCES

AWS Marketplace Overview Sell in AWS Marketplace

DISTRACTOR BREAKDOWN

****A. Sell unused EC2 Spot Instances **** Incorrect. Spot Instances are non-transferable — AWS manages the bidding and lifecycle.

****D. Purchase AWS security documents **** Incorrect. Compliance reports are accessed via AWS Artifact, not Marketplace.

****E. Order AWS Snowball **** Incorrect. Snowball is ordered via AWS Console, not Marketplace.

REAL-WORLD TIE-IN

If your Nairobi-based team builds a SaaS analytics tool: List it on AWS Marketplace with metered billing Reach enterprise customers with integrated procurement Let users deploy it directly into their AWS environments That's how AWS Marketplace enables scalable software monetization and frictionless adoption.

Question 105

****What are the possible uses for AWS edge locations? (Select two.)**

OPTIONS

A. Hosting applications

B. Delivering content closer to users

C. Running NoSQL database caching services

D. Reducing traffic on the server by caching responses

E. Sending notification messages to end users

[!TIP]

Correct Answer:

B. Delivering content closer to users and D. Reducing traffic on the server by caching responses

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS edge infrastructure — especially how edge locations support low-latency content delivery and caching. The key phrase is “possible uses.”

 Why B and D Are Correct

B.

Delivering content closer to users  Correct. Edge locations are part of Amazon CloudFront, which: Distributes content via a global network of Points of Presence (PoPs) Serves data from the nearest edge location to reduce latency

D.

Reducing traffic on the server by caching responses  Correct. CloudFront caches content at edge locations, which: Minimizes repeated requests to origin servers Improves performance and reduces backend load

SOURCES

AWS Edge Locations Overview CloudFront and Edge Location Use Cases

DISTRACTOR BREAKDOWN

****A. Hosting applications **** Incorrect. Applications are typically hosted in Regions, not edge locations.

****C. NoSQL database caching services **** Incorrect. Services like Amazon Elasti Cache run in Regions — not at edge locations.

****E. Sending notification messages **** Incorrect. Messaging services like SNS and SES are not tied to edge locations.

REAL-WORLD TIE-IN

If your Nairobi-based team serves video content globally: Use CloudFront to cache and deliver assets from edge locations Reduce latency for users in Frankfurt, Tokyo, and São Paulo Offload origin servers and scale efficiently That's how AWS edge locations enable fast, scalable content delivery with intelligent caching.

Question 106

****Amazon Elastic Container Service (Amazon ECS) and Amazon DynamoDB are used by a firm to execute its mission-critical web application. Multiple times during the day, the workload jumps to up to ten times the regular level. Which AWS Cloud feature helps the business to adapt to these demand changes?**

OPTIONS

A. Agility

B. Global reach

C. Scalability

D. Security

[!TIP]

Correct Answer:

C. Scalability

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cloud elasticity and auto-scaling — especially how AWS enables workloads to adapt to fluctuating demand. The key phrase is “workload jumps to up to ten times the regular level.”

WHY C IS CORRECT

Scalability in AWS means: Automatically adjusting resources to meet demand ECS can scale tasks based on CPU/memory metrics DynamoDB offers on-demand capacity mode and auto scaling Ensures performance without overprovisioning It's the backbone of cost-efficient, performance-resilient cloud architecture.

SOURCES

AWS Scalability Overview DynamoDB Auto Scaling ECS Service Auto Scaling

DISTRACTOR BREAKDOWN

****A. Agility** **** Incorrect.** Agility refers to speed of innovation — not dynamic resource scaling.

****B. Global reach** **** Incorrect.** Global reach enables geographic distribution — not workload adaptation.

****D. Security** **** Incorrect.** Security protects data and infrastructure — not related to scaling.

REAL-WORLD TIE-IN

If your Nairobi-based team sees traffic spikes during product launches: ECS scales containers to handle load DynamoDB adjusts throughput automatically You maintain uptime and responsiveness without manual intervention That's how AWS delivers scalable infrastructure that flexes with demand.

Question 107

****A** company wants to improve the overall availability and performance of its applications that are hosted on AWS. Which AWS service should the company use?

OPTIONS

A. Amazon Connect

B. Amazon Lightsail

C. AWS Global Accelerator

D. AWS Storage Gateway

[!TIP]

Correct Answer:

C. AWS Global Accelerator

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of network-level performance optimization — especially how AWS enables global traffic routing for high availability and low latency. The key phrase is “improve availability and performance.”

WHY C IS CORRECT

AWS Global Accelerator provides: Static IP addresses for global entry points Intelligent routing to the nearest healthy endpoint Automatic failover across Regions and Availability Zones Enhanced performance using the AWS global network backbone It's the backbone of resilient, latency-optimized global application delivery.

SOURCES

AWS Global Accelerator Overview Global Accelerator Use Cases

DISTRACTOR BREAKDOWN

****A. Amazon Connect **** Incorrect. Used for contact center solutions — not application performance.

****B. Amazon Lightsail **** Incorrect. Simplified hosting — lacks advanced routing and availability features.

****D. AWS Storage Gateway **** Incorrect. Hybrid storage integration — not relevant to application performance or availability.

REAL-WORLD TIE-IN

If your Nairobi-based team serves users in Frankfurt, Tokyo, and São Paulo: Use Global Accelerator to route traffic via AWS's edge network Ensure fast failover and low-latency access Maintain uptime even during Regional disruptions That's how AWS delivers global resilience and performance at the network edge.

Question 108

****Which of the following tasks is the customer's duty under the shared responsibility model? (Select two.)**

OPTIONS

A. Maintaining the underlying Amazon EC2 hardware

B. Managing the VPC network access control lists

C. Encrypting data in transit and at rest

D. Replacing failed hard disk drives

E. Deploying hardware in different Availability Zones

[!TIP]

Correct Answer:

B. Managing the VPC network access control lists and C. Encrypting data in transit and at rest

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model — especially how customers are responsible for security in the cloud, while AWS handles security of the cloud. 🔍 Why B and C Are Correct

B.

VPC network ACLs ✅ Correct. Customers configure: Inbound/outbound rules Subnet-level access controls Firewall-like behavior for traffic filtering

C.

Data encryption ✅ Correct. Customers must: Encrypt sensitive data before transmission or storage Manage encryption keys (unless using AWS-managed KMS) Ensure compliance with data protection standards

SOURCES

AWS Shared Responsibility Model VPC Security Best Practices

DISTRACTOR BREAKDOWN

****A. EC2 hardware maintenance **** Incorrect. AWS handles physical server upkeep.

****D. Replacing hard drives **** Incorrect. AWS manages hardware failures and replacements.

****E. Deploying hardware in AZs **** Incorrect. AWS provisions and maintains infrastructure across Availability Zones.

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 and S3 workloads: You configure VPC ACLs to restrict traffic You encrypt customer data using TLS and KMS AWS ensures the physical servers and disks are secure and operational That's how AWS enables secure, scalable cloud operations with clear boundaries.

Question 109

****Which of the following are AWS obligations, according to the AWS shared responsibility model? (Select two.)**

OPTIONS

A. Network infrastructure and virtualization of infrastructure

B. Security of application data

C. Guest operating systems

D. Physical security of hardware

[!TIP]

Correct Answer:

A. Network infrastructure and virtualization of infrastructure and D. Physical security of hardware

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model — especially the boundary between AWS-managed infrastructure and customer-managed configurations and data security. The key phrase is “AWS obligations.” 🔍 Why A and D Are Correct

A.

Network infrastructure and virtualization ☒ Correct. AWS is responsible for: Securing the underlying network fabric Managing hypervisors and virtualization layers Ensuring isolation between tenants

D.

Physical security of hardware ☒ Correct. AWS handles: Data center access controls Environmental safeguards (power, cooling, fire suppression) Hardware lifecycle and disposal

SOURCES

AWS Shared Responsibility Model AWS Security Whitepaper

DISTRACTOR BREAKDOWN

****B. Security of application data **** Incorrect. Customers must secure their data — including encryption, access controls, and backups.

****C. Guest operating systems **** Incorrect. Customers patch and configure OS on EC2 and other unmanaged services.

****E. Credentials and policies **** Incorrect. IAM users, roles, and policies are customer-managed.

REAL-WORLD TIE-IN

If your Nairobi-based team deploys EC2 and S3 workloads: AWS secures the physical servers and network layers You manage IAM, OS patches, and data encryption Together, you maintain a secure, scalable cloud environment with clear boundaries That's how AWS delivers compliance-ready infrastructure with customer-controlled security layers.

Question 110

****What charges are included in the comparison of AWS vs on-premises Total Cost of Ownership (TCO)?**

OPTIONS

A. Data center security

B. Business analysis

C. Project management

D. Operating system administration

[!TIP]

Correct Answer:

A. Data center security and D. Operating system administration

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of TCO analysis in cloud migration — especially how AWS compares against on-premises infrastructure in terms of operational and capital expenses. The key phrase is “charges included in the comparison.” 🔍 Why A and D Are Correct

A.

Data center security ☒ Correct. On-premises TCO includes: Physical security staff and systems Access control, surveillance, and compliance costs AWS absorbs these costs in its managed infrastructure

D.

Operating system administration ☒ Correct. TCO accounts for: System administrator salaries Patch management and OS lifecycle support AWS offers managed services that reduce this overhead

SOURCES

NetApp on AWS TCO AWS Cloud Financial Management Guide

DISTRACTOR BREAKDOWN

****B. Business analysis **** Incorrect. Considered a strategic or planning cost, not typically included in TCO comparisons.

****C. Project management **** Incorrect. Often treated as a one-time migration or implementation cost, not recurring infrastructure expense.

REAL-WORLD TIE-IN

If your Nairobi-based team is evaluating cloud migration: Include costs like power, cooling, hardware refresh, and admin labor in your on-prem TCO Compare against AWS's managed services and pay-as-you-go pricing Use AWS's TCO calculator to model savings over 3–5 years That's how AWS enables data-driven migration decisions with transparent cost modeling.

Question 111

****Which AWS service or feature identifies whether an Amazon S3 bucket or an IAM role has been shared with an external entity?**

OPTIONS

A. AWS Service Catalog

B. AWS Systems Manager

C. AWS IAM Access Analyzer

D. AWS Organizations

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of access visibility and external sharing detection — especially how AWS helps identify unintended exposure of resources. The key phrase is “shared with an external entity.”

WHY C IS CORRECT

IAM Access Analyzer provides: Automated analysis of resource policies (S3, IAM, KMS, Lambda, etc.) Detection of external access — including cross-account, public, or third-party sharing Alerts for unintended exposure and actionable insights It’s the backbone of policy-aware, trust boundary enforcement in AWS.

SOURCES

IAM Access Analyzer Overview Detecting External Access with Access Analyzer

DISTRACTOR BREAKDOWN

****A. AWS Service Catalog **** Incorrect. Manages approved products — not access analysis.

****B. AWS Systems Manager **** Incorrect. Manages infrastructure — not policy inspection.

****D. AWS Organizations **** Incorrect. Manages multi-account structure — not resource-level access visibility.

REAL-WORLD TIE-IN

If your Nairobi-based team stores sensitive data in S3: Use Access Analyzer to detect public buckets or cross-account role sharing Get alerts when policies allow external access Tighten permissions before exposure becomes a breach That's how AWS delivers automated access intelligence with trust boundary enforcement.

Question 112

**Which AWS service supports MySQL and PostgreSQL as relational databases?

**

OPTIONS

A. Amazon Redshift

B. Amazon DynamoDB

C. Amazon Aurora

D. Amazon Neptune

[!TIP]

Correct Answer:

C. Amazon Aurora

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of relational database services in AWS — especially which ones support open-source engines like MySQL and PostgreSQL. The key phrase is “supports MySQL and PostgreSQL.”

WHY C IS CORRECT

Amazon Aurora is a fully managed relational database that: Offers MySQL and PostgreSQL compatibility Delivers up to 5x performance over MySQL and 3x over PostgreSQL Supports replication, failover, and autoscaling Integrates with RDS, IAM, and VPC for security and access control

SOURCES

Aurora Overview Aurora Engine Compatibility

DISTRACTOR BREAKDOWN

****A. Amazon Redshift **** Incorrect. Redshift is a data warehouse, not a transactional relational database.

****B. Amazon DynamoDB **** Incorrect. Dynamo DB is a No SQL key-value store — no support for SQL engines.

****D. Amazon Neptune **** Incorrect. Neptune is a graph database — uses Gremlin and SPARQL, not SQL.

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a legacy MySQL app or building a PostgreSQL analytics engine: Aurora gives you open-source compatibility with cloud-native resilience You get automatic backups, failover, and scaling without managing the underlying infrastructure That’s how AWS delivers relational power with cloud simplicity and performance.

Question 113

**Which AWS service is natively supported by AWS Snowball Edge?

**

OPTIONS

A. AWS Server Migration Service (AWS SMS)

B. Amazon Aurora

C. AWS Trusted Advisor

D. Amazon EC2

[!TIP]

Correct Answer:

D. Amazon EC2

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of edge computing capabilities in AWS — especially how Snowball Edge supports local compute workloads. The key phrase is “natively supported.”

WHY D IS CORRECT

AWS Snowball Edge is a rugged device designed for: Data migration and edge computing
Running Amazon EC2 instances locally on the device Supporting S3-compatible storage,

Lambda functions, and machine learning inference It's ideal for disconnected environments or low-latency edge processing.

SOURCES

AWS Snowball Edge Developer Guide Snowball Edge FAQs

DISTRACTOR BREAKDOWN

****A. AWS SMS **** Incorrect. Used for migrating VMs — not supported on Snowball Edge.

****B. Amazon Aurora **** Incorrect. Aurora is a managed relational database — not deployable on Snowball Edge.

****C. AWS Trusted Advisor **** Incorrect. A cloud-based recommendation tool — not available on edge devices.

REAL-WORLD TIE-IN

If your Nairobi-based team needs to process drone footage or sensor data in remote areas: Deploy EC2 instances on Snowball Edge Run ML inference or analytics locally Sync results to AWS when connectivity resumes That's how AWS delivers compute power at the edge with cloud-native tooling.

Question 114

****Which AWS shared responsibility controls are shared? (Select two.)**

OPTIONS

- A. Awareness and training
- B. Patching of Amazon RDS
- C. Configuration management
- D. Physical and environmental controls
- E. Service and communications protection or security

[!TIP]

Correct Answer:

C. Configuration management and E. Service and communications protection or security

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of shared controls in the AWS shared responsibility model — especially where both AWS and the customer contribute to security outcomes. The key phrase is “controls are shared.” 🔍 Why C and E Are Correct

C.

Configuration management ☒ Shared. AWS configures infrastructure defaults and APIs Customers configure OS, applications, and services Both parties influence the security posture

E.

Service and communications protection or security ☒ Shared. AWS secures the underlying network and services Customers secure their data, endpoints, and encryption Together they ensure secure communication and service boundaries

SOURCES

AWS Shared Responsibility Model AWS Security Best Practices

DISTRACTOR BREAKDOWN

****A. Awareness and training **** Incorrect. Fully customer responsibility — AWS doesn't train your team.

****B. Patching of Amazon RDS **** Incorrect. AWS handles patching for managed services like RDS — not shared.

****D. Physical and environmental controls **** Incorrect. Fully AWS responsibility — customers never touch the data center.

REAL-WORLD TIE-IN

If your Nairobi-based team deploys EC2 and RDS: You configure EC2 security groups and OS patches AWS patches RDS and secures the physical infrastructure You both contribute to configuration and communication security That's how AWS delivers layered, collaborative security with clear boundaries and shared accountability.

Question 115

****A** company does not want to rely on elaborate forecasting to determine its usage of compute resources. Instead, the company wants to pay only for the resources that it uses. The company also needs the ability to increase or decrease its resource usage to meet business requirements. Which pillar of the AWS Well-Architected Framework aligns with these requirements?

OPTIONS

A. Operational excellence

B. Security

C. Reliability

D. Cost optimization

[!TIP]

Correct Answer:

D. Cost optimization

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Well-Architected Framework, especially the Cost Optimization pillar, which focuses on pay-as-you-go efficiency and elasticity. The key phrases are “pay only for the resources” and “increase or decrease usage.”

WHY D IS CORRECT

Cost Optimization pillar emphasizes: Avoiding overprovisioning by using autoscaling and serverless models
Using pricing models like Spot, Savings Plans, and Reserved Instances
Monitoring usage to eliminate waste and optimize spend

SOURCES

AWS Well-Architected Framework Cost Optimization Pillar

DISTRACTOR BREAKDOWN

****A. Operational excellence **** Incorrect. Focuses on operations and monitoring — not cost or elasticity.

****B. Security **** Incorrect. Covers identity, data protection, and incident response — not resource usage.

****C. Reliability **** Incorrect. Ensures workloads recover and scale — but doesn't address cost efficiency directly.

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 and Lambda workloads: Use autoscaling and serverless to match demand Pay only for actual usage — no idle capacity Monitor with Cost Explorer and Budgets That's how AWS delivers elastic infrastructure aligned with cost-aware architecture principles.

Question 116

****After a single Availability Zone service disruption, a corporation must guarantee that the endpoint for a database instance stays the same. The program must continue database operations without human intervention from an administrator. How are these stipulations to be met?**

OPTIONS

A. Use multiple Amazon Route 53 routes to the standby database instance endpoint hosted on AWS Storage Gateway

B. Configure Amazon RDS Multi-Availability Zone deployments with automatic failover to the standby

C. Add multiple Application Load Balancers and deploy the database instance with AWS Elastic Beanstalk

D. Deploy a single Network Load Balancer to distribute incoming traffic across multiple Amazon CloudFront origins

[!TIP]

Correct Answer:

B. Configure Amazon RDS Multi-Availability Zone deployments with automatic failover to the standby

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of high availability and automatic failover in AWS database services — especially how RDS Multi-AZ deployments maintain continuity during AZ disruptions. The key phrases are “endpoint stays the same” and “without human intervention.”

WHY B IS CORRECT

Amazon RDS Multi-AZ provides: Synchronous replication to a standby in a different AZ
Automatic failover when the primary becomes unavailable
Stable endpoint — no change required in application configuration
No manual intervention — failover is handled by AWS

SOURCES

RDS Multi-AZ Deployments High Availability with RDS

DISTRACTOR BREAKDOWN

****A. Route 53 + Storage Gateway **** Incorrect. Storage Gateway is for hybrid storage — not database failover.

****C. Elastic Beanstalk + ALBs **** Incorrect. Beanstalk is for app deployment — not database failover or endpoint stability.

****D. NLB + CloudFront origins **** Incorrect. Cloud Front is for content delivery — not database traffic or failover.

REAL-WORLD TIE-IN

If your Nairobi-based team runs a mission-critical PostgreSQL database: Deploy RDS in Multi-AZ mode AWS handles replication and failover Your app keeps using the same endpoint — no downtime, no manual switch That's how AWS delivers resilient database operations with seamless failover and endpoint consistency.

Question 117

****Which costs must be addressed when comparing AWS Cloud vs on-premises Total Cost of Ownership? (Select two.)**

OPTIONS

A. Software development

B. Project management

C. Storage hardware

D. Physical servers

E. Antivirus software license

[!TIP]

Correct Answer:

C. Storage hardware and D. Physical servers

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of infrastructure-level cost modeling — especially how AWS TCO comparisons focus on capital and operational expenses that AWS absorbs and customers would otherwise manage on-premises. 🔍 Why C and D Are Correct

C.

Storage hardware ✅ Correct. On-premises setups require: Purchasing and maintaining SAN/NAS devices Managing lifecycle, upgrades, and power/cooling AWS replaces this with S3, EBS, and Glacier

D.

Physical servers ✅ Correct. On-premises TCO includes: Rack-mounted servers, blades, and chassis Maintenance, depreciation, and provisioning AWS abstracts this via EC2 and managed services

SOURCES

AWS TCO Calculator TerraZone TCO Breakdown LinkedIn TCO Comparison

DISTRACTOR BREAKDOWN

****A. Software development **** Incorrect. Considered a business function, not infrastructure — excluded from TCO.

****B. Project management **** Incorrect. Typically a one-time or strategic cost, not recurring infrastructure expense.

****E. Antivirus software license **** Incorrect. Often bundled or variable — not a core infrastructure cost in TCO modeling.

REAL-WORLD TIE-IN

If your Nairobi-based team is evaluating cloud migration: Include costs like server racks, storage arrays, and cooling systems in your on-prem TCO Compare against AWS's pay-as-you-go model with managed infrastructure Use the AWS TCO calculator to model savings over 3–5 years That's how AWS enables transparent cost modeling and strategic infrastructure decisions.

Question 118

****A company is migrating to the AWS Cloud. The company requires consultative review and guidance for its applications during the migration. After the migration is complete, the company requires a response within 30 minutes if business-critical systems go down. Which AWS Support plans meet these requirements? (Choose two.)**

OPTIONS

A. AWS Enterprise Support

B. AWS Enterprise On-Ramp Support

C. AWS Developer Support

D. AWS Basic Support

E. AWS Business Support

[!TIP]

Correct Answer:

A. AWS Enterprise Support and B. AWS Enterprise On-Ramp Support

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS support tiers — especially which plans offer consultative guidance during migration and fast response SLAs for critical issues. The key phrases are “consultative review” and “30-minute response for business-critical systems.” 🔍

Why A and B Are Correct

A.

Enterprise Support ✅ Correct. Includes a Technical Account Manager (TAM) Offers consultative architecture reviews Provides 15-minute response time for critical issues Designed for mission-critical workloads

B.

Enterprise On-Ramp Support ✅ Correct. Tailored for mid-sized businesses migrating to AWS Includes consultative guidance and 30-minute response for urgent cases Offers access to AWS Support Concierge

SOURCES

AWS Support Plans Enterprise On-Ramp Details

DISTRACTOR BREAKDOWN

****C. Developer Support **** Incorrect. No consultative guidance or fast SLA — response time is 12–24 hours.

****D. Basic Support **** Incorrect. Free tier — no technical support or SLA guarantees.

****E. Business Support **** Tempting but incorrect. Offers 1-hour response for urgent cases
No consultative migration guidance — lacks TAM and architecture reviews

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a payment platform to AWS: Use Enterprise On-Ramp for migration guidance and 30-minute SLA Upgrade to Enterprise Support for full TAM access and strategic reviews Ensure business continuity with proactive architecture and incident response That's how AWS delivers support plans aligned with mission-critical migration and operational resilience.

Question 119

****Which AWS products anticipate future AWS expenses automatically?**

OPTIONS

A. AWS Support Center

B. AWS Total Cost of Ownership (TCO) Calculator

C. AWS Simple Monthly Calculator

D. Cost Explorer

[!TIP]

Correct Answer:

D. Cost Explorer

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS cost management tools — especially which ones offer automated forecasting based on usage trends. The key phrase is “anticipate future AWS expenses automatically.”

WHY D IS CORRECT

AWS Cost Explorer provides: Forecasting of future costs based on historical usage
Visualizations for daily, monthly, and service-level spend Filtering by service, account, and tags
Budget alerts and anomaly detection It’s the backbone of proactive cloud financial planning.

SOURCES

AWS Cost Explorer Overview Forecasting with Cost Explorer

DISTRACTOR BREAKDOWN

****A. AWS Support Center **** Incorrect. Manages support cases — no cost forecasting.

****B. TCO Calculator **** Incorrect. Compares on-prem vs AWS — not usage-based

forecasting.

****C. Simple Monthly Calculator **** Incorrect. Estimates costs manually — no automatic forecasting.

REAL-WORLD TIE-IN

If your Nairobi-based team wants to predict EC2 and S3 costs for Q1: Use Cost Explorer to forecast based on past usage Set budget thresholds and alerts Optimize spend with Reserved Instances or Savings Plans That's how AWS delivers automated cost insights for strategic cloud budgeting.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

Question 120

****Which functionality may be utilized to prevent inadvertent overwrites or deletions of Amazon S3 buckets?**

OPTIONS

A. Lifecycle policy

B. Object versioning

C. Server-side encryption

D. Bucket ACL

[!TIP]

Correct Answer:

B. Object versioning

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of S3 data durability and protection mechanisms — especially how AWS prevents accidental overwrites or deletions. The key phrase is “prevent inadvertent overwrites or deletions.”

WHY B IS CORRECT

Object versioning in S3: Maintains multiple versions of an object Prevents accidental overwrites — new uploads create new versions Protects against accidental deletions — delete markers can be removed to restore prior versions Enables audit trails and rollback

SOURCES

S3 Versioning Overview Protecting Data with S3 Versioning

DISTRACTOR BREAKDOWN

****A. Lifecycle policy **** Incorrect. Automates object transitions and deletions — not protection against them.

****C. Server-side encryption **** Incorrect. Secures data at rest — doesn't prevent overwrites or deletions.

****D. Bucket ACL **** Incorrect. Controls access — doesn't preserve object history or prevent deletion.

REAL-WORLD TIE-IN

If your Nairobi-based team stores critical logs or backups in S3: Enable versioning to preserve every change Recover deleted or overwritten files easily Combine with MFA Delete for extra protection That's how AWS delivers durable, rollback-ready storage with overwrite resilience.

Question 121

****Amazon DynamoDB is used by a business in its AWS Cloud architecture. Which of the following is a duty of the organization, according to the AWS shared responsibility model? (Select two.)**

OPTIONS

- A. Operating system patching and upgrades
- B. Application of appropriate permissions with IAM tools
- C. Configuration of data encryption options
- D. Creation of DynamoDB endpoints
- E. Infrastructure provisioning and maintenance

[!TIP]

Correct Answer:

B. Application of appropriate permissions with IAM tools and C. Configuration of data encryption options

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of customer responsibilities in managed services like DynamoDB — especially around access control and data protection. The key phrase is “duty of the organization.” 🔍 Why B and C Are Correct

B.

IAM permissions ☒ Correct. Customers define who can access DynamoDB Use IAM policies, roles, and least privilege principles Critical for securing data and operations

C.

Data encryption configuration ☒ Correct. Customers choose encryption settings (AWS-managed or customer-managed keys) Responsible for key rotation and access control Ensures compliance with data protection standards

SOURCES

AWS Shared Responsibility Model DynamoDB Security Best Practices

DISTRACTOR BREAKDOWN

****A. OS patching **** Incorrect. AWS handles patching for fully managed services like Dynamo DB.

****D. Endpoint creation **** Incorrect. Dynamo DB endpoints are managed by AWS — customers don't provision them.

****E. Infrastructure provisioning **** Incorrect. AWS provisions and maintains the underlying infrastructure for Dynamo DB.

REAL-WORLD TIE-IN

If your Nairobi-based team uses DynamoDB for session storage: You define IAM roles for read/write access You configure encryption with KMS keys AWS handles the hardware, patching, and scaling That's how AWS delivers managed database power with customer-controlled security and access boundaries. When comparing Total Cost of Ownership (TCO) between AWS and on-premises infrastructure, the focus is on capital and operational costs that AWS absorbs and customers would otherwise manage. These include:

C.

Storage hardware On-premises setups require purchasing and maintaining storage arrays (SAN/NAS) AWS replaces this with services like Amazon S3, EBS, and Glacier Eliminates costs for upgrades, power, cooling, and physical space

D.

Physical servers On-premises environments require buying, provisioning, and maintaining physical servers AWS abstracts this with EC2 instances, Lambda, and other compute services Reduces hardware lifecycle costs and improves scalability ❌ Why the Others Are Incorrect
Option Reason It's Not Included in TCOA. Software development Business function cost, not infrastructure-related B. Project management Strategic or one-time cost, not recurring infrastructure E. Antivirus software license Often bundled or variable, not a core infrastructure cost 🔍 Source Insight The AWS TCO Calculator and Well-Architected Framework emphasize comparing: Server and storage hardware Power and cooling Data center space IT labor and maintenance Licensing and support contracts These are the core cost categories AWS helps eliminate or reduce.

SOURCES

Question 123

**Which AWS service should a company use to create a NoSQL database?

**

OPTIONS

A. Amazon Aurora

B. Amazon DynamoDB

C. Amazon Redshift

D. Amazon Neptune

[!TIP]

Correct Answer:

B. Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS database service types — especially which one is designed for NoSQL workloads. The key phrase is “create a NoSQL database.”

WHY B IS CORRECT

Amazon DynamoDB is: A fully managed NoSQL database service Supports key-value and document data models Offers single-digit millisecond latency at scale Ideal for web apps, gaming, IoT, and serverless architectures

SOURCES

DynamoDB Overview NoSQL on AWS

DISTRACTOR BREAKDOWN

A. Amazon Aurora Relational database — supports MySQL and PostgreSQL

C. Amazon Redshift Data warehouse — optimized for analytics, not No SQL

D. Amazon Neptune Graph database — uses Gremlin and SPARQL, not No SQL

REAL-WORLD TIE-IN

If your Nairobi-based team is building a real-time leaderboard or session store: Use DynamoDB for scalable, low-latency NoSQL storage. Integrate with Lambda, API Gateway, and IAM. Avoid schema constraints and scale horizontally. That's how AWS delivers NoSQL power with serverless simplicity and global scalability.

Question 124

**Which AWS service or functionality is utilized by distributed applications to send text and email messages?

**

OPTIONS

A. Amazon Simple Notification Service (Amazon SNS)

B. Amazon Simple Email Service (Amazon SES)

C. Amazon CloudWatch alerts

D. Amazon Simple Queue Service (Amazon SQS)

[!TIP]

Correct Answer:

A. Amazon SNS

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of messaging services in distributed AWS architectures — especially which one supports multi-protocol delivery including SMS and email. The key phrase is “send text and email messages.”

WHY A IS CORRECT

Amazon SNS (Simple Notification Service): Supports SMS, email, mobile push, and HTTP/S endpoints Ideal for event-driven architectures and fan-out messaging Integrates with Lambda, SQS, CloudWatch, and more Enables topic-based pub/sub for scalable notifications

SOURCES

Amazon SNS Overview SNS Use Cases

DISTRACTOR BREAKDOWN

B. Amazon SES Only supports email — no SMS or multi-protocol delivery

C. Cloud Watch alerts Triggers alarms — doesn't send messages directly

D. Amazon SQS Queues messages — doesn't push notifications to users

REAL-WORLD TIE-IN

If your Nairobi-based team builds a distributed app for order tracking: Use SNS to notify users via SMS and email when order status changes Integrate with Lambda for custom logic Scale across regions with topic-based delivery That's how AWS delivers multi-channel messaging for distributed, event-driven applications.

REFERENCES

<https://aws.amazon.com/sns/>

Question 125

**To achieve high availability, how many Availability Zones should computing resources be provided across?

**

OPTIONS

A. A minimum of one

B. A minimum of two

C. A minimum of three

D. A minimum of four or more

[!TIP]

Correct Answer:

B. A minimum of two

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS high availability architecture — especially how Availability Zones (AZs) are used to ensure fault tolerance and uptime. The key phrase is “achieve high availability.”

WHY B IS CORRECT

High availability requires: Redundant compute resources across at least two AZs Automatic failover and load balancing Isolation from localized failures (power, network, hardware) AWS regions typically have 3+ AZs, but two is the minimum for HA design

SOURCES

AWS Well-Architected Reliability Pillar Multi-AZ Best Practices

DISTRACTOR BREAKDOWN

A. One AZ No redundancy — single point of failure

C. Three AZs Valid for higher fault tolerance, but not minimum

D. Four or more Overkill for basic HA — adds cost and complexity

REAL-WORLD TIE-IN

If your Nairobi-based team runs a payment API: Deploy EC2 instances or RDS across two AZs
Use Elastic Load Balancing and Auto Scaling Ensure zero downtime during AZ disruptions
That's how AWS delivers resilient infrastructure with zone-aware failover and uptime guarantees.

Question 126

**Which of the following is AWS's obligation under the AWS shared responsibility model?

**

OPTIONS

A. Data encryption in transit

B. Firmware updates on hardware

C. Operating system patching on Amazon EC2 instances

D. Data encryption at rest

[!TIP]

Correct Answer:

B. Firmware updates on hardware

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS’s responsibilities in the shared responsibility model — especially around infrastructure-level maintenance and physical security. The key phrase is “AWS’s obligation.”

WHY B IS CORRECT

Firmware updates on hardware ☒ Correct. AWS owns and operates the physical infrastructure Responsible for hardware lifecycle, including firmware, BIOS, and hypervisor patches Ensures security and performance of the underlying compute

SOURCES

AWS Shared Responsibility Model Security Pillar – Well-Architected Framework

DISTRACTOR BREAKDOWN

A. Data encryption in transit Shared — AWS provides tools, but customers configure and enforce

C. OS patching on EC2 Customer responsibility — EC2 is IaaS, not managed

D. Data encryption at rest Shared — AWS offers encryption, but customers choose and manage keys

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 workloads: You patch the OS and configure encryption AWS handles the firmware, hypervisor, and physical security Together, you maintain a secure and resilient environment That’s how AWS delivers layered infrastructure security with clear boundaries and operational trust.

Question 127

**To boost availability, a user intends to create two more Amazon EC2 instances. What should the user do?

**

OPTIONS

- A. Launch the instances across multiple Availability Zones in a single AWS Region
- B. Launch the instances as EC2 Reserved Instances in the same AWS Region and the same Availability Zone
- C. Launch the instances in multiple AWS Regions, but in the same Availability Zone
- D. Launch the instances as EC2 Spot Instances in the same AWS Region, but in different Availability Zones

[!TIP]

Correct Answer:

- A. Launch the instances across multiple Availability Zones in a single AWS Region

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of high availability architecture using EC2 — especially how Availability Zones (AZs) provide fault isolation and redundancy. The key phrase is “boost availability.”

WHY A IS CORRECT

Multiple AZs in one region ensures: Redundancy across physically isolated data centers Low-latency failover within the same region Consistent endpoint and region-level services Ideal for load balancing and auto scaling

SOURCES

AWS EC2 Placement Best Practices High Availability on AWS

DISTRACTOR BREAKDOWN

B. Reserved Instances in same AZ Reserved pricing doesn't affect availability — same AZ = single point of failure

C. Multiple regions, same AZs are region-specific — this setup is invalid

D. Spot Instances in different AZs Spot pricing is cost-driven — not guaranteed availability or persistence

REAL-WORLD TIE-IN

If your Nairobi-based team runs a web app on EC2: Deploy instances in us-east-1a and us-east-1b Use Elastic Load Balancer to distribute traffic Ensure zero downtime during AZ disruptions That's how AWS delivers resilient compute with zone-aware placement and seamless failover.

Question 128

****What are the customer's duties under the AWS shared responsibility model? (Select two.)**

OPTIONS

- A. Physical and environmental security
- B. Physical network devices including firewalls
- C. Storage device decommissioning
- D. Security of data in transit
- E. Data integrity authentication

[!TIP]

Correct Answer:

D. Security of data in transit and E. Data integrity authentication

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of customer responsibilities in cloud security — especially around data protection and integrity. The key phrase is “customer’s duties.” 🔍 Why D and E Are Correct

D.

Security of data in transit ✅ Customer responsibility. You must configure TLS/SSL, VPNs, and encryption protocols AWS provides tools, but you enforce and monitor them

E.

Data integrity authentication ✅ Customer responsibility. You ensure that data hasn’t been tampered with Use hashing, digital signatures, and validation mechanisms

SOURCES

AWS Shared Responsibility Model Security Pillar – Well-Architected Framework

DISTRACTOR BREAKDOWN

Option Why It's AWS's Responsibility
A. Physical and environmental security AWS secures data centers, power, HVAC, and access controls
B. Physical network devices AWS owns and maintains routers, switches, and firewalls
C. Storage device decommissioning AWS handles secure disposal and sanitization of hardware

REAL-WORLD TIE-IN

If your Nairobi-based team stores customer data in S3: You encrypt data in transit using HTTPS
You validate data integrity with checksums or signatures AWS secures the physical infrastructure and networking That's how AWS delivers layered security with clear boundaries and customer-controlled data protection.

Question 129

**A large enterprise with multiple VPCs in several AWS Regions around the world needs to connect and centrally manage network connectivity between its VPCs. Which AWS service or feature meets these requirements?

**

OPTIONS

A. AWS Direct Connect

B. AWS Transit Gateway

C. AWS Site-to-Site VPN

D. VPC endpoints

[!TIP]

Correct Answer:

B. AWS Transit Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of centralized network management across VPCs and regions — especially how AWS enables scalable, hub-and-spoke routing. The key phrase is “centrally manage network connectivity.”

WHY B IS CORRECT

AWS Transit Gateway provides: Centralized routing between thousands of VPCs and on-prem networks Inter-region peering for global connectivity Simplified management with route tables and attachments Scales to enterprise-grade architectures

SOURCES

Transit Gateway Overview Global Network Architecture

DISTRACTOR BREAKDOWN

A. Direct Connect Dedicated link to AWS — doesn’t manage VPC-to-VPC routing

C. Site-to-Site VPN Point-to-point — not scalable or centralized

D. VPC endpoints Private access to AWS services — not for inter-VPC connectivity

REAL-WORLD TIE-IN

If your Nairobi-based team manages VPCs in us-east-1, eu-west-1, and af-south-1: Use Transit Gateway to peer VPCs across regions Route traffic centrally with fine-grained control Integrate with Direct Connect and VPNs for hybrid workloads That's how AWS delivers global, scalable network management with centralized routing and enterprise-grade performance.

Question 130

**Which AWS service should be used to monitor Amazon EC2 instances for CPU and network utilization?

**

OPTIONS

A. Amazon Inspector

B. AWS CloudTrail

C. Amazon CloudWatch

D. AWS Config

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS monitoring and observability tools — especially which one tracks performance metrics like CPU and network usage. The key phrase is “monitor EC2 instances.”

WHY C IS CORRECT

Amazon CloudWatch provides: Real-time metrics for EC2: CPU, network I/O, disk usage
Custom dashboards and alarms Log collection and analysis Integration with Auto Scaling and Lambda

SOURCES

CloudWatch Metrics Overview Monitoring EC2 with CloudWatch

DISTRACTOR BREAKDOWN

A. Amazon Inspector Security assessment tool — not for performance metrics

B. AWS Cloud Trail Tracks API calls — not resource utilization

D. AWS Config Tracks configuration changes — not runtime metrics

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 workloads for a web app: Use CloudWatch to monitor CPU spikes and network throughput Set alarms for threshold breaches Trigger Auto Scaling or notifications based on metrics That's how AWS delivers observability and performance insights for scalable, resilient compute.

Question 131

**How can customers minimize the amount of time they spend patching their operating systems by migrating to the AWS Cloud? (Select two.)

**

OPTIONS

- A. Users can take advantage of managed services on AWS.
- B. Users can outsource operating system patching to the AWS Support team.
- C. AWS Professional Services will upgrade instances to the latest operating system versions.
- D. Users have the ability to use license-included Amazon EC2 instances.
- E. Users can take advantage of AWS Systems Manager features.

[!TIP]

Correct Answer:

A. Users can take advantage of managed services on AWS and E. Users can take advantage of AWS Systems Manager features

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of operational efficiency in cloud environments — especially how AWS helps reduce manual OS patching overhead. The key phrase is “minimize time spent patching.” 🔍 Why A and E Are Correct

A.

Managed services ✅ Correct. Services like RDS, DynamoDB, Lambda, and Fargate abstract away OS management AWS handles patching, scaling, and availability Customers focus on application logic, not infrastructure

E.

AWS Systems Manager ✅ Correct. Automates patch management across EC2 and hybrid environments Includes Patch Manager, State Manager, and Automation documents Enables compliance reporting and scheduling

SOURCES

AWS Systems Manager Patch Manager Managed Services Overview

DISTRACTOR BREAKDOWN

B. AWS Support team Support provides guidance — not operational patching

C. AWS Professional Services Offers consulting — not ongoing patching or upgrades

D. License-included EC2 Covers OS licensing — not patch automation or reduction

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 and RDS workloads: Use RDS for database hosting — no patching required Use Systems Manager to automate EC2 patching across environments

Reduce manual effort and improve compliance That's how AWS delivers operational excellence with automation and managed infrastructure.

REFERENCES

<https://aws.amazon.com/systems-manager/>

Question 132

**In the case of an environmental disruption, a company needs to make sure its infrastructure is structured for fault tolerance and business continuity. Which parts of the AWS architecture should the organization replicate?

**

OPTIONS

A. Edge locations

B. Availability Zones

C. Regions

D. Amazon Route 53

[!TIP]

Correct Answer:

C. Regions

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of disaster recovery and fault isolation in AWS architecture — especially how Regions provide geographic redundancy. The key phrase is “environmental disruption,” which implies regional-level failure (e.g., natural disasters, power grid collapse).

WHY C IS CORRECT

Regions are: Geographically isolated — each with multiple AZs
Designed to contain and isolate failures
Replicating across Regions ensures business continuity during large-scale disruptions
Enables active-active or active-passive DR strategies

SOURCES

AWS Global Infrastructure Disaster Recovery on AWS

DISTRACTOR BREAKDOWN

- A. Edge locations Used for content delivery — not infrastructure replication
- B. Availability Zones Good for high availability, but not sufficient for regional disruptions
- D. Amazon Route 53 DNS routing — doesn't replicate infrastructure or data

REAL-WORLD TIE-IN

If your Nairobi-based team runs a critical healthcare platform: Replicate workloads in af-south-1 and eu-west-1 Use Route 53 for failover routing Sync data with Cross-Region Replication or Global Datastore That's how AWS delivers resilient, region-aware architecture for fault tolerance and business continuity.

Question 133

**According to which AWS cloud design guideline, systems should minimize their interdependence?

**

OPTIONS

A. Scalability

B. Services, not servers

C. Removing single points of failure

D. Loose coupling

[!TIP]

Correct Answer:

D. Loose coupling

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cloud-native design principles — especially how loose coupling improves resilience, scalability, and maintainability. The key phrase is “minimize interdependence.”

WHY D IS CORRECT

Loose coupling means: Systems communicate through well-defined interfaces Components can fail, scale, or evolve independently Promotes fault isolation and modular architecture Enables event-driven and microservices patterns

SOURCES

AWS Well-Architected Framework Design Principles for Cloud-Native Apps

DISTRACTOR BREAKDOWN

A. Scalability Focuses on performance under load — not interdependence

B. Services, not servers Encourages managed services — doesn't address system coupling

C. Removing single points of failure Improves availability — but doesn't imply modular independence

REAL-WORLD TIE-IN

If your Nairobi-based team builds a microservices app: Use SNS + SQS for decoupled messaging Deploy services independently with Lambda or ECS Monitor and retry failures without cascading impact That's how AWS delivers modular, fault-tolerant architecture with loose coupling and graceful degradation.

Question 134

**Which tasks need the root user credentials for an AWS account? (Select two.)

OPTIONS

- A. Creating an Amazon EC2 key pair
- B. Removing an IAM user from the administrators group
- C. Changing the AWS Support plan
- D. Creating an Amazon CloudFront key pair
- E. Granting an IAM user full administrative access

[!TIP]

Correct Answer:

C. Changing the AWS Support plan and E. Granting an IAM user full administrative access

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of root-only operations in AWS — especially tasks that affect account-level configuration, billing, and access control. The key phrase is “need the root user credentials.” 🔍 Why C and E Are Correct

C.

Changing the AWS Support plan ☒ Root-only task. Involves billing and subscription tier changes IAM users cannot access the support plan console Requires root access to modify support levels (Basic, Developer, Business, Enterprise)

E.

Granting an IAM user full administrative access ☒ Root-only when bootstrapping. Initial IAM setup and full admin delegation require root credentials Especially critical when no existing

admin IAM user is available Ensures secure control over policy boundaries and access escalation

SOURCES

AWS Root User Tasks Support Plan Management

DISTRACTOR BREAKDOWN

A. Creating EC2 key pair IAM users with EC2 permissions can do this

B. Removing IAM user from admin group IAM users with group management rights can perform this

D. Creating Cloud Front key pair Deprecated — replaced by public key AP Is, no longer root-only

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding a new AWS account: Use root credentials to set up IAM roles and support plans Delegate access securely with least privilege policies Enable MFA and CloudTrail logging for root activity That's how AWS delivers secure account initialization with root-only boundaries and delegated operational control.

Question 135

**A user wants to deploy a service to the AWS Cloud by using infrastructure-as-code (IaC) principles. Which AWS service can be used to meet this requirement?

OPTIONS

A. AWS Systems Manager

B. AWS CloudFormation

C. AWS CodeCommit

D. AWS Config

[!TIP]

Correct Answer:

B. AWS CloudFormation

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of IaC tooling in AWS — especially which service enables declarative infrastructure deployment. The key phrase is “infrastructure-as-code.”

WHY B IS CORRECT

AWS CloudFormation: Allows users to define infrastructure using JSON or YAML templates
Supports automated provisioning of EC2, VPCs, IAM roles, RDS, Lambda, and more
Enables version-controlled, repeatable deployments
Integrates with CI/CD pipelines and drift detection

SOURCES

CloudFormation Overview IaC on AWS

DISTRACTOR BREAKDOWN

A. Systems Manager Manages resources — doesn't define infrastructure declaratively

C. Code Commit Git repository — stores code, not deploys infrastructure

D. AWS Config Tracks configuration changes — doesn't provision resources

REAL-WORLD TIE-IN

If your Nairobi-based team wants to deploy a multi-tier app: Use CloudFormation to define VPC, EC2, RDS, and IAM roles Store templates in CodeCommit or GitHub Automate deployments with CodePipeline or CDK That's how AWS delivers scalable, repeatable infrastructure with declarative templates and audit-friendly workflows.

Question 136

**A company that has multiple business units wants to centrally manage and govern its AWS Cloud environments. The company wants to automate the creation of AWS accounts, apply service control policies (SCPs), and simplify billing processes. Which AWS service or tool should the company use to meet these requirements?

**

OPTIONS

A. AWS Organizations

B. Cost Explorer

C. AWS Budgets

[!TIP]

Correct Answer:

A. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of multi-account governance and automation in AWS — especially how to manage SCPs, account creation, and consolidated billing. The key phrase is “centrally manage and govern.”

WHY A IS CORRECT

AWS Organizations enables: Automated account creation via APIs or CloudFormation Service Control Policies (SCPs) to enforce permission boundaries Consolidated billing across all accounts Organizational Units (OUs) for structured governance

SOURCES

AWS Organizations Overview SCPs and Account Automation

DISTRACTOR BREAKDOWN

B. Cost Explorer Visualizes spend — doesn’t manage accounts or policies

C. AWS Budgets Sets cost thresholds — no governance or account automation

D. Trusted Advisor Provides recommendations — doesn't manage accounts or SC Ps

REAL-WORLD TIE-IN

If your Nairobi-based team supports multiple departments: Use AWS Organizations to create accounts per unit Apply SCPs to restrict risky services View combined billing and optimize spend That's how AWS delivers enterprise-grade governance with automation, policy enforcement, and financial clarity.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 137

**An administrator must fast install and begin utilizing a popular IT product. What resources are available to the administrator?

**

OPTIONS

A. AWS Well-Architected Framework documentation

B. Amazon CloudFront

C. AWS CodeCommit

D. AWS Quick Start reference deployments

[!TIP]

Correct Answer:

D. AWS Quick Start reference deployments

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of accelerated infrastructure deployment — especially how AWS enables rapid provisioning of popular IT solutions using prebuilt templates and automation. The key phrase is “fast install and begin utilizing.”

WHY D IS CORRECT

AWS Quick Start reference deployments: Provide automated CloudFormation templates for popular IT products Include step-by-step guides and architecture diagrams Support fast, secure, and compliant deployments Cover solutions like Active Directory, SAP, Jenkins, Splunk, and more

SOURCES

AWS Quick Start Catalog Quick Start Reference Deployments

DISTRACTOR BREAKDOWN

A. Well-Architected Framework Offers design guidance — not deployment resources

B. Cloud Front Content delivery network — not for installing IT products

C. Code Commit Git repository — stores code, doesn't deploy infrastructure or products

REAL-WORLD TIE-IN

If your Nairobi-based team needs to deploy Active Directory or Jenkins: Use Quick Start to launch production-ready stacks in minutes Customize templates for scaling, security, and compliance Avoid manual setup and reduce time-to-value That's how AWS delivers accelerated, reference-driven deployments with audit-friendly infrastructure-as-code.

Question 138

**Which solution enables users in various AWS Regions to have the FASTEST application response times for frequently requested data?

**

OPTIONS

- A. AWS CloudTrail across multiple Availability Zones
- B. Amazon CloudFront to edge locations
- C. AWS CloudFormation in multiple regions
- D. A virtual private gateway over AWS Direct Connect

[!TIP]

Correct Answer:

B. Amazon CloudFront to edge locations

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of content delivery and latency optimization — especially how AWS accelerates access to frequently requested data across global regions. The key phrase is “fastest response times.”

WHY B IS CORRECT

Amazon CloudFront: Is a Content Delivery Network (CDN) Caches content at edge locations close to users Reduces latency and origin load Supports static and dynamic content, including APIs and video

SOURCES

CloudFront Overview Global Edge Network

DISTRACTOR BREAKDOWN

- A. Cloud Trail Logs API activity — doesn't serve data or reduce latency
- C. Cloud Formation Deploys infrastructure — not a delivery or caching service
- D. Direct Connect Improves hybrid connectivity — not global content acceleration

REAL-WORLD TIE-IN

If your Nairobi-based team serves media or APIs to users in Tokyo, Frankfurt, and São Paulo: Use CloudFront to cache content at local edge nodes Reduce round-trip latency and improve user experience Integrate with S3, EC2, or Lambda@Edge for dynamic delivery That's how AWS delivers low-latency, globally distributed access to frequently requested data.

Question 139

**Which qualities make AWS Cloud computing advantageous? (Select two.)

**

OPTIONS

A. A 100% service level agreement (SLA) for all AWS services

B. Compute capacity that is adjusted on demand

C. Availability of AWS Support for code development

D. Enhanced security

E. Increases in cost and complexity

[!TIP]

Correct Answer:

B. Compute capacity that is adjusted on demand and D. Enhanced security

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cloud-native benefits — especially how AWS delivers elasticity, security, and operational efficiency. The key phrase is “advantageous qualities.” 🔍

Why B and D Are Correct

B.

Compute capacity that is adjusted on demand  Elasticity and scalability. AWS lets you scale resources up or down based on demand Avoids overprovisioning and reduces cost Supports Auto Scaling, Lambda, and EC2 burstable instances

D.

Enhanced security  Built-in protection. AWS offers encryption, IAM, VPC isolation, DDoS protection, and compliance tooling Security is a shared responsibility, but AWS provides strong infrastructure-level safeguards

SOURCES

AWS Cloud Advantages AWS Benefits Overview

DISTRACTOR BREAKDOWN

A. 100% SLA for all services No AWS service offers a universal 100% SLA — most offer 99.9% or lower

C. AWS Support for code development Support helps troubleshoot — doesn't write or develop code

E. Increases in cost and complexity AWS reduces both via pay-as-you-go and managed services

REAL-WORLD TIE-IN

If your Nairobi-based team runs a seasonal e-commerce platform: Use Auto Scaling to handle traffic spikes Rely on AWS Shield and IAM for secure access Pay only for what you use — no upfront infrastructure costs That's how AWS delivers scalable, secure, and cost-efficient cloud computing for dynamic workloads.

Question 140

******A corporation anticipates a brief increase in internet traffic for their application. The program cannot be interrupted during the traffic spike. In addition, the organization must reduce costs while increasing flexibility. To achieve these needs, which Amazon EC2 instance type should the organization use?

OPTIONS

A. On-Demand Instances

B. Spot Instances

C. Reserved Instances

D. Dedicated Hosts

[!TIP]

Correct Answer:

A. On-Demand Instances

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of EC2 purchasing models — especially how to balance availability, cost, and flexibility during short-term traffic surges. The key phrases are “brief increase,” “cannot be interrupted,” and “reduce costs.”

WHY A IS CORRECT

On-Demand Instances: Provide instant provisioning with no long-term commitment Ideal for short-term, unpredictable workloads Guaranteed availability (within capacity limits) — no risk of termination Cost-effective when used briefly, without upfront payment

SOURCES

AWS EC2 Instance Types Explained Cost-Efficient EC2 Selection Guide

DISTRACTOR BREAKDOWN

B. Spot Instances Cheapest, but can be interrupted anytime — not suitable for critical workloads

C. Reserved Instances Cost-effective for long-term predictable usage — not flexible for short spikes

D. Dedicated Hosts Provide physical isolation — expensive and inflexible for short-term scaling

REAL-WORLD TIE-IN

If your Nairobi-based team expects a holiday traffic spike: Use On-Demand EC2 to scale up instantly Avoid long-term commitments or upfront costs Maintain uninterrupted service with full control over instance lifecycle That's how AWS delivers flexible, reliable compute for dynamic workloads with cost-aware scaling.

Question 141

**Which IT controls do AWS and the customer share, according to the AWS shared responsibility model? (Choose two.)

**

OPTIONS

A. Physical and environmental controls

B. Patch management

C. Cloud awareness and training

D. Zone security

E. Application data encryption

[!TIP]

Correct Answer:

B. Patch management and E. Application data encryption

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of shared security responsibilities in AWS — especially where both AWS and the customer contribute to protection and compliance. The key phrase is “shared IT controls.” 🔍 Why B and E Are Correct

B.

Patch management ☒ Shared depending on service type. For IaaS (e.g., EC2): customers patch the OS and applications For PaaS (e.g., RDS, Lambda): AWS handles underlying infrastructure patching Shared responsibility varies by service model

E.

Application data encryption ☒ Shared control. AWS provides encryption tools (KMS, TLS, etc.) Customers configure, manage keys, and enforce encryption policies Both parties ensure data is protected in transit and at rest

SOURCES

AWS Shared Responsibility Model Digital Cloud Training Cheat Sheet

DISTRACTOR BREAKDOWN

A. Physical and environmental controls Fully AWS's responsibility — includes data center security, HVAC, and power

C. Cloud awareness and training Fully customer's responsibility — AWS doesn't train your staff

D. Zone security Refers to physical AZ boundaries — AWS manages this entirely

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 and RDS workloads: You patch EC2 OS manually AWS patches RDS infrastructure You configure encryption policies using KMS and IAM roles That's how AWS delivers layered security with shared operational controls and service-aware boundaries.

Question 142

******A company is launching an application in the AWS Cloud. The application will use Amazon S3 storage. A large team of researchers will have shared access to the data. The company must be able to recover data that is accidentally overwritten or deleted. Which S3 feature should the company turn on to meet this requirement?

OPTIONS

A. Server access logging

B. S3 Versioning

C. S3 Lifecycle rules

D. Encryption in transit and at rest

[!TIP]

Correct Answer:

B. S3 Versioning

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of data protection and recovery in S3 — especially how to guard against accidental deletion or overwrites. The key phrase is “recover data.”

WHY B IS CORRECT

S3 Versioning: Preserves multiple versions of an object Allows recovery of previous versions after overwrite or deletion Works with MFA Delete for enhanced protection Essential for collaborative environments with shared access

SOURCES

S3 Versioning Overview Protecting Data in S3

DISTRACTOR BREAKDOWN

A. Server access logging Tracks requests — doesn't preserve or recover data

C. Lifecycle rules Automate transitions/deletions — not for recovery

D. Encryption Secures data — doesn't prevent or undo overwrites

REAL-WORLD TIE-IN

If your Nairobi-based research team stores datasets in S3: Enable Versioning to recover overwritten files Use Lifecycle rules to manage storage costs Combine with MFA Delete for tamper-resistant protection That's how AWS delivers resilient, version-aware storage for collaborative and audit-sensitive workloads.

Question 143

**An Amazon RDS database instance is deployed across several Availability Zones. Which pillar of the AWS Well-Architected Framework is included in this strategy?

**

OPTIONS

A. Performance efficiency

B. Reliability

C. Cost optimization

D. Security

[!TIP]

Correct Answer:

B. Reliability

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Well-Architected Framework, especially how multi-AZ deployments contribute to fault tolerance and recovery. The key phrase is “deployed across several Availability Zones.”

WHY B IS CORRECT

Reliability pillar focuses on: Recovering from failures Minimizing downtime Designing for high availability Multi-AZ RDS: Automatically replicates data to a standby in another AZ Performs automatic failover during disruptions Ensures business continuity without manual intervention

SOURCES

AWS Well-Architected Reliability Pillar RDS Multi-AZ Deployment

DISTRACTOR BREAKDOWN

A. Performance efficiency Focuses on selecting optimal resources and scaling — not fault tolerance or availability

C. Cost optimization Involves rightsizing, pricing models, and usage efficiency — multi-AZ increases cost for availability

D. Security Covers access control, encryption, and compliance — doesn't address availability or failover strategies

REAL-WORLD TIE-IN

If your Nairobi-based team runs a critical analytics platform: Deploy RDS in multi-AZ mode for automatic failover. Ensure zero downtime during AZ outages. Align with the Reliability pillar for audit and compliance. That's how AWS delivers resilient, fault-tolerant infrastructure aligned with Well-Architected best practices.

Question 144

**Amazon EC2, an Elastic Load Balancer, and Amazon RDS are all components of an architectural design. What is the BEST method for estimating the monthly cost of this architecture?

**

OPTIONS

- A. Open an AWS Support case, provide the architecture proposal, and ask for a monthly cost estimation
- B. Collect the published prices of the AWS services and calculate the monthly estimate
- C. Use the AWS Simple Monthly Calculator to estimate the monthly cost
- D. Use the AWS Total Cost of Ownership (TCO) Calculator to estimate the monthly cost

[!TIP]

Correct Answer:

- C. Use the AWS Simple Monthly Calculator to estimate the monthly cost

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cost estimation tools in AWS — especially how to forecast monthly spend based on service usage. The key phrase is “monthly cost of this architecture.”

WHY C IS CORRECT

AWS Simple Monthly Calculator: Designed to estimate monthly costs for AWS services. Lets you input usage parameters for EC2, ELB, RDS, S3, etc. Provides granular breakdowns by region, instance type, storage, and data transfer. Ideal for pre-deployment budgeting and cost modeling.

SOURCES

AWS Pricing Calculator
Cost Estimation Best Practices

DISTRACTOR BREAKDOWN

A. AWS Support case Support doesn't provide cost estimates — they guide usage and troubleshooting

B. Manual price collection Time-consuming and error-prone — lacks automation and regional pricing logic

D. TCO Calculator Compares AWS vs on-premises costs — not for estimating monthly AWS architecture spend

REAL-WORLD TIE-IN

If your Nairobi-based team is planning a multi-tier web app: Use the Simple Monthly Calculator to model EC2, RDS, and ELB costs Adjust parameters for instance size, uptime, and data transfer Export estimates for budget approval and stakeholder review That's how AWS delivers transparent, customizable cost modeling for cloud-native architectures.

Question 144

**Amazon EC2, an Elastic Load Balancer, and Amazon RDS are all components of an architectural design. What is the BEST method for estimating the monthly cost of this architecture?

**

OPTIONS

- A. Open an AWS Support case, provide the architecture proposal, and ask for a monthly cost estimation
- B. Collect the published prices of the AWS services and calculate the monthly estimate
- C. Use the AWS Simple Monthly Calculator to estimate the monthly cost
- D. Use the AWS Total Cost of Ownership (TCO) Calculator to estimate the monthly cost

[!TIP]

Correct Answer:

- B. Collect the published prices of the AWS services and calculate the monthly estimate

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of manual cost estimation — especially when you want direct control over pricing inputs. The key phrase is “BEST method,” which in some contexts favors precision over convenience.

WHY B IS CORRECT

Manual price collection: Ensures real-time accuracy from AWS pricing pages
Allows for custom calculations based on usage patterns
Useful when modeling complex or non-standard architectures
Avoids assumptions baked into calculators

SOURCES

AWS Pricing Overview EC2 Pricing RDS Pricing ELB Pricing

DISTRACTOR BREAKDOWN

A. AWS Support case Support won't calculate costs — they guide usage and troubleshooting

C. Simple Monthly Calculator Helpful, but may be deprecated or replaced by the Pricing Calculator — not always precise for custom setups

D. TCO Calculator Compares AWS vs on-premises — not for estimating monthly AWS spend

REAL-WORLD TIE-IN

If your Nairobi-based team is modeling a hybrid workload with EC2, RDS, and ELB: Pull current pricing from AWS service pages
Calculate based on uptime, data transfer, and storage tiers
Validate against budget constraints and scaling forecasts
That's how AWS enables transparent, customizable cost modeling with direct control over pricing inputs.

Question 145

**According to the AWS shared responsibility model, which job is the customer's duty?

**

OPTIONS

- A. Maintain the security of the AWS Cloud
- B. Configure firewalls and networks
- C. Patch the operating system of Amazon RDS instances
- D. Implement physical and environmental controls

[!TIP]

Correct Answer:

- B. Configure firewalls and networks

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model, which defines who secures what in the cloud. The key phrase is “customer’s duty.”

WHY B IS CORRECT

Configure firewalls and networks: Customers manage security groups, NACLs, and VPC configurations. They define inbound/outbound rules, subnet segmentation, and routing. AWS provides the infrastructure — customers secure their workloads.

SOURCES

Shared Responsibility Model Security Best Practices

DISTRACTOR BREAKDOWN

A. Maintain the security of the AWS Cloud AWS handles infrastructure security — customers secure their workloads in the cloud, not of the cloud

C. Patch the OS of Amazon RDS instances AWS manages RDS patching — customers patch EC2, not managed services

D. Physical and environmental controls AWS owns and secures data centers — customers never touch physical infrastructure

REAL-WORLD TIE-IN

If your Nairobi-based team deploys EC2 and RDS: You configure VPC firewalls and IAM roles AWS patches RDS and secures the physical layer You monitor network traffic and access policies That's how AWS delivers shared security boundaries with customer-managed controls and infrastructure-level protections.

Question 146

**On Amazon EC2, a business hosts a web application in a Docker container. Which of the following duties is AWS in charge of?

**

OPTIONS

- A. Scaling the web application and services developed with Docker
- B. Provisioning or scheduling containers to run on clusters and maintain their availability
- C. Performing hardware maintenance in the AWS facilities that run the AWS Cloud
- D. Managing the guest operating system, including updates and security patches

[!TIP]

Correct Answer:

- C. Performing hardware maintenance in the AWS facilities that run the AWS Cloud

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model in the context of EC2-hosted containers. The key phrase is “AWS in charge of.”

WHY C IS CORRECT

Hardware maintenance: AWS is responsible for physical infrastructure: servers, networking, power, cooling, and data center operations This includes replacing failed hardware, maintaining physical security, and ensuring availability zones remain operational Customers never touch or manage the underlying hardware

SOURCES

AWS Shared Responsibility Model EC2 Hosting Responsibilities

DISTRACTOR BREAKDOWN

A. Scaling the web application and services developed with Docker Customers manage scaling unless using ECS or EKS — EC2 doesn't auto-scale containers

B. Provisioning or scheduling containers to run on clusters and maintain their availability That's the job of Amazon ECS, EKS, or Docker Swarm — not EC2 itself

D. Managing the guest operating system, including updates and security patches On EC2, the customer manages the OS — AWS only handles the hypervisor and hardware layer

REAL-WORLD TIE-IN

If your Nairobi-based team runs Docker containers on EC2: You manage OS patching, Docker runtime, and container orchestration AWS ensures the hardware stays healthy and secure For automated container scheduling, consider ECS or EKS That's how AWS delivers infrastructure-level reliability while customers retain full control over the operating system and application stack.

Question 147

**What are the AWS Cloud's advantages? (Select two.)

**

OPTIONS

A. Fixed rate monthly cost

B. No need to guess capacity requirements

C. Increased speed to market

D. Increased upfront capital expenditure

E. Physical access to cloud data centers

[!TIP]

Correct Answer:

B. No need to guess capacity requirements and C. Increased speed to market

EXPLANATION

CONCEPTUAL FRAMING

This question tests your grasp of cloud-native benefits — especially how AWS enables elasticity, agility, and cost efficiency. The key phrase is “advantages.” 🔍 Why B and C Are Correct

B.

No need to guess capacity requirements ☒ Elastic scaling. AWS lets you scale resources up/down based on demand Avoids overprovisioning and underutilization Supports Auto Scaling, Lambda, and pay-as-you-go pricing Cited as a top benefit in AWS’s official documentation

C.

Increased speed to market ☒ Rapid provisioning. AWS enables launching infrastructure in minutes Eliminates hardware procurement delays Supports CI/CD pipelines, serverless apps, and global reach Drives faster innovation and deployment cycles

DISTRACTOR BREAKDOWN

A. Fixed rate monthly cost AWS uses variable pricing — based on usage, not fixed rates

D. Increased upfront capital expenditure AWS reduces upfront costs — shifts Cap Ex to Op Ex with on-demand billing

E. Physical access to cloud data centers Customers never access AWS data centers — AWS handles physical security

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics platform: Use Auto Scaling and Lambda to handle unpredictable traffic Deploy globally in minutes with CloudFormation or CDK Avoid hardware delays and upfront costs That's how AWS delivers agile, scalable infrastructure with rapid deployment and cost-aware elasticity.

Question 148

**An Elastic Load Balancer, numerous Amazon EC2 instances, and Amazon RDS are used to run a web application on AWS. Which security measures are AWS's responsibility? (Select two.)

**

OPTIONS

A. Running a virus scan on EC2 instances

B. Protecting against IP spoofing and packet sniffing

C. Installing the latest security patches on the RDS instance

D. Encrypting communication between the EC2 instances and the Elastic Load Balancer

E. Configuring a security group and a network access control list (NACL) for EC2 instances

[!TIP]

Correct Answer:

B. Protecting against IP spoofing and packet sniffing and C. Installing the latest security patches on the RDS instance

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model, especially how AWS handles infrastructure-level and managed service security, while customers manage application-level and configuration controls. 🔍 Why B and C Are Correct

B.

Protecting against IP spoofing and packet sniffing ✅ Infrastructure-level security. AWS secures its network fabric against spoofing, sniffing, and other low-level attacks This includes hypervisor isolation, VPC traffic filtering, and data center protections

C.

Installing the latest security patches on the RDS instance ✅ Managed service responsibility. AWS handles OS patching and database engine updates for Amazon RDS Customers configure backups and access, but AWS maintains the underlying infrastructure

SOURCES

AWS Shared Responsibility Model RDS Maintenance and Patching

DISTRACTOR BREAKDOWN

A. Running a virus scan on EC2 instances Customers manage EC2 OS and security — AWS doesn't scan your instances

D. Encrypting communication between EC2 and ELB Customers configure TLS certificates and listener settings — AWS provides the tools, not the config

E. Configuring security groups and NACLs Customers define these rules — AWS enforces them but doesn't create or manage them

REAL-WORLD TIE-IN

If your Nairobi-based team runs a web app on EC2 behind an ELB with RDS: AWS secures the network layer and patches RDS. You configure firewalls, TLS, and EC2 antivirus. Use IAM roles and CloudTrail to monitor access and changes. That's how AWS delivers layered security with clear boundaries between infrastructure protections and customer-managed configurations.

Question 149

**A manufacturing company has a critical application that runs at a remote site with a slow internet connection. The application is sensitive to latency and interruptions. Which AWS service or feature should the company use to meet these requirements?

**

OPTIONS

A. Availability Zones

B. AWS Local Zones

C. AWS Wavelength

D. AWS Outposts

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS edge infrastructure — especially how to support latency-sensitive workloads in remote or underserved areas. The key phrases are “slow internet,” “critical application,” and “minimum latency.”

WHY B IS CORRECT

AWS Local Zones: Extend AWS infrastructure closer to end users in metro areas Reduce latency by hosting compute, storage, and database services outside core regions Ideal for real-time applications like manufacturing control systems, media rendering, and healthcare diagnostics Operate independently of internet quality — traffic doesn't need to traverse long distances to reach a core region

SOURCES

AWS Local Zones Overview AWS Wavelength vs Local Zones

DISTRACTOR BREAKDOWN

A. Availability Zones Exist within AWS Regions — not close enough to remote sites to reduce latency

C. AWS Wavelength Designed for mobile edge computing via telecom networks — not suitable for fixed-site manufacturing workloads

D. AWS Outposts Brings AWS to on-premises data centers — requires stable connectivity to AWS Region, not ideal for poor internet conditions

REAL-WORLD TIE-IN

If your Nairobi-based manufacturing site runs latency-sensitive control software: Deploy workloads in a Local Zone nearest to the site Avoid long-haul traffic to core regions Maintain low-latency access even with limited internet bandwidth That's how AWS delivers edge-native infrastructure for remote, latency-critical workloads with minimal connectivity dependencies.

Question 150

**Which AWS service or feature facilitates the purchase and deployment of third-party software by providing an online, managed software catalog?

**

OPTIONS

A. AWS Support

B. AWS Marketplace

C. Amazon EC2 private Amazon Machine Images (AMIs)

D. AWS reseller programs

[!TIP]

Correct Answer:

B. AWS Marketplace

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of third-party software procurement in AWS — especially how AWS enables automated deployment and licensing via a curated catalog. The key phrase is “purchase and deployment... managed software catalog.”

WHY B IS CORRECT

AWS Marketplace: Offers a digital catalog of third-party software Includes SaaS, AMIs, containers, and data products Supports one-click deployment into your AWS environment Handles billing, licensing, and updates Used by enterprises to streamline procurement and compliance

SOURCES

AWS Marketplace Overview Deploying Marketplace Products

DISTRACTOR BREAKDOWN

A. AWS Support Provides technical guidance — doesn't sell or deploy third-party software

C. EC2 private AM Is Used for internal deployment — not a catalog or procurement tool

D. AWS reseller programs Focused on partner sales — not customer-facing software deployment or catalog access

REAL-WORLD TIE-IN

If your Nairobi-based team needs to deploy Splunk, Trend Micro, or Tableau: Use AWS Marketplace to purchase and launch pre-configured AMIs or SaaS Integrate with CloudFormation or Service Catalog for governance Track usage and billing via Consolidated

Billing and Cost Explorer That's how AWS delivers frictionless third-party software deployment with licensing, billing, and compliance built in.

Question 151

**A business wishes to improve its capacity for infrastructure recovery in the event of a natural catastrophe. This capability corresponds to which pillar of the AWS Well-Architected Framework?

**

OPTIONS

A. Cost optimization

B. Performance efficiency

C. Reliability

D. Security

[!TIP]

Correct Answer:

C. Reliability

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Well-Architected Framework, especially how it guides resilient infrastructure design. The key phrase is “infrastructure recovery in the event of

a natural catastrophe.”

WHY C IS CORRECT

Reliability pillar focuses on: Recovering from failures Designing for high availability and fault tolerance Automating recovery and failover Ensuring workloads can survive disasters and continue operating

SOURCES

AWS Reliability Pillar Disaster Recovery Strategies

DISTRACTOR BREAKDOWN

A. Cost optimization Focuses on minimizing spend — not on recovery or fault tolerance

B. Performance efficiency Optimizes resource selection and scaling — doesn’t address disaster recovery

D. Security Protects data and access — doesn’t ensure infrastructure recovery or availability during disasters

REAL-WORLD TIE-IN

If your Nairobi-based team runs a critical analytics platform: Use multi-AZ deployments, backups, and failover automation Align with the Reliability pillar to ensure continuity during floods, power outages, or fiber cuts Combine with CloudWatch alarms and Route 53 health checks for proactive failover That’s how AWS delivers resilient, disaster-ready infrastructure aligned with Well-Architected best practices.

Question 151

**A business wishes to improve its capacity for infrastructure recovery in the event of a natural catastrophe. This capability corresponds to which pillar of the AWS Well-Architected Framework?

**

OPTIONS

A. Cost optimization

B. Performance efficiency

C. Reliability

D. Security

[!TIP]

Correct Answer:

C. Reliability

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Well-Architected Framework, especially how it guides resilient infrastructure design. The key phrase is “infrastructure recovery in the event of a natural catastrophe.”

WHY C IS CORRECT

Reliability pillar focuses on: Recovering from failures Designing for high availability and fault tolerance Automating recovery and failover Ensuring workloads can survive disasters and

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SOURCES

AWS Reliability Pillar Disaster Recovery Strategies

DISTRACTOR BREAKDOWN

A. Cost optimization Focuses on minimizing spend — not on recovery or fault tolerance

B. Performance efficiency Optimizes resource selection and scaling — doesn't address disaster recovery

D. Security Protects data and access — doesn't ensure infrastructure recovery or availability during disasters

REAL-WORLD TIE-IN

If your Nairobi-based team runs a critical analytics platform: Use multi-AZ deployments, backups, and failover automation Align with the Reliability pillar to ensure continuity during floods, power outages, or fiber cuts Combine with CloudWatch alarms and Route 53 health checks for proactive failover That's how AWS delivers resilient, disaster-ready infrastructure aligned with Well-Architected best practices.

Question 153

**For security reasons, a business demands an isolated environment inside AWS. Which course of action is necessary to achieve this?

OPTIONS

- A. Create a separate Availability Zone to host the resources
- B. Create a separate VPC to host the resources
- C. Create a placement group to host the resources
- D. Create an AWS Direct Connect connection between the company and AWS

[!TIP]

Correct Answer:

- B. Create a separate VPC to host the resources

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of network isolation in AWS — especially how to create logically separated environments for security and compliance. The key phrase is “isolated environment.”

WHY B IS CORRECT

Amazon VPC (Virtual Private Cloud): Provides a logically isolated section of the AWS Cloud. You control IP ranges, subnets, route tables, firewalls, and gateways. Ideal for multi-tenant isolation, compliance zones, and secure app segmentation. Supports private-only workloads, hybrid connectivity, and granular IAM policies.

SOURCES

DISTRACTOR BREAKDOWN

A. Availability Zone AZs are physical fault domains, not isolation boundaries — workloads in the same VPC can span AZs

C. Placement group Used for performance optimization (e.g., low-latency clustering) — not for security or isolation

D. AWS Direct Connect Provides private network connectivity — doesn't isolate workloads inside AWS itself

REAL-WORLD TIE-IN

If your Nairobi-based team needs to isolate workloads for financial compliance or multi-tenant SaaS: Create a dedicated VPC per tenant or environment Use security groups, NACLs, and IAM boundaries Optionally connect via Direct Connect or VPN for hybrid isolation That's how AWS delivers secure, logically isolated environments with full control over networking and access boundaries.

REFERENCES

<https://aws.amazon.com/vpc/>

Question 154

**How do Amazon's massive economies of scale help customers?

**

OPTIONS

- A. Periodic price reductions as the result of Amazon's operational efficiencies
- B. New Amazon EC2 instance types providing the latest hardware
- C. The ability to scale up and down when needed
- D. Increased reliability in the underlying hardware of Amazon EC2 instances

[!TIP]

Correct Answer:

- A. Periodic price reductions as the result of Amazon's operational efficiencies

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cloud economics — especially how AWS’s global scale translates into customer benefits. The key phrase is “economies of scale.”

WHY A IS CORRECT

Economies of scale: AWS operates at massive global scale — buying hardware, bandwidth, and energy in bulk. These efficiencies allow AWS to reduce prices over time. Customers benefit from lower costs without sacrificing performance. AWS has reduced prices over 100 times since launch.

SOURCES

AWS Cloud Economics AWS Pricing Reductions History

DISTRACTOR BREAKDOWN

B. New EC2 instance types Driven by innovation, not economies of scale — reflects tech evolution, not cost reduction

C. Scale up/down when needed A core cloud benefit — but tied to elasticity, not economies of scale

D. Increased reliability of EC2 hardware Reliability is a design goal, not a direct result of economic scale — handled via AZs and instance health checks

REAL-WORLD TIE-IN

If your Nairobi-based team runs a cost-sensitive analytics platform: You benefit from AWS's bulk purchasing and operational efficiency Your EC2, S3, and Lambda costs drop over time You scale globally without absorbing infrastructure overhead That's how AWS delivers cost savings through scale, passing operational efficiencies directly to customers.

Question 156

**Which AWS service uses edge locations?

**

OPTIONS

A. Amazon Aurora

B. AWS Global Accelerator

C. Amazon Connect

D. AWS Outposts

[!TIP]

Correct Answer:

B. AWS Global Accelerator

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS edge services — especially how AWS uses global edge locations to optimize latency, routing, and availability. The key phrase is “uses edge locations.”

WHY B IS CORRECT

AWS Global Accelerator: Uses the AWS global network of edge locations to route traffic
Improves performance and availability for global applications
Automatically directs users to the optimal endpoint based on health and proximity
Ideal for multi-region apps, gaming platforms, and latency-sensitive workloads

SOURCES

AWS Global Accelerator Overview Edge Location Use Cases

DISTRACTOR BREAKDOWN

A. Amazon Aurora A regional database service — doesn't use edge locations for traffic routing

C. Amazon Connect A contact center service — runs in AWS Regions, not edge locations

D. AWS Outposts Brings AWS to on-premises data centers — not part of the edge location network

REAL-WORLD TIE-IN

If your Nairobi-based team runs a global multiplayer game or real-time analytics dashboard: Use AWS Global Accelerator to route traffic through edge locations Reduce latency and improve failover resilience Combine with CloudFront and Route 53 for full edge-native delivery That's how AWS delivers low-latency, globally optimized traffic routing using its edge location infrastructure.

Question 157

**When utilizing the AWS Command Line Interface (AWS CLI), which of the following Identity and Access Management (IAM) entities is connected with an access key ID and secret access key?

**

OPTIONS

A. IAM group

B. IAM user

C. IAM role

D. IAM policy

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of IAM credential boundaries — especially how access keys are tied to identity entities for CLI and API access. The key phrase is “connected with an access key ID and secret access key.”

WHY B IS CORRECT

IAM user: Represents a human or application identity Can be assigned access key ID and secret access key for CLI/API use Credentials are stored in ~/.aws/credentials for CLI access IAM users can also have passwords for AWS Management Console

SOURCES

IAM User Credentials AWS CLI Authentication

DISTRACTOR BREAKDOWN

A. IAM group A container for users — cannot have credentials or be used directly for authentication

C. IAM role Used for temporary credentials via STS — not directly tied to access keys unless assumed

D. IAM policy Defines permissions — not an identity and cannot hold credentials

REAL-WORLD TIE-IN

If your Nairobi-based team uses the AWS CLI for deployment automation: Create IAM users with scoped permissions Generate access keys for CLI use Rotate keys regularly and monitor with CloudTrail and IAM Access Analyzer That's how AWS delivers secure, identity-bound access for CLI and API operations with credential hygiene and auditability.

Question 158

**What is AWS's sole obligation under the AWS shared responsibility model?

**

OPTIONS

A. Application security

B. Edge location management

C. Patch management

D. Client-side data

[!TIP]

Correct Answer:

B. Edge location management

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS’s infrastructure responsibilities under the shared responsibility model. The key phrase is “sole obligation,” which points to what AWS exclusively manages — not shared or customer-controlled elements.

WHY B IS CORRECT

Edge location management: AWS is solely responsible for operating and securing its global infrastructure, including edge locations used by services like CloudFront and Global Accelerator. This includes hardware maintenance, physical security, and network integrity. Customers never interact directly with edge infrastructure — AWS owns and operates it end-to-end.

SOURCES

AWS Shared Responsibility Model Edge Location Overview

DISTRACTOR BREAKDOWN

A. Application security: Customers secure their own applications — AWS provides tools, not enforcement.

C. Patch management: Shared responsibility — AWS patches managed services (e.g., RDS), but customers patch EC2 and OS-level components.

D. Client-side data: Customers are responsible for encrypting, securing, and managing their own data — AWS doesn’t touch it.

REAL-WORLD TIE-IN

If your Nairobi-based team deploys a global app using CloudFront: AWS manages the edge locations that serve content. You manage IAM, encryption, and application logic. AWS ensures availability and security of the infrastructure, while you control data and access. That’s how AWS

delivers infrastructure-level reliability while customers retain control over workloads and data security.

Question 159

**Which component of the AWS architecture permits global computing and storage deployment?

**

OPTIONS

A. Availability Zones

B. Regions

C. Tags

D. Resource groups

[!TIP]

Correct Answer:

B. Regions

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS global infrastructure — especially how AWS enables geographically distributed deployments. The key phrase is “global computing and storage deployment.”

WHY B IS CORRECT

AWS Regions: Are geographically isolated areas where AWS clusters data centers Each Region contains multiple Availability Zones (AZs) Customers choose Regions to deploy workloads close to users, meet compliance, or optimize latency Enables global reach — from Nairobi to Tokyo to São Paulo

SOURCES

AWS Global Infrastructure Guide AWS Architecture Center

DISTRACTOR BREAKDOWN

A. Availability Zones AZs are within Regions — they support high availability, not global deployment

C. Tags Used for resource labeling and cost tracking — not for deployment or architecture

D. Resource groups Help organize resources — don't affect geographic deployment or infrastructure layout

REAL-WORLD TIE-IN

If your Nairobi-based team wants to deploy a latency-sensitive app for users in Europe and Asia: Choose Regions like Frankfurt and Singapore Use multi-region S3 replication and EC2 deployments Optimize for latency, compliance, and disaster recovery That's how AWS delivers global infrastructure with regional control and scalable deployment options.

Question 161

****Which activity is entirely the user's responsibility while executing workloads on AWS?**

OPTIONS

- A. Patching the infrastructure components
- B. Implementing controls to route application traffic
- C. Maintaining physical and environmental controls
- D. Maintaining the underlying infrastructure components

[!TIP]

Correct Answer:

B. Implementing controls to route application traffic

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS shared responsibility model, especially the application-layer responsibilities that customers retain. The key phrase is “entirely the user's responsibility.”

WHY B IS CORRECT

Routing application traffic: Customers configure load balancers, DNS records, routing rules, and firewall policies AWS provides the tools (e.g., ELB, Route 53), but you define how traffic flows This includes: Failover logic Geo-routing Path-based routing Security group and NACL configurations

SOURCES

AWS Shared Responsibility Model Traffic Routing with Route 53

DISTRACTOR BREAKDOWN

A. Patching infrastructure components AWS handles patching for managed services — customers patch EC2 and OS-level components

C. Physical and environmental controls AWS owns and secures data centers — customers never touch physical infrastructure

D. Underlying infrastructure components AWS maintains servers, networking, and storage — customers operate above the virtualization layer

REAL-WORLD TIE-IN

If your Nairobi-based team runs a multi-tenant SaaS platform: You define routing logic for each tenant Use Route 53, ALB, and WAF to control traffic flow AWS ensures infrastructure availability — you control how users reach your app That's how AWS delivers infrastructure-level reliability while customers retain full control over traffic routing and application behavior.

Question 162

**Which statement best describes the AWS Cloud's agility?

**

OPTIONS

A. Agility gives users the ability to host applications in multiple AWS Regions around the world

B. Agility gives users the ability to pay upfront to reduce cost

C. Agility provides customizable physical hardware at the lowest possible cost

D. Agility provides the means for users to provision resources in minutes

[!TIP]

Correct Answer:

D. Agility provides the means for users to provision resources in minutes

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cloud agility — especially how AWS enables rapid experimentation, deployment, and scaling. The key phrase is “best describes... agility.”

WHY D IS CORRECT

Agility in AWS: Refers to the ability to quickly provision and de-provision resources Supports rapid iteration, testing, and scaling Enables teams to move from idea to deployment in minutes, not weeks Core to DevOps, CI/CD, and innovation workflows

SOURCES

AWS Cloud Benefits Well-Architected Framework – Agility

DISTRACTOR BREAKDOWN

A. Hosting in multiple Regions Describes global reach, not agility — agility is about speed and responsiveness

B. Pay upfront to reduce cost Describes cost optimization, not agility — agility favors on-demand flexibility

C. Customizable physical hardware AWS abstracts hardware — agility comes from virtualized, scalable infrastructure, not physical customization

REAL-WORLD TIE-IN

If your Nairobi-based team wants to launch a new analytics dashboard: Use CloudFormation or CDK to spin up infrastructure in minutes Test multiple configurations without long procurement cycles Iterate quickly with Lambda, EC2, and S3 — all provisioned on demand That's how AWS delivers agility through rapid provisioning, enabling faster innovation and reduced time-to-market.

Question 163

****Which AWS product or service enables businesses to monitor and classify their expenditure at a precise level? ****

OPTIONS

A. Cost allocation tags

B. Consolidated billing

C. AWS Budgets

[!TIP]

Correct Answer:

A. Cost allocation tags

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS cost management tools, especially how to classify and track spending across teams, projects, or environments. The key phrase is “monitor and classify... at a precise level.”

WHY A IS CORRECT

Cost allocation tags: Allow you to label AWS resources with metadata (e.g., Project=Analytics, Team=DevOps) Enable granular cost tracking in the AWS Cost Explorer and billing reports Help businesses attribute costs to departments, environments, or workloads Support chargeback models and budget accountability

SOURCES

AWS Cost Allocation Tags Cost Explorer Tag Filtering

DISTRACTOR BREAKDOWN

B. Consolidated billing Combines bills across accounts — doesn’t classify or tag individual resource costs

C. AWS Budgets Sets spending thresholds — doesn't classify or label costs at a granular level

D. AWS Marketplace Used for third-party software purchases — not for cost classification or tracking

REAL-WORLD TIE-IN

If your Nairobi-based team runs multi-project workloads: Tag resources with Environment=Prod, Project=Onboarding, Owner=Leonard Use Cost Explorer to break down spend by tag Align with chargeback models and budget alerts That's how AWS delivers precise cost classification with tagging, enabling accountability and financial transparency across cloud operations.

Question 164

****A company needs to install an application in a Docker container. Which AWS service eliminates the need to provision and manage the container hosts? ****

OPTIONS

A. AWS Fargate

B. Amazon FSx for Windows File Server

C. Amazon Elastic Container Service (Amazon ECS)

D. Amazon EC2

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of serverless container orchestration — especially how AWS abstracts host management for Docker workloads. The key phrase is “eliminates the need to provision and manage the container hosts.”

WHY A IS CORRECT

AWS Fargate: A serverless compute engine for containers Works with Amazon ECS and EKS You define the container specs — Fargate handles: Provisioning and scaling Patching and securing the infrastructure Resource isolation and billing per container Ideal for teams that want containerized deployments without managing EC2 instances

SOURCES

AWS Fargate Overview Fargate vs EC2 Launch Types

DISTRACTOR BREAKDOWN

B. Amazon FSx for Windows File Server A managed file system — not related to container hosting

C. Amazon ECS Requires a launch type — ECS alone doesn’t eliminate host management unless paired with Fargate

D. Amazon EC2 You must provision, patch, and manage the instances — opposite of host abstraction

REAL-WORLD TIE-IN

If your Nairobi-based team wants to deploy a microservice in Docker: Use ECS with Fargate launch type Skip EC2 provisioning and focus on task definitions and scaling policies Monitor with CloudWatch and IAM roles per task That's how AWS delivers containerized agility with serverless infrastructure management via Fargate.

Question 165

****Which AWS service is responsible for monitoring the health of your applications automatically? ****

OPTIONS

A. Amazon API Gateway

B. AWS Elastic Beanstalk

C. AWS Lambda

D. AWS Config

[!TIP]

Correct Answer:

B. AWS Elastic Beanstalk

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of application-level health monitoring in AWS — especially services that offer automated orchestration and diagnostics. The key phrase is “monitoring the health... automatically.”

WHY B IS CORRECT

AWS Elastic Beanstalk: A PaaS offering that deploys and manages applications Automatically monitors: Instance health Application responsiveness Environment status Provides health dashboards, alerts, and auto-recovery Ideal for teams that want managed infrastructure with built-in health checks

SOURCES

Elastic Beanstalk Health Monitoring Elastic Beanstalk Overview

DISTRACTOR BREAKDOWN

A. Amazon API Gateway Routes API traffic — doesn’t monitor application health automatically

C. AWS Lambda Executes code — doesn’t provide health dashboards or environment monitoring

D. AWS Config Tracks configuration changes — not designed for application health monitoring

REAL-WORLD TIE-IN

If your Nairobi-based team deploys a web app with Elastic Beanstalk: AWS handles deployment, scaling, and health checks You get real-time health metrics and alerts You focus on code — AWS handles the ops That’s how AWS delivers automated health monitoring with minimal operational overhead using Elastic Beanstalk.

Question 166

****A user is developing a service that adheres to the AWS Well-Architected Framework's operational excellence pillar. Which design concept is the user to adhere to? ****

OPTIONS

- A. Anticipate failure
- B. Make large-scale changes
- C. Perform manual operations
- D. Create static operational procedures

[!TIP]

Correct Answer:

A. Anticipate failure

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the Operational Excellence pillar, which focuses on continuous improvement, automation, and proactive risk management. The key phrase is “design concept... operational excellence.”

WHY A IS CORRECT

Anticipate failure is a core design principle under Operational Excellence: Encourages building systems that expect and tolerate faults Promotes automated recovery, monitoring, and alerting

Supports chaos engineering, game days, and fault injection to validate resilience Helps teams learn from incidents and improve continuously

SOURCES

AWS Operational Excellence Design Principles Well-Architected Labs – Operational Excellence

DISTRACTOR BREAKDOWN

B. Make large-scale changes Violates change control best practices — Operational Excellence favors small, reversible changes

C. Perform manual operations Manual steps are error-prone — the framework promotes automation and infrastructure as code

D. Create static operational procedures Static procedures don't adapt — Operational Excellence requires continuous improvement and agility

REAL-WORLD TIE-IN

If your Nairobi-based team is building a payment service: Design with failure scenarios in mind — simulate outages and latency spikes Use CloudWatch, X-Ray, and Lambda Destinations to monitor and recover Document lessons learned and refine runbooks after each incident That's how AWS enables operational excellence through proactive failure anticipation and continuous improvement.

Question 167

****When comparing AWS Total Cost of Ownership (TCO) to on-premises TCO, what charges are included? ****

OPTIONS

A. Project management

B. Antivirus software licensing

C. Data center security

D. Software development

[!TIP]

Correct Answer:

C. Data center security

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of TCO comparisons between AWS and traditional on-premises infrastructure. The key phrase is “charges included,” which refers to costs AWS absorbs that customers would otherwise manage themselves.

WHY C IS CORRECT

Data center security: AWS owns and operates its global infrastructure, including physical security, surveillance, access control, and environmental safeguards. These costs are included in AWS pricing and factored into TCO comparisons. On-premises setups require businesses to build, secure, and maintain their own facilities, which adds significant cost.

SOURCES

NetApp on AWS TCO AWS TCO Whitepaper

DISTRACTOR BREAKDOWN

A. Project management A business-side cost — not included in AWS's infrastructure TCO comparison

B. Antivirus software licensing Depends on the OS and workload — not universally included in AWS TCO

D. Software development A customer responsibility — AWS doesn't include dev effort in its infrastructure TCO

REAL-WORLD TIE-IN

If your Nairobi-based team is evaluating cloud migration: AWS absorbs data center security, hardware maintenance, and power costs You compare TCO against on-prem expenses like guards, HVAC, and access control systems Use the AWS TCO Calculator to model savings across compute, storage, and facilities That's how AWS delivers cost transparency by bundling infrastructure-level expenses into its pricing, simplifying TCO analysis.

Question 168

**Which AWS service or feature checks access policies and offers actionable recommendations to help users set secure and functional policies? **

OPTIONS

A. AWS Systems Manager

B. AWS IAM Access Analyzer

C. AWS Trusted Advisor

D. Amazon GuardDuty

[!TIP]

Correct Answer:

B. AWS IAM Access Analyzer

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of IAM policy validation and security tooling — especially services that provide automated analysis and recommendations. The key phrase is “checks access policies... actionable recommendations.”

WHY B IS CORRECT

IAM Access Analyzer: Analyzes resource-based policies (S3, IAM, KMS, Lambda, etc.)
Identifies unintended public or cross-account access Provides recommendations to tighten permissions
Helps teams validate least privilege and secure sharing

SOURCES

IAM Access Analyzer Overview Policy Validation and Recommendations

DISTRACTOR BREAKDOWN

A. AWS Systems Manager Manages infrastructure — doesn't analyze IAM policies

C. AWS Trusted Advisor Offers general best practices — not focused on policy-level analysis

D. Amazon GuardDuty Detects threats — doesn't validate or recommend IAM policy changes

REAL-WORLD TIE-IN

If your Nairobi-based team shares S3 buckets across accounts: Use IAM Access Analyzer to detect risky policies Get recommendations to restrict access Integrate with CloudTrail and Config for audit-ready workflows That's how AWS delivers policy-level visibility and actionable security insights with IAM Access Analyzer.

Question 169

****Which design concept is fulfilled by adhering to the AWS Well-Architected Framework's dependability pillar? ****

OPTIONS

A. Vertical scaling

B. Manual failure recovery

C. Testing recovery procedures

D. Changing infrastructure manually

[!TIP]

Correct Answer:

C. Testing recovery procedures

EXPLANATION

CONCEPTUAL FRAMING

This question targets the Reliability pillar of the AWS Well-Architected Framework, which is often referred to as the dependability pillar. It focuses on building systems that can recover from failures and continue to function.

WHY C IS CORRECT

Testing recovery procedures: Ensures that systems can recover gracefully from disruptions
Validates backup and restore mechanisms, failover strategies, and incident response plans
Encourages game days and chaos engineering to simulate failures
Builds confidence in the system's ability to meet RTO (Recovery Time Objective) and RPO (Recovery Point Objective)

SOURCES

AWS Well-Architected Framework Pillars [GeeksforGeeks](#) AWS Well-Architected Overview

DISTRACTOR BREAKDOWN

A. Vertical scaling Increases capacity but doesn't address fault tolerance or recovery

B. Manual failure recovery Slows response time — the framework promotes automated, tested recovery

D. Changing infrastructure manually Violates principles of automation and repeatability — leads to inconsistency and errors

REAL-WORLD TIE-IN

If your Nairobi-based team runs a critical analytics platform: Schedule game days to simulate EC2 failures or S3 access loss Use CloudFormation rollback and Route 53 failover to test recovery Document outcomes and refine your incident response playbooks That's how AWS enables dependable architecture through proactive testing and automated recovery procedures.

Question 170

****Which AWS service or functionality can assist a business in determining if it has publicly accessible Amazon S3 buckets? ****

OPTIONS

A. AWS Service Health Dashboard

B. Amazon CloudWatch Logs

C. AWS Trusted Advisor

D. AWS Service Catalog

[!TIP]

Correct Answer:

C. AWS Trusted Advisor

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS security visibility tools, especially those that audit configurations and flag risky exposure. The key phrase is “publicly accessible Amazon S3 buckets.”

WHY C IS CORRECT

AWS Trusted Advisor: Scans your AWS environment for security, cost, fault tolerance, and performance issues Includes a check for publicly accessible S3 buckets Flags buckets with PublicAccess enabled or overly permissive ACLs Offers actionable recommendations to restrict access and improve security posture

SOURCES

Trusted Advisor Security Checks S3 Bucket Permissions Check

DISTRACTOR BREAKDOWN

A. AWS Service Health Dashboard Monitors AWS service status — doesn’t inspect your resources or configurations

B. Amazon CloudWatch Logs Tracks events — doesn’t analyze S3 bucket policies or access settings

D. AWS Service Catalog Manages approved service templates — not related to S3 access visibility

REAL-WORLD TIE-IN

If your Nairobi-based team stores sensitive onboarding data in S3: Use Trusted Advisor to detect public exposure Pair with IAM Access Analyzer for policy-level insights Automate remediation with Config rules and Lambda triggers That's how AWS delivers visibility and actionable guidance to secure S3 buckets and prevent data leaks.

Question 171

**Seasonal sales surges occur many times a year for an online retailer, most notably during the holidays. At other times of year, demand is lower. The corporation has difficulty forecasting the seasonal increase in infrastructure demand. Which benefits of migrating to the AWS Cloud would be the most beneficial to the business? (Select two.) **

OPTIONS

A. Global footprint

B. Elasticity

C. AWS service quotas

D. AWS shared responsibility model

E. Pay-as-you-go pricing

[!TIP]

Correct Answer:

B. Elasticity and E. Pay-as-you-go pricing

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of cloud-native scalability and cost efficiency, especially for businesses with variable workloads. The key phrases are “seasonal sales surges” and “difficulty forecasting.” 🔍 Why B and E Are Correct Elasticity: AWS automatically scales resources up during peak demand and down during quiet periods Supports auto scaling groups, Lambda concurrency, and Aurora serverless Eliminates the need for overprovisioning or manual forecasting Pay-as-you-go pricing: You only pay for what you use, when you use it Ideal for seasonal businesses — no sunk costs during low-demand periods Enables cost optimization without sacrificing performance

SOURCES

AWS Cloud Benefits Elasticity and Cost Optimization

DISTRACTOR BREAKDOWN

A. Global footprint Helps with latency and compliance — not directly tied to seasonal demand or cost forecasting

C. AWS service quotas Limits resource usage — not a benefit for scaling or pricing

D. Shared responsibility model Defines security boundaries — not a direct benefit for seasonal infrastructure challenges

REAL-WORLD TIE-IN

If your Nairobi-based retail team sees holiday spikes: Use Auto Scaling and Lambda to handle traffic bursts Monitor spend with AWS Budgets and Cost Explorer Avoid idle infrastructure with on-demand pricing and spot instances That’s how AWS delivers scalable, cost-efficient infrastructure tailored for unpredictable seasonal demand.

Question 172

****Which concepts of AWS Cloud architecture may assist boost reliability? (Select two) ****

OPTIONS

- A. Using monolithic architecture
- B. Measuring overall efficiency
- C. Testing recovery procedures
- D. Adopting a consumption model
- E. Automatically recovering from failure

[!TIP]

Correct Answer:

C. Testing recovery procedures and E. Automatically recovering from failure

EXPLANATION

CONCEPTUAL FRAMING

This question targets the Reliability pillar of the AWS Well-Architected Framework, which emphasizes resilience, fault tolerance, and recovery. The goal is to ensure workloads can recover quickly and continue operating under stress or failure. 🔍 **Why C and E Are Correct**

Testing recovery procedures: Validates your system's ability to recover from disruptions
Encourages game days, chaos engineering, and simulated outages
Helps teams refine incident response and rollback strategies

Automatically recovering from failure: Uses services like Auto Scaling, Route 53 failover, and Elastic Load Balancing
Enables self-healing infrastructure that detects and replaces unhealthy components
Reduces downtime and manual intervention

SOURCES

AWS Reliability Pillar AWS Architecture Design Principles

DISTRACTOR BREAKDOWN

A. Using monolithic architecture Reduces fault isolation — modern architectures favor microservices and loose coupling

B. Measuring overall efficiency Belongs to the Performance Efficiency pillar — not directly tied to reliability

D. Adopting a consumption model Related to Cost Optimization — doesn't improve fault tolerance or recovery

REAL-WORLD TIE-IN

If your Nairobi-based team runs a multi-region analytics platform: Simulate EC2 failures and validate automated recovery Use CloudFormation rollback and Route 53 health checks Document lessons learned and refine runbooks and RTO/RPO targets That's how AWS enables reliable architecture through proactive testing and automated recovery mechanisms.

Question 173

****Which sort of storage does Amazon Elastic File System (Amazon EFS) and Amazon FSx provide? ****

OPTIONS

A. File storage

B. Object storage

C. Block storage

D. Instance store

[!TIP]

Correct Answer:

A. File storage

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS storage types, especially services that support file-based access protocols like NFS and SMB.

WHY A IS CORRECT

Amazon EFS: Provides scalable, elastic file storage for Linux-based workloads Uses NFS protocol Ideal for shared access across multiple EC2 instances Amazon FSx: Offers managed file systems like: FSx for Windows File Server (uses SMB) FSx for Lustre (high-performance file system for HPC) Supports Windows and Linux workloads

SOURCES

[Amazon EFS Overview](#) [Amazon FSx Overview](#)

DISTRACTOR BREAKDOWN

B. Object storage Refers to Amazon S3 — stores data as objects, not files

C. Block storage Refers to Amazon EBS — used for low-level disk access

D. Instance store Temporary block storage tied to EC2 — not file-based or persistent

REAL-WORLD TIE-IN

If your Nairobi-based team runs a shared analytics platform: Use Amazon EFS for Linux-based shared file access Use Amazon FSx for Windows File Server for SMB-based workloads Scale storage elastically without managing file servers That's how AWS delivers fully managed, scalable file storage for diverse workloads across operating systems.

REFERENCES

<https://aws.amazon.com/efs/>

<https://aws.amazon.com/fsx/>

Question 174

****Which AWS service or functionality can assist a business in determining if it has publicly accessible Amazon S3 buckets? ****

OPTIONS

A. AWS Config

B. Amazon GuardDuty

C. AWS Trusted Advisor

D. AWS Artifact

[!TIP]

Correct Answer:

D. AWS Artifact

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of compliance and audit tooling in AWS — especially services that provide self-service access to regulatory documentation and control reports.

WHY D IS CORRECT

AWS Artifact: Provides on-demand access to AWS compliance documentation Includes SOC reports, ISO certifications, PCI reports, and more Enables businesses to review AWS's control environment and support internal audits Ideal for teams managing regulated workloads or preparing for compliance assessments

SOURCES

AWS Artifact Overview Compliance Reports Access

DISTRACTOR BREAKDOWN

A. AWS Config Tracks configuration changes — doesn't provide compliance reports

B. Amazon GuardDuty Detects threats — not focused on compliance documentation

C. AWS Trusted Advisor Offers best practice checks — doesn't deliver formal compliance reports

REAL-WORLD TIE-IN

If your Nairobi-based team is preparing for a PCI audit: Use AWS Artifact to download the latest PCI DSS Attestation of Compliance Share reports with auditors and map them to your shared responsibility model. Combine with AWS Config and Security Hub for continuous compliance monitoring. That's how AWS delivers audit-ready transparency with Artifact, empowering teams to meet regulatory obligations efficiently.

REFERENCES

<https://aws.amazon.com/artifact/>

Question 175

****What may aid in the evaluation of a cloud-based application? (Select two) ****

OPTIONS

A. AWS Trusted Advisor

B. AWS Professional Services

C. AWS Systems Manager

D. AWS Partner Network (APN)

E. AWS Secrets Manager

[!TIP]

Correct Answer:

A. AWS Trusted Advisor and D. AWS Partner Network (APN)

EXPLANATION

CONCEPTUAL FRAMING

This question targets evaluation tools and expert ecosystems that help assess cloud applications for security, performance, cost, and reliability. 🔍 **Why A and D Are Correct**

AWS Trusted Advisor: Provides automated checks across five categories: security, fault tolerance, performance, service limits, and cost optimization. Flags misconfigurations and offers actionable recommendations. Ideal for ongoing evaluation and improvement.

AWS Partner Network (APN): A global community of consulting and technology partners. Offers third-party evaluations, audits, and migration support. Helps businesses assess architecture, compliance, and optimization strategies.

SOURCES

Trusted Advisor Overview AWS Partner Network

DISTRACTOR BREAKDOWN

B. AWS Professional Services Offers implementation support, not evaluation tools — typically used post-assessment

C. AWS Systems Manager Manages infrastructure — not designed for application evaluation

E. AWS Secrets Manager Manages credentials — unrelated to application evaluation

REAL-WORLD TIE-IN

If your Nairobi-based team is assessing a multi-tenant SaaS platform: Use Trusted Advisor to flag security gaps and cost inefficiencies Engage an APN partner for architecture review and compliance validation Combine findings into a refinement roadmap That's how AWS enables continuous evaluation through automated insights and expert partner support.

REFERENCES

<https://aws.amazon.com/partners/>

Question 175

**What may aid in the evaluation of a cloud-based application? (Select two) **

OPTIONS

A. AWS Trusted Advisor

B. AWS Professional Services

C. AWS Systems Manager

D. AWS Partner Network (APN)

E. AWS Secrets Manager

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question focuses on tools and ecosystems that help assess cloud applications for performance, security, cost, and architecture quality. 🔍 Why A and D Are Correct AWS Trusted Advisor: Offers automated checks across five categories: cost optimization, performance, security, fault tolerance, and service limits Helps identify misconfigurations, inefficiencies, and vulnerabilities Provides actionable recommendations for improvement AWS Partner Network (APN): Includes certified consulting and technology partners Offers third-party evaluations, audits, and migration support Helps businesses assess architecture, compliance, and optimization strategies

SOURCES

Trusted Advisor AWS Partner Network

DISTRACTOR BREAKDOWN

B. AWS Professional Services Focuses on implementation and migration, not evaluation or assessment

C. AWS Systems Manager Manages infrastructure — not designed for application evaluation

E. AWS Secrets Manager Manages credentials — unrelated to application evaluation

REAL-WORLD TIE-IN

If your Nairobi-based team is evaluating a multi-tenant SaaS platform: Use Trusted Advisor to flag security gaps and cost inefficiencies Engage an APN partner for architecture review and compliance validation Combine findings into a refinement roadmap I can also help you draft an evaluation checklist using Trusted Advisor categories and APN partner capabilities.

REFERENCES

<https://aws.amazon.com/partners/>

Question 176

****Which AWS service enables expense control across numerous AWS accounts effectively? ****

OPTIONS

A. AWS Organizations

B. AWS Trusted Advisor

C. AWS Direct Connect

D. Amazon Connect

[!TIP]

Correct Answer:

A. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account management and cost governance, especially for enterprises with distributed teams or business units.

WHY A IS CORRECT

AWS Organizations: Enables centralized billing and consolidated cost tracking Supports Service Control Policies (SCPs) for governance Allows businesses to group accounts into Organizational Units (OUs) Integrates with AWS Budgets, Cost Explorer, and tagging strategies

SOURCES

AWS Organizations Overview Consolidated Billing Features

DISTRACTOR BREAKDOWN

B. AWS Trusted Advisor Offers best practice checks — doesn't manage multi-account billing or expense control

C. AWS Direct Connect Provides dedicated network links — unrelated to cost management

D. Amazon Connect A contact center service — not used for expense control

REAL-WORLD TIE-IN

If your Nairobi-based team operates across multiple departments: Use AWS Organizations to group accounts by function (e.g., Dev, QA, Prod) Apply SCPs to enforce budget limits and access boundaries Monitor spend with Cost Explorer and tagging That's how AWS delivers scalable expense control and governance across complex cloud environments.

REFERENCES

Question 177

****A company's on-premises servers and the AWS Cloud need a dedicated network connection. Which Amazon Web Services (AWS) service should be used? ****

OPTIONS

A. AWS VPN

B. AWS Direct Connect

C. Amazon API Gateway

D. Amazon Connect

[!TIP]

Correct Answer:

B. AWS Direct Connect

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid networking architecture, especially for enterprises that require reliable, high-bandwidth connectivity between their data centers and AWS.

WHY B IS CORRECT

AWS Direct Connect: Establishes a dedicated physical connection to AWS Bypasses the public internet for lower latency and consistent performance Supports private VIFs (Virtual Interfaces) for VPC access and public VIFs for AWS services Ideal for data-intensive workloads, compliance-sensitive environments, and hybrid cloud setups

SOURCES

AWS Direct Connect Overview Hybrid Cloud Networking

DISTRACTOR BREAKDOWN

A. AWS VPN Uses the public internet — not a dedicated connection

C. Amazon API Gateway Manages AP Is — not used for network connectivity

D. Amazon Connect A contact center service — unrelated to networking

REAL-WORLD TIE-IN

If your Nairobi-based team runs latency-sensitive workloads: Use AWS Direct Connect to link your data center to AWS Pair with Transit Gateway for scalable routing Monitor with CloudWatch and Network Manager That's how AWS delivers dedicated, secure, and performant hybrid connectivity with Direct Connect.

Question 178

**On-premises resources have been underused by a user. Which AWS Cloud idea is optimal for resolving this issue? **

OPTIONS

A. High availability

B. Elasticity

C. Security

D. Loose coupling

[!TIP]

Correct Answer:

B. Elasticity

EXPLANATION

CONCEPTUAL FRAMING

This question targets resource optimization and scalability, especially for users transitioning from static on-prem infrastructure to dynamic cloud environments.

WHY B IS CORRECT

Elasticity: Allows AWS resources to scale up or down automatically based on demand Prevents overprovisioning and underutilization Supports cost efficiency and performance optimization Ideal for workloads with variable or unpredictable usage patterns

SOURCES

AWS Cloud Benefits Elasticity in AWS

DISTRACTOR BREAKDOWN

A. High availability Ensures uptime — doesn't address resource utilization or scaling

C. Security Protects data and infrastructure — unrelated to resource efficiency

D. Loose coupling Improves system design — not focused on resource optimization

REAL-WORLD TIE-IN

If your Nairobi-based team runs a seasonal analytics platform: Use Auto Scaling and Lambda to match compute to demand Monitor usage with CloudWatch and Cost Explorer Avoid idle infrastructure and reduce spend That's how AWS delivers elastic infrastructure that adapts to real-world usage, eliminating waste and boosting efficiency.

Question 179

****Which of the following advantages does Amazon Relational Database Service (Amazon RDS) provide over conventional database management? ****

OPTIONS

A. AWS manages the data stored in Amazon RDS tables

B. AWS manages the maintenance of the operating system

C. AWS automatically scales up instance types on demand

D. AWS manages the database type

[!TIP]

Correct Answer:

B. AWS manages the maintenance of the operating system

EXPLANATION

CONCEPTUAL FRAMING

This question targets managed service benefits, especially how RDS simplifies database operations by abstracting infrastructure and OS-level tasks.

WHY B IS CORRECT

Amazon RDS: Handles OS patching, backups, monitoring, and failover
Frees developers from manual maintenance and infrastructure management
Lets teams focus on schema design, query optimization, and application logic

SOURCES

Amazon RDS Features RDS Maintenance Tasks

DISTRACTOR BREAKDOWN

A. AWS manages the data stored in Amazon RDS tables Data ownership and management remain with the customer — AWS doesn't control table contents

C. AWS automatically scales up instance types on demand RDS supports manual scaling and Aurora auto-scaling, but not automatic instance type upgrades

D. AWS manages the database type Customers choose the engine (e.g., MySQL, PostgreSQL) — AWS doesn't decide this

REAL-WORLD TIE-IN

If your Nairobi-based team runs a multi-tenant SaaS platform: Use Amazon RDS to offload OS patching and backups Focus on performance tuning and schema evolution Pair with CloudWatch and IAM roles for secure, observable operations That's how AWS delivers managed database infrastructure that reduces operational overhead and boosts developer productivity.

REFERENCES

<https://aws.amazon.com/rds/>

Question 180

**Service control policies (SCPs) manage permissions for which of the following? **

OPTIONS

A. Availability Zones

B. AWS Regions

C. AWS Organizations

D. Edge locations

[!TIP]

Correct Answer:

C. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of multi-account governance in AWS, especially how SCPs enforce permission boundaries across organizational units.

WHY C IS CORRECT

Service Control Policies (SCPs): Are part of AWS Organizations Define maximum available permissions for accounts within an organization Apply to organizational units (OUs), root, or individual accounts Do not grant permissions directly, but limit what IAM policies can allow

SOURCES

SCPs in AWS Organizations [AWS Organizations Overview](#)

DISTRACTOR BREAKDOWN

A. Availability Zones SC Ps don't manage infrastructure placement — AZs are physical locations

B. AWS Regions SC Ps don't control geographic access — IAM and service settings handle that

D. Edge locations Used for CDN (Cloud Front) — not governed by SC Ps

REAL-WORLD TIE-IN

If your Nairobi-based team wants to restrict S3 access in a dev account: Use SCPs in AWS Organizations to block s3:PutObject Apply the policy to the Dev OU IAM users in that account

can't override the restriction That's how AWS delivers centralized permission control across accounts using SCPs within Organizations.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 181

****Which AWS service can be used to encrypt data at rest? ****

OPTIONS

A. Amazon GuardDuty

B. AWS Shield

C. AWS Security Hub

D. AWS Key Management Service (AWS KMS)

[!TIP]

Correct Answer:

D. AWS Key Management Service (AWS KMS)

EXPLANATION

CONCEPTUAL FRAMING

This question targets data protection and encryption, especially for workloads that require secure storage and compliance.

WHY D IS CORRECT

AWS KMS: Manages encryption keys used to protect data at rest and in transit Integrates with services like S3, EBS, RDS, Redshift, and Lambda Supports customer-managed and AWS-managed keys Enables automatic key rotation, audit logging, and fine-grained access control

SOURCES

AWS KMS Overview Encrypting Data at Rest

DISTRACTOR BREAKDOWN

A. Amazon GuardDuty Detects threats — doesn't handle encryption

B. AWS Shield Protects against DDoS attacks — unrelated to data encryption

C. AWS Security Hub Aggregates security findings — doesn't manage encryption keys

REAL-WORLD TIE-IN

If your Nairobi-based team stores sensitive customer data in S3: Use AWS KMS to encrypt objects with customer-managed keys Monitor key usage with CloudTrail Rotate keys periodically for compliance That's how AWS delivers secure, scalable encryption for data at rest using KMS.

Question 182

****A cloud practitioner has a seldom run data analysis job that can be stopped without causing damage. Which Amazon EC2 purchase option should be utilized to maximize cost savings? ****

OPTIONS

A. On-Demand Instances

B. Reserved Instances

C. Spot Instances

D. Dedicated Hosts

[!TIP]

Correct Answer:

C. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization strategies for EC2, especially for non-critical, interruptible workloads.

WHY C IS CORRECT

Spot Instances: Offer up to 90% cost savings compared to On-Demand Use unused EC2 capacity Can be interrupted by AWS with two minutes' notice Perfect for batch jobs, data analysis, CI/CD pipelines, and stateless workloads

SOURCES

Spot Instances Overview Best Practices for Spot

DISTRACTOR BREAKDOWN

A. On-Demand Instances Flexible but expensive — not ideal for cost savings in intermittent jobs

B. Reserved Instances Require long-term commitment — best for predictable workloads

D. Dedicated Hosts Provide physical isolation — costly and unnecessary for this use case

REAL-WORLD TIE-IN

If your Nairobi-based team runs monthly analytics jobs: Use Spot Instances with EC2 Auto Scaling and interruption handling Monitor job progress with CloudWatch and S3 logging Save significantly while maintaining performance That's how AWS enables cost-efficient compute for fault-tolerant workloads using Spot Instances.

Question 183

****What are the advantages of AWS Cloud service billing consolidation? (Select two) ****

OPTIONS

A. Volume discounts

B. A minimal additional fee for use

C. One bill for multiple accounts

D. Installment payment options

E. Custom cost and usage budget creation

[!TIP]

Correct Answer:

A. Volume discounts and C. One bill for multiple accounts

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost management and billing efficiency in AWS, especially for organizations using multiple linked accounts. 🔍 Why A and C Are Correct
Volume discounts: AWS aggregates usage across accounts in an organization Helps you reach pricing tiers faster, reducing overall spend Applies to services like EC2, S3, and data transfer
One bill for multiple accounts: Simplifies financial tracking and reconciliation Centralizes billing under a payer account Enables cost allocation tagging and budget enforcement

SOURCES

AWS Consolidated Billing AWS Pricing Benefits

DISTRACTOR BREAKDOWN

B. A minimal additional fee for use AWS doesn't charge extra for billing consolidation

D. Installment payment options Not offered — AWS bills monthly based on usage

E. Custom cost and usage budget creation Available via AWS Budgets, but not exclusive to billing consolidation

REAL-WORLD TIE-IN

If your Nairobi-based team operates across Dev, QA, and Prod accounts: Use AWS Organizations to consolidate billing Track spend with Cost Explorer and tagging Unlock volume discounts for shared services like EC2 and S3 That's how AWS delivers cost efficiency and simplified financial management through billing consolidation.

Question 184

**Which Amazon EC2 instance type is necessary when a user wishes to use their current per-socket, per-core, or per-virtual machine software licenses on a Microsoft Windows server operating on AWS? **

OPTIONS

A. Spot Instances

B. Dedicated Instances

C. Dedicated Hosts

D. Reserved Instances

[!TIP]

Correct Answer:

C. Dedicated Hosts

EXPLANATION

CONCEPTUAL FRAMING

This question targets licensing compliance and infrastructure control, especially for users bringing their own Microsoft licenses (BYOL) to AWS.

WHY C IS CORRECT

Dedicated Hosts: Provide physical server-level visibility and control Required for BYOL scenarios involving per-socket, per-core, or per-VM licensing models Support Windows Server and SQL Server licensing under Microsoft terms Enable license tracking and enforcement via AWS License Manager

SOURCES

Microsoft Licensing on AWS EC2 Licensing FAQ CLF-C02 Licensing Guide

DISTRACTOR BREAKDOWN

A. Spot Instances Cost-effective but lack physical host control — unsuitable for license-bound deployments

B. Dedicated Instances Provide instance-level isolation — but don't expose host-level details required for licensing

D. Reserved Instances Offer billing discounts — not tied to physical host control or licensing flexibility

REAL-WORLD TIE-IN

If your Seattle-based team wants to migrate licensed Windows workloads: Use Dedicated Hosts to comply with Microsoft's licensing terms Track license usage with AWS License Manager Pair with AWS Systems Manager for patching and compliance That's how AWS enables license-compliant Windows deployments with host-level control via Dedicated Hosts.

Question 185

****When users connect to a website with a worldwide consumer base, they report experiencing delay. Which Amazon Web Services (AWS) offering will enhance the user experience by lowering latency? ****

OPTIONS

A. Amazon CloudFront

B. AWS Direct Connect

C. Amazon EC2 Auto Scaling

D. AWS Transit Gateway

[!TIP]

Correct Answer:

A. Amazon CloudFront

EXPLANATION

CONCEPTUAL FRAMING

This question targets global performance optimization, especially for websites serving distributed users who experience latency due to geographic distance.

WHY A IS CORRECT

Amazon CloudFront: Distributes content via a global network of edge locations Caches static and dynamic content closer to users Reduces round-trip time and latency Supports HTTPS, custom SSL, and Lambda@Edge for dynamic personalization

SOURCES

Amazon CloudFront Overview Latency Optimization with CloudFront

DISTRACTOR BREAKDOWN

B. AWS Direct Connect Improves private connectivity — not designed for global web latency

C. Amazon EC2 Auto Scaling Manages compute capacity — doesn't address content delivery or latency

D. AWS Transit Gateway Connects VP Cs — unrelated to user-facing latency reduction

REAL-WORLD TIE-IN

If your Nairobi-based team runs a global e-commerce site: Use CloudFront to cache product images and pages at edge locations Pair with S3 and Lambda@Edge for dynamic content and personalization Monitor performance with CloudWatch metrics and Real User Monitoring That's how AWS delivers low-latency, high-performance content delivery for global audiences using CloudFront.

Question 186

**Which capabilities or services are available for monitoring an AWS account's charges and expenses? (Select two) **

OPTIONS

A. AWS Cost and Usage Report

B. AWS product pages

C. AWS Simple Monthly Calculator

D. Billing alerts and Amazon CloudWatch alarms

E. AWS Price List API

[!TIP]

Correct Answer:

A. AWS Cost and Usage Report and D. Billing alerts and Amazon CloudWatch alarms

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost visibility and proactive monitoring, essential for managing cloud spend and avoiding budget surprises. 🔍 Why A and D Are Correct AWS Cost and Usage Report: Provides detailed billing data at the account and service level Includes usage metrics, pricing, and resource IDs Can be exported to Amazon S3 and analyzed with Athena or QuickSight Billing alerts and Amazon CloudWatch alarms: Enable real-time monitoring of charges Trigger alerts when spending exceeds thresholds Help teams stay within budget and respond quickly to anomalies

SOURCES

Cost and Usage Report Billing Alerts with CloudWatch

DISTRACTOR BREAKDOWN

B. AWS product pages Provide service descriptions — not usage or billing data

C. AWS Simple Monthly Calculator Estimates costs — doesn't monitor actual charges

E. AWS Price List API Offers pricing data — not tied to account-specific usage or alerts

REAL-WORLD TIE-IN

If your Seattle-based team wants to monitor monthly EC2 spend: Use Cost and Usage Report to break down instance types and regions Set CloudWatch alarms for budget thresholds Visualize trends with QuickSight dashboards That's how AWS enables granular cost tracking and proactive budget control across cloud environments.

Question 187

****Which service enables users to store data in the Amazon Web Services cloud? ****

OPTIONS

A. Amazon EFS

B. Amazon Redshift

C. Amazon RDS

D. Amazon VPC

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of AWS storage services, especially those designed for file-level access and cloud-native scalability.

WHY A IS CORRECT

Amazon EFS (Elastic File System): Provides scalable, elastic file storage for use with AWS services and on-premises resources Supports NFS protocol for Linux-based workloads Automatically grows and shrinks as files are added or removed Ideal for shared access, big data, and container workloads

SOURCES

Amazon EFS Overview AWS Storage Options

DISTRACTOR BREAKDOWN

B. Amazon Redshift A data warehouse — optimized for analytics, not general-purpose storage

C. Amazon RDS A managed database service — stores structured data, not general files

D. Amazon VPC A networking service — doesn't store data

REAL-WORLD TIE-IN

If your Seattle-based team runs a multi-tenant analytics platform: Use Amazon EFS to store shared datasets across EC2 and container workloads Pair with IAM and encryption for secure access Monitor usage with CloudWatch and lifecycle policies That's how AWS enables flexible, scalable file storage directly in the cloud with Amazon EFS.

REFERENCES

<https://aws.amazon.com/efs/>

Question 188

**Which AWS service or product enables an organization to automate the delivery of application changes? **

OPTIONS

A. Amazon AppFlow

B. AWS CodeDeploy

C. AWS PrivateLink

D. Amazon EKS Distro

[!TIP]

Correct Answer:

B. AWS CodeDeploy

EXPLANATION

CONCEPTUAL FRAMING

This question targets DevOps automation, especially how AWS supports continuous delivery and deployment of application updates.

WHY B IS CORRECT

AWS CodeDeploy: Automates deployments to Amazon EC2, AWS Lambda, ECS, and on-premises servers Supports blue/green, canary, and rolling deployments Minimizes downtime and reduces deployment errors Integrates with CodePipeline, GitHub, and CI/CD workflows

SOURCES

AWS CodeDeploy Overview Deployment Strategies

DISTRACTOR BREAKDOWN

A. Amazon AppFlow Automates data transfer between SaaS apps — not code deployment

C. AWS PrivateLink Provides private connectivity — unrelated to application delivery

D. Amazon EKS Distro A Kubernetes distribution — doesn't automate deployments itself

REAL-WORLD TIE-IN

If your Seattle-based team deploys microservices across EC2 and Lambda: Use CodeDeploy to automate updates with rollback support Monitor deployments with CloudWatch and CodeDeploy events Integrate with CodePipeline for full CI/CD automation That's how AWS enables safe, scalable, and automated application delivery with CodeDeploy.

Question 189

****A corporation wishes to minimize the physical footprint of the computing resources used by developers to execute programs. Which service would enable serverless architectures to address this need? ****

OPTIONS

A. Amazon Elastic Compute Cloud (Amazon EC2)

B. AWS Lambda

C. Amazon DynamoDB

D. AWS CodeCommit

[!TIP]

Correct Answer:

B. AWS Lambda

EXPLANATION

CONCEPTUAL FRAMING

This question targets serverless architecture principles, especially how AWS enables event-driven, infrastructure-free execution environments.

WHY B IS CORRECT

AWS Lambda: Runs code in response to events (e.g., HTTP requests, file uploads, database changes) Requires no server provisioning or management Scales automatically and charges only for compute time used Supports multiple languages and integrates with services like API Gateway, S3, and DynamoDB

SOURCES

AWS Lambda Overview Serverless Architectures

DISTRACTOR BREAKDOWN

A. Amazon EC2 Requires server provisioning — not serverless

C. Amazon DynamoDB A serverless database — doesn't execute code

D. AWS CodeCommit A source control service — not used for code execution

REAL-WORLD TIE-IN

If your Seattle-based team wants to run lightweight data processing jobs: Use AWS Lambda to trigger code on S3 uploads or API calls Avoid managing EC2 instances or containers Monitor performance with CloudWatch and X-Ray That's how AWS delivers scalable, serverless execution with zero infrastructure footprint using Lambda.

REFERENCES

<https://aws.amazon.com/lambda/>

Question 190

**A Load Balancer Elastic enables online traffic to be distributed across multiple: **

OPTIONS

A. AWS Regions

B. Availability Zones

C. Dedicated Hosts

D. Amazon S3 buckets

[!TIP]

Correct Answer:

B. Availability Zones

EXPLANATION

CONCEPTUAL FRAMING

This question targets high availability and fault tolerance, especially how AWS Load Balancers ensure resilient traffic distribution within a region.

WHY B IS CORRECT

Elastic Load Balancing (ELB): Automatically distributes traffic across multiple Availability Zones within a region Supports Application Load Balancer (ALB), Network Load Balancer (NLB), and Gateway Load Balancer (GLB) Enhances redundancy, scalability, and fault isolation Integrates with Auto Scaling and health checks to route traffic only to healthy targets

SOURCES

Elastic Load Balancing Overview Availability Zone Distribution

DISTRACTOR BREAKDOWN

A. AWS Regions ELB operates within a single region — cross-region traffic requires other services like Route 53

C. Dedicated Hosts Used for licensing and isolation — not traffic distribution

D. Amazon S3 buckets Object storage — not involved in load balancing

REAL-WORLD TIE-IN

If your Seattle-based team runs a web app across multiple AZs: Use ALB to route traffic based on path or host headers Ensure high availability even if one AZ fails Monitor performance with CloudWatch metrics and access logs That's how AWS delivers resilient, scalable traffic distribution using Elastic Load Balancing across Availability Zones.

REFERENCES

<https://aws.amazon.com/elasticloadbalancing/>

Question 191

****What is AWS Storage Gateway's purpose? ****

OPTIONS

A. It ensures on-premises data storage is 99.999999999% durable.

B. It transports petabytes of data to and from AWS.

C. It connects to multiple Amazon EC2 instances.

D. It connects on-premises data storage to the AWS Cloud.

[!TIP]

Correct Answer:

D. It connects on-premises data storage to the AWS Cloud

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud integration, especially how AWS enables seamless data movement and access between local infrastructure and cloud services.

WHY D IS CORRECT

AWS Storage Gateway: Provides file, volume, and tape-based interfaces Enables on-premises applications to use AWS cloud storage Supports caching, backup, and disaster recovery Integrates with Amazon S3, Glacier, and EBS Snapshots

SOURCES

AWS Storage Gateway Overview Hybrid Cloud Storage Use Cases

DISTRACTOR BREAKDOWN

A. 99.999999999% durability Refers to Amazon S3, not Storage Gateway

B. Transports petabytes of data Describes AWS Snowball or Snowmobile, not Storage Gateway

C. Connects to EC2 instances Not its primary function — Storage Gateway connects on-prem storage to AWS

REAL-WORLD TIE-IN

If your Seattle-based team wants to back up local file servers to AWS: Use Storage Gateway File Gateway to sync with Amazon S3 Enable lifecycle policies for archival to Glacier Monitor with CloudWatch and AWS Backup That's how AWS enables hybrid storage solutions that extend on-premises infrastructure into the cloud with Storage Gateway.

Question 192

****How can a client anticipate future expenses associated with the operation of a new web application? ****

OPTIONS

A. Amazon Aurora Backtrack

B. Amazon CloudWatch Billing Alarms

C. AWS Simple Monthly Calculator

D. AWS Cost and Usage report

[!TIP]

Correct Answer:

C. AWS Simple Monthly Calculator

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost forecasting, especially how AWS enables users to plan budgets before launching workloads.

WHY C IS CORRECT

AWS Simple Monthly Calculator: Allows users to model service usage and pricing Estimates monthly costs based on selected services, regions, and usage patterns Helps teams compare configurations and optimize budgets before deployment

SOURCES

AWS Pricing Calculator Cost Estimation Best Practices

DISTRACTOR BREAKDOWN

A. Amazon Aurora Backtrack Enables point-in-time recovery — unrelated to cost estimation

B. CloudWatch Billing Alarms Monitors actual spend — doesn't forecast future costs

D. Cost and Usage Report Tracks historical usage — not predictive or planning-focused

REAL-WORLD TIE-IN

If your Seattle-based team is planning a new analytics platform: Use the AWS Pricing Calculator to model EC2, S3, and Lambda usage Compare costs across regions and instance types Share estimates with finance for budget approval That's how AWS empowers teams to anticipate and plan cloud expenses with precision before deployment.

Question 193

****The term "fault tolerance" relates to the following: ****

OPTIONS

- A. the ability of an application to accommodate growth without changing design
- B. how well and how quickly an application's environment can have lost data restored
- C. how secure your application is
- D. the built-in redundancy of an application's components

[!TIP]

Correct Answer:

- D. the built-in redundancy of an application's components

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience engineering, especially how AWS designs systems to withstand failures without service disruption.

WHY D IS CORRECT

Fault tolerance: Ensures systems continue functioning despite hardware, software, or network failures Achieved through redundant components, such as multiple EC2 instances across Availability Zones Often paired with load balancers, auto scaling, and failover mechanisms

SOURCES

AWS Well-Architected Framework – Reliability Pillar Designing Fault-Tolerant Applications

DISTRACTOR BREAKDOWN

A. Accommodating growth Describes scalability, not fault tolerance

B. Restoring lost data Refers to backup and disaster recovery, not fault tolerance

C. Application security Focuses on protection, not resilience

REAL-WORLD TIE-IN

If your Nairobi-based team runs a mission-critical API: Deploy across multiple AZs with ELB and Auto Scaling Use Route 53 health checks for DNS failover Monitor with CloudWatch and AWS Backup That's how AWS enables fault-tolerant architectures that survive component failures without impacting user experience.

Question 194

****Which AWS service enables conventional SQL queries against stored datasets straight from Amazon S3? ****

OPTIONS

A. AWS Glue

B. AWS Data Pipeline

C. Amazon CloudSearch

D. Amazon Athena

[!TIP]

Correct Answer:

D. Amazon Athena

EXPLANATION

CONCEPTUAL FRAMING

This question targets serverless analytics, especially how AWS enables ad hoc querying of S3 data using familiar SQL syntax.

WHY D IS CORRECT

Amazon Athena: Is a serverless query service Uses Presto under the hood to run SQL queries on S3 data Supports multiple formats: CSV, JSON, Parquet, ORC, Avro Charges per query or per scanned data volume, making it cost-efficient for exploratory analysis

SOURCES

Amazon Athena Overview Querying S3 with Athena

DISTRACTOR BREAKDOWN

A. AWS Glue Used for ETL and data cataloging — not direct SQL querying

B. AWS Data Pipeline Orchestrates data movement — doesn't support SQL queries

C. Amazon CloudSearch Provides search indexing, not SQL-based querying

REAL-WORLD TIE-IN

If your Nairobi-based team stores logs in S3: Use Athena to run SQL queries for error patterns
Integrate with QuickSight for dashboards Avoid provisioning Redshift or EMR clusters That's
how AWS delivers fast, serverless data exploration directly from S3 using Amazon Athena.

Question 195

**A startup wants to launch a new application immediately and adapt to changing requirements in the near future. Which AWS Cloud feature best supports this need?

**

OPTIONS

A. Elasticity

B. Reliability

C. Performance

D. Agility

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud-native development principles, especially how AWS enables rapid innovation and responsiveness — key traits for startups and fast-moving teams.

WHY D IS CORRECT

Agility in AWS refers to the ability to: Quickly provision resources and deploy applications Iterate and experiment without long procurement cycles Respond to changing business needs with minimal friction Scale teams and infrastructure as the product evolves As the explanation below the question states: “Cloud infrastructure provides more agility and responsiveness than traditional IT environments. This requires organizations to think differently about how they design, build, and manage applications.” This means startups can: Launch MVPs without upfront hardware investment Pivot features based on user feedback Use services like AWS Lambda, ECS, and CodePipeline to automate and accelerate delivery

SOURCES

AWS Cloud Benefits Agility in the Cloud

DISTRACTOR BREAKDOWN

A. Elasticity Helps scale resources up/down — but doesn’t address speed of launching or adapting

B. Reliability Ensures uptime and fault tolerance — not focused on speed or adaptability

C. Performance Refers to speed and efficiency — not organizational responsiveness or

deployment velocity

REAL-WORLD TIE-IN

If your Nairobi-based startup is launching a mobile app: Use AWS Amplify to deploy quickly. Integrate CI/CD pipelines for rapid iteration. Leverage serverless backends to stay lean and responsive. That's how AWS delivers startup-grade agility, empowering teams to move fast, test ideas, and adapt without infrastructure bottlenecks.

Question 196

**Which characteristics are advantages of using the AWS Cloud? (Choose two)

**

OPTIONS

A. A 100% service level agreement (SLA) for all AWS services

B. Compute capacity that is adjusted on demand

C. Availability of AWS Support for code development

D. Enhanced security

E. Increases in cost and complexity

[!TIP]

Correct Answer:

B. Compute capacity that is adjusted on demand and D. Enhanced security

EXPLANATION

CONCEPTUAL FRAMING

This question targets the core benefits of cloud computing, especially how AWS enables scalable, secure, and cost-efficient infrastructure for organizations of all sizes. 🔍 Why B and D Are Correct

B.

Compute capacity that is adjusted on demand This reflects elasticity, a key AWS feature that allows you to: Scale resources up or down automatically based on demand Avoid overprovisioning and underutilization Pay only for what you use Ideal for workloads with variable traffic, such as seasonal apps or batch processing

D.

Enhanced security AWS provides robust security controls, including: Encryption at rest and in transit IAM roles and policies for fine-grained access Network isolation via VPCs Compliance certifications (e.g., ISO, SOC, HIPAA) Security is a shared responsibility model, where AWS secures the infrastructure and users secure their data and configurations

SOURCES

AWS Cloud Benefits AWS Security Overview

DISTRACTOR BREAKDOWN

A. 100% SLA for all services AWS offers high availability but not 100% SLA across all services — SLAs vary by service

C. AWS Support for code development AWS Support helps with service usage, not direct code development

E. Increases in cost and complexity Cloud computing reduces cost and complexity through automation and managed services

REINFORCING WITH THE SIX CLOUD ADVANTAGES YOU LISTED

Let's map your six cloud advantages to the correct answers and broader AWS benefits: -

Cloud Advantage

Description Related Correct Option -

Trade fixed expense for variable expense

Pay-as-you-go pricing model B -

Benefit from massive economies of scale

Shared infrastructure lowers per-unit cost B -

Stop guessing capacity

Elastic scaling removes overprovisioning B -

Increase speed and agility

Rapid provisioning and deployment B & D -

Stop spending money running and maintaining data centers

AWS handles infrastructure B & D -

Go global in minutes

Deploy across global regions instantly D

REAL-WORLD TIE-IN

If your Nairobi-based startup is launching a new SaaS platform: Use Auto Scaling and Lambda to adjust compute capacity dynamically Secure your environment with IAM, KMS, and VPC configurations Monitor spend with Cost Explorer and CloudWatch Billing Alarms That's how AWS delivers scalable, secure, and cost-efficient infrastructure that adapts to your needs and protects your data.

Question 197

**A company wants to convert video files and audio files from their source format into a format that will play on smartphones, tablets, and web browsers. Which AWS service will meet these requirements?

**

OPTIONS

A. Amazon Elastic Transcoder

B. Amazon Comprehend

C. AWS Glue

D. Amazon Rekognition

[!TIP]

Correct Answer:

A. Amazon Elastic Transcoder

EXPLANATION

CONCEPTUAL FRAMING

This question targets media processing in the cloud, especially how AWS enables format conversion for cross-device compatibility.

WHY A IS CORRECT

Amazon Elastic Transcoder: A fully managed media transcoding service Converts media files stored in Amazon S3 into formats optimized for: Smartphones (e.g., iOS, Android) Tablets Web browsers Connected TVs Supports multiple formats including MP4, H.264, AAC, WebM, and MPEG-2 TS Automatically adjusts bitrate, resolution, and container format based on presets or custom settings  Explanation from the image: “Amazon Elastic Transcoder lets you convert media files that you have stored in Amazon Simple Storage Service (Amazon S3) into media files in the formats required by consumer playback devices. For example, you can convert large, high-quality digital media files into formats that users can play back on mobile devices, tablets, web browsers, and connected televisions.” This confirms that Elastic Transcoder is purpose-built for device-aware media conversion, making it ideal for companies distributing content across diverse platforms.

SOURCES

Amazon Elastic Transcoder Documentation AWS Media Services Overview

DISTRACTOR BREAKDOWN

B. Amazon Comprehend Used for natural language processing — not media conversion

C. AWS Glue A data integration and ETL service — irrelevant to audio/video formats

D. Amazon Rekognition Performs image and video analysis (e.g., facial recognition) — not transcoding

REAL-WORLD TIE-IN

If your Nairobi-based media startup stores raw video in S3: Use Elastic Transcoder to convert files into mobile-friendly formats Automate workflows with Lambda triggers and SNS notifications Optimize for bandwidth and device compatibility using preset pipelines That’s how AWS enables scalable, device-aware media conversion directly from cloud storage using Elastic Transcoder.

Question 198

**A company has several departments. Each department has its own AWS accounts for its applications. The company wants all AWS costs on a single invoice to simplify payment, but the company wants to know the costs that each department is incurring. Which AWS tool or feature will provide this functionality?

**

OPTIONS

A. AWS Cost and Usage Reports

B. Consolidated billing

C. Savings Plans

D. AWS Budgets

[!TIP]

Correct Answer:

B. Consolidated billing

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account financial governance, especially how AWS Organizations enables centralized billing while preserving department-level cost visibility.

WHY B IS CORRECT

Consolidated billing is a feature of AWS Organizations that: Allows a payer account to receive a single invoice for all linked accounts Aggregates usage to unlock volume discounts and Savings Plans sharing Maintains individual account-level cost tracking via: Cost Explorer Tags and linked account filters AWS Budgets and CUR (Cost and Usage Reports)  Explanation from the image: “Consolidated billing is a feature of AWS Organizations that allows a single AWS account to pay the bills for multiple AWS accounts. This can be useful for companies that have multiple AWS accounts, as it allows them to see all of their costs on a single invoice, while still being able to track the costs of each department separately.” This confirms that consolidated billing solves both goals: Centralized payment Department-level cost visibility

SOURCES

AWS Organizations – Consolidated Billing Cost Allocation Best Practices

DISTRACTOR BREAKDOWN

A. AWS Cost and Usage Reports Provides granular usage data — but doesn’t consolidate invoices

C. Savings Plans Offers cost savings — doesn’t manage billing or account grouping

D. AWS Budgets Tracks spend against thresholds — doesn’t unify billing across accounts

REINFORCING WITH REAL-WORLD STRATEGY

Let’s map this to a practical multi-account billing strategy: -

Feature

Role in Cost Management -

Consolidated Billing

Centralizes invoices and enables discount sharing -

Linked Accounts

Isolate department spend for tagging and reporting -

Cost Allocation Tags

Attribute costs to teams, projects, or environments -

Cost Explorer

Visualize spend trends across accounts and services -

AWS Budgets

Set thresholds and alerts for each department -

CUR (Cost and Usage Report)

Export detailed billing data for analysis in Athena or Quick Sight

REAL-WORLD TIE-IN

If your Nairobi-based company has Dev, QA, and Prod accounts: Use AWS Organizations to link accounts under a payer Enable cost allocation tags for team-level tracking Set department-specific budgets and alerts Export CUR to S3 and analyze with Athena for monthly breakdowns That's how AWS delivers centralized billing with granular cost visibility across departments using consolidated billing.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 199

**A company wants to eliminate the need to guess infrastructure capacity before deployments. The company also wants to spend its budget on cloud resources only as the company uses the resources. Which advantage of the AWS Cloud matches the company's requirements?

**

OPTIONS

- A. Reliability
- B. Global reach
- C. Economies of scale
- D. Pay-as-you-go pricing

[!TIP]

Correct Answer:

D. Pay-as-you-go pricing

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud financial agility, especially how AWS enables on-demand resource provisioning and usage-based billing — critical for startups and enterprises alike.

WHY D IS CORRECT

Pay-as-you-go pricing means: You only pay for what you use, when you use it No need to pre-purchase or overprovision infrastructure Enables cost alignment with actual usage, reducing waste Supports auto scaling and serverless models that dynamically adjust to demand 

Explanation from the image: “By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay-as-you-go prices.” This confirms that AWS pricing is: Usage-based Economically efficient due to shared infrastructure Flexible, allowing teams to scale without upfront commitments

SOURCES

AWS Pricing Overview AWS Cloud Economics

DISTRACTOR BREAKDOWN

A. Reliability Refers to uptime and fault tolerance — not pricing or capacity planning

B. Global reach Enables global deployment — doesn't address cost or capacity guessing

C. Economies of scale A benefit of cloud pricing — but not the mechanism that eliminates capacity guessing

REINFORCING WITH AWS COST STRATEGY

Let's map this to a practical cost management workflow: -

Feature

Role in Cost Optimization -

Pay-as-you-go pricing

Aligns spend with actual usage — no upfront investment -

Auto Scaling

Dynamically adjusts compute capacity based on demand -

AWS Lambda

Serverless execution — charges per request and duration -

AWS Budgets

Tracks spend and sets alerts for cost thresholds -

Cost Explorer

Visualizes usage trends and forecasts future spend

REAL-WORLD TIE-IN

If your Nairobi-based startup is launching a new web app: Use EC2 Auto Scaling or Lambda to avoid guessing capacity Monitor spend with CloudWatch Billing Alarms Forecast future costs with Cost Explorer and AWS Pricing Calculator That's how AWS delivers flexible, usage-based infrastructure that eliminates capacity guessing and aligns spend with actual demand.

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**

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REAL-WORLD TIE-IN

If your Nairobi-based startup is launching a new web app: Use EC2 Auto Scaling or Lambda to avoid guessing capacity Monitor spend with CloudWatch Billing Alarms Forecast future costs with Cost Explorer and AWS Pricing Calculator That's how AWS delivers flexible, usage-based infrastructure that eliminates capacity guessing and aligns spend with actual demand. I can also help you scaffold a cost-aware architecture using Auto Scaling, Lambda, and budget alerts to match your deployment strategy. Absolutely, Leonard — let's build a fully enriched breakdown of this question by integrating the correct answer, your provided explanation, and deeper AWS governance context.

Question 200

****Which AWS service must be enabled in order for the AWS Management Console to monitor all user account changes?**

OPTIONS

- A. AWS CloudTrail
- B. Amazon Simple Notification Service (Amazon SNS)
- C. VPC Flow Logs
- D. AWS CloudHSM

[!TIP]

Correct Answer:

A. AWS CloudTrail

EXPLANATION

CONCEPTUAL FRAMING

This question targets account-level auditing and governance, especially how AWS enables visibility into user actions across services and interfaces.

WHY A IS CORRECT

AWS CloudTrail: Records API calls and console actions across AWS services Captures events from: AWS Management Console AWS SDKs CLI tools Other AWS services Provides a comprehensive event history for: Security analysis Resource change tracking Operational troubleshooting Integrates with Amazon S3, CloudWatch Logs, and EventBridge for storage, monitoring, and automation ■ Explanation from the image: “AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of your AWS account. With CloudTrail, you can log, continuously monitor, and retain account activity related to actions across your AWS infrastructure. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS SDKs, command line

tools, and other AWS services. This event history simplifies security analysis, resource change tracking, and troubleshooting. In addition, you can use CloudTrail to detect unusual activity in your AWS accounts.” This confirms that CloudTrail is the central audit log for AWS environments, enabling both real-time monitoring and historical analysis.

SOURCES

AWS CloudTrail Documentation Monitoring Account Activity

DISTRACTOR BREAKDOWN

B. Amazon SNS Used for notifications and messaging — doesn’t log user actions

C. VPC Flow Logs Captures network traffic metadata — not user account changes

D. AWS CloudHSM Provides hardware-based key storage — unrelated to activity monitoring

REINFORCING WITH GOVERNANCE STRATEGY

Let’s map CloudTrail into a broader governance and security workflow: -

Feature

Role in Governance -

CloudTrail Logs

Record all API and console activity -

EventBridge Rules

Trigger alerts or automation on suspicious actions -

CloudWatch Logs

Store and analyze log data for trends -

S3 Buckets

Archive logs for long-term retention and compliance -

AWS Config

Track resource configuration changes alongside Cloud Trail events

REAL-WORLD TIE-IN

If your Nairobi-based team manages multiple AWS accounts: Enable CloudTrail across all accounts via AWS Organizations Store logs in a centralized S3 bucket with encryption and lifecycle policies Use EventBridge and Lambda to respond to unauthorized changes or access attempts That's how AWS delivers full-stack visibility and auditability of user actions through CloudTrail, empowering teams to maintain compliance and security.

REFERENCES

<https://aws.amazon.com/cloudtrail/>

Question 201

**Recently, an ecommerce firm began using the AWS Cloud. Which security-related responsibilities fall within the purview of the business? (Select two)

**

OPTIONS

A. Restrict who is allowed physical access to the hosts that run the company's Amazon EC2 instances

B. Install security patches on Amazon EC2 Linux instances

C. Choose to encrypt data at rest that is stored on Amazon S3

D. Wipe Amazon Elastic Block Store (Amazon EBS) volumes clean before they are decommissioned

E. Conduct database patching for Amazon RDS instances

[!TIP]

Correct Answer:

B. Install security patches on Amazon EC2 Linux instances and C. Choose to encrypt data at rest that is stored on Amazon S3

EXPLANATION

CONCEPTUAL FRAMING

This question tests your understanding of the AWS Shared Responsibility Model, which divides security duties between AWS and the customer. AWS secures the infrastructure, while customers secure their data, configurations, and operating systems. 🔍 Why B and C Are Correct

B.

Install security patches on Amazon EC2 Linux instances EC2 is an Infrastructure-as-a-Service (IaaS) offering Customers are responsible for: OS-level maintenance Security patching Firewall configuration and access controls This ensures that vulnerabilities in the guest OS are addressed promptly

C.

Choose to encrypt data at rest that is stored on Amazon S3 S3 is a managed service, but customers control: Data classification Encryption settings (e.g., SSE-S3, SSE-KMS, client-side encryption) IAM policies and bucket permissions AWS provides the tools, but customers must choose and enforce encryption policies These responsibilities reflect the customer's role in managing data security, access control, and system-level hygiene. For abstracted services like Amazon S3 and DynamoDB, AWS manages the infrastructure, OS, and platform. Customers

are responsible for managing their data (including encryption), classifying assets, and applying IAM permissions.

SOURCES

AWS Shared Responsibility Model Security Best Practices for EC2 S3 Encryption Options

DISTRACTOR BREAKDOWN

A. Restrict physical access to EC2 hosts AWS handles physical security at data centers — not the customer's responsibility

D. Wipe EBS volumes before decommissioning AWS automatically sanitizes storage before reuse — customers don't need to manually wipe

E. Conduct database patching for Amazon RDS RDS is a managed service — AWS handles patching and maintenance

REAL-WORLD TIE-IN

If your Nairobi-based ecommerce firm runs EC2-based web servers and stores customer data in S3: You must regularly patch EC2 instances to prevent exploits You should enable S3 encryption and enforce access policies via IAM AWS will handle data center security, hardware maintenance, and managed service patching That's how AWS empowers businesses to focus on data and application-level security, while AWS secures the underlying infrastructure.

Question 202

******A business needs to guarantee that users of the AWS Management Console adhere to password complexity guidelines. How can a business customize the difficulty of its passwords?

OPTIONS

- A. Using an AWS IAM user policy
- B. Using an AWS Organizations service control policy (SCP)
- C. Using an AWS IAM account password policy
- D. Using an AWS Security Hub managed insight

[!TIP]

Correct Answer:

- C. Using an AWS IAM account password policy

EXPLANATION

CONCEPTUAL FRAMING

This question targets identity and access management hygiene, especially how AWS enables organizations to enforce strong password policies for IAM users accessing the Management Console.

WHY C IS CORRECT

AWS IAM account password policy allows administrators to:

- Define minimum password length
- Require uppercase, lowercase, numbers, and non-alphanumeric characters
- Enforce password expiration and reuse prevention
- Set mandatory rotation periods for IAM user passwords

These settings apply to console access only, not programmatic access via access keys. This ensures that IAM users comply with organizational security standards, reducing the risk of weak credentials and unauthorized access. You can set a custom password policy on your AWS

account to specify complexity requirements and mandatory rotation periods for your IAM users' passwords. If you don't set a custom password policy, IAM user passwords must meet the default Amazon password policy.

SOURCES

IAM Password Policy Documentation AWS Security Best Practices

DISTRACTOR BREAKDOWN

A. IAM user policy Controls permissions, not password complexity

B. SCP (Service Control Policy) Used to limit actions across accounts — doesn't manage password rules

D. Security Hub managed insight Provides security findings and recommendations — not enforcement of password policies

REAL-WORLD TIE-IN

If your Nairobi-based team manages multiple IAM users: Use IAM account password policy to enforce strong credentials Combine with MFA (Multi-Factor Authentication) for added protection Monitor compliance via AWS Config and Security Hub That's how AWS empowers businesses to enforce secure, customizable password standards for console access using IAM account password policies.

Question 203

****Which AWS service would determine if a security group has granted unlimited access to a resource?**

OPTIONS

A. AWS Trusted Advisor

B. Amazon CloudWatch

C. VPC Flow Logs

D. AWS CloudTrail

[!TIP]

Correct Answer:

A. AWS Trusted Advisor

EXPLANATION

CONCEPTUAL FRAMING

This question targets security posture assessment, especially how AWS helps identify misconfigured access controls that could expose resources to the public internet.

WHY A IS CORRECT

AWS Trusted Advisor provides real-time guidance to help: Optimize performance and cost
Improve fault tolerance Strengthen security posture One of its key security checks is: Security Groups — Specific Ports Unrestricted: Flags rules that allow unrestricted access (0.0.0.0/0) to sensitive ports like SSH (22) and RDP (3389) Helps prevent malicious activity, such as brute-force attacks or unauthorized remote access This makes Trusted Advisor a proactive auditing tool for identifying risky configurations before they lead to compromise. Trusted Advisor includes controls for security configurations of AWS resources. For example, the “Security Groups —

Specific Ports Unrestricted” check identifies rules that allow unrestricted access to ports like SSH and RDP, which increases the risk of hacking, denial-of-service attacks, or data loss.

SOURCES

AWS Trusted Advisor Security Checks Trusted Advisor Best Practices

DISTRACTOR BREAKDOWN

B. Amazon CloudWatch Monitors metrics and logs — doesn’t audit security group configurations

C. VPC Flow Logs Captures network traffic metadata — doesn’t flag misconfigured access rules

D. AWS CloudTrail Logs API activity — useful for tracking changes, but not for evaluating exposure risks

REAL-WORLD TIE-IN

If your Nairobi-based team manages EC2 instances with public-facing services: Use Trusted Advisor to detect open ports and overly permissive security groups Combine with AWS Config for compliance tracking Automate remediation with EventBridge and Lambda That’s how AWS empowers teams to audit and secure network access configurations using Trusted Advisor’s built-in checks.

REFERENCES

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Question 204

**Which features are available to users while using AWS KMS?

**

OPTIONS

- A. Create and manage AWS access keys for the AWS account root user
- B. Create and manage AWS access keys for an AWS account IAM user
- C. Create and manage keys for encryption and decryption of data
- D. Create and manage keys for multi-factor authentication

[!TIP]

Correct Answer:

- C. Create and manage keys for encryption and decryption of data

EXPLANATION

CONCEPTUAL FRAMING

This question targets data protection and cryptographic control, especially how AWS enables secure key management for encryption workflows across services and applications.

WHY C IS CORRECT

AWS Key Management Service (KMS) is a fully managed encryption and key management service designed for: Creating, rotating, disabling, and deleting cryptographic keys Encrypting and decrypting data across AWS services and custom applications Integrating with services like S3, EBS, RDS, Lambda, and CloudTrail Supporting envelope encryption, automatic key rotation, and audit logging via CloudTrail This empowers users to protect sensitive data at rest

and in transit, while maintaining centralized control over key usage and access policies. AWS KMS is an encryption and key management service scaled for the cloud. KMS keys and functionality are used by other AWS services, and you can use them to protect data in your own applications that use AWS.

SOURCES

AWS KMS Overview KMS Developer Guide

DISTRACTOR BREAKDOWN

A. Access keys for root user Managed via IAM — not KMS functionality

B. Access keys for IAM users IAM handles access credentials — KMS handles encryption keys

D. Keys for MFA MFA uses virtual or hardware tokens, not KMS-managed encryption keys

REINFORCING WITH ENCRYPTION STRATEGY

Let's map KMS into a broader security architecture: -

Feature

Role in Data Protection -

Customer Master Keys (CMKs)

Centralized key control for encryption workflows -

Automatic Key Rotation

Reduces risk of long-lived keys -

Envelope Encryption

Efficient encryption using data keys wrapped by CM Ks -

CloudTrail Integration

Logs all key usage for audit and compliance -

IAM Policies & Grants

Fine-grained access control over key usage

REAL-WORLD TIE-IN

If your Nairobi-based ecommerce firm stores customer data in S3 and logs in CloudTrail: Use KMS to encrypt S3 buckets and CloudTrail logs Enable automatic key rotation for long-term security Monitor key usage with CloudTrail events and IAM policies That's how AWS empowers businesses to secure sensitive data with scalable, auditable encryption using AWS KMS.

Question 205

**On Amazon EC2 instances, a business has installed various relational databases. Each month, the database software manufacturer publishes new security updates for databases that must be deployed. Which method is the MOST EFFECTIVE for applying security patches?

**

OPTIONS

A. Connect to each database instance on a monthly basis, and download and apply the necessary security patches from the vendor

B. Enable automatic patching for the instances using the Amazon RDS console

C. In AWS Config, configure a rule for the instances and the required patch level

D. Use AWS Systems Manager to automate database patching according to a schedule

[!TIP]

Correct Answer:

D. Use AWS Systems Manager to automate database patching according to a schedule

EXPLANATION

CONCEPTUAL FRAMING

This question targets operational efficiency and security hygiene, especially how AWS enables automated patch management for EC2-hosted applications and databases.

WHY D IS CORRECT

AWS Systems Manager Patch Manager: Automates patching for EC2 instances, on-prem servers, and edge devices Supports both security updates and application patches Allows you to: Define patch baselines Schedule patching windows Apply patches across fleets of instances Monitor compliance via Systems Manager Compliance Dashboard Reduces manual effort and ensures consistent patching across environments Patch Manager, a capability of AWS Systems Manager, automates the process of patching managed nodes with both security-related and other types of updates. You can patch fleets of Amazon EC2 instances, edge devices, or your on-premises servers and virtual machines (VMs) by operating system type.

SOURCES

AWS Systems Manager Patch Manager Automating Patch Management

DISTRACTOR BREAKDOWN

A. Manual patching Labor-intensive, error-prone, and not scalable for multiple instances

B. RDS console patching Applies only to Amazon RDS, not EC2-hosted databases

C. AWS Config rule Tracks compliance — doesn't apply patches or automate updates

REINFORCING WITH PATCH STRATEGY

Let's map Patch Manager into a broader security and ops workflow: -

Feature

Role in Patch Management -

Patch Baselines

Define approved patches and exceptions -

Maintenance Windows

Schedule patching during low-impact periods -

Compliance Reporting

Monitor which instances are patched or out-of-date -

Automation Documents (SSM Documents)

Customize patch workflows and remediation steps -

Integration with CloudWatch and EventBridge

Trigger alerts or automation based on patch status

REAL-WORLD TIE-IN

If your Nairobi-based team runs PostgreSQL and MySQL on EC2: Use Patch Manager to schedule monthly updates Define OS-specific baselines for Linux and Windows Monitor patch compliance with Systems Manager dashboards and CloudWatch metrics That's how AWS empowers businesses to automate and scale secure patching workflows across EC2-hosted databases using Systems Manager.

REFERENCES

Question 206

**A business is releasing a new application on AWS. The application will be hosted on an Amazon Elastic Compute Cloud (EC2) instance. Additional EC2 instances will be required as the demand grows. Which AWS service or technology can the business utilize to deploy the required number of EC2 instances?

**

OPTIONS

A. Elastic Load Balancing

B. Amazon EC2 Auto Scaling

C. AWS App2Container (A2C)

D. AWS Systems Manager

[!TIP]

Correct Answer:

B. Amazon EC2 Auto Scaling

EXPLANATION

CONCEPTUAL FRAMING

This question targets scalability and elasticity, especially how AWS enables applications to respond to changing demand automatically without manual provisioning.

WHY B IS CORRECT

Amazon EC2 Auto Scaling: Automatically adjusts the number of EC2 instances in response to: Traffic spikes CPU/memory thresholds Custom CloudWatch metrics Uses Auto Scaling Groups (ASGs) to: Define minimum, maximum, and desired capacity Maintain availability and performance Replace unhealthy instances automatically Supports scheduled scaling and predictive scaling for proactive resource management Amazon EC2 Auto Scaling helps ensure you have the correct number of EC2 instances available to handle load. You create Auto Scaling groups with defined min/max instance counts, and EC2 Auto Scaling ensures your group stays within those bounds.

SOURCES

Amazon EC2 Auto Scaling Documentation Scaling Strategies

DISTRACTOR BREAKDOWN

A. Elastic Load Balancing Distributes traffic — doesn't launch or scale EC2 instances

C. AWS App2Container Containerizes legacy apps — not used for scaling EC2 fleets

D. AWS Systems Manager Manages infrastructure — doesn't handle instance provisioning or scaling

REINFORCING WITH SCALING STRATEGY

Let's map EC2 Auto Scaling into a broader deployment architecture: -

Component

Role -

Auto Scaling Group (ASG)

Manages EC2 fleet size based on demand -

Launch Template

Defines instance configuration (AMI, type, security groups) -

CloudWatch Alarms

Trigger scaling actions based on metrics -

Elastic Load Balancer (ELB)

Distributes traffic across scaled instances -

Lifecycle Hooks

Customize actions during instance launch/termination

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a web app with unpredictable traffic: Use EC2 Auto Scaling to maintain performance during peak hours Pair with Application Load Balancer for traffic distribution Monitor scaling events with CloudWatch and EventBridge That's how AWS delivers automated, responsive infrastructure scaling using EC2 Auto Scaling, ensuring performance and cost-efficiency.

REFERENCES

<https://aws.amazon.com/ec2/>

Question 207

**What does it mean to provide AWS IAM users the fewest possible privileges?

**

OPTIONS

A. It is granting permissions to a single user only

- B. It is granting permissions using AWS IAM policies only
- C. It is granting AdministratorAccess policy permissions to trustworthy users
- D. It is granting only the permissions required to perform a given task

[!TIP]

Correct Answer:

- D. It is granting only the permissions required to perform a given task

EXPLANATION

CONCEPTUAL FRAMING

This question targets the principle of least privilege, a foundational concept in cloud security. It ensures that users, roles, and services have only the access they need — and nothing more.

WHY D IS CORRECT

Least privilege means: Granting minimal permissions necessary to complete a specific task
Avoiding broad or overly permissive policies like AdministratorAccess unless absolutely required
Regularly reviewing and refining IAM policies to match evolving responsibilities
Using IAM roles, conditions, and scoped policies to enforce granular access
When you create IAM policies, follow the standard security advice of granting least privilege — or granting only the permissions required to perform a task. Determine what users (and roles) need to do and then craft policies that allow them to perform only those tasks. This approach reduces the attack surface, limits accidental misuse, and strengthens compliance posture.

SOURCES

IAM Best Practices Least Privilege Principle

DISTRACTOR BREAKDOWN

A. Single user only Least privilege applies to all users and roles, not just one

B. IAM policies only Policies are the mechanism — not the definition of least privilege

C. AdministratorAccess for trustworthy users Violates least privilege by granting full access unnecessarily

REINFORCING WITH IAM STRATEGY

Let's map least privilege into a secure IAM workflow: -

Practice

Description -

Scoped Policies

Use Action, Resource, and Condition to limit access -

Role-Based Access Control (RBAC)

Assign permissions based on job function -

IAM Access Analyzer

Detect overly permissive policies -

Policy Versioning

Track changes and roll back if needed -

Regular Audits

Review and refine permissions periodically

REAL-WORLD TIE-IN

If your Nairobi-based team manages developers and analysts: Grant read-only access to S3 for analysts Allow EC2 start/stop permissions for developers — but not termination Use IAM roles

with session policies for temporary elevated access That's how AWS empowers teams to securely manage access by enforcing least privilege across IAM users and roles.

Question 208

**When building an Amazon Relational Database Service (Amazon RDS) instance in Multiple Availability Zone mode, which architectural concept is followed?

**

OPTIONS

A. Implement loose coupling

B. Design for failure

C. Automate everything that can be automated

D. Use services, not servers

[!TIP]

Correct Answer:

B. Design for failure

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilient architecture design, especially how AWS enables high availability and fault tolerance through Multi-AZ deployments.

WHY B IS CORRECT

Design for failure means: Anticipating that hardware, network, or service disruptions will occur
Architecting systems to recover automatically and gracefully Minimizing downtime and data loss through redundancy and failover mechanisms In Amazon RDS Multi-AZ mode: AWS automatically provisions a standby replica in a different Availability Zone Synchronous replication ensures data consistency between primary and standby In case of failure (e.g., AZ outage, instance crash), automatic failover redirects traffic to the standby This enhances availability, durability, and disaster recovery Amazon RDS Multi-AZ deployments enhance availability and durability for database instances. They support automatic failover across Availability Zones, ensuring reliability even during infrastructure failures — embodying the principle of designing for failure.

SOURCES

Amazon RDS Multi-AZ Deployments AWS Well-Architected Framework – Reliability Pillar

DISTRACTOR BREAKDOWN

A. Loose coupling Refers to minimizing dependencies between components — not relevant to RDS failover

C. Automate everything Valuable principle, but not the core concept behind Multi-AZ resilience

D. Use services, not servers Encourages managed services — RDS is one, but this doesn't explain Multi-AZ behavior

REINFORCING WITH RESILIENT ARCHITECTURE STRATEGY

Let's map Multi-AZ into a broader reliability workflow: -

Feature

Role in Resilience -

Multi-AZ Deployment

Provides automatic failover and data durability -

Synchronous Replication

Ensures data consistency across zones -

Health Monitoring

Detects failures and triggers failover -

Backup & Restore

Complements failover with point-in-time recovery -

Read Replicas (Single-AZ)

Used for scaling reads — not for failover

REAL-WORLD TIE-IN

If your Nairobi-based ecommerce firm runs PostgreSQL on RDS: Enable Multi-AZ to ensure uptime during AZ outages Use CloudWatch alarms to monitor failover events Combine with automated backups and snapshots for full recovery coverage That's how AWS empowers teams to build fault-tolerant database architectures by designing for failure with Multi-AZ RDS deployments.

REFERENCES

<https://aws.amazon.com/rds/>

Question 209

**What is a user's responsibility while using the AWS Cloud to execute an application?

**

OPTIONS

- A. Managing physical hardware
- B. Updating the underlying hypervisor
- C. Providing a list of users approved for data center access
- D. Managing application software updates

[!TIP]

Correct Answer:

D. Managing application software updates

EXPLANATION

CONCEPTUAL FRAMING

This question targets the AWS Shared Responsibility Model, which defines the boundary between what AWS secures and what the customer must manage. When running applications on EC2 or other compute services, customers retain control over the guest operating system, application stack, and configurations.

WHY D IS CORRECT

Customers are responsible for: Managing the guest OS (e.g., Linux, Windows) Applying security patches and updates to application software Configuring firewalls and IAM policies Ensuring compliance with internal policies and external regulations AWS handles: Physical infrastructure Network and hypervisor security Facility access controls and hardware maintenance The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software, as well as the configuration of the AWS-provided security group firewall. Responsibilities vary depending on the services used and how they're integrated into your IT environment.

SOURCES

AWS Shared Responsibility Model Security Best Practices for EC2

DISTRACTOR BREAKDOWN

A. Managing physical hardware AWS owns and operates the physical infrastructure — customers never touch the hardware

B. Updating the hypervisor AWS manages the virtualization layer — customers don't have access to it

C. Data center access approvals AWS controls physical access — customers don't manage facility-level permissions

REINFORCING WITH RESPONSIBILITY MAPPING

Let's map responsibilities across service types: -

AWS Service Type

Customer Responsibilities -

EC2 (IaaS)

OS patching, app updates, firewall config -

RDS (PaaS)

Data encryption, access control, backups -

Lambda (FaaS)

Code security, IAM roles, event triggers -

S3 (Storage)

Data classification, encryption, bucket policies

REAL-WORLD TIE-IN

If your Nairobi-based team deploys a Node.js app on EC2: You must update Node.js and dependencies regularly Configure security groups and IAM roles Monitor for vulnerabilities using Amazon Inspector or Systems Manager That's how AWS empowers customers to secure and maintain their application stack while AWS handles the infrastructure layer.

Question 210

**Which statement is true about AWS's worldwide infrastructure?

**

OPTIONS

- A. Availability Zones can span multiple AWS Regions
- B. A VPC can have different subnets in different AWS Regions
- C. AWS Regions consist of multiple Availability Zones
- D. A single subnet can span multiple Availability Zones

[!TIP]

Correct Answer:

C. AWS Regions consist of multiple Availability Zones

EXPLANATION

CONCEPTUAL FRAMING

This question targets AWS's global infrastructure design, especially how it balances fault isolation, scalability, and geographic reach through a hierarchy of Regions and Availability Zones.

WHY C IS CORRECT

AWS Regions are: Geographically isolated areas (e.g., us-east-1, eu-west-1) Each Region contains multiple Availability Zones (AZs) AZs are physically separate data centers with independent power, cooling, and networking This design enables high availability and fault tolerance across AZs within a Region AWS has a more extensive global footprint than other cloud providers. It rapidly opens new Regions to support global customer needs. Each Region consists of multiple Availability Zones, enabling customers to deploy resilient applications across isolated fault domains.

SOURCES

AWS Global Infrastructure Regions and Availability Zones

DISTRACTOR BREAKDOWN

A. AZs spanning multiple Regions AZs are confined to a single Region — they never span across Regions

B. VPC subnets across Regions A VPC is Region-specific — subnets must reside within the same Region

D. Subnet spanning AZs A subnet is confined to one AZ — it cannot span multiple zones

REINFORCING WITH INFRASTRUCTURE STRATEGY

Let's map AWS's infrastructure hierarchy: -

Layer

Description -

Region

Geographic area with multiple AZs -

Availability Zone (AZ)

Isolated data center within a Region -

VPC

Region-scoped virtual network -

Subnet

AZ-scoped segment of a VPC -

Service Deployment

Can span AZs for resilience (e.g., RDS Multi-AZ, EC2 Auto Scaling)

REAL-WORLD TIE-IN

If your Nairobi-based team deploys a web app in af-south-1 (Cape Town): You can place EC2 instances in multiple AZs within that Region Use Elastic Load Balancing to distribute traffic across AZs Design for failover and redundancy without crossing Region boundaries That's how AWS empowers teams to build globally distributed, fault-tolerant applications using Regions and Availability Zones.

Question 211

**A business wants to link AWS to its corporate network through a private network connection. Which AWS service or functionality will satisfy this requirement?

**

OPTIONS

A. Amazon Connect

B. Amazon Route 53

C. AWS Direct Connect

D. VPC peering

[!TIP]

Correct Answer:

C. AWS Direct Connect

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud connectivity, especially how AWS enables dedicated, private network links between customer data centers and AWS infrastructure.

WHY C IS CORRECT

AWS Direct Connect: Establishes a dedicated physical connection between your premises and AWS. Bypasses the public internet for lower latency, higher bandwidth, and consistent performance. Supports private VIFs (Virtual Interfaces) to access VPCs and public VIFs for AWS services. Often used for: Hybrid cloud architectures, Data-intensive workloads, Regulated environments requiring private links. AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. It enables private connectivity between AWS and your datacenter, office, or colocation environment, often reducing network costs and improving performance compared to internet-based connections.

SOURCES

AWS Direct Connect Overview Hybrid Connectivity Options

DISTRACTOR BREAKDOWN

A. Amazon Connect A contact center service — not related to network connectivity

B. Amazon Route 53 A DNS service — doesn't establish private network links

D. VPC peering Connects two VPCs — not suitable for linking on-prem networks to AWS

REINFORCING WITH HYBRID ARCHITECTURE STRATEGY

Let's map Direct Connect into a broader hybrid cloud setup: -

Component

Role -

AWS Direct Connect

Dedicated private link to AWS -

Virtual Private Gateway

Terminates Direct Connect into a VPC -

Router on Premises

Connects to AWS via fiber or Ethernet -

BGP (Border Gateway Protocol)

Manages routing between networks -

CloudHub or Transit Gateway

Enables multi-VPC or multi-site connectivity

REAL-WORLD TIE-IN

If your Nairobi-based enterprise runs workloads in both AWS and an on-prem data center: Use Direct Connect for secure, high-throughput data transfer Pair with Transit Gateway to route traffic across multiple VPCs Monitor link health and usage with CloudWatch and NetFlow tools

That's how AWS empowers businesses to build secure, high-performance hybrid networks using Direct Connect.

Question 212

**Which of the following may be used to restrict certain users' access to Amazon Simple Storage Service (Amazon S3) buckets?

**

OPTIONS

A. A public and private key-pair

B. Amazon Inspector

C. AWS Identity and Access Management (IAM) policies

D. Security Groups

[!TIP]

Correct Answer:

C. AWS Identity and Access Management (IAM) policies

EXPLANATION

CONCEPTUAL FRAMING

This question targets fine-grained access control, especially how AWS enables user-level permissions for managing access to S3 buckets and objects.

WHY C IS CORRECT

IAM policies are the primary mechanism for: Granting or denying access to AWS resources
Defining who can perform what actions on which resources Using conditions like IP address, VPC endpoint, MFA, or tags to restrict access For S3, IAM policies can: Allow or deny actions like s3:GetObject, s3:PutObject, s3:ListBucket Be attached to users, groups, or roles Work alongside bucket policies for layered control To allow users to perform S3 actions on the bucket from VPC endpoints or IP addresses, you must explicitly grant those user-level permissions. You can grant user-level permissions using IAM policies or statements in the bucket policy.

SOURCES

IAM Policies for S3 S3 Bucket Policy Examples

DISTRACTOR BREAKDOWN

A. Public/private key-pair Used for encryption and authentication, not access control policies

B. Amazon Inspector Scans for security vulnerabilities — doesn't manage access permissions

D. Security Groups Control network-level access — not applicable to S3, which is a global service

REINFORCING WITH ACCESS CONTROL STRATEGY

Let's map IAM policies into a broader S3 security framework: -

Control Type

Role -

IAM Policies

Define user-level access to S3 actions -

Bucket Policies

Set resource-level permissions for buckets -

Access Control Lists (ACLs)

Legacy method for object-level permissions -

VPC Endpoint Policies

Restrict access via private network paths -

Encryption Keys (KMS)

Control access to encrypted data

REAL-WORLD TIE-IN

If your Nairobi-based team stores customer data in S3: Use IAM policies to restrict access to specific users or roles Combine with bucket policies to enforce organization-wide rules Apply conditions like IP address or VPC endpoint for extra control That's how AWS empowers teams to secure S3 buckets with precise, user-level access controls using IAM policies.

Question 213

**Which of the following is an AWS Well-Architected Framework design principle?

**

OPTIONS

A. Reduce downtime by making infrastructure changes infrequently and in large increments

B. Invest the time to configure infrastructure manually

C. Learn to improve from operational failures

D. Use monolithic application design for centralization

[!TIP]

Correct Answer:

C. Learn to improve from operational failures

EXPLANATION

CONCEPTUAL FRAMING

This question targets the Operational Excellence pillar of the AWS Well-Architected Framework, which emphasizes continuous improvement, automation, and resilience through structured learning from failures.

WHY C IS CORRECT

AWS encourages teams to: Treat failures as learning opportunities Conduct post-incident analysis and share lessons across teams Use failures to drive architectural and process improvements Build mechanisms to detect, respond, and recover quickly Learn from all operational failures: Drive improvement through lessons learned from all operational events and failures. Share what is learned across teams and through the entire organization. This principle fosters a culture of resilience and accountability, where every incident becomes a catalyst for better design and stronger operations.

SOURCES

AWS Well-Architected Framework Operational Excellence Pillar

DISTRACTOR BREAKDOWN

A. Large, infrequent changes Violates the principle of small, reversible deployments — increases risk and downtime

B. Manual configuration Contradicts the principle of automating everything that can be automated

D. Monolithic design Opposes the principle of loose coupling and microservices for agility and scalability

REINFORCING WITH DESIGN PRINCIPLES

Let's highlight key AWS design principles: -

Principle

Description -

Stop guessing capacity

Use auto scaling and serverless to match demand -

Test recovery procedures

Simulate failures to validate resilience -

Automate to improve consistency

Reduce human error and accelerate delivery -

Make frequent, small, reversible changes

Minimize blast radius and enable agility -

Learn from failures

Institutionalize postmortems and continuous improvement

REAL-WORLD TIE-IN

If your Nairobi-based team experiences a Lambda timeout: Conduct a blameless postmortem
Share findings across dev and ops teams Update monitoring thresholds and retry logic
Document the fix in your runbooks and architecture notes That's how AWS empowers teams to build resilient systems by learning from operational failures and continuously improving.

Question 214

**Which AWS service enables a business to identify and reroute customers to other servers in the event of a website server outage?

**

OPTIONS

A. Amazon CloudFront

B. Amazon GuardDuty

C. Amazon Route 53

D. AWS Trusted Advisor

[!TIP]

Correct Answer:

C. Amazon Route 53

EXPLANATION

CONCEPTUAL FRAMING

This question targets high availability and fault tolerance, especially how AWS enables automated DNS-level failover to reroute traffic during outages.

WHY C IS CORRECT

Amazon Route 53 is a highly available and scalable Domain Name System (DNS) web service that: Routes end-user requests to infrastructure in a globally distributed, resilient manner Supports routing policies such as: Failover routing: Automatically redirects traffic to healthy endpoints Latency-based routing: Sends users to the lowest-latency Region Geolocation routing: Routes based on user location Integrates with health checks to detect outages and trigger failover Amazon Route 53 allows you to set routing policies to pre-determine and automate responses in case of failure, like redirecting traffic to alternative Availability Zones or Regions.

SOURCES

Amazon Route 53 Routing Policies Route 53 Health Checks and Failover

DISTRACTOR BREAKDOWN

A. Amazon CloudFront Distributes content via edge locations — doesn't reroute based on server health

B. Amazon GuardDuty Detects threats — not used for routing or failover

D. AWS Trusted Advisor Provides recommendations — doesn't actively reroute traffic

REINFORCING WITH ROUTING STRATEGY

Let's map Route 53 into a resilient web architecture: -

Component

Role -

Route 53 DNS

Resolves domain names to IP addresses or endpoints -

Health Checks

Monitor endpoint availability -

Failover Routing Policy

Redirect traffic to standby endpoints during outages -

Latency-Based Routing

Optimize performance by routing to closest Region -

Integration with ELB and CloudFront

Enhance global delivery and resilience

REAL-WORLD TIE-IN

If your Nairobi-based team runs a web app across multiple Regions: Use Route 53 failover routing to redirect users if the primary Region goes down Combine with CloudWatch alarms to monitor health check failures Ensure standby endpoints are warm and ready for traffic That's how AWS empowers businesses to maintain uptime and customer experience by rerouting traffic using Route 53 during server outages.

REFERENCES

<https://aws.amazon.com/route53/>

Question 215

**An organization with a Developer Support plan has established an Amazon RDS database but is unable to connect to it. To get this degree of help, who should the developer contact?

**

OPTIONS

A. AWS Support using a support case

B. AWS Professional Services

C. AWS technical account manager

D. AWS consulting partners

[!TIP]

Correct Answer:

A. AWS Support using a support case

EXPLANATION

CONCEPTUAL FRAMING

This question targets AWS support plan capabilities, especially how Developer Support enables technical troubleshooting through direct engagement with AWS Support engineers.

WHY A IS CORRECT

The Developer Support plan includes: Best practice guidance Client-side diagnostic tools Building-block architecture support Ability to open unlimited support cases via the AWS Support Center Access to AWS documentation and forums, plus email support during business hours For issues like RDS connectivity, the developer can: Open a support case describing the problem Receive guidance on network configuration, security groups, IAM roles, and endpoint setup Customers with a Developer Support plan can open unlimited support cases through the AWS Support Center. These cases are handled by AWS engineers who provide guidance on using AWS services, diagnosing issues, and applying best practices.

SOURCES

AWS Support Plans Opening a Support Case

DISTRACTOR BREAKDOWN

B. AWS Professional Services Focuses on long-term strategic projects and enterprise migrations — not day-to-day troubleshooting

C. Technical Account Manager (TAM) Available only with Enterprise Support, not Developer Support

D. AWS Consulting Partners Third-party firms — not part of AWS Support and not included in support plans

REINFORCING WITH SUPPORT PLAN COMPARISON

Plan Key Features

Basic Billing and account support only

Developer Email support, best practices, unlimited cases from one contact

Business 24/7 access, phone/chat, full case management, access to TAMs

Enterprise Dedicated TAM, concierge support, white-glove architecture reviews

REAL-WORLD TIE-IN

If your Nairobi-based team hits a connectivity issue with RDS: Open a support case via the AWS Console Provide error messages, instance details, and security group settings Receive tailored guidance from AWS engineers to resolve the issue That's how AWS empowers

developers to resolve technical issues efficiently through support cases under the Developer Support plan.

Question 216

****Which AWS service enables you to monitor and debug distributed applications end-to-end?**

OPTIONS

A. AWS Cloud9

B. AWS CodeStar

C. AWS Cloud Map

D. AWS X-Ray

[!TIP]

Correct Answer:

D. AWS X-Ray

EXPLANATION

CONCEPTUAL FRAMING

This question targets application observability, especially how AWS enables developers to trace requests across microservices and identify performance bottlenecks.

WHY D IS CORRECT

AWS X-Ray is designed for: End-to-end tracing of requests across distributed systems
Visualizing service maps to see how components interact Identifying latency spikes, errors, and root causes Supporting custom annotations and filtering by trace ID, service, or error type It integrates with: Lambda, EC2, ECS, EKS, and API Gateway SDKs for Node.js, Python, Java, Go, .NET, Ruby AWS X-Ray is a powerful tool that enables developers to debug production and distributed applications, especially in microservices architectures. It analyzes performance and helps identify root causes of issues quickly.

SOURCES

AWS X-Ray Documentation X-Ray Service Map

DISTRACTOR BREAKDOWN

A. AWS Cloud9 Cloud-based IDE — doesn't provide tracing or observability

B. AWS CodeStar Manages CI/CD pipelines — not used for debugging runtime issues

C. AWS Cloud Map Service discovery tool — doesn't trace or monitor application performance

REINFORCING WITH OBSERVABILITY STRATEGY

Let's map X-Ray into a broader debugging and monitoring workflow: -

Tool

Role -

AWS X-Ray

Traces requests and visualizes service interactions -

Amazon CloudWatch

Collects logs, metrics, and alarms -

AWS Distro for OpenTelemetry (ADOT)

Standardized telemetry collection for X-Ray and Cloud Watch -

AWS Lambda Insights

Deep visibility into serverless performance -

Amazon DevOps Guru

Detects anomalies and suggests remediation

REAL-WORLD TIE-IN

If your Nairobi-based team runs a microservices app across Lambda and API Gateway: Use X-Ray to trace requests from entry point to backend Identify slow services or failed calls Visualize the service map to understand dependencies and latency That's how AWS empowers developers to monitor, trace, and debug distributed applications using X-Ray for full-stack visibility.

Question 217

**Which AWS shared responsibility model duties are the customer's responsibility? (Select two.)

**

OPTIONS

- A. Infrastructure facilities access management
- B. Cloud infrastructure hardware lifecycle management
- C. Configuration management of user's applications
- D. Networking infrastructure protection

[!TIP]

Correct Answer:

C. Configuration management of user's applications and E. Security groups configuration

EXPLANATION

CONCEPTUAL FRAMING

This question targets the AWS Shared Responsibility Model, which defines the boundary between AWS's responsibilities and those of the customer. It's foundational for understanding cloud security and compliance. 🔍 Why C and E Are Correct

C.

Configuration management of user's applications Customers are responsible for: Installing, updating, and securing their application stack Managing runtime configurations, dependencies, and patching Ensuring secure coding practices and access controls

E.

Security groups configuration Security groups act as virtual firewalls for EC2 instances Customers define: Allowed inbound/outbound traffic Port ranges and IP sources Rules that enforce least privilege and segmentation The customer is responsible for the security configuration or firewall (like security groups), Identity and Access Management (IAM), client and server-side encryption, and customer data. AWS handles the infrastructure, but customers must secure what they deploy on it.

SOURCES

AWS Shared Responsibility Model Security Best Practices

DISTRACTOR BREAKDOWN

A. Infrastructure facilities access management AWS controls physical access to data centers

B. Hardware lifecycle management AWS owns and maintains the physical servers and networking gear

D. Networking infrastructure protection AWS secures the underlying network — customers secure their own configurations (e.g., VPC, SGs)

REINFORCING WITH RESPONSIBILITY MAPPING

Layer	AWS Responsibility	Customer Responsibility
Physical Infrastructure	Data center security, hardware maintenance —	
Networking	Backbone routing, DDoS protection VPC, security groups, NACLs	
Compute	Hypervisor, host OS Guest OS, patching, app config	
Storage	Disk encryption, durability Data classification, access policies	
IAM	Root account protection User roles, policies, MFA enforcement	

REAL-WORLD TIE-IN

If your Nairobi-based team deploys a Django app on EC2: You must configure the app securely, manage updates, and patch vulnerabilities Define security group rules to allow only necessary traffic (e.g., port 443 from trusted IPs) Use IAM roles to restrict access to S3 or RDS That's how AWS empowers customers to secure their cloud workloads by owning configuration and access controls under the Shared Responsibility Model.

Question 217

**Which AWS shared responsibility model duties are the customer's responsibility? (Select two.)

**

OPTIONS

- A. Infrastructure facilities access management
- B. Cloud infrastructure hardware lifecycle management
- C. Configuration management of user's applications
- D. Networking infrastructure protection
- E. Security groups configuration

[!TIP]

Correct Answer:

C. Configuration management of user's applications and E. Security groups configuration

EXPLANATION

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and patching Ensuring secure coding practices and access controls

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AWS Shared Responsibility Model Security Best Practices

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B. Hardware lifecycle management AWS owns and maintains the physical servers and networking gear

D. Networking infrastructure protection AWS secures the underlying network — customers secure their own configurations (e.g., VPC, SGs)

REINFORCING WITH RESPONSIBILITY MAPPING

Layer	AWS Responsibility	Customer Responsibility
Physical Infrastructure	Data center security, hardware maintenance —	
Networking	Backbone routing, DDoS protection VPC, security groups, NACLs	

Compute Hypervisor, host OS Guest OS, patching, app config

Storage Disk encryption, durability Data classification, access policies

IAM Root account protection User roles, policies, MFA enforcement

REAL-WORLD TIE-IN

If your Nairobi-based team deploys a Django app on EC2: You must configure the app securely, manage updates, and patch vulnerabilities Define security group rules to allow only necessary traffic (e.g., port 443 from trusted IPs) Use IAM roles to restrict access to S3 or RDS That's how AWS empowers customers to secure their cloud workloads by owning configuration and access controls under the Shared Responsibility Model.

Question 219

**Which variables impact AWS Cloud costs? (Select two.)

**

OPTIONS

A. The number of unused AWS Lambda functions

B. The number of configured Amazon S3 buckets

C. Inbound data transfers without acceleration

D. Outbound data transfers without acceleration

E. Compute resources that are currently in use

[!TIP]

Correct Answer:

D. Outbound data transfers without acceleration and E. Compute resources that are currently in use

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost drivers in AWS, especially how usage patterns and resource consumption directly influence billing. 🔍 Why D and E Are Correct

D.

Outbound data transfers without acceleration AWS charges for data transferred out of AWS to the internet or other Regions Pricing varies by: Region Volume of data Destination (e.g., internet vs. inter-region) Inbound data transfers are generally free, but outbound traffic is a major cost factor

E.

Compute resources that are currently in use EC2, Lambda, Fargate, and other compute services bill based on: Instance type and size Duration of usage Provisioned capacity (e.g., vCPU, memory) Idle or underutilized compute resources still incur charges unless stopped or terminated Charges may apply if there is data transfer between different components of your workload. These charges vary depending on where the components are deployed.

SOURCES

AWS Pricing Calculator AWS Data Transfer Pricing AWS Compute Services Pricing

DISTRACTOR BREAKDOWN

A. Unused Lambda functions No cost unless invoked — storage of function code is negligible

B. Number of S3 buckets Buckets themselves are free — charges apply for stored data and requests

C. Inbound data transfers Typically free — not a cost driver unless using acceleration services

REINFORCING WITH COST STRATEGY

Let's map key AWS cost drivers: -

Compute

Instance type, usage time, autoscaling -

Storage

Volume stored, request frequency, retrieval tier -

Data Transfer

Outbound traffic, inter-region movement -

Support Plans

Tiered pricing based on business needs -

Reserved vs. On-Demand

Long-term commitments reduce cost

REAL-WORLD TIE-IN

If your Nairobi-based team runs a web app with EC2 and S3: Monitor EC2 usage and consider Auto Scaling or Spot Instances Review CloudWatch metrics for outbound traffic spikes Use Trusted Advisor to identify underutilized resources That's how AWS empowers teams to optimize cloud costs by managing compute usage and outbound data transfers.

Question 220

**A corporation is required by law to track and assess configuration changes to AWS resources, as well as to conduct corrective steps. Which AWS service should the business use?

**

OPTIONS

A. AWS Config

B. AWS Secrets Manager

C. AWS CloudTrail

D. AWS Trusted Advisor

[!TIP]

Correct Answer:

A. AWS Config

EXPLANATION

CONCEPTUAL FRAMING

This question targets compliance, auditability, and configuration drift detection, especially how AWS enables businesses to track, evaluate, and remediate changes to cloud resources.

WHY A IS CORRECT

AWS Config is a configuration management and compliance service that: Continuously records resource configurations and changes over time Enables automated evaluation against custom or managed rules Supports remediation actions when resources drift from desired states Provides timeline views and relationship maps for forensic analysis AWS Config continuously

monitors and records your AWS resource configurations. You can automate evaluation and remediation against desired configurations and review historical changes and relationships.

SOURCES

AWS Config Overview Compliance and Governance with AWS Config

DISTRACTOR BREAKDOWN

B. AWS Secrets Manager Manages secrets (e.g., API keys, passwords) — not configuration tracking

C. AWS CloudTrail Logs API calls — useful for auditing, but doesn't evaluate or remediate configuration changes

D. AWS Trusted Advisor Provides best practice checks — not continuous configuration monitoring or rule enforcement

REINFORCING WITH COMPLIANCE STRATEGY

Let's map AWS Config into a broader governance workflow: -

Component

Role -

Configuration Recorder

Tracks changes to supported AWS resources -

Config Rules

Define compliance checks (e.g., encryption, tagging, public access) -

Remediation Actions

Automatically correct non-compliant resources -

Timeline & Relationships

Visualize change history and dependencies -

Integration with CloudTrail

Correlate config changes with API activity

REAL-WORLD TIE-IN

If your Nairobi-based team must comply with GDPR or ISO 27001: Use AWS Config to monitor S3 bucket policies, IAM roles, and EC2 security groups Define custom rules for encryption, tagging, and access control Automate remediation using Systems Manager documents That's how AWS empowers businesses to maintain compliance and auditability by tracking and remediating configuration changes with AWS Config.

REFERENCES

<https://aws.amazon.com/config/>

Question 221

**Which AWS service should be utilized to store data backups for an extended period of time at a reasonable cost?

**

OPTIONS

A. Amazon RDS

B. Amazon Glacier

C. AWS Snowball

[!TIP]

Correct Answer:

B. Amazon Glacier

EXPLANATION

CONCEPTUAL FRAMING

This question targets long-term data retention and cost optimization, especially how AWS enables secure, durable, and low-cost archival storage for backups and compliance.

WHY B IS CORRECT

Amazon S3 Glacier (now part of the S3 storage classes) is designed for: Archiving infrequently accessed data Long-term backup retention for compliance or disaster recovery Low-cost storage — as little as \$0.004/GB/month Retrieval options ranging from minutes to hours depending on urgency Ideal for: Database snapshots Log archives Compliance records Media archives Amazon S3 Glacier is a secure, durable, and low-cost storage class for data archiving and long-term backup. Customers can store large or small amounts of data for as little as \$0.004 per gigabyte per month.

SOURCES

Amazon S3 Glacier Storage Classes S3 Glacier Pricing

DISTRACTOR BREAKDOWN

A. Amazon RDS Manages relational databases — not a storage service for backups

C. AWS Snowball Used for physical data transfer — not for long-term cloud storage

D. AWS EBS Provides block storage for EC2 — higher cost, not ideal for archival

REINFORCING WITH STORAGE STRATEGY

Let's map S3 Glacier into a broader backup and archival workflow: -

Storage Class

Use Case Cost Efficiency -

S3 Standard

Frequent access High -

S3 Intelligent-Tiering

Variable access Adaptive -

S3 Glacier Instant Retrieval

Archive with fast access Moderate -

S3 Glacier Flexible Retrieval

Archive with hours retrieval Low -

S3 Glacier Deep Archive

Rarely accessed, long-term Lowest

REAL-WORLD TIE-IN

If your Nairobi-based team needs to retain customer logs and compliance records for 7+ years: Use S3 Glacier Deep Archive for ultra-low-cost storage Set up lifecycle policies to transition data from S3 Standard to Glacier Enable object lock for write-once-read-many (WORM) compliance That's how AWS empowers teams to store backups affordably and securely using Glacier for long-term retention.

REFERENCES

<https://aws.amazon.com/s3/>

Question 222

**What does the AWS Cloud bring clients in terms of increased execution speed and agility?
(Select two.)

**

OPTIONS

A. Readily available resources with low provisioning times

B. Scalable compute capacity

C. Free Tier services usage

D. Access to AWS data centers

E. Lower resource provisioning cost

[!TIP]

Correct Answer:

A. Readily available resources with low provisioning times and B. Scalable compute capacity

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud agility and operational velocity, especially how AWS enables teams to experiment, iterate, and deploy faster by abstracting infrastructure complexity. 🔍 Why A and B Are Correct

A.

Readily available resources with low provisioning times AWS offers on-demand provisioning of compute, storage, and networking Developers can spin up EC2 instances, Lambda functions, or databases in minutes This eliminates traditional delays from hardware procurement and setup

B.

Scalable compute capacity AWS provides elastic scalability — resources grow or shrink based on demand Services like Auto Scaling, Lambda, and ECS enable rapid response to traffic spikes This supports agile development and continuous delivery In a cloud computing environment, new IT resources are only a click away, reducing provisioning time from weeks to minutes. This dramatically increases agility and lowers the cost and time to experiment and develop.

SOURCES

AWS Cloud Benefits AWS Well-Architected Framework – Operational Excellence

DISTRACTOR BREAKDOWN

C. Free Tier services usage Helpful for cost savings — but not a driver of agility or execution speed

D. Access to AWS data centers Customers don't directly access data centers — AWS abstracts this layer

E. Lower resource provisioning cost Cost is a benefit, but speed and agility come from automation and scalability

REINFORCING WITH AGILITY STRATEGY

Let's map AWS agility into a modern DevOps workflow: -

Feature

Role in Speed & Agility -

Elastic Compute (EC2, Lambda)

Rapid provisioning and scaling -

Infrastructure as Code (CloudFormation, CDK)

Automates deployments and rollback -

CI/CD Pipelines (CodePipeline, CodeBuild)

Speeds up testing and delivery -

Global Availability

Deploy close to users for low latency -

Monitoring & Feedback (CloudWatch, X-Ray)

Enables fast iteration and debugging

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics dashboard: Use Lambda and API Gateway for instant backend deployment Provision Aurora Serverless for scalable database access Deploy via CloudFormation templates to reduce setup time That's how AWS empowers teams to move fast, iterate often, and scale seamlessly with cloud-native agility.

Question 223

**A retailer wishes to supply just the resources required to meet current demand. Which cloud advantage is the organization attempting to accomplish with this objective?

**

OPTIONS

A. Reliability

B. Global reach

C. Scalability

D. High availability

[!TIP]

Correct Answer:

C. Scalability

EXPLANATION

CONCEPTUAL FRAMING

This question targets elastic resource provisioning, especially how AWS enables businesses to scale up or down based on real-time demand — a cornerstone of cloud efficiency.

WHY C IS CORRECT

Scalability means: Dynamically adjusting resource capacity to match workload demand
Avoiding over-provisioning (waste) and under-provisioning (performance bottlenecks)
Supporting both vertical scaling (more power per instance) and horizontal scaling (more instances)
AWS services that support scalability: Auto Scaling Groups for EC2 Lambda for event-driven scaling Aurora Serverless for database elasticity Elastic Load Balancing to distribute traffic across scaled resources
AWS Auto Scaling lets you build scaling plans that automate how groups of different resources respond to changes in demand.

SOURCES

DISTRACTOR BREAKDOWN

A. Reliability Refers to system uptime and fault tolerance — not dynamic resource provisioning

B. Global reach Describes AWS's geographic footprint — not demand-based scaling

D. High availability Ensures uptime across failures — but doesn't imply elastic resource adjustment

REINFORCING WITH ELASTIC ARCHITECTURE STRATEGY

Let's map scalability into a modern cloud-native workflow: -

Component

Role in Scalability -

Auto Scaling Groups

Adjust EC2 fleet size based on metrics -

Lambda Functions

Scale automatically with event volume -

DynamoDB On-Demand

Scales read/write throughput seamlessly -

Elastic Load Balancer

Routes traffic across scaled instances -

CloudWatch Alarms

Trigger scaling actions based on thresholds

REAL-WORLD TIE-IN

If your Nairobi-based retail team runs a seasonal ecommerce site: Use Auto Scaling to handle traffic surges during holiday sales Deploy Lambda for backend tasks like order processing Monitor CloudWatch metrics to fine-tune scaling thresholds That's how AWS empowers retailers to match resource supply with demand using scalable cloud infrastructure.

REFERENCES

<https://aws.amazon.com/autoscaling/>

Question 224

**What storage capabilities does Amazon S3 Intelligent-Tiering provide?

**

OPTIONS

A. Payment flexibility by reserving storage capacity

B. Long-term retention of data by copying the data to an encrypted Amazon Elastic Block Store (Amazon EBS) volume

C. Automatic cost savings by moving objects between tiers based on access pattern changes

D. Secure, durable, and lowest cost storage for data archival

[!TIP]

Correct Answer:

C. Automatic cost savings by moving objects between tiers based on access pattern changes

EXPLANATION

CONCEPTUAL FRAMING

This question targets automated storage optimization, especially how AWS enables cost-efficient data management without sacrificing performance or durability.

WHY C IS CORRECT

Amazon S3 Intelligent-Tiering is designed for: Automatic tiering based on object access frequency No retrieval fees or performance penalties for frequent access Optional archive tiers (Archive and Deep Archive) for infrequent access Ideal for datasets with unpredictable or changing access patterns Key benefits: Cost savings without manual intervention Low latency for frequently accessed data Durability and security consistent with other S3 classes The S3 Intelligent-Tiering storage class delivers automatic storage cost savings in three low latency and high throughput access tiers. It also offers optional archive capabilities to help you get the lowest storage costs in the cloud for data that can be accessed in minutes to hours.

SOURCES

S3 Intelligent-Tiering Overview S3 Intelligent-Tiering FAQs

DISTRACTOR BREAKDOWN

A. Reserved storage capacity Not applicable to S3 — applies to services like EC2 or RDS

B. Copying to EBS EBS is block storage for EC2 — not used for S3 backups or tiering

REINFORCING WITH TIERING STRATEGY

Let's map Intelligent-Tiering into a broader storage lifecycle: -

Tier

Access Pattern Use Case -

Frequent Access

Regularly accessed Active datasets -

Infrequent Access

Occasionally accessed Logs, backups -

Archive

Rarely accessed Compliance data -

Deep Archive

Almost never accessed Long-term retention (e.g., 7+ years)

REAL-WORLD TIE-IN

If your Nairobi-based team stores analytics logs and customer reports: Use Intelligent-Tiering to automatically shift cold data to cheaper tiers Enable archive tiers for regulatory documents Set lifecycle policies to transition data based on age or access That's how AWS empowers teams to optimize storage costs dynamically with S3 Intelligent-Tiering's automated tier transitions.

Question 225

****A retailer wants to migrate business applications to an AWS VPC while maintaining access to on-premises resources. Which two activities will help achieve this objective?**

OPTIONS

- A. Use the AWS Service Catalog to identify a list of on-premises resources that can be migrated
- B. Build a VPN connection between an on-premises device and a virtual private gateway in the new VPC
- C. Use Amazon Athena to query data from the on-premises database servers
- D. Connect the company's on-premises data center to AWS using AWS Direct Connect
- E. Leverage Amazon CloudFront to restrict access to static web content provided through the company's on-premises web servers

[!TIP]

Correct Answer:

B. Build a VPN connection... and D. Connect via AWS Direct Connect

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud architecture, especially how AWS enables secure, performant connectivity between on-premises environments and cloud-based VPCs. 🔍 Why B and D Are Correct

B.

VPN connection to virtual private gateway Establishes a secure IPsec tunnel between your data center and AWS Ideal for quick setup, lower bandwidth, and encrypted traffic Uses Virtual Private Gateway on AWS side and Customer Gateway on-prem

D.

AWS Direct Connect Provides a dedicated, private network link between AWS and your data center Offers higher bandwidth, lower latency, and consistent performance Often used for data-intensive workloads and regulated environments AWS Direct Connect links your AWS and on-premises networks to build applications that span environments without compromising performance.

SOURCES

AWS Hybrid Connectivity Options VPN and Direct Connect Comparison

DISTRACTOR BREAKDOWN

A. AWS Service Catalog Manages approved AWS resources — doesn't discover or migrate on-prem assets

C. Amazon Athena Queries S3 data — not designed for querying on-prem databases

E. Amazon CloudFront Distributes web content — doesn't manage access to on-prem servers

REINFORCING WITH HYBRID ARCHITECTURE STRATEGY

Let's map hybrid connectivity into a resilient deployment model: -

Component

Role -

VPN Gateway

Secure tunnel for encrypted traffic -

Direct Connect

Dedicated link for high-throughput workloads -

Transit Gateway

Central hub for multi-VPC and multi-site routing -

Route Tables

Control traffic flow between AWS and on-prem -

CloudWatch & VPC Flow Logs

Monitor traffic and troubleshoot connectivity

REAL-WORLD TIE-IN

If your Nairobi-based retail team runs inventory systems on-prem and wants to migrate analytics to AWS: Use VPN for initial connectivity and testing Transition to Direct Connect for production-grade performance Route traffic through Transit Gateway for multi-VPC access That's how AWS empowers businesses to build hybrid applications that span cloud and on-prem environments using VPN and Direct Connect.

Question 226

**What is the most effective approach to link an on-premises network to numerous VPCs located in separate AWS Regions?

**

OPTIONS

A. Use AWS Direct Connect

B. Use AWS VPN

C. Use AWS Client VPN

D. Use an AWS Transit Gateway

[!TIP]

Correct Answer:

D. Use an AWS Transit Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question targets scalable hybrid networking, especially how AWS enables centralized routing across multiple VPCs and Regions while maintaining connectivity to on-premises environments.

WHY D IS CORRECT

AWS Transit Gateway acts as a cloud router that: Connects multiple VPCs, VPNs, and Direct Connect gateways Simplifies network architecture by eliminating complex peering meshes Supports inter-region peering, enabling global VPC connectivity Scales to thousands of VPC attachments with centralized control AWS Transit Gateway connects your Amazon VPCs and on-premises networks through a central hub. This simplifies your network and ends complex peering relationships. Each new connection is made only once.

SOURCES

AWS Transit Gateway Overview Inter-Region Peering with Transit Gateway

DISTRACTOR BREAKDOWN

A. AWS Direct Connect Provides a dedicated link — but doesn't manage multi-VPC or multi-Region routing

B. AWS VPN Creates point-to-point tunnels — not scalable for many VP Cs or Regions

C. AWS Client VPN Designed for end-user access — not for linking networks or VP Cs

REINFORCING WITH HYBRID ARCHITECTURE STRATEGY

Let's map Transit Gateway into a global hybrid network: -

Component

Role -

Transit Gateway

Central hub for VPC and on-prem connectivity -

Direct Connect Gateway

Connects Direct Connect to Transit Gateway -

Inter-Region Peering

Links Transit Gateways across Regions -

Route Tables

Control traffic flow between attachments -

CloudWatch & Flow Logs

Monitor and troubleshoot network traffic

REAL-WORLD TIE-IN

If your Nairobi-based enterprise runs workloads in af-south-1 and eu-west-1: Use Transit Gateway peering to link VPCs across Regions Connect your on-prem data center via Direct Connect Gateway Centralize routing and monitoring with Transit Gateway route tables and CloudWatch That's how AWS empowers businesses to build scalable, global hybrid networks using Transit Gateway as the backbone.

Question 227

**Multiple Regions of the AWS Cloud are an example of:

**

OPTIONS

A. Agility

B. Global infrastructure

C. Elasticity

D. Pay-as-you-go pricing

[!TIP]

Correct Answer:

B. Global infrastructure

EXPLANATION

CONCEPTUAL FRAMING

This question targets AWS's physical and geographic footprint, especially how its global network of Regions and Availability Zones supports resilience, latency optimization, and compliance.

WHY B IS CORRECT

Global infrastructure refers to: 31 AWS Regions and 99 Availability Zones (with more announced) Each Region is a separate geographic area with multiple isolated Availability Zones Enables customers to: Deploy applications close to users Meet data residency requirements Build multi-Region failover architectures The AWS Cloud spans 99 Availability Zones within 31

geographic regions around the world, with announced plans for 12 more Availability Zones and 4 more AWS Regions in Canada, Israel, New Zealand, and Thailand.

SOURCES

AWS Global Infrastructure Regions and Availability Zones

DISTRACTOR BREAKDOWN

A. Agility Refers to rapid deployment and experimentation — not geographic distribution

C. Elasticity Describes scaling resources up/down — not physical infrastructure

D. Pay-as-you-go pricing Refers to billing model — not infrastructure layout

REINFORCING WITH DEPLOYMENT STRATEGY

Let's map AWS global infrastructure into a resilient architecture: -

Component

Role -

Region

Geographic deployment zone -

Availability Zone (AZ)

Isolated data center cluster within a Region -

Edge Location

CDN endpoint for low-latency delivery -

Local Zone

Extension of Region for ultra-low latency workloads -

Wavelength Zone

5G edge computing for mobile apps

REAL-WORLD TIE-IN

If your Nairobi-based team serves users in Africa and Europe: Deploy workloads in Cape Town (af-south-1) and Frankfurt (eu-central-1) Use Route 53 latency-based routing to direct traffic Replicate data across Regions for disaster recovery and compliance That's how AWS empowers global teams to build resilient, low-latency applications using its expansive global infrastructure.

Question 228

**Which AWS service does Chef and Puppet utilize to automate configuration management?

**

OPTIONS

A. AWS Config

B. AWS OpsWorks

C. AWS CloudFormation

D. AWS Systems Manager

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets infrastructure automation and configuration management, especially how AWS supports third-party tools like Chef and Puppet for managing server state and compliance.

WHY B IS CORRECT

AWS OpsWorks is a configuration management service that: Supports Chef Automate and Puppet Enterprise as managed offerings Automates software installation, configuration, and updates across EC2 instances Handles backups, patching, and lifecycle management of Chef/Puppet masters Enables scalable, repeatable infrastructure deployments using declarative recipes and manifests AWS OpsWorks for Chef Automate and Puppet Enterprise lets you launch managed configuration servers and focus on core tasks like enforcing desired state, applying patches, and orchestrating deployments.

SOURCES

[AWS OpsWorks Overview](#) [OpsWorks for Chef Automate](#) [OpsWorks for Puppet Enterprise](#)

DISTRACTOR BREAKDOWN

A. AWS Config Tracks resource configurations — doesn't automate server setup or software deployment

C. AWS CloudFormation Manages AWS infrastructure as code — not third-party config tools like Chef/Puppet

D. AWS Systems Manager Provides patching, automation, and parameter management

— but doesn't integrate Chef/Puppet directly

REINFORCING WITH CONFIGURATION STRATEGY

Let's map OpsWorks into a broader DevOps workflow: -

Tool

Role -

OpsWorks for Chef/Puppet

Enforce desired state across servers -

CloudFormation

Provision AWS resources declaratively -

Systems Manager

Patch, automate, and manage parameters -

CodeDeploy

Deploy application code to EC2 or Lambda -

Config

Audit and evaluate resource configurations

REAL-WORLD TIE-IN

If your Nairobi-based team manages EC2 fleets with custom app stacks: Use OpsWorks for Chef to enforce package versions and file configurations Define recipes for app deployment and service restarts Integrate with CloudWatch and Systems Manager for monitoring and automation That's how AWS empowers teams to automate configuration management using Chef and Puppet via OpsWorks.

Question 229

******A corporation wants to eliminate the need for pre-deployment infrastructure capacity estimation and only pay for resources when used. Which AWS Cloud feature best meets this need?

OPTIONS

A. Reliability

B. Global reach

C. Economies of scale

D. Pay-as-you-go pricing

[!TIP]

Correct Answer:

D. Pay-as-you-go pricing

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud cost flexibility and operational efficiency, especially how AWS enables businesses to avoid upfront commitments and scale spending with actual usage.

WHY D IS CORRECT

Pay-as-you-go pricing means: No upfront costs or long-term contracts Charges based on actual usage of compute, storage, and data transfer Supports agile experimentation and dynamic scaling without financial risk Ideal for workloads with variable or unpredictable demand Examples: EC2 instances billed per second or hour Lambda functions billed per request and execution time S3 storage billed per GB stored and retrieved The on-demand pricing model is

the true embodiment of Amazon's pay-as-you-go philosophy. You're charged only for the compute capacity you use, with no upfront payments and flexible scaling.

SOURCES

AWS Pricing Overview AWS Free Tier and On-Demand Pricing

DISTRACTOR BREAKDOWN

A. Reliability Refers to uptime and fault tolerance — not cost or billing

B. Global reach Describes AWS's geographic footprint — not pricing flexibility

C. Economies of scale Refers to AWS's ability to offer lower prices due to scale — not usage-based billing

REINFORCING WITH COST STRATEGY

Let's map AWS pricing models: -

Model

Description Use Case -

On-Demand

Pay per use Variable workloads -

Reserved Instances

Commit for 1–3 years Steady-state usage -

Spot Instances

Bid for unused capacity Fault-tolerant, flexible jobs -

Savings Plans

REAL-WORLD TIE-IN

If your Nairobi-based team launches a new app with unpredictable traffic: Use on-demand EC2 and Lambda to avoid overprovisioning Monitor usage with AWS Cost Explorer Transition to Savings Plans or Reserved Instances once patterns stabilize That's how AWS empowers businesses to scale infrastructure without upfront cost using pay-as-you-go pricing.

Question 230

**SQL injection attacks are being launched against an application from a variety of external locations. Which AWS service or functionality can assist in automating response to these attacks?

**

OPTIONS

A. AWS WAF

B. Security groups

C. Elastic Load Balancer

D. Network ACL

[!TIP]

Correct Answer:

A. AWS WAF (Web Application Firewall)

EXPLANATION

CONCEPTUAL FRAMING

This question targets application-layer threat mitigation, especially how AWS enables automated protection against common web exploits like SQL injection and XSS.

WHY A IS CORRECT

AWS WAF is designed to: Protect web applications from Layer 7 attacks (e.g., SQL injection, cross-site scripting) Use rule-based filtering to inspect HTTP/S requests Automate responses by: Blocking malicious IPs Rate-limiting suspicious traffic Logging and alerting via CloudWatch and AWS Firewall Manager Key features: Managed rule groups for OWASP Top 10 threats Custom rules for app-specific patterns Integration with CloudFront, ALB, and API Gateway AWS WAF makes it easy to create rules that block common web exploits like SQL injection and cross-site scripting. It allows centralized rule deployment across multiple websites.

SOURCES

AWS WAF Overview AWS WAF Security Automations

DISTRACTOR BREAKDOWN

B. Security groups Operate at Layer 4 — control IP/port access, not application-layer threats

C. Elastic Load Balancer Distributes traffic — doesn't inspect or block malicious payloads

D. Network ACL Stateless firewall at subnet level — not suitable for detecting SQL injection

REINFORCING WITH THREAT MITIGATION STRATEGY

Let's map AWS WAF into a broader security posture: -

Layer

Tool Role -

Network

Security Groups, NACLs IP/port filtering -

Transport

TLS via ACM Encrypt traffic -

Application

AWS WAF Block web exploits -

Monitoring

Cloud Watch, Guard Duty Detect anomalies -

Automation

Firewall Manager Centralize WAF rule management

REAL-WORLD TIE-IN

If your Nairobi-based team runs a public-facing API: Deploy AWS WAF in front of API Gateway or ALB Use managed rule groups to block SQL injection and XSS Set up CloudWatch alarms for rule matches and traffic anomalies That's how AWS empowers teams to automate protection against web exploits using WAF's rule-based filtering and centralized control.

REFERENCES

<https://aws.amazon.com/waf/>

Question 231

**Which feature enables Amazon EC2 instances to be more elastic in response to changing workload demand?

OPTIONS

A. Resource groups

B. Lifecycle policies

C. Application Load Balancer

D. Amazon EC2 Auto Scaling

[!TIP]

Correct Answer:

D. Amazon EC2 Auto Scaling

EXPLANATION

CONCEPTUAL FRAMING

This question targets elasticity, one of the core tenets of cloud computing, especially how AWS enables EC2 instances to scale dynamically based on demand.

WHY D IS CORRECT

Amazon EC2 Auto Scaling: Automatically adjusts the number of EC2 instances based on CloudWatch metrics, such as CPU utilization or request count Ensures high availability by replacing unhealthy instances Supports scaling policies like target tracking, step scaling, and scheduled scaling Integrates with Elastic Load Balancer and target groups for traffic distribution and health checks Attaching a target group to an Auto Scaling group enables you to scale each service dynamically based on demand. Health checks and CloudWatch metrics ensure each service is monitored independently.

SOURCES

EC2 Auto Scaling Documentation Scaling Policies

DISTRACTOR BREAKDOWN

A. Resource groups Used for organizing AWS resources — not for scaling or elasticity

B. Lifecycle policies Manage data lifecycle (e.g., S3 object transitions) — not EC2 instance scaling

C. Application Load Balancer Distributes traffic — doesn't scale EC2 instances automatically

REINFORCING WITH ELASTIC ARCHITECTURE STRATEGY

Let's map EC2 Auto Scaling into a responsive infrastructure: -

Component

Role -

Auto Scaling Group

Manages EC2 fleet size -

Launch Template

Defines instance configuration -

Scaling Policy

Triggers scale-in/out actions -

Target Group

Routes traffic to healthy instances -

CloudWatch Alarms

Monitor metrics and trigger scaling

REAL-WORLD TIE-IN

If your Nairobi-based team runs a retail app with fluctuating traffic: Use Auto Scaling to add EC2 instances during peak hours. Define target tracking policies based on CPU or request count. Integrate with ALB to distribute traffic and monitor health. That's how AWS empowers teams to build elastic, resilient applications that scale automatically with EC2 Auto Scaling.

Question 232

**What time-saving benefit does Amazon Rekognition provide?

**

OPTIONS

- A. Automatic watermarking of images
- B. Automatic detection of objects appearing in pictures
- C. Ability to resize millions of images automatically
- D. Use of Amazon Mechanical Turk for object detection jobs

[!TIP]

Correct Answer:

- B. Automatic detection of objects appearing in pictures

EXPLANATION

CONCEPTUAL FRAMING

This question targets computer vision automation, especially how AWS enables developers to extract meaningful insights from images and videos using deep learning.

WHY B IS CORRECT

Amazon Rekognition is a fully managed image and video analysis service that: Detects objects, scenes, and faces in images Identifies text, celebrities, and inappropriate content Supports face comparison and search across datasets Enables real-time and batch processing for scalable workflows Time-saving benefits include: Eliminating manual tagging and annotation Automating moderation and compliance checks Accelerating media indexing and search Amazon Rekognition detects objects, scenes, and faces; extracts text; recognizes celebrities; and identifies inappropriate content. It's built on scalable deep learning technology used to analyze billions of images daily for Prime Photos.

SOURCES

Amazon Rekognition Overview Rekognition Use Cases

DISTRACTOR BREAKDOWN

A. Watermarking Rekognition doesn't modify images — it analyzes them

C. Resizing images Handled by services like Lambda or Image Magick — not Rekognition

D. Mechanical Turk Not used by Rekognition — it's a separate crowdsourcing platform

REINFORCING WITH VISION WORKFLOW STRATEGY

Let's map Rekognition into a media processing pipeline: -

Step

Tool Role -

Image ingestion

S3 Store raw media -

Trigger analysis

Lambda Invoke Rekognition -

Object detection

Rekognition Identify scenes, faces, text -

Metadata storage

Dynamo DB Index results -

Search & moderation

Elasticsearch, SNS Enable querying and alerts

REAL-WORLD TIE-IN

If your Nairobi-based team manages a photo-sharing app: Use Rekognition to tag images with detected objects and scenes Automate moderation by flagging inappropriate content Enable face search to group user albums That's how AWS empowers teams to accelerate image analysis and metadata generation using Rekognition's deep learning capabilities.

Question 233

**A corporation wants to migrate petabytes of data from on-premises sites to the AWS Cloud as rapidly as feasible. Which AWS service should the business use?

**

OPTIONS

A. AWS Snowball

B. AWS Global Accelerator

C. Amazon S3 Transfer Acceleration

D. Amazon Connect

[!TIP]

Correct Answer:

A. AWS Snowball

EXPLANATION

CONCEPTUAL FRAMING

This question targets large-scale data migration, especially how AWS enables secure, high-throughput transfer of petabyte-scale datasets from on-premises environments to the cloud.

WHY A IS CORRECT

AWS Snowball is a physical edge device designed for: Offline data transfer at faster-than-Internet speeds Moving archives, data lakes, backups, and media libraries Supporting encryption, tamper-resistance, and chain-of-custody tracking Seamless integration with Amazon S3, Glacier, Redshift, and EMR Key benefits: Petabyte-scale capacity per device No bandwidth bottlenecks Secure and ruggedized hardware Automated import into S3 buckets The Snowball appliance allows you to move archives, data lakes, and whatever data you have at faster-than-Internet speeds right into Amazon S3 buckets. From there, data can be archived or analyzed.

SOURCES

AWS Snowball Overview Snowball Edge Device Specs

DISTRACTOR BREAKDOWN

B. AWS Global Accelerator Optimizes routing for live traffic — not used for bulk data migration

C. S3 Transfer Acceleration Speeds up small-to-medium uploads over the internet — not suitable for petabyte-scale

D. Amazon Connect Cloud contact center service — unrelated to data migration

REINFORCING WITH MIGRATION STRATEGY

Let's map Snowball into a hybrid ingestion workflow: -

Step

Tool Role -

Data export

On-prem systems Prepare datasets for transfer -

Snowball device

AWS Snowball Secure, high-speed transport -

S3 ingestion

Amazon S3 Store and organize data -

Archival

Glacier / Deep Archive Long-term retention -

Analysis

Redshift / EMR Data warehousing and processing

REAL-WORLD TIE-IN

If your Nairobi-based enterprise needs to migrate 500 TB of customer logs: Order Snowball Edge devices Use Snowball client software to load data securely Ship devices back to AWS for automated ingestion into S3 Archive into Glacier or analyze with Athena or EMR That's how

AWS empowers organizations to migrate massive datasets quickly and securely using Snowball's edge-based transfer model.

Question 234

**Where should a business go to locate, test, purchase, and deploy software that works on AWS?

**

OPTIONS

A. AWS Marketplace

B. Amazon Lumberyard

C. AWS Artifact

D. Amazon CloudSearch

[!TIP]

Correct Answer:

A. AWS Marketplace

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud-native software procurement, especially how AWS enables businesses to discover, evaluate, and deploy third-party solutions directly within their cloud ecosystem.

WHY A IS CORRECT

AWS Marketplace is a digital catalog offering: Thousands of listings from independent software vendors (ISVs) Pre-configured AMIs, SaaS, and container-based solutions Free trials, hourly pricing, and annual subscriptions Seamless deployment into EC2, EKS, Lambda, and other AWS services Key benefits: Accelerates procurement cycles Supports license management and billing integration Enables governance and budget controls via AWS Organizations AWS Marketplace is a digital catalog with thousands of software listings from independent software vendors that make it easy to find, test, buy, and deploy software that runs on AWS.

SOURCES

AWS Marketplace Overview Procurement Governance in AWS Marketplace

DISTRACTOR BREAKDOWN

B. Amazon Lumberyard Game engine for 3D development — not a software catalog

C. AWS Artifact Provides compliance reports — not software procurement

D. Amazon CloudSearch Search service for app data — not a marketplace

REINFORCING WITH PROCUREMENT STRATEGY

Let's map AWS Marketplace into a cloud-native acquisition workflow: -

Step

Tool Role -

Discovery

AWS Marketplace Browse and filter software -

Evaluation

Free trials, test environments Assess functionality -

Purchase

AWS billing integration Pay via consolidated invoice -

Deployment

EC2, EKS, Lambda Launch directly into AWS -

Governance

AWS Organizations Control access and spend

REAL-WORLD TIE-IN

If your Nairobi-based team needs a secure file-sharing solution: Search AWS Marketplace for SaaS or container-based options Test using free trial or hourly pricing Deploy into EKS or EC2 with IAM and VPC controls That's how AWS empowers businesses to streamline software acquisition and deployment using Marketplace's integrated ecosystem.

Question 235

**A business's user base is worldwide in scope. The organization needs a highly available application with reduced latency for end users. Which AWS architecture approach will meet these criteria the MOST EFFECTIVELY?

**

OPTIONS

A. Single-Region, Multi-AZ architecture

B. Multi-Region, active-active architecture

C. Multi-Region, active-passive architecture

D. Single-Region, Single-AZ architecture

[!TIP]

Correct Answer:

B. Multi-Region, active-active architecture

EXPLANATION

CONCEPTUAL FRAMING

This question targets global application design, especially how AWS enables businesses to deliver low-latency, highly available services to users across continents.

WHY B IS CORRECT

Multi-Region, active-active architecture: Deploys identical stacks in multiple AWS Regions Uses Route 53 latency-based routing to direct users to the nearest Region Ensures high availability by distributing traffic and load Supports real-time data replication across Regions for consistency Key benefits: Reduced latency by serving users from geographically closer endpoints Resilience against Regional outages Scalability across global demand A multi-region, active-active architecture gets all the services on the client request path deployed across multiple AWS Regions. Data replication between regions must be fast and reliable.

SOURCES

AWS Global Architecture Best Practices Multi-Region Deployment Patterns

DISTRACTOR BREAKDOWN

A. Single-Region, Multi-AZ Improves availability within one Region — doesn't reduce global latency

C. Multi-Region, active-passive Improves failover — but only one Region serves traffic at a time

D. Single-Region, Single-AZ Least resilient — vulnerable to outages and high latency for distant users

REINFORCING WITH GLOBAL ARCHITECTURE STRATEGY

Let's map active-active architecture into a resilient deployment model: -

Component

Role -

Multiple Regions

Serve users close to their location -

Route 53

Latency-based DNS routing -

Global DynamoDB Tables / Aurora Global DB

Real-time data replication -

CloudFront

Edge caching for static assets -

S3 Cross-Region Replication

Durable object sync across Regions

REAL-WORLD TIE-IN

If your Nairobi-based team serves users in Africa, Europe, and Asia: Deploy workloads in Cape Town, Frankfurt, and Singapore Use Route 53 latency routing to direct traffic Replicate data

using Aurora Global Database or DynamoDB Global Tables That's how AWS empowers global businesses to deliver fast, resilient applications using multi-Region active-active architecture.

Question 236

**Which duty is the customer's responsibility under the AWS shared responsibility model?

**

OPTIONS

- A. Maintaining the infrastructure needed to run AWS Lambda
- B. Updating the operating system of Amazon DynamoDB instances
- C. Maintaining Amazon S3 infrastructure
- D. Updating the guest operating system on Amazon EC2 instances

[!TIP]

Correct Answer:

- D. Updating the guest operating system on Amazon EC2 instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets the boundary of responsibility between AWS and the customer, especially in Infrastructure-as-a-Service (IaaS) scenarios like EC2.

WHY D IS CORRECT

Amazon EC2 provides virtual machines (instances) where: AWS manages the underlying physical infrastructure, hypervisor, and host OS. The customer is responsible for the guest OS (e.g., Ubuntu, Windows Server) inside the instance. This includes: Security patches, OS updates, Firewall configuration (e.g., security groups), Application software and data. The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software, as well as the configuration of the AWS-provided security group firewall.

SOURCES

AWS Shared Responsibility Model EC2 Security Best Practices

DISTRACTOR BREAKDOWN

A. AWS Lambda infrastructure Fully managed — AWS handles all infrastructure and runtime

B. DynamoDB OS updates Dynamo DB is serverless — AWS manages all backend operations

C. S3 infrastructure AWS handles durability, availability, and scaling — customers manage data and access policies

REINFORCING WITH RESPONSIBILITY MAPPING

Layer AWS Responsibility Customer Responsibility

Physical Infrastructure Data center, hardware, networking —

Virtualization Hypervisor, host OS —

Compute (EC2) Instance provisioning Guest OS, patching, app config

Storage (S3) Infrastructure, durability Data classification, access policies

IAM Root account protection User roles, MFA, policy design

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2 instances for a Django app: You must update Ubuntu or Amazon Linux regularly Apply security patches and configure firewall rules via security groups Monitor with CloudWatch and automate patching via Systems Manager That's how AWS empowers customers to secure and manage their workloads while AWS handles the underlying infrastructure.

Question 237

**How can a user safeguard against AWS service outages in the event of a widespread natural disaster?

**

OPTIONS

A. Deploy applications across multiple Availability Zones within an AWS Region

B. Use a hybrid cloud computing deployment model within the geographic area

C. Deploy applications across multiple AWS Regions

D. Store application artifacts using AWS Artifact and replicate them across multiple AWS Regions

[!TIP]

Correct Answer:

C. Deploy applications across multiple AWS Regions

EXPLANATION

CONCEPTUAL FRAMING

This question targets disaster recovery and geographic fault tolerance, especially how AWS enables businesses to maintain availability and continuity even during regional disruptions.

WHY C IS CORRECT

Multiple AWS Regions provide: Geographic isolation — each Region is independent and physically separated Redundant infrastructure — multiple Availability Zones per Region Global failover capability — using Route 53, Global Accelerator, or custom DNS Data replication — via S3 Cross-Region Replication, Aurora Global DB, or DynamoDB Global Tables An AWS Region is a geographic location where AWS provides multiple, physically separated and isolated Availability Zones connected with low latency, high throughput, and highly redundant networking.

SOURCES

AWS Disaster Recovery Strategies Multi-Region Architecture Patterns

DISTRACTOR BREAKDOWN

A. Multi-AZ within one Region Protects against AZ-level failures, not Region-wide disasters

B. Hybrid cloud in same area Doesn't mitigate geographic risk — still vulnerable to local disasters

D. AWS Artifact replication Artifact is for compliance reports, not application deployment or DR

REINFORCING WITH RESILIENCE STRATEGY

Let's map multi-Region deployment into a robust DR plan: -

Component

Role -

Primary Region

Main workload execution -

Secondary Region

Failover or active-active replica -

Route 53

DNS-based failover and latency routing -

Global DBs

Real-time data replication -

S3 CRR

Object replication across Regions -

CloudFormation StackSets

Deploy infrastructure across Regions

REAL-WORLD TIE-IN

If your Nairobi-based team serves users in Africa and Europe: Deploy workloads in Cape Town and Frankfurt Use Route 53 health checks for failover Replicate data using Aurora Global

Database or S3 CRR That's how AWS empowers organizations to build resilient, disaster-tolerant applications using multi-Region architecture.

Question 238

**What is an example of a cloud-based application that is decoupled and scalable?

**

OPTIONS

A. A mail and log application that runs on a single Amazon EC2 instance

B. A webpage that is hosted on Amazon S3 and uses AWS Lambda to update an Amazon DynamoDB database

C. An Application Load Balancer, web server, and database server that support a monolithic application

D. A legacy database server that is running on the maximum instance size supported by its license

[!TIP]

Correct Answer:

B. A webpage hosted on Amazon S3 and using AWS Lambda to update DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question targets modern cloud-native design principles, especially how AWS enables serverless, event-driven, and loosely coupled architectures that scale automatically.

WHY B IS CORRECT

This architecture is: Decoupled: Each component (S3, Lambda, DynamoDB) operates independently Scalable: Lambda and DynamoDB scale automatically based on demand Serverless: No infrastructure management required — AWS handles provisioning and scaling Event-driven: Lambda triggers based on user actions or S3 events Key benefits: Resilience: Failures in one component don't cascade Agility: Easy to update or replace individual services Cost-efficiency: Pay only for usage — no idle resources As applications grow in complexity, monolithic designs become fragile and slow development. Decoupled architectures allow teams to move faster and scale more efficiently.

SOURCES

AWS Serverless Application Model Decoupled Architectures on AWS

DISTRACTOR BREAKDOWN

A. Single EC2 instance Tightly coupled and non-scalable — single point of failure

C. Monolithic app Hard to scale and maintain — tightly bound components

D. Legacy DB on max instance Vertical scaling only — not decoupled or cloud-native

REINFORCING WITH DECOUPLED DESIGN STRATEGY

Let's map the correct architecture into a scalable workflow: -

Component

Role -

Amazon S3

Static web hosting -

AWS Lambda

Backend logic triggered by events -

Amazon DynamoDB

Scalable No SQL database -

Amazon API Gateway

Optional — expose Lambda as RESTful API -

CloudWatch Logs

Monitor and debug Lambda execution

REAL-WORLD TIE-IN

If your Nairobi-based team builds a feedback form: Host the form on S3 static website Use Lambda to process submissions Store results in DynamoDB Add SNS notifications for alerts That's how AWS empowers teams to build scalable, decoupled applications using serverless components like S3, Lambda, and DynamoDB.

Question 239

**A client has many AWS accounts, each with its own billing. How can the client benefit from bulk savings while minimizing the effect on AWS resources?

**

OPTIONS

A. Create one global AWS account and move all AWS resources to that account

B. Sign up for three years of Reserved Instance pricing up front

C. Use the consolidated billing feature from AWS Organizations

D. Sign up for the AWS Enterprise support plan to get volume discounts

[!TIP]

Correct Answer:

C. Use the consolidated billing feature from AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization and financial governance, especially how AWS enables organizations to aggregate usage across accounts for volume discounts and centralized visibility.

WHY C IS CORRECT

Consolidated Billing via AWS Organizations: Combines usage from all member accounts under a single management account Enables volume discounts on services like EC2, S3, and data transfer Centralizes billing, cost analysis, and budget enforcement Integrates with Cost Explorer, Budgets, and CUR (Cost and Usage Reports) Key benefits: No need to migrate resources — accounts remain independent Improved visibility into spend and usage patterns Scalable governance for large organizations Consolidated billing is a feature of AWS Organizations. The management account can consolidate and pay for all member accounts, access billing data, and use tools like Cost Explorer to improve cost performance.

SOURCES

AWS Organizations and Consolidated Billing Cost Optimization Best Practices

DISTRACTOR BREAKDOWN

A. One global account Centralizing resources increases risk and complexity — not needed for billing optimization

B. Reserved Instances Offers savings — but doesn't aggregate usage across accounts

D. Enterprise Support Plan Provides premium support — not a billing consolidation feature

REINFORCING WITH BILLING STRATEGY

Let's map consolidated billing into a cost governance workflow: -

Component

Role -

Management Account

Pays consolidated bill -

Member Accounts

Operate independently -

Cost Explorer

Analyze usage and trends -

Budgets

Set thresholds and alerts -

Savings Plans / RI Sharing

Apply discounts across accounts

REAL-WORLD TIE-IN

If your Nairobi-based enterprise has separate accounts for dev, staging, and production: Use AWS Organizations to link them under one management account Enable RI and Savings Plan

sharing for compute discounts Monitor spend with Cost Explorer and Budgets That's how AWS empowers organizations to maximize savings and maintain account autonomy using consolidated billing.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 240

**A business must keep its data near to its core consumers. Which AWS Cloud advantage satisfies this requirement?

**

OPTIONS

A. Security

B. High availability

C. Elasticity

D. Global footprint

[!TIP]

Correct Answer:

D. Global footprint

EXPLANATION

CONCEPTUAL FRAMING

This question targets geographic infrastructure reach, especially how AWS enables businesses to deploy applications and store data close to users for reduced latency and compliance alignment.

WHY D IS CORRECT

Global footprint refers to: AWS's presence in multiple geographic Regions, each with Availability Zones Enables data residency, low-latency access, and regional failover Supports compliance with local regulations and performance optimization Current scale: AWS spans 99 Availability Zones across 31 Regions, with more announced Includes Local Zones, Wavelength Zones, and Edge Locations for ultra-low latency A successful global footprint depends on how you use Regions and their Availability Zones. The AWS Global Infrastructure is comprised of 69 Availability Zones within 22 geographic Regions.

SOURCES

AWS Global Infrastructure Regions and Availability Zones

DISTRACTOR BREAKDOWN

A. Security Refers to data protection — not geographic proximity

B. High availability Ensures uptime — doesn't guarantee proximity to users

C. Elasticity Refers to scaling resources — not location-based deployment

REINFORCING WITH DEPLOYMENT STRATEGY

Let's map global footprint into a proximity-aware architecture: -

Component

Role -

Regions

Deploy workloads near users -

Availability Zones

Isolate failures within a Region -

Local Zones

Extend Region closer to metro areas -

Edge Locations

Serve cached content via Cloud Front -

Route 53

Latency-based DNS routing

REAL-WORLD TIE-IN

If your Nairobi-based team serves users in Africa and Europe: Deploy workloads in Cape Town (af-south-1) and Frankfurt (eu-central-1) Use Route 53 latency routing to direct users to the nearest Region Store data in Region-specific S3 buckets to meet residency requirements That's how AWS empowers businesses to serve global users efficiently by leveraging its expansive infrastructure footprint.

Question 242

**A company's managed IAM policy does not allow users the rights essential to do needed activities. How is this situation to be resolved?

**

OPTIONS

A. Enable AWS Shield Advanced

B. Create a custom IAM policy

C. Use a third-party web application firewall (WAF) managed rule from the AWS Marketplace

D. Use AWS Key Management Service (AWS KMS) to create a customer-managed key

[!TIP]

Correct Answer:

B. Create a custom IAM policy

EXPLANATION

CONCEPTUAL FRAMING

This question targets fine-grained access control, especially how AWS enables organizations to tailor permissions to match operational needs while enforcing least privilege.

WHY B IS CORRECT

Custom IAM policies allow: Precise control over actions, resources, and conditions Alignment with business-specific workflows and compliance requirements Integration with IAM Access Analyzer for policy validation and refinement JSON-based policy documents that can be versioned and audited Key benefits: Flexibility to grant only the needed permissions Security by avoiding over-permissioning Scalability across teams and roles When you create or edit IAM policies, AWS can automatically perform policy validation to help you create an effective policy with least privilege in mind. IAM Access Analyzer provides additional checks and recommendations.

SOURCES

IAM Policy Creation Guide IAM Access Analyzer

DISTRACTOR BREAKDOWN

A. AWS Shield Advanced Protects against DDoS — unrelated to IAM permissions

C. Third-party WAF rule Applies to web traffic filtering — not IAM access control

D. AWS KMS key Manages encryption — doesn't resolve IAM policy limitations

REINFORCING WITH IAM STRATEGY

Let's map IAM customization into a secure access workflow: -

Step

Tool Role -

Policy authoring

IAM Console / CLI Define custom permissions -

Validation

IAM Access Analyzer Detect overly broad access -

Testing

Policy Simulator Preview effects of changes -

Deployment

IAM Roles / Groups Assign policies to users -

Monitoring

Cloud Trail / Access Analyzer Audit usage and anomalies

REAL-WORLD TIE-IN

If your Nairobi-based team needs developers to access S3 buckets but not delete objects:

Create a custom IAM policy with s3:GetObject, s3:ListBucket Exclude s3:DeleteObject Attach to

a developer IAM group Validate with Access Analyzer and test with Policy Simulator That's how AWS empowers teams to grant precise access using custom IAM policies aligned with least privilege principles. ? Question 244 Which scenarios warrant the utilization of Amazon EC2 Spot Instances?

A.

Migrating a main website to AWS B. Application services with 99.999% SLA C. Heavily used legacy database on-premises D. Infrequent, interruptible jobs currently using On-Demand Instances

CONCEPTUAL FRAMING

This question targets cost optimization for fault-tolerant workloads, especially how AWS enables teams to leverage unused EC2 capacity at steep discounts when workloads can tolerate interruptions.

WHY D IS CORRECT

EC2 Spot Instances: Offer up to 90% discount compared to On-Demand pricing Are ideal for stateless, batch, or fault-tolerant jobs Can be interrupted with two minutes' notice when AWS reclaims capacity Support Auto Scaling groups and EC2 Fleet for flexible provisioning Best suited for: Data analysis, image processing, CI/CD pipelines Web crawling, simulations, and rendering jobs Machine learning training tasks A Spot Instance is an unused Amazon EC2 instance available for less than the On-Demand price. You can lower EC2 costs significantly for interruptible workloads.

SOURCES

Amazon EC2 Spot Instances Spot Instance Best Practices

DISTRACTOR BREAKDOWN

A. Main website migration Requires high availability — Spot interruptions could cause downtime

B. SLA-bound services 99.999% uptime demands predictable availability — Spot is too volatile

C. Legacy database Stateful and performance-sensitive — not suitable for interruption-prone instances

REINFORCING WITH SPOT STRATEGY

Let's map Spot Instances into a cost-efficient compute model: -

Component

Role -

Spot Instances

Run fault-tolerant workloads at low cost -

Auto Scaling Groups

Replace interrupted instances automatically -

EC2 Fleet

Mix Spot, On-Demand, and Reserved Instances -

Spot Instance Advisor

Evaluate availability and pricing trends -

CloudWatch / SNS

Monitor and alert on interruptions

REAL-WORLD TIE-IN

If your Nairobi-based team runs nightly data processing jobs: Switch from On-Demand to Spot Instances Use Auto Scaling with interruption handling Monitor with CloudWatch alarms Save up to 90% on compute costs That's how AWS empowers teams to optimize spend for interruptible workloads using EC2 Spot Instances.

REFERENCES

Question 245

******A business must transmit time-sensitive communications to a large number of subscribers using a push technique. Which AWS service should the business use?

OPTIONS

A. Amazon Kinesis

B. Amazon MQ

C. Amazon Simple Queue Service (Amazon SQS)

D. Amazon Simple Notification Service (Amazon SNS)

[!TIP]

Correct Answer:

D. Amazon Simple Notification Service (Amazon SNS)

EXPLANATION

CONCEPTUAL FRAMING

This question targets real-time, fan-out messaging, especially how AWS enables applications to broadcast time-critical updates to multiple endpoints using push-based delivery.

WHY D IS CORRECT

Amazon SNS is a fully managed pub/sub messaging service that: Sends push notifications to multiple subscribers simultaneously Supports HTTP/S, email, SMS, Lambda, SQS, and mobile push endpoints Enables topic-based broadcasting for scalable fan-out Integrates with CloudWatch, EventBridge, and IAM for monitoring and access control Key benefits: Low latency delivery High throughput for large subscriber bases Flexible protocol support Decoupled architecture for modular design Amazon SNS allows applications to send time-critical messages to multiple subscribers through a “Push” mechanism.

SOURCES

Amazon SNS Overview SNS vs SQS Comparison

DISTRACTOR BREAKDOWN

A. Amazon Kinesis Streams real-time data — not designed for push notifications

B. Amazon MQ Manages message brokers — suited for legacy apps, not scalable push

C. Amazon SQS Pull-based queue — not suitable for time-sensitive push delivery

REINFORCING WITH MESSAGING STRATEGY

Let’s map SNS into a scalable notification pipeline: -

Component

Role -

SNS Topic

Central broadcast channel -

Subscribers

Email, SMS, Lambda, SQS, HTTP endpoints -

IAM Policies

Control who can publish -

CloudWatch Logs

Monitor delivery success -

EventBridge

Trigger downstream workflows

REAL-WORLD TIE-IN

If your Nairobi-based team needs to alert users about system outages: Create an SNS topic for “System Alerts” Subscribe email, SMS, and Lambda endpoints Publish alerts via CloudWatch Alarms or custom apps That’s how AWS empowers teams to deliver time-sensitive messages instantly using SNS’s push-based architecture.

REFERENCES

<https://aws.amazon.com/sns/>

Question 247

**Which AWS Cloud best practice takes advantage of cloud computing's flexibility and agility?

**

OPTIONS

A. Provision capacity based on past usage and theoretical peaks

B. Dynamically and predictively scale to meet usage demands

C. Build the application and infrastructure in a data center that grants physical access

D. Break apart the application into loosely coupled components

[!TIP]

Correct Answer:

B. Dynamically and predictively scale to meet usage demands

EXPLANATION

CONCEPTUAL FRAMING

This question targets elasticity and agility, especially how AWS enables businesses to scale resources in real time based on actual demand rather than static forecasts.

WHY B IS CORRECT

Dynamic scaling leverages: Auto Scaling Groups for EC2 Lambda concurrency for serverless workloads Aurora and DynamoDB autoscaling for databases Predictive scaling using CloudWatch metrics and machine learning Key benefits: Cost efficiency — pay only for what you use Performance optimization — meet demand without overprovisioning Operational agility — respond to traffic spikes or seasonal loads In traditional environments, provisioning for peak usage leads to idle resources or outages. Cloud computing allows dynamic scaling to meet actual demand, reducing waste and improving responsiveness.

SOURCES

AWS Auto Scaling Elasticity in the Cloud

DISTRACTOR BREAKDOWN

A. Past usage provisioning Leads to overprovisioning and idle resources — not agile

C. Physical data center Opposes cloud principles — lacks flexibility and remote scalability

D. Loosely coupled components Improves modularity — but doesn't directly address dynamic scaling

REINFORCING WITH ELASTIC ARCHITECTURE STRATEGY

Let's map dynamic scaling into a responsive workload model: -

Component

Role -

Auto Scaling Groups

Scale EC2 instances based on demand -

Lambda

Automatically scale with request volume -

DynamoDB

Adjust read/write capacity units dynamically -

CloudWatch Alarms

Trigger scaling actions -

Predictive Scaling

Forecast future demand using ML

REAL-WORLD TIE-IN

If your Nairobi-based team runs a retail app with seasonal spikes: Use Auto Scaling for EC2 web servers Enable DynamoDB autoscaling for cart and inventory tables Apply predictive scaling to anticipate Black Friday traffic That's how AWS empowers teams to scale with precision and agility using dynamic, demand-driven provisioning.

REFERENCES

<https://aws.amazon.com/autoscaling/>

Question 248

**What are users of Amazon Route 53 able to do?

**

OPTIONS

- A. Encrypt data in transit
- B. Register DNS domain names
- C. Generate and manage SSL certificates
- D. Establish a dedicated network connection to AWS

[!TIP]

Correct Answer:

B. Register DNS domain names

EXPLANATION

CONCEPTUAL FRAMING

This question targets domain management and DNS routing, especially how AWS enables businesses to connect user requests to applications via scalable, globally distributed infrastructure.

WHY B IS CORRECT

Amazon Route 53 is a highly available and scalable DNS web service that: Allows users to register domain names directly through AWS Provides DNS resolution and routing policies (e.g., latency-based, failover, geolocation) Integrates with health checks to route traffic away from unhealthy endpoints Supports domain transfers, renewals, and WHOIS privacy protection Amazon Route 53 connects user requests to internet applications running on AWS or on-premises. It also allows domain registration and DNS management.

SOURCES

Amazon Route 53 Overview Registering Domains with Route 53

DISTRACTOR BREAKDOWN

A. Encrypt data in transit Handled by TLS/SSL, not Route 53 — use ACM or ELB

C. Manage SSL certificates Done via AWS Certificate Manager (ACM) — not Route 53

D. Dedicated network connection Provided by AWS Direct Connect — unrelated to DNS

REINFORCING WITH DNS STRATEGY

Let's map Route 53 into a global routing workflow: -

Component

Role -

Domain Registration

Acquire and manage domain names -

Hosted Zones

Define DNS records for domains -

Routing Policies

Control traffic flow (e.g., latency, failover) -

Health Checks

Monitor endpoint availability -

Integration

Works with ELB, Cloud Front, S3, EC2, etc.

REAL-WORLD TIE-IN

If your Nairobi-based team launches a new app: Register myapp.ke via Route 53 Create a hosted zone with A and CNAME records Use latency-based routing to direct users to the nearest Region Add health checks to ensure uptime That's how AWS empowers teams to manage domains and route traffic intelligently using Route 53's DNS capabilities.

REFERENCES

<https://aws.amazon.com/route53/>

Question 249

**Which characteristic of Amazon Virtual Private Cloud (Amazon VPC) allows customers to link two VPCs?

**

OPTIONS

A. Amazon VPC endpoints

B. Amazon EC2 ClassicLink

C. Amazon VPC peering

D. AWS Direct Connect

[!TIP]

Correct Answer:

C. Amazon VPC peering

EXPLANATION

CONCEPTUAL FRAMING

This question targets private network connectivity, especially how AWS enables organizations to establish secure, low-latency communication between isolated VPCs across accounts or Regions.

WHY C IS CORRECT

Amazon VPC Peering: Creates a direct, private connection between two VPCs Supports IPv4 and IPv6 routing Works within the same account or across accounts Supports inter-Region peering for global workloads No need for VPNs, gateways, or NAT traversal Key benefits: Low latency, high bandwidth No single point of failure Secure routing via private IPs Simplified architecture for microservices or shared services A VPC peering connection enables routing traffic between VPCs using private IPs. Instances in either VPC can communicate as if they're in the same network.

SOURCES

VPC Peering Guide Inter-Region VPC Peering

DISTRACTOR BREAKDOWN

A. VPC endpoints Connect to AWS services privately — not for VPC-to-VPC communication

B. EC2 ClassicLink Deprecated — only links EC2-Classic to VPC, not VPC-to-VPC

D. AWS Direct Connect Establishes on-prem to AWS links — not VPC-to-VPC

REINFORCING WITH NETWORK ARCHITECTURE STRATEGY

Let's map VPC peering into a multi-VPC topology: -

Component

Role -

VPC A

Hosts shared services (e.g., logging, monitoring) -

VPC B

Hosts application workloads -

Peering Connection

Enables private routing between VPCs -

Route Tables

Define traffic flow across peered VPCs -

Security Groups / NACLs

Control access between instances

REAL-WORLD TIE-IN

If your Nairobi-based team runs separate VPCs for dev and prod: Use VPC peering to allow secure communication Share centralized logging or database services Avoid public exposure or

VPN complexity That's how AWS empowers teams to link VPCs securely and efficiently using VPC peering for modular, scalable architectures.

Question 250

**Multiple Amazon EC2 instances are used to host an application. The program sends messages using Amazon Simple Notification Service (Amazon SNS). Which AWS service or feature grants authorization for the application to access needed AWS services?

**

OPTIONS

A. AWS Certificate Manager (ACM)

B. IAM roles

C. AWS Security Hub

D. Amazon GuardDuty

[!TIP]

Correct Answer:

B. IAM roles

EXPLANATION

CONCEPTUAL FRAMING

This question targets secure service-to-service authorization, especially how AWS enables EC2 instances and applications to access other AWS services without embedding credentials.

WHY B IS CORRECT

IAM roles: Define a set of permissions that trusted entities (like EC2 instances) can assume. Allow applications to interact with services like SNS, S3, DynamoDB, etc. Avoid hardcoding credentials — use temporary security tokens via instance metadata. Can be scoped with least privilege policies and resource-level conditions. Key benefits: Secure and scalable access control. Auditable via CloudTrail. Flexible — roles can be assumed by EC2, Lambda, ECS, and federated identities. An IAM role is an entity that defines permissions for making AWS service requests. Trusted entities like EC2 instances can assume roles to access services securely.

SOURCES

IAM Roles for EC2 Best Practices for IAM

DISTRACTOR BREAKDOWN

A. AWS Certificate Manager Manages SSL/TLS certs — not used for service authorization

C. AWS Security Hub Aggregates security findings — doesn't grant permissions

D. Amazon GuardDuty Detects threats — not involved in access control

REINFORCING WITH IAM ROLE STRATEGY

Let's map IAM roles into a secure EC2-to-SNS workflow: -

Component

Role -

IAM Role

Grants sns:Publish permission -

EC2 Instance Profile

Binds role to EC2 instance -

Instance Metadata Service

Provides temporary credentials -

SNS Topic

Receives published messages -

CloudTrail

Logs access and usage

REAL-WORLD TIE-IN

If your Nairobi-based team runs EC2-based alerting apps: Create an IAM role with sns:Publish permissions Attach it as an instance profile to EC2 Use AWS SDK to publish messages without storing credentials Monitor access with CloudTrail and IAM Access Analyzer That's how AWS empowers teams to securely authorize applications using IAM roles for seamless service integration.

Question 251

**A business is contemplating migrating to the AWS Cloud. The firm wishes to be able to scale its computing capacity in response to changing demand conditions. Which AWS Cloud advantage does this case illustrate?

**

OPTIONS

A. Global deployments in minutes

B. Cost savings

C. Agility

D. Elasticity

[!TIP]

Correct Answer:

D. Elasticity

EXPLANATION

CONCEPTUAL FRAMING

This question targets dynamic resource provisioning, especially how AWS enables businesses to scale infrastructure up or down automatically based on real-time demand.

WHY D IS CORRECT

Elasticity in AWS means: Provisioning resources only when needed Releasing resources when demand drops Avoiding overprovisioning and idle capacity Paying only for what you use Key services that support elasticity: Auto Scaling Groups for EC2 Lambda for serverless scaling DynamoDB and Aurora autoscaling Elastic Load Balancing to distribute traffic In AWS, elasticity means dynamically acquiring and releasing resources based on actual demand — growing or shrinking infrastructure as needed.

SOURCES

AWS Elasticity Explained Well-Architected Framework: Cost Optimization

DISTRACTOR BREAKDOWN

A. Global deployments Refers to geographic reach — not dynamic scaling

B. Cost savings A benefit of elasticity — but not the mechanism itself

C. Agility Refers to speed of innovation — not automatic scaling

REINFORCING WITH ELASTIC ARCHITECTURE STRATEGY

Let's map elasticity into a responsive workload model: -

Component

Role -

Auto Scaling Groups

Scale EC2 instances based on demand -

Lambda

Automatically scale with invocation volume -

DynamoDB

Adjust read/write capacity units dynamically -

CloudWatch Alarms

Trigger scaling actions -

Elastic Load Balancer

Distribute traffic across healthy targets

REAL-WORLD TIE-IN

If your Nairobi-based team runs a ticketing app with unpredictable traffic: Use Auto Scaling for EC2 web servers Enable DynamoDB autoscaling for user sessions Add CloudWatch alarms to trigger scale-in and scale-out events That's how AWS empowers teams to scale infrastructure elastically in response to real-world usage patterns.

Question 252

**Which of the following is a NoSQL database service that is both quick and dependable?

**

OPTIONS

A. Amazon Redshift

B. Amazon RDS

C. Amazon DynamoDB

D. Amazon S3

[!TIP]

Correct Answer:

C. Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question targets database architecture and performance, especially how AWS enables applications to store and retrieve structured data with low latency and high reliability using a serverless NoSQL model.

WHY C IS CORRECT

Amazon DynamoDB is: A fully managed NoSQL database Designed for key-value and document data models Offers single-digit millisecond latency at any scale Automatically scales throughput and storage Supports global tables, point-in-time recovery, and fine-grained access control Key benefits: Highly available and durable across multiple AZs Serverless — no

provisioning or patching required Integrated with Lambda, API Gateway, and EventBridge for event-driven architectures Amazon DynamoDB provides fast and predictable performance with seamless scalability — ideal for modern NoSQL workloads.

SOURCES

Amazon DynamoDB Overview NoSQL vs SQL on AWS

DISTRACTOR BREAKDOWN

A. Amazon Redshift Data warehouse — optimized for OLAP, not No SQL

B. Amazon RDS Relational database — supports SQL engines like MySQL, PostgreSQL

D. Amazon S3 Object storage — not a database service

REINFORCING WITH NOSQL STRATEGY

Let's map DynamoDB into a scalable data model: -

Component

Role -

Table

Stores items with partition and sort keys -

Item

JSON-like document with attributes -

Global Secondary Index (GSI)

Enables flexible querying -

DAX (DynamoDB Accelerator)

In-memory caching for read-heavy workloads -

Streams

Capture item-level changes for event processing

REAL-WORLD TIE-IN

If your Nairobi-based team builds a mobile app with real-time user sessions: Use DynamoDB to store session data Enable DynamoDB Streams to trigger Lambda functions Add DAX for low-latency reads Apply fine-grained IAM policies for secure access That's how AWS empowers teams to build fast, scalable, and dependable NoSQL applications using DynamoDB.

REFERENCES

<https://aws.amazon.com/dynamodb/>

Question 253

**A client wants to develop and construct a new workload on AWS Cloud but lacks the necessary technical skills in AWS-related technologies. Which of the following AWS programs may a client use to accomplish that goal?

**

OPTIONS

A. AWS Partner Network Technology Partners

B. AWS Marketplace

C. AWS Partner Network Consulting Partners

D. AWS Service Catalog

[!TIP]

Correct Answer:

C. AWS Partner Network Consulting Partners

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud adoption support, especially how AWS enables clients to leverage certified experts for architecture, migration, and implementation when internal skills are limited.

WHY C IS CORRECT

AWS Partner Network (APN) Consulting Partners: Are vetted third-party firms with deep AWS expertise Help clients design, build, migrate, and optimize workloads Provide training, certification, and hands-on implementation Often specialize in industry verticals, compliance, and modernization Key benefits: Accelerated cloud adoption Reduced risk and misconfiguration Access to innovation credits and sandbox environments Visibility to AWS field teams and go-to-market support APN Consulting Partners offer technical guidance, training, and implementation services to help clients build on AWS even without in-house expertise.

SOURCES

AWS Partner Network Overview Find a Consulting Partner

DISTRACTOR BREAKDOWN

A. Technology Partners Provide software solutions — not consulting or implementation help

B. AWS Marketplace Offers third-party tools — not hands-on guidance or architecture support

D. AWS Service Catalog Manages approved resources — doesn't help clients build workloads

REINFORCING WITH ENABLEMENT STRATEGY

Let's map APN Consulting Partners into a cloud onboarding workflow: -

Phase

Partner Role -

Assessment

Evaluate current environment and goals -

Design

Architect scalable, secure AWS workloads -

Migration

Move data and applications to AWS -

Optimization

Tune performance, cost, and compliance -

Enablement

Train internal teams and transfer knowledge

REAL-WORLD TIE-IN

If your Nairobi-based client wants to launch a fintech platform on AWS: Engage an APN Consulting Partner with financial services experience Architect a secure, compliant multi-tier workload Migrate legacy systems and integrate with AWS-native services Train internal staff on IAM, EC2, Lambda, and CloudWatch That's how AWS empowers clients to build confidently on the cloud with expert guidance from certified consulting partners.

REFERENCES

<https://aws.amazon.com/partners/>

Question 253

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REINFORCING WITH ENABLEMENT STRATEGY

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REFERENCES

<https://aws.amazon.com/partners/>

Question 255

****Which operation needs the usage of the root account user credentials for the AWS account?**

OPTIONS

- A. Closing an AWS account
- B. Creating a log file
- C. Modifying IAM user permissions
- D. Deleting IAM users

[!TIP]

Correct Answer:

- A. Closing an AWS account

EXPLANATION

CONCEPTUAL FRAMING

This question targets account-level security boundaries, especially how AWS enforces root-only access for high-impact operations that affect the entire account lifecycle.

WHY A IS CORRECT

Closing an AWS account: Is a permanent and irreversible action
Requires root user authentication to ensure full ownership
Involves terminating all resources and disabling billing
Can only be initiated via the AWS Management Console by the root user
Root user credentials are also required for:
Changing account name or email
Enabling MFA
Delete on S3 buckets
Restoring IAM access if all admins are locked out
To close your AWS account, sign in as the root user. From the navigation bar, choose your account name, then select “Account.”

SOURCES

DISTRACTOR BREAKDOWN

B. Creating a log file Can be done by any IAM user with appropriate permissions

C. Modifying IAM permissions Requires IAM privileges — not root access

D. Deleting IAM users IAM admins can perform this — root not required

REINFORCING WITH ACCOUNT GOVERNANCE STRATEGY

Let's map root-only actions into a secure account lifecycle: -

Action

Root Required? -

Close AWS account

Yes -

Change root email or password

Yes -

Enable MFA Delete on S3

Yes -

Create IAM users or roles

No -

Modify billing preferences

No (if IAM access to billing is enabled)

REAL-WORLD TIE-IN

If your Nairobi-based team is decommissioning a sandbox account: Sign in as the root user
Review all active resources and billing Use the Account Settings page to initiate closure
Confirm via email and console prompts That's how AWS ensures only the root user can perform high-impact operations like account closure for security and accountability.

Question 256

**A business operates Amazon EC2 instances with its own Docker environment. It seeks help managing cluster size, scheduling, and environment management. Which AWS service satisfies these criteria?

**

OPTIONS

A. AWS Lambda

B. Amazon RDS

C. AWS Fargate

D. Amazon Athena

[!TIP]

Correct Answer:

C. AWS Fargate

EXPLANATION

CONCEPTUAL FRAMING

This question targets container orchestration and serverless compute, especially how AWS enables teams to run containers without managing EC2 clusters, scaling policies, or infrastructure provisioning.

WHY C IS CORRECT

AWS Fargate: Is a serverless compute engine for containers Works with Amazon ECS and EKS Eliminates the need to provision, scale, or patch EC2 instances Automatically handles cluster sizing, task scheduling, and resource isolation Key benefits: Simplified operations — no cluster management Cost efficiency — pay per vCPU and memory used Security isolation — each task runs in its own kernel Scalability — tasks scale with demand, no manual intervention With AWS Fargate, you no longer have to provision, configure, or scale clusters of virtual machines to run containers.

SOURCES

AWS Fargate Overview Fargate vs EC2 Comparison

DISTRACTOR BREAKDOWN

A. AWS Lambda Runs code — not suitable for long-running containerized workloads

B. Amazon RDS Manages relational databases — not for container orchestration

D. Amazon Athena Queries data in S3 — unrelated to container management

REINFORCING WITH CONTAINER STRATEGY

Let's map Fargate into a modern container workflow: -

Component

Role -

ECS/EKS

Orchestrates container tasks and services -

Fargate

Runs containers without managing servers -

Task Definitions

Define container specs and resource needs -

CloudWatch Logs

Monitor container output and performance -

IAM Roles

Secure access to AWS services from containers

REAL-WORLD TIE-IN

If your Nairobi-based team runs a microservice architecture: Define services in ECS with Fargate launch type Avoid EC2 provisioning and patching Use CloudWatch and X-Ray for observability Scale tasks automatically based on CPU or memory thresholds That's how AWS empowers teams to run containerized workloads with minimal operational overhead using Fargate's serverless model.

Question 256

**Which of the following best describes the idea of elasticity? (Select two.)

**

OPTIONS

A. Scaling the number of Amazon EC2 instances based on traffic

B. Resizing Amazon RDS instances as business needs change

C. Automatically directing traffic to less-utilized Amazon EC2 instances

D. Using AWS compliance documents to accelerate the compliance process

E. Having the ability to create and govern environments using code

[!TIP]

Correct Answer:

A and B

EXPLANATION

CONCEPTUAL FRAMING

This question targets elasticity, one of the core tenets of cloud computing, especially how AWS enables businesses to dynamically adjust resource capacity in response to real-time demand.

Elasticity is not just about scaling up — it's about scaling down too, ensuring that resources match usage patterns to avoid waste and optimize cost. 🔍 Why A and B Are Correct ✅ A.

Scaling EC2 instances based on traffic This is a textbook example of horizontal elasticity. AWS

Auto Scaling can: Increase EC2 instances during traffic spikes Reduce instances during quiet periods Use metrics like CPU, memory, or custom CloudWatch alarms This ensures

performance stability and cost efficiency. ✅ B. Resizing RDS instances as business needs

change This is vertical elasticity — changing the instance type or storage size. You can:

Upgrade to a larger instance for more throughput Downgrade to save costs during low usage

Use RDS storage autoscaling for dynamic disk growth This supports flexible database

performance tuning. ❌ Why the Others Are Incorrect Option Why It's Not Elasticity

C.

Traffic redirection That's load balancing, not resource scaling

D.

Compliance documents That's governance and audit acceleration, not elasticity

E.

Infrastructure as Code (IaC) That's automation and governance, not dynamic scaling 

Elasticity Implementation Strategy Let's map elasticity into a real-world AWS architecture:

Component Elastic Role EC2 Auto Scaling Group Adjusts instance count based on demand

RDS Instance Class Resized manually or via scheduled events Lambda Scales automatically

with invocation volume DynamoDB Adjusts read/write capacity units dynamically CloudWatch

Alarms Triggers scaling actions based on metrics Elastic Load Balancer Distributes traffic —

complements elasticity, but not part of it

REAL-WORLD TIE-IN

If your Nairobi-based team runs a ticketing app: Use Auto Scaling for EC2 web servers Enable RDS instance resizing for seasonal traffic Add CloudWatch alarms to trigger scale-in and scale-out events Monitor elasticity with AWS Trusted Advisor and Cost Explorer That's how AWS empowers teams to scale infrastructure elastically in response to real-world usage patterns, optimizing both performance and cost. Question 258 Which of the following are advantages of using the AWS Cloud to host infrastructure? (Select two.)

A.

There are no upfront commitments.

B.

AWS manages all security in the cloud.

C.

Users have the ability to provision resources on demand.

D.

Users have access to free and unlimited storage.

E.

Users have control over the physical infrastructure. Why A and C are correct No upfront commitments: AWS's pay-as-you-go and subscription models let you start without capital expenditure. You can commit later for discounts (e.g., Savings Plans and Reserved Instances) but aren't forced to prepay to begin. This shifts spend from CapEx to OpEx and reduces risk when experimenting or scaling. Provision resources on demand: You can spin up compute, storage, and databases in minutes, then scale elastically based on demand. This enables rapid iteration, faster time-to-value, and alignment of capacity with real workload needs. Quick comparison: commitment and provisioning options On-Demand: No commitment; pay per

hour/second for usage. Savings Plans / Reserved Instances: Optional commitments for 1–3 years to reduce cost; still benefit from on-demand provisioning where applicable. Autoscaling & serverless: Automatically adjusts capacity (EC2 Auto Scaling, Lambda concurrency, DynamoDB autoscaling), minimizing manual intervention. Why the others are incorrect

B.

AWS manages all security in the cloud: Security is shared. AWS secures the underlying infrastructure (“of” the cloud). Customers secure workloads, data, identities, configurations, and network controls (“in” the cloud).

D.

Free and unlimited storage: Storage is metered and billed (S3, EBS, EFS). While some tiers offer free allowances, there’s no unlimited free storage.

E.

Control over physical infrastructure: Customers don’t manage or access AWS’s physical data centers or hardware. Control is provided at the logical layer via APIs, IAM, and configuration. Practical takeaways Cost flexibility: Start without commitments; add Savings Plans later to reduce steady-state compute costs. Operational agility: Use on-demand provisioning and autoscaling to match capacity to demand, preventing both overprovisioning and bottlenecks. Governance and security: Apply IAM, guardrails, and configuration baselines to meet “in the cloud” responsibilities while benefiting from AWS’s “of the cloud” security posture.

Question 259

**Which of the following are benefits of Amazon Web Services' cloud computing platform? (Select two)

**

OPTIONS

A. AWS manages the maintenance of the cloud infrastructure

B. AWS manages the security of applications built on AWS

C. AWS manages capacity planning for physical servers

D. AWS manages the development of applications on AWS

E. AWS manages cost planning for virtual servers

[!TIP]

Correct Answer:

A and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets shared responsibility and infrastructure abstraction, especially how AWS enables businesses to offload undifferentiated heavy lifting like hardware maintenance and capacity planning while retaining control over application logic and data. 🔍 Why A and C Are Correct ✅ A. AWS manages the maintenance of the cloud infrastructure AWS handles: Data center operations (power, cooling, physical security) Hardware lifecycle (server provisioning, patching, replacement) Virtualization stack (hypervisors, host OS updates) This frees customers to focus on application-level concerns. ✅ C. AWS manages capacity planning for physical servers AWS abstracts: Server sizing and provisioning Rack density and availability zone distribution Scaling infrastructure to meet aggregate demand Customers use Auto Scaling, Fargate, Lambda, and managed services without worrying about physical server limits. AWS manages all infrastructure layers including data centers, hardware, virtualization, and networking — enabling elastic, scalable, and resilient workloads.

SOURCES

AWS Shared Responsibility Model AWS Global Infrastructure ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

Security of applications Customers are responsible for securing their apps, data, and configurations — AWS secures the infrastructure only

D.

Application development AWS provides tools (e.g., Lambda, Amplify), but does not develop apps for customers

E.

Cost planning AWS provides billing tools (e.g., Cost Explorer, Budgets), but customers manage their own cost strategies

REINFORCING WITH INFRASTRUCTURE STRATEGY

Let's map AWS responsibilities vs customer responsibilities: -

Layer

AWS Responsibility Customer Responsibility -

Physical Infrastructure

Yes No -

Virtualization & Networking

Yes No -

OS & Runtime

(for managed services) (for EC2, containers) -

Application Code

No Yes -

Data & Access Control

No Yes -

Cost Optimization

No Yes

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a multi-tenant SaaS platform: Let AWS handle hardware, scaling, and patching Focus your team on application logic, IAM policies, and cost governance Use Auto Scaling and managed services to avoid manual capacity planning That's how AWS empowers teams to build resilient, scalable applications while offloading infrastructure complexity and physical resource management.

Question 260

**Which AWS service enables infrastructure as code management?

**

OPTIONS

A. AWS CodePipeline

B. AWS CodeDeploy

C. AWS Direct Connect

D. AWS CloudFormation

[!TIP]

Correct Answer:

D. AWS CloudFormation

EXPLANATION

CONCEPTUAL FRAMING

This question targets infrastructure as code (IaC), especially how AWS enables teams to define, provision, and manage cloud resources using declarative templates that serve as a single source of truth.

WHY D IS CORRECT

AWS CloudFormation: Uses YAML or JSON templates to define infrastructure Supports version control, auditing, and repeatable deployments Automates provisioning of EC2, S3, IAM, VPC, RDS, Lambda, etc. Integrates with Change Sets, StackSets, and Drift Detection for governance Key benefits: Consistency — same template across environments Speed — rapid provisioning and teardown Security — IAM roles and rollback protection Scalability — deploy across accounts and Regions AWS CloudFormation provides a common language to describe and provision all infrastructure resources in your cloud environment.

SOURCES

AWS CloudFormation Documentation IaC Best Practices

DISTRACTOR BREAKDOWN

A. CodePipeline Orchestrates CI/CD workflows — doesn't define infrastructure

B. CodeDeploy Automates application deployment — not infrastructure provisioning

C. Direct Connect Establishes private network links — unrelated to Ia C

REINFORCING WITH IAC STRATEGY

Let's map CloudFormation into a secure, scalable deployment model: -

Component

Role -

Template File

Defines resources and configurations -

Stack

Deployed instance of the template -

Parameters

Customize deployments across environments -

Outputs

Share values across stacks -

Change Sets

Preview updates before applying -

StackSets

Deploy across multiple accounts and Regions

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a multi-tier web app: Use CloudFormation templates to define VPC, EC2, RDS, and IAM roles Store templates in Git for version control Deploy via StackSets across dev, staging, and prod accounts Monitor drift and rollback failures with CloudFormation console That's how AWS empowers teams to codify infrastructure for repeatable, secure, and scalable cloud deployments using CloudFormation.

REFERENCES

<https://aws.amazon.com/cloudformation/>

Question 261

**How might AWS help a business manage costs when an application's consumption varies unpredictably?

OPTIONS

- A. AWS will refund the cost difference if a customer moves to larger servers
- B. The application can be built to scale up or down automatically as resources are needed
- C. Spot instances will automatically be used if the price is lower than on-demand instances
- D. Amazon CloudWatch will automatically predict what resources are needed

[!TIP]

Correct Answer:

- B. The application can be built to scale up or down automatically as resources are needed

EXPLANATION

CONCEPTUAL FRAMING

This question targets elasticity and cost control, especially how AWS enables businesses to dynamically adjust resource usage based on real-time demand, ensuring optimal performance without overspending.

WHY B IS CORRECT

AWS Auto Scaling: Monitors usage metrics (CPU, memory, throughput) Automatically adjusts capacity up or down Works with EC2, ECS, DynamoDB, Aurora, and more Ensures you only pay for what you use Prevents overprovisioning during low demand and underprovisioning during spikes Key benefits: Cost efficiency — eliminate idle resources Performance assurance — meet demand without manual intervention Operational agility — respond to unpredictable traffic patterns AWS Auto Scaling helps optimize utilization and cost performance by scaling resources dynamically based on demand.

SOURCES

AWS Auto Scaling Overview Elasticity in the Cloud

DISTRACTOR BREAKDOWN

A. Refunds for server changes AWS doesn't offer refunds for instance resizing — cost is based on usage

C. Spot instance substitution Spot usage must be explicitly configured — not automatic based on price

D. CloudWatch prediction Cloud Watch monitors and alerts — it doesn't predict or provision resources

REINFORCING WITH ELASTIC COST STRATEGY

Let's map dynamic scaling into a responsive workload model: -

Component

Role -

Auto Scaling Groups

Adjust EC2 instance count based on demand -

Lambda

Scales automatically with invocation volume -

DynamoDB

Adjusts read/write capacity units dynamically -

CloudWatch Alarms

Trigger scaling actions -

Elastic Load Balancer

Distributes traffic across healthy targets

REAL-WORLD TIE-IN

If your Nairobi-based team runs a ride-hailing app with unpredictable traffic: Use Auto Scaling for EC2 backend services Enable DynamoDB autoscaling for session and location data Add CloudWatch alarms to trigger scale-in and scale-out events Monitor usage with Cost Explorer and Budgets That's how AWS empowers teams to scale infrastructure elastically and manage costs intelligently for variable workloads.

REFERENCES

<https://aws.amazon.com/autoscaling/>

Question 262

**What are the benefits of deploying an application across various Availability Zones using Amazon EC2 instances? (Select two)

**

OPTIONS

- A. Preventing a single point of failure
- B. Reducing the operational costs of the application
- C. Allowing the application to serve cross-region users with low latency
- D. Increasing the availability of the application
- E. Increasing the load of the application

[!TIP]

Correct Answer:

A and D

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience and availability, especially how AWS enables applications to withstand infrastructure failures and maintain uptime by distributing workloads across isolated fault domains. 🔍 Why A and D Are Correct ✅ A. Preventing a single point of failure Availability Zones (AZs) are physically separated data centers within a Region. Deploying across multiple AZs ensures that: Failure in one AZ doesn't affect the entire application Load balancers can reroute traffic to healthy instances You avoid infrastructure bottlenecks or outages ✅ D. Increasing the availability of the application Multi-AZ deployments support: High availability architectures Automatic failover with Elastic Load Balancing Redundant compute and storage layers This ensures 99.99%+ uptime for mission-critical workloads Deploying across multiple AZs improves fault tolerance and availability by isolating failure domains and enabling seamless failover.

SOURCES

AWS Regions and Availability Zones High Availability on AWS ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

Reducing operational costs Multi-AZ deployments may increase cost due to redundancy

C.

Cross-region latency AZs are within a Region — cross-region latency requires Global Accelerator or Route 53

E.

Increasing load AZs distribute load — they don't inherently increase it

REINFORCING WITH MULTI-AZ STRATEGY

Let's map multi-AZ deployment into a resilient architecture: -

Component

Role -

EC2 Instances

Deployed across AZs for redundancy -

Elastic Load Balancer

Routes traffic to healthy instances -

Auto Scaling Group

Maintains instance count across AZs -

RDS Multi-AZ

Provides synchronous replication and automatic failover -

CloudWatch Alarms

Detect failures and trigger recovery actions

REAL-WORLD TIE-IN

If your Nairobi-based team runs a payment gateway: Deploy EC2 instances in at least two AZs
Use Elastic Load Balancer to distribute traffic
Enable Auto Scaling to maintain capacity
Monitor with CloudWatch and AWS Health Dashboard
That's how AWS empowers teams to build fault-tolerant, highly available applications using multi-AZ deployments.

Question 263

**Which AWS service supports a hybrid architecture that gives users the ability to extend AWS infrastructure, AWS services, APIs, and tools to data centers, co-location environments, or on-premises facilities?

OPTIONS

A. AWS Snowmobile

B. AWS Local Zones

C. AWS Outposts

D. AWS Fargate

[!TIP]

Correct Answer:

C. AWS Outposts

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud enablement, especially how AWS empowers organizations to run native AWS services on-premises with consistent APIs, tools, and security models.

WHY C IS CORRECT

AWS Outposts: Delivers fully managed AWS infrastructure to your on-premises location
Supports services like EC2, EBS, ECS, EKS, RDS, S3, and EMR locally Uses the same APIs, IAM policies, and monitoring tools as AWS Regions Enables low-latency workloads, data residency compliance, and edge processing Key benefits: Unified management across cloud and on-prem Consistent developer experience Integrated with AWS Organizations, Control Tower, and Systems Manager Supports hybrid use cases like manufacturing, healthcare, and financial services AWS Outposts extends AWS infrastructure, services, APIs, and tools to on-premises environments for hybrid cloud deployments.

SOURCES

AWS Outposts Overview Hybrid Cloud Architectures

DISTRACTOR BREAKDOWN

A. Snowmobile Used for mass data transfer — not hybrid infrastructure

B. Local Zones Extend AWS closer to users — not into customer data centers

D. Fargate Serverless containers — no on-premises deployment capability

REINFORCING WITH HYBRID STRATEGY

Let's map Outposts into a hybrid deployment model: -

Component

Role -

Outposts Rack

Physical AWS hardware installed on-prem -

Local EC2/EBS

Run compute and storage workloads locally -

AWS APIs

Same interface as cloud-based services -

IAM & CloudWatch

Unified access control and monitoring -

Direct Connect

Optional link to AWS Region for hybrid networking

REAL-WORLD TIE-IN

If your Nairobi-based enterprise needs to run latency-sensitive workloads in a local data center: Deploy AWS Outposts to host EC2 and RDS locally Use IAM and CloudWatch for unified governance Integrate with AWS Region services for backup and analytics Maintain data residency compliance while leveraging AWS tools That's how AWS empowers teams to build hybrid architectures with consistent tooling and governance using Outposts.

Question 264

**A company has a physical tape library to store data backups. The tape library is running out of space. The company needs to extend the tape library's capacity to the AWS Cloud. Which AWS service should the company use to meet this requirement?

**

OPTIONS

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon S3

C. Amazon Elastic File System (Amazon EFS)

D. AWS Storage Gateway

[!TIP]

Correct Answer:

D. AWS Storage Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid storage integration, especially how AWS enables organizations to extend on-premises backup infrastructure into the cloud using familiar interfaces and minimal disruption.

WHY D IS CORRECT

AWS Storage Gateway: Offers a Tape Gateway mode that emulates a physical tape library
Allows backup applications to write to virtual tapes, stored in Amazon S3 and archived to Amazon Glacier
Supports industry-standard iSCSI interface, so existing backup software works without modification
Enables scalable, cost-effective archival with lifecycle transitions
Key benefits: Seamless extension of existing tape workflows
No need to rearchitect backup software
Durable, low-cost storage with Glacier and Deep Archive
Hybrid architecture — local caching with cloud-backed storage
AWS Storage Gateway provides tape-based, file-based, and block-based hybrid storage options to extend on-premises environments to AWS.

SOURCES

AWS Storage Gateway Overview
Tape Gateway Use Cases

DISTRACTOR BREAKDOWN

A. Amazon EBS Block storage for EC2 — not suitable for tape emulation or backup workflows

B. Amazon S3 Backend for virtual tapes — but not directly usable by tape libraries

C. Amazon EFS Network file system — not designed for backup tape integration

REINFORCING WITH HYBRID BACKUP STRATEGY

Let's map Tape Gateway into a hybrid archival model: -

Component

Role -

On-prem Tape Library

Existing backup infrastructure -

Tape Gateway Appliance

Virtual tape interface via iSCSI -

Virtual Tapes

Stored in Amazon S3, archived to Glacier -

Backup Software

Writes to virtual tapes as if physical -

Lifecycle Policies

Transition tapes to Deep Archive for cost savings

REAL-WORLD TIE-IN

If your Nairobi-based enterprise is running out of tape capacity: Deploy Tape Gateway as a VM or hardware appliance Connect via iSCSI to existing backup software Store virtual tapes in S3, archive to Glacier Deep Archive Monitor usage and transitions with AWS Backup and CloudWatch That's how AWS empowers teams to extend legacy backup systems into the cloud with minimal disruption using Storage Gateway.

Question 265

******A business wishes to downsize its infrastructure in order to save money. At what stages should a business downsize? (Select two)

OPTIONS

- A. Rightsize before a migration occurs to the cloud
- B. Rightsize continuously after the cloud onboarding process
- C. Rightsize when AWS Support calls and explains that rightsizing is needed
- D. Rightsize when seasonal workloads are at their peak
- E. Rightsize after purchasing all Reserved Instances

[!TIP]

Correct Answer:

A and B

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization and lifecycle planning, especially how AWS enables businesses to align resource provisioning with actual workload needs through proactive and continuous rightsizing. 🔍 Why A and B Are Correct ✓ A. Rightsize before migration Pre-migration rightsizing ensures: You don't lift-and-shift overprovisioned workloads You map legacy resources to optimal cloud equivalents You avoid wasteful instance types and storage tiers Tools like AWS Migration Evaluator and Compute Optimizer help assess baseline needs. ✓ B. Rightsize continuously after onboarding Post-migration rightsizing ensures: You adapt to changing usage patterns You eliminate idle or underutilized resources You optimize for performance and cost using real-time metrics Use Cost Explorer, Trusted Advisor, and Compute Optimizer for ongoing insights. Rightsizing is the process of matching instance types and sizes to workload performance and capacity needs at the lowest possible cost — both before and after migration.

SOURCES

AWS Cost Optimization Pillar AWS Compute Optimizer ❌ Why the Others Are Incorrect Option
Why It's Incorrect

C.

AWS Support calls Rightsizing is proactive — not reactive to support outreach

D.

Seasonal peak You scale up during peaks — rightsizing happens before or after, not during

E.

After Reserved Instances Rightsizing should precede RI purchases to avoid locking in waste

REINFORCING WITH RIGHTSIZING STRATEGY

Let's map rightsizing into a cost-aware cloud lifecycle: -

Stage

Rightsizing Action -

Pre-Migration

Assess current workloads, eliminate waste, map to optimal cloud resources -

Onboarding

Monitor usage, adjust instance families and sizes -

Ongoing Ops

Use Cloud Watch, Cost Explorer, and Compute Optimizer to refine continuously -

Before RI Purchase

Confirm steady-state usage to avoid overcommitment -

Post-Event Scaling

Scale down after peak demand subsides

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a legacy CRM system: Use Migration Evaluator to assess current resource needs Rightsize EC2 instances and RDS storage before migration Monitor usage with CloudWatch and Trusted Advisor Continuously optimize with Compute Optimizer recommendations That's how AWS empowers teams to reduce costs and improve efficiency through strategic rightsizing at every stage of the cloud journey.

REFERENCES

<https://aws.amazon.com/compute-optimizer/>

Question 266

**AWS Budgets may be used for the following purposes:

**

OPTIONS

- A. Prevent a given user from creating a resource
- B. Send an alert when the utilization of Reserved Instances drops below a certain percentage
- C. Set resource limits in AWS accounts to prevent overspending
- D. Split an AWS bill across multiple forms of payment

[!TIP]

Correct Answer:

- B. Send an alert when the utilization of Reserved Instances drops below a certain percentage

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost monitoring and proactive alerting, especially how AWS enables organizations to track usage, forecast spending, and respond to underutilized commitments like Reserved Instances and Savings Plans.

WHY B IS CORRECT

AWS Budgets: Allows you to set custom thresholds for cost, usage, RI utilization, and coverage
Sends alerts via email or SNS when thresholds are breached
Supports actual and forecasted metrics
Helps identify underutilized commitments, enabling corrective action
Key benefits:
Proactive cost control
Visibility into RI/Savings Plan efficiency
Automated notifications for finance and ops teams
Integration with AWS Cost Explorer and Organizations
AWS Budgets helps track cost and usage, and alerts you when RI utilization drops below your target threshold.

SOURCES

AWS Budgets Documentation RI Utilization Alerts  Why the Others Are Incorrect Option Why It's Incorrect

A.

Prevent user from creating resources IAM policies and SCPs control access — Budgets only monitor and alert

C.

Set hard resource limits Budgets don't enforce limits — they notify when thresholds are exceeded

D.

Split bills across payments Billing is managed via Consolidated Billing and payment methods, not Budgets

REINFORCING WITH BUDGET STRATEGY

Let's map AWS Budgets into a cost governance model: -

Budget Type

Purpose -

Cost Budget

Track total spend across services or linked accounts -

Usage Budget

Monitor usage of specific services (e.g., EC2 hours, S3 storage) -

RI Utilization Budget

Alert when RI usage drops below target -

Savings Plan Coverage Budget

Track how much usage is covered by Savings Plans -

Forecast Budget

Predict future spend and alert early

REAL-WORLD TIE-IN

If your Nairobi-based team has committed to EC2 Reserved Instances: Set a RI utilization budget at 90% Receive SNS alerts when usage drops Investigate idle instances or misaligned workloads Adjust instance types or consolidate workloads to improve coverage That's how AWS empowers teams to monitor cost efficiency and respond to underutilized commitments using Budgets.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-budgets/>

Question 267

**An online retail company has seasonal sales spikes several times a year, primarily around holidays. Demand is lower at other times. The company finds it difficult to predict the increasing infrastructure demand for each season. Which advantages of moving to the AWS Cloud would MOST benefit the company? (Choose two)

**

OPTIONS

A. Global footprint

B. Elasticity

C. AWS service quotas

D. AWS shared responsibility model

E. Pay-as-you-go pricing

[!TIP]

Correct Answer:

B and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud scalability and cost control, especially how AWS enables businesses to respond to unpredictable demand patterns without overprovisioning or overspending. 🔍 Why B and E Are Correct ✅ B. Elasticity AWS allows dynamic scaling of: Compute (EC2, Lambda) Storage (S3, EBS) Databases (Aurora Serverless, DynamoDB) You can scale up during peak seasons and down during quiet periods Avoids both

underprovisioning (performance risk) and overprovisioning (cost waste)  E. Pay-as-you-go pricing You pay only for what you use — no upfront infrastructure investment Ideal for seasonal or unpredictable workloads Supports cost optimization with: On-Demand pricing Savings Plans and Spot Instances Auto Scaling to match usage Cloud elasticity enables rapid, dynamic allocation of resources. Pay-as-you-go pricing ensures cost aligns with actual usage — perfect for seasonal demand.

SOURCES

AWS Elasticity Explained AWS Pricing Models  Why the Others Are Incorrect Option Why It's Incorrect

A.

Global footprint Helps with geographic reach — not seasonal scaling or cost control

C.

Service quotas Prevent overuse — not a benefit for scaling or cost optimization

D.

Shared responsibility model Defines security roles — not related to elasticity or pricing

REINFORCING WITH SEASONAL STRATEGY

Let's map AWS elasticity and pricing into a seasonal workload model: -

Component

Role -

Auto Scaling Groups

Scale EC2 instances based on traffic spikes -

Lambda

Automatically scale with invocation volume -

DynamoDB Autoscaling

Adjust read/write capacity units dynamically -

CloudWatch Alarms

Trigger scaling actions -

Cost Explorer & Budgets

Monitor spend and forecast seasonal costs

REAL-WORLD TIE-IN

If your Nairobi-based retail team runs holiday campaigns: Use Auto Scaling for EC2 web servers Enable Lambda for serverless checkout flows Monitor usage with CloudWatch and AWS Budgets Pay only for what you use — no idle infrastructure between seasons That's how AWS empowers teams to scale elastically and pay efficiently for seasonal workloads using cloud-native architecture.

Question 268

**What are the advantages of unified billing for Amazon Web Services accounts?

**

OPTIONS

A. Access to AWS Personal Health Dashboard

B. Combined usage volume discounts

C. Improved account security

D. Centralized AWS IAM

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account cost management, especially how AWS enables organizations to aggregate usage across accounts to unlock pricing efficiencies and simplify billing operations.

WHY B IS CORRECT

Consolidated Billing via AWS Organizations: Combines usage from all linked accounts under a single paying account Enables volume-based discounts on services like EC2, S3, and data transfer Provides a unified bill with detailed breakdowns per account Supports centralized budget tracking and forecasting Key benefits: Cost savings through aggregated usage Simplified financial reporting Visibility into per-account spend No additional charge for using consolidated billing AWS combines usage from all accounts to qualify for volume pricing discounts, reducing overall spend.

SOURCES

AWS Consolidated Billing AWS Organizations ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Personal Health Dashboard Available per account — not tied to billing consolidation

C.

Account security Managed via IAM and SCPs — not a billing feature

D.

Centralized IAM IAM is scoped per account — not unified across accounts via billing

REINFORCING WITH BILLING STRATEGY

Let's map consolidated billing into a multi-account governance model: -

Feature

Benefit -

One Bill

Unified view of all charges -

Detailed Tracking

Per-account cost breakdowns -

Volume Discounts

Aggregated usage unlocks lower pricing tiers -

Budgeting & Forecasting

Centralized control with AWS Budgets and Cost Explorer -

Free Tier

Shared across accounts — not multiplied

REAL-WORLD TIE-IN

If your Nairobi-based team runs separate AWS accounts for dev, staging, and prod: Link them under AWS Organizations Use Consolidated Billing to combine usage Monitor spend with Cost Explorer and Budgets Unlock tiered discounts for EC2, S3, and data transfer That's how AWS empowers teams to optimize costs and simplify financial operations across multiple accounts using consolidated billing.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 269

**A corporation wants to connect to AWS from a distant office using a private, low-latency connection. Which strategy is best?

**

OPTIONS

- A. Create a VPN tunnel
- B. Connect across the public internet
- C. Use VPC peering to create a connection
- D. Use AWS Direct Connect

[!TIP]

Correct Answer:

D. Use AWS Direct Connect

EXPLANATION

CONCEPTUAL FRAMING

This question targets enterprise-grade connectivity, especially how AWS enables organizations to establish dedicated, high-throughput, low-latency links between their data centers and AWS infrastructure.

WHY D IS CORRECT

AWS Direct Connect: Provides a dedicated network connection from your premises to AWS Bypasses the public internet, reducing latency and jitter Offers consistent performance for workloads like video streaming, financial transactions, and large data transfers Supports 1

Gbps to 100 Gbps links with optional redundancy and failover Key benefits: Private connectivity — traffic stays on AWS backbone Low latency — ideal for real-time applications Secure and predictable throughput Integrated with VPCs, Transit Gateway, and hybrid architectures AWS Direct Connect is the shortest path to your AWS resources, keeping traffic off the public internet and reducing latency.

SOURCES

AWS Direct Connect Overview Hybrid Connectivity Patterns ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

VPN tunnel Uses public internet — less reliable and higher latency

B.

Public internet Unpredictable latency and security risks

C.

VPC peering Connects VPCs — not suitable for on-premises to AWS links

REINFORCING WITH HYBRID NETWORKING STRATEGY

Let's map Direct Connect into a secure enterprise architecture: -

Component

Role -

Direct Connect Gateway

Connects multiple VPCs across Regions -

Virtual Interface (VIF)

Establishes Layer 3 routing to AWS -

Redundant Links

Ensures high availability -

BGP Routing

Enables dynamic path selection -

CloudWatch Metrics

Monitors link health and throughput

REAL-WORLD TIE-IN

If your Nairobi-based enterprise needs to stream telemetry data to AWS: Provision Direct Connect at a local colocation facility Establish private VIFs to your VPCs Use Transit Gateway to route across accounts Monitor performance with CloudWatch and Network Manager That's how AWS empowers teams to build secure, low-latency hybrid networks using Direct Connect for enterprise-grade connectivity.

Question 270

**Which AWS service can be used to turn text into lifelike speech?

**

OPTIONS

A. Amazon Polly

B. Amazon Kendra

C. Amazon Rekognition

D. Amazon Connect

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets natural language processing and voice synthesis, especially how AWS enables developers to convert written content into realistic spoken audio using deep learning models.

WHY A IS CORRECT

Amazon Polly: Uses neural text-to-speech (NTTS) to generate natural-sounding voices Supports dozens of languages and voices, including neural and standard options Offers SSML (Speech Synthesis Markup Language) for fine-grained control over pronunciation, pauses, and intonation Enables real-time streaming or file-based output for applications like: Accessibility tools Voice-enabled apps Audiobooks IVR systems Key benefits: Lifelike voice quality Scalable and cost-effective Custom lexicons for domain-specific pronunciation Secure integration with other AWS services (e.g., S3, Lambda, Transcribe) Amazon Polly uses advanced deep learning to synthesize speech that sounds natural and lifelike, converting text into spoken language.

SOURCES

Amazon Polly Overview Polly Use Cases

DISTRACTOR BREAKDOWN

B. Amazon Kendra Enterprise search — not related to speech synthesis

C. Amazon Rekognition Image and video analysis — no text-to-speech capabilities

D. Amazon Connect Cloud contact center — may use Polly, but doesn't perform TTS itself

REINFORCING WITH VOICE STRATEGY

Let's map Polly into a voice-enabled architecture: -

Component

Role -

Polly API

Converts text to speech -

S3

Stores generated audio files -

Lambda

Orchestrates TTS workflows -

CloudFront

Delivers audio to global users -

SSML Tags

Controls speech tone, rate, and pauses

REAL-WORLD TIE-IN

If your Nairobi-based team is building an educational app: Use Amazon Polly to narrate lessons in multiple languages Store audio in S3, stream via CloudFront Customize pronunciation with SSML and lexicons Integrate with Transcribe for bi-directional voice-text workflows That's how AWS empowers teams to build accessible, voice-enabled applications using Polly's lifelike speech synthesis.

Question 271

****Which Amazon EC2 pricing model is dynamically adjusted in response to EC2 instance availability and demand?**

OPTIONS

- A. On-Demand Instances
- B. Reserved Instances
- C. Spot Instances
- D. Convertible Reserved Instances

[!TIP]

Correct Answer:

C. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization and capacity-based pricing, especially how AWS enables businesses to leverage unused EC2 capacity at steep discounts through a dynamic market model.

WHY C IS CORRECT

Spot Instances: Offer up to 90% discount compared to On-Demand pricing Prices are determined by supply and demand for spare EC2 capacity AWS no longer uses bidding — prices are predictable and updated less frequently Ideal for fault-tolerant, flexible workloads like: Batch processing Big data analytics CI/CD pipelines Containerized workloads Key benefits: Massive cost savings Scalable access to unused capacity Integration with Auto Scaling and EC2 Fleet Interruptible model — AWS can reclaim instances with 2-minute warning Spot pricing is now based on EC2 availability and demand, not user bids — making it more predictable and cost-effective.

SOURCES

Amazon EC2 Spot Instances Spot Instance Pricing Model ❌ Why the Others Are Incorrect
Option Why It's Incorrect

A.

On-Demand Fixed hourly/second pricing — not dynamic or capacity-based

B.

Reserved Instances Prepaid or committed pricing — not adjusted dynamically

D.

Convertible RIs Allow instance type changes — but pricing is fixed by commitment

REINFORCING WITH EC2 PRICING STRATEGY

Let's map EC2 pricing models into a workload optimization framework: -

Model

Use Case Pricing Behavior -

On-Demand

Short-term, unpredictable workloads Fixed per hour/second -

Reserved Instances

Steady-state workloads Discounted with commitment -

Spot Instances

Flexible, fault-tolerant workloads Dynamic based on spare capacity -

Savings Plans

Broad compute usage Flexible across services and instance families

REAL-WORLD TIE-IN

If your Nairobi-based team runs nightly data processing jobs: Use Spot Instances for batch workloads Integrate with EC2 Auto Scaling and Spot Fleet Monitor interruptions with CloudWatch Events Save up to 90% compared to On-Demand pricing That's how AWS empowers teams to optimize compute costs by dynamically leveraging unused EC2 capacity with Spot Instances.

REFERENCES

<https://aws.amazon.com/ec2/>

Question 272

**Which service should a client utilize to integrate and manage several Amazon Web Services accounts centrally?

**

OPTIONS

A. AWS IAM

B. AWS Organizations

C. AWS Schema Conversion Tool

D. AWS Config

[!TIP]

Correct Answer:

B. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account governance and centralized management, especially how AWS enables businesses to structure, control, and monitor multiple accounts under a unified framework.

WHY B IS CORRECT

AWS Organizations: Allows creation of organizational units (OUs) to group accounts by function, team, or compliance level Supports Service Control Policies (SCPs) to enforce permissions boundaries across accounts Enables centralized billing via consolidated billing Integrates with AWS Control Tower, IAM Identity Center, and CloudTrail for full-stack governance Key benefits: Centralized policy enforcement Scalable account structure for legal, project, or team boundaries Resource sharing via AWS RAM Audit-ready architecture with unified logging and tagging AWS Organizations enables centralized management of multiple AWS accounts with shared policies, billing, and governance controls.

SOURCES

AWS Organizations Overview Multi-Account Strategy

DISTRACTOR BREAKDOWN

A. IAM Manages identities within a single account — not designed for multi-account integration

C. Schema Conversion Tool Used for database migration — unrelated to account management

D. AWS Config Tracks resource configurations — doesn't manage accounts or policies

REINFORCING WITH MULTI-ACCOUNT STRATEGY

Let's map AWS Organizations into a secure governance model: -

Component

Role -

Management Account

Root of the organization — controls billing and policy -

Organizational Units (OUs)

Logical groupings of accounts -

Service Control Policies (SCPs)

Enforce permissions boundaries -

Consolidated Billing

Aggregates usage and unlocks volume discounts -

Tag Policies

Standardize resource tagging across accounts

REAL-WORLD TIE-IN

If your Nairobi-based enterprise has separate AWS accounts for dev, staging, prod, and audit: Use AWS Organizations to group them under OUs Apply SCPs to restrict risky actions in prod Enable consolidated billing for cost optimization Share resources like VPCs and KMS keys via AWS RAM That's how AWS empowers teams to manage multi-account environments with centralized control, security, and cost efficiency using AWS Organizations.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 273

****Which AWS service or tool can be used to capture information about inbound and outbound traffic in an Amazon VPC?**

OPTIONS

A. VPC Flow Logs

B. Amazon Inspector

C. VPC endpoint services

D. NAT gateway

[!TIP]

Correct Answer:

A. VPC Flow Logs

EXPLANATION

CONCEPTUAL FRAMING

This question targets network observability and security auditing, especially how AWS enables teams to monitor IP-level traffic across VPC interfaces for diagnostics, compliance, and threat detection.

WHY A IS CORRECT

VPC Flow Logs: Capture IP traffic metadata for network interfaces in your VPC Log details like source/destination IP, ports, protocol, packet count, and accept/deny status Can be published to Amazon CloudWatch Logs or S3 Useful for: Security auditing Troubleshooting connectivity

issues Analyzing traffic patterns Detecting anomalies or unauthorized access Key benefits:
Granular visibility into VPC traffic Low overhead — logs are metadata, not payloads Integration
with SIEMs and analytics tools Supports subnet, ENI, and VPC-level logging VPC Flow Logs
enable you to capture information about IP traffic going to and from network interfaces in your
VPC.

SOURCES

VPC Flow Logs Documentation Monitoring Best Practices ❌ Why the Others Are Incorrect
Option Why It's Incorrect

B.

Amazon Inspector Scans for vulnerabilities — doesn't monitor traffic

C.

VPC endpoint services Enable private connectivity — not traffic logging

D.

NAT gateway Facilitates outbound traffic — doesn't log or monitor it

REINFORCING WITH OBSERVABILITY STRATEGY

Let's map VPC Flow Logs into a secure monitoring architecture: -

Component

Role -

VPC Flow Logs

Capture traffic metadata -

CloudWatch Logs

Store and visualize flow data -

S3

Archive logs for long-term analysis -

Athena

Query flow logs for insights -

GuardDuty

Detect threats using flow log data

REAL-WORLD TIE-IN

If your Nairobi-based team is investigating intermittent connectivity issues: Enable VPC Flow Logs at the subnet level Stream logs to CloudWatch Logs for real-time analysis Use Athena to query traffic patterns Integrate with GuardDuty for anomaly detection That's how AWS empowers teams to gain deep visibility into VPC traffic using Flow Logs for diagnostics, security, and compliance.

Question 274

**Which choices do users have when contacting AWS Support?

**

OPTIONS

A. Create an email case in the AWS Support Center

B. Visit a local AWS Support Center

C. Use live chat functionality

D. Call the customer service phone number

E. Use the video conference functionality of the AWS Support console

[!TIP]

Correct Answer:

C. Use live chat functionality

EXPLANATION

CONCEPTUAL FRAMING

This question targets support access channels, especially how AWS enables users to engage with technical and account support through digital interfaces based on their support plan tier.

WHY C IS CORRECT

Live chat is available to users with Business or Enterprise Support plans. It provides: Real-time interaction with AWS support engineers. Faster resolution for urgent issues. Convenient access via the AWS Support Center. Chat is ideal for: Technical troubleshooting. Billing inquiries. Service limit increases. AWS Support offers live chat functionality for eligible support plans, enabling real-time assistance for technical and account issues.

SOURCES

AWS Support Plans Contacting AWS Support  Why the Others Are Incorrect Option Why It's Incorrect

A.

Email case AWS uses web-based case creation, not direct email submission

B.

Local Support Center AWS does not operate physical support centers for walk-ins

D.

Phone support Available only with Business or Enterprise plans, not universal

E.

Video conferencing Not a feature of the AWS Support console

REINFORCING WITH SUPPORT STRATEGY

Let's map AWS Support access by plan tier: -

Plan

Access Channels -

Basic

Documentation, forums, Trusted Advisor (limited) -

Developer

Case creation for technical support -

Business

Case creation, live chat, phone support, Trusted Advisor -

Enterprise

All Business features + Technical Account Manager (TAM), Concierge support

REAL-WORLD TIE-IN

If your Nairobi-based team is troubleshooting EC2 connectivity: Use live chat for real-time guidance Create a support case for deeper investigation Access Trusted Advisor for configuration checks Upgrade to Business Support for phone and chat access That's how AWS empowers teams to resolve issues quickly and optimize cloud operations through tailored support channels.

Question 275

******A business must track and anticipate AWS expenditures and usage. Additionally, the organization must configure event-driven alerts that are triggered when expenditure limitations are exceeded. Which AWS offering or technology should the business employ to achieve these requirements?

OPTIONS

A. AWS Budgets

B. Amazon CloudWatch

C. AWS Config

D. AWS Service Catalog

[!TIP]

Correct Answer:

A. AWS Budgets

EXPLANATION

CONCEPTUAL FRAMING

This question targets financial observability and proactive cost control, especially how AWS enables organizations to monitor spend, forecast usage, and trigger alerts when thresholds are breached.

WHY A IS CORRECT

AWS Budgets: Tracks actual and forecasted cost and usage across linked accounts, services, and tags Supports event-driven alerts via email or SNS when thresholds are exceeded Monitors: RI and Savings Plan utilization Service-specific usage Linked account spend Integrates with Cost Explorer, Organizations, and CloudWatch Events Key benefits: Customizable thresholds for cost, usage, and commitment efficiency Automated notifications for finance and ops teams Multi-account visibility with consolidated billing Supports budget actions to restrict IAM roles or service access when limits are breached AWS Budgets enables businesses to track spend, anticipate usage, and trigger alerts when limits are exceeded — ideal for cost governance.

SOURCES

AWS Budgets Documentation Budget Actions and Alerts ❌ Why the Others Are Incorrect
Option Why It's Incorrect

B.

CloudWatch Monitors service metrics — not designed for cost tracking or budget alerts

C.

AWS Config Tracks resource configurations — not financial metrics

D.

Service Catalog Manages approved service portfolios — not cost or usage monitoring

REINFORCING WITH BUDGET STRATEGY

Let's map AWS Budgets into a cost governance framework: -

Budget Type

Purpose -

Cost Budget

Track total spend across services or accounts -

Usage Budget

Monitor usage of specific services (e.g., EC2 hours, S3 storage) -

RI Utilization Budget

Alert when RI usage drops below target -

Forecast Budget

Predict future spend and alert early -

Budget Actions

Trigger IAM or service restrictions when limits are breached

REAL-WORLD TIE-IN

If your Nairobi-based team manages multiple AWS accounts: Set monthly cost budgets per team Enable SNS alerts for threshold breaches Track RI utilization to avoid waste Use budget actions to restrict non-essential services when nearing limits That's how AWS empowers teams to monitor, forecast, and control cloud spend with precision using AWS Budgets.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-budgets/>

Question 276

**Which of the following IT functions does AWS perform to relieve a business of its responsibility for managing IT resources? (Select two)

**

OPTIONS

A. Configuring operating system firewalls

B. Setting up access controls for data

C. Backing up databases

D. Configuring database user accounts

E. Installing operating systems

[!TIP]

Correct Answer:

C and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets the shared responsibility model, especially how AWS offloads infrastructure-level tasks while customers retain control over application-level configurations and data governance. 🔍 Why C and E Are Correct ✅ C. Backing up databases AWS automates backups for managed services like: Amazon RDS (automated snapshots, PITR) Amazon DynamoDB (on-demand and continuous backups) Amazon Aurora (cluster-wide backups) These backups are: Encrypted Durable (stored in S3) Regionally replicated (optional) ✅ E. Installing operating systems For managed services (e.g., RDS, Lambda, ECS Fargate), AWS: Installs and maintains the underlying OS Applies security patches Ensures compatibility and performance For EC2, AWS provides prebuilt AMIs with OS installed — customers can customize further AWS handles infrastructure tasks like OS installation and database backups, freeing customers to focus on application logic and data governance.

SOURCES

AWS Shared Responsibility Model RDS Backup Features ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

OS firewalls Customers configure security groups and NACLs — AWS doesn't manage guest OS firewalls

B.

Access controls for data Customers define IAM policies, KMS keys, and encryption settings

D.

Database user accounts Customers manage users, roles, and permissions within the database engine

REINFORCING WITH RESPONSIBILITY MAPPING

Let's map AWS vs customer responsibilities: -

Layer

AWS Responsibility Customer Responsibility -

Physical Infrastructure

Yes No -

OS Installation & Patching

Yes (for managed services) Yes (for EC2) -

Database Backups

Yes (for RDS, Aurora) Optional (custom DBs) -

Firewall Rules

No Yes (SGs, NACLs, OS-level) -

Access Controls

No Yes (IAM, DB roles) -

Application Logic

No Yes

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a multi-tenant SaaS platform: Use Amazon RDS for automated backups Choose managed services to offload OS maintenance Configure IAM policies and database roles for tenant isolation Monitor backup status via CloudWatch and AWS Backup That's how AWS empowers teams to delegate infrastructure tasks while retaining control over security and application logic.

Question 277

**Which AWS technologies aid in cost estimation? (Select three)

**

OPTIONS

- A. Detailed billing report
- B. Cost allocation tags
- C. AWS Simple Monthly Calculator
- D. AWS Total Cost of Ownership (TCO) Calculator
- E. Cost Estimator

[!TIP]

Correct Answer:

B, C, D

EXPLANATION

CONCEPTUAL FRAMING

This question targets pre-deployment cost modeling and post-deployment cost attribution, especially how AWS enables organizations to forecast, analyze, and allocate cloud spend across services, teams, and projects. 🔍 Why B, C, and D Are Correct ☒ B. Cost Allocation Tags Tags are key-value pairs applied to AWS resources Enable granular tracking of spend by: Department Project Environment (e.g., dev, staging, prod) Integrated with Cost Explorer, Budgets, and Billing Reports ☒ C. AWS Simple Monthly Calculator Legacy tool for estimating monthly costs based on: Service usage Region Instance types and quantities Helps forecast spend before deployment ☒ D. AWS Total Cost of Ownership (TCO) Calculator Compares on-premises vs AWS cloud costs Models: Hardware Software Facilities Labor Ideal for migration planning and executive buy-in These tools help forecast cloud costs, allocate spend by tag, and compare AWS vs traditional infrastructure.

SOURCES

AWS Cost Allocation Tags AWS Pricing Calculator AWS TCO Calculator ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Detailed Billing Report Useful for analysis — but not designed for estimation

E.

Cost Estimator Not an official AWS tool — may refer to third-party or outdated terminology

REINFORCING WITH COST ESTIMATION STRATEGY

Let's map AWS cost tools into a financial planning framework: -

Tool

Purpose -

Cost Allocation Tags

Attribute spend to teams/projects -

Pricing Calculator

Estimate monthly costs pre-deployment -

TCO Calculator

Compare AWS vs on-prem infrastructure -

Budgets

Monitor and alert on actual spend -

Cost Explorer

Visualize and analyze historical usage

REAL-WORLD TIE-IN

If your Nairobi-based team is planning a cloud migration: Use TCO Calculator to justify the move Model monthly spend with the Pricing Calculator Apply cost allocation tags to EC2, S3,

and RDS resources Monitor spend with Budgets and Cost Explorer That's how AWS empowers teams to plan, forecast, and attribute cloud costs with precision using integrated cost estimation tools.

Question 278

**A company wants to ensure that two Amazon EC2 instances are in separate data centers with minimal communication latency between the data centers. How can the company meet this requirement?

**

OPTIONS

A. Place the EC2 instances in two separate AWS Regions connected with a VPC peering connection

B. Place the EC2 instances in two separate Availability Zones within the same AWS Region

C. Place one EC2 instance on premises and the other in an AWS Region. Then connect them by using an AWS VPN connection

D. Place both EC2 instances in a placement group for dedicated bandwidth

[!TIP]

Correct Answer:

B. Place the EC2 instances in two separate Availability Zones within the same AWS Region

EXPLANATION

CONCEPTUAL FRAMING

This question targets high availability and low-latency architecture, especially how AWS enables organizations to deploy fault-tolerant applications across physically isolated data centers with minimal latency.

WHY B IS CORRECT

Availability Zones (AZs): Each AZ is a physically distinct data center within an AWS Region AZs are connected via high-bandwidth, low-latency links Deploying EC2 instances across AZs ensures: Fault isolation High availability Minimal latency for inter-instance communication Key benefits: Redundancy without geographic spread Fast failover in case of AZ outage Meets SLA requirements for service continuity Ideal for multi-tier apps, databases, and load-balanced services Deploying EC2 instances in separate AZs within the same Region ensures physical separation and low-latency communication.

SOURCES

AWS Regions and Availability Zones High Availability Best Practices ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Separate Regions + VPC Peering Regions are geographically distant — introduces higher latency

C.

On-prem + AWS VPN VPN adds latency and complexity — not ideal for low-latency needs

D.

Placement Group Used for same-AZ bandwidth optimization — doesn't separate instances across data centers

REINFORCING WITH DEPLOYMENT STRATEGY

Let's map EC2 placement into a high-availability model: -

Strategy

Use Case Latency -

Same AZ + Placement Group

HPC, tightly coupled workloads 🔥 Ultra-low -

Cross-AZ

HA, fault tolerance ⚡ Low -

Cross-Region

Disaster recovery, geo-redundancy 🌐 Higher -

On-prem + VPN

Hybrid workloads 🐢 Moderate to high

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a payment gateway: Place web and app tiers in separate AZs Use Elastic Load Balancing across AZs Enable Auto Scaling for resilience Monitor latency with CloudWatch metrics That's how AWS empowers teams to build resilient, low-latency architectures using Availability Zones for fault isolation and performance.

Question 279

**Compared to conventional and virtualized data center pricing, what does AWS offer in terms of cost structure?

**

OPTIONS

A. Greater variable costs and greater upfront costs

B. Fixed usage costs and lower upfront costs

C. Lower variable costs and greater upfront costs

D. Lower variable costs and lower upfront costs

[!TIP]

Correct Answer:

D. Lower variable costs and lower upfront costs

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud economics and financial agility, especially how AWS enables organizations to shift from capital-intensive infrastructure to flexible, usage-based pricing.

WHY D IS CORRECT

Lower upfront costs: No need for hardware purchases, data center leases, or long-term contracts Services like EC2, S3, and Lambda are provisioned on demand Ideal for startups, seasonal businesses, and agile teams Lower variable costs: Pay only for actual usage — compute hours, storage GBs, API calls Optimize spend with: Auto Scaling Spot Instances Savings Plans Serverless architectures Key benefits: No minimum spend commitments No multi-year licensing Elastic pricing that scales with demand Predictable billing via Budgets and Cost Explorer AWS replaces large upfront expenses with low variable payments tied directly to usage — freeing businesses from rigid infrastructure costs.

SOURCES

AWS Pricing Overview Cloud Financial Management  Why the Others Are Incorrect Option Why It's Incorrect

A.

Greater variable + upfront costs AWS reduces both — not increases

B.

Fixed usage costs AWS pricing is usage-based, not fixed

C.

Greater upfront costs AWS eliminates upfront infrastructure investments

REINFORCING WITH CLOUD COST STRATEGY

Let's map AWS pricing into a financial agility model: -

Model

Description -

On-Demand

Pay per hour/second — no commitment -

Spot Instances

Up to 90% discount — uses spare capacity -

Savings Plans

Commit to usage for 1–3 years — flexible across services -

Free Tier

Entry-level usage at no cost -

Budgets & Alerts

Monitor and control spend proactively

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics platform: Start with On-Demand EC2 and S3 Use Spot Instances for batch jobs Monitor spend with AWS Budgets Scale down during quiet periods to reduce variable costs That's how AWS empowers teams to build scalable, cost-efficient infrastructure with minimal upfront investment and dynamic pricing.

Question 280

**Which situations should the AWS Abuse team be notified about?

**

OPTIONS

- A. An Availability Zone has a service disruption
- B. An intrusion attempt is made from an AWS IP address
- C. A user has trouble accessing an Amazon S3 bucket from an AWS IP address
- D. A user needs to change payment methods due to a compromise

[!TIP]

Correct Answer:

- B. An intrusion attempt is made from an AWS IP address

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud security and incident response, especially how AWS enables external parties to report malicious or abusive behavior originating from AWS infrastructure.

WHY B IS CORRECT

The AWS Trust & Safety (Abuse) team handles reports of: Intrusion attempts from AWS-owned IPs Spam originating from AWS resources Port scanning and denial-of-service (DoS) attacks Credential stuffing or brute-force login attempts These incidents typically involve: Unauthorized

access attempts Malicious traffic patterns Security violations affecting third-party systems
Reporting is done via: AWS Abuse Report Form Email: abuse@amazonaws.com If your logs show intrusion attempts from AWS IPs, notify the AWS Abuse team to investigate and mitigate.

SOURCES

AWS Abuse Reporting AWS Trust & Safety ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

AZ disruption Handled by AWS Support and Service Health Dashboard — not Abuse team

C.

S3 access issues Likely a configuration or IAM issue — not abuse-related

D.

Payment method changes Managed via AWS Billing Support — not Trust & Safety

REINFORCING WITH ABUSE RESPONSE STRATEGY

Let's map AWS abuse handling into a security escalation model: -

Incident Type

Report To -

Intrusion from AWS IPs

AWS Abuse team -

Phishing or spam from AWS

AWS Abuse team -

Service outage or disruption

AWS Support -

Billing or account compromise

AWS Billing Support -

IAM misconfigurations

REAL-WORLD TIE-IN

If your Nairobi-based team detects login attempts from AWS IPs: Capture IP addresses, timestamps, and logs Submit a report to abuse@amazonaws.com Monitor for continued activity Consider blocking offending IPs and enabling GuardDuty That's how AWS empowers teams to respond to malicious activity originating from its infrastructure through the Abuse team's dedicated channels.

Question 281

**Which AWS feature should a client exploit to ensure an application's high availability?

**

OPTIONS

A. AWS Direct Connect

B. Availability Zones

C. Data centers

D. Amazon Virtual Private Cloud (Amazon VPC)

[!TIP]

Correct Answer:

B. Availability Zones

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience and fault tolerance, especially how AWS enables organizations to deploy applications across isolated infrastructure zones to ensure uptime and continuity.

WHY B IS CORRECT

Availability Zones (AZs): Each AZ is a physically separate data center within a Region AZs are connected via low-latency, high-throughput links Deploying across AZs ensures: Redundancy Failover capability Minimal impact during outages or maintenance Key benefits: High availability for web apps, databases, and APIs Auto Scaling across AZs for load balancing Zero-downtime deployments with blue/green strategies Service-level agreement (SLA) compliance for uptime guarantees Availability Zones are the backbone of AWS's high availability strategy — enabling fault isolation and seamless failover.

SOURCES

AWS High Availability Architecture Regions and AZs  Why the Others Are Incorrect Option Why It's Incorrect

A.

AWS Direct Connect Provides private connectivity — not designed for availability or failover

C.

Data centers Generic term — AWS abstracts this into AZs for managed availability

D.

Amazon VPC Provides network isolation — not inherently tied to high availability

REINFORCING WITH HA STRATEGY

Let's map Availability Zones into a resilient architecture: -

Component

Role -

Multi-AZ EC2 Deployment

Ensures app uptime during AZ failure -

RDS Multi-AZ

Synchronous replication for database failover -

Elastic Load Balancer

Distributes traffic across AZs -

Auto Scaling Groups

Dynamically adjust capacity across AZs -

Route 53 Health Checks

Redirect traffic during failures

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying SwiftCode: Launch EC2 instances in multiple AZs Use Elastic Load Balancer to route traffic Enable Auto Scaling for demand spikes Deploy RDS with Multi-AZ for database resilience Monitor uptime with CloudWatch and Route 53 That's how AWS empowers teams to build highly available, fault-tolerant applications using Availability Zones as the foundation.

Question 282

**In which situations should a company create an IAM user instead of an IAM role? (Choose two)

**

OPTIONS

A. When an application that runs on Amazon EC2 instances requires access to other AWS services

B. When the company creates AWS access credentials for individuals

C. When the company creates an application that runs on a mobile phone that makes requests to AWS

D. When the company needs to add users to IAM groups

E. When users are authenticated in the corporate network and want to be able to use AWS without having to sign in a second time

[!TIP]

Correct Answer:

B and D

EXPLANATION

CONCEPTUAL FRAMING

This question targets identity management and access control, especially how AWS enables organizations to assign long-term credentials to human users and short-term credentials to workloads via roles. 🔍 Why B and D Are Correct ✓ B. Creating AWS access credentials for individuals IAM users represent human identities Assigned long-term credentials

(username/password, access keys) Ideal for: Developers Admins Finance or audit personnel

✓ D. Adding users to IAM groups IAM groups are collections of IAM users Used to assign permissions at scale Common for: Role-based access control (RBAC) Departmental policies Onboarding/offboarding automation IAM users are best for long-term human access, while IAM roles are designed for temporary, assumed access by applications or federated identities.

SOURCES

IAM Users vs Roles IAM Groups ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

EC2 apps accessing AWS Use IAM roles attached to EC2 — not IAM users

C.

Mobile apps accessing AWS Use Cognito or IAM roles — not IAM users

E.

Corporate SSO access Use IAM roles with SAML federation — not IAM users

REINFORCING WITH IDENTITY STRATEGY

Let's map IAM users vs roles into a secure access model: -

Identity Type

Use Case Credential Type -

IAM User

Human access Long-term credentials -

IAM Role

EC2, Lambda, federated users Temporary credentials -

IAM Group

Permission management Group-based policies -

SAML Role

Corporate SSO Assumed role via federation -

Cognito Identity

Mobile/web apps Token-based access via roles

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding developers: Create IAM users with MFA and access keys Assign them to IAM groups for permission boundaries Use IAM roles for EC2 instances

and Lambda functions Enable SAML federation for corporate SSO That's how AWS empowers teams to securely manage human and machine identities using IAM users, roles, and groups with clear separation of responsibilities.

Question 283

**How might an AWS client implement common access restrictions to a big group of users easily?

**

OPTIONS

A. Apply an IAM policy to an IAM group

B. Apply an IAM policy to an IAM role

C. Apply the same IAM policy to all IAM users with access to the same workload

D. Apply an IAM policy to an Amazon Cognito user pool

[!TIP]

Correct Answer:

A. Apply an IAM policy to an IAM group

EXPLANATION

CONCEPTUAL FRAMING

This question targets scalable access management, especially how AWS enables organizations to assign permissions efficiently across large user bases using group-based policies.

WHY A IS CORRECT

IAM Groups: Logical containers for IAM users Allow centralized policy management Ideal for: Role-based access control (RBAC) Departmental segmentation (e.g., devs, admins, auditors) Onboarding/offboarding automation Key benefits: Single point of policy control Inheritance of permissions by all group members Simplified audits and updates Dynamic user management — move users between groups as roles change IAM groups let you apply policies once and affect many users — perfect for scalable, maintainable access control.

SOURCES

IAM Groups and Policies Best Practices for IAM ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

IAM Role Designed for temporary access — not for managing large user groups

C.

Apply same policy to each user Manual and error-prone — lacks scalability

D.

Cognito User Pool Used for app-level identity federation — not IAM-based access control

REINFORCING WITH IAM STRATEGY

Let's map IAM group-based policy into a scalable access model: -

Component

Role -

IAM Group

Container for users with shared permissions -

IAM Policy

Defines allowed/denied actions -

IAM User

Assigned to groups for inherited access -

CloudTrail

Audits policy usage and changes -

Access Analyzer

Detects unintended access paths

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding 50 developers: Create an IAM group called DevTeam Attach a policy granting access to EC2, S3, and CloudWatch Add users to the group during onboarding Modify the group policy as roles evolve — no need to touch individual users That's how AWS empowers teams to manage access at scale using IAM groups and centralized policy control.

Question 284

**Which AWS service is used to provide encryption for Amazon EBS?

**

OPTIONS

A. AWS Certificate Manager

B. AWS Systems Manager

C. AWS KMS

D. AWS Config

[!TIP]

Correct Answer:

C. AWS KMS (Key Management Service)

EXPLANATION

CONCEPTUAL FRAMING

This question targets data-at-rest encryption and key management, especially how AWS enables organizations to secure block storage volumes using managed cryptographic keys integrated with audit and compliance tooling.

WHY C IS CORRECT

AWS Key Management Service (KMS): Manages encryption keys for services like EBS, S3, RDS, Lambda, and more Supports customer-managed keys (CMKs) and AWS-managed keys Uses FIPS 140-2 validated HSMs for secure key storage Integrated with CloudTrail for full auditability of key usage For Amazon EBS: Encryption is transparent to applications Covers data at rest, disk I/O, and snapshots Can be enabled at volume creation or enforced via launch templates and policies AWS KMS provides secure, scalable key management for encrypting EBS volumes, snapshots, and associated data flows.

SOURCES

EBS Encryption Overview AWS KMS Documentation  Why the Others Are Incorrect Option Why It's Incorrect

A.

AWS Certificate Manager Manages SSL/TLS certificates — not block storage encryption

B.

AWS Systems Manager Provides automation and patching — not key management

D.

REINFORCING WITH ENCRYPTION STRATEGY

Let's map EBS encryption into a secure architecture: -

Component

Role -

EBS Volume

Stores encrypted block data -

KMS Key (CMK)

Encrypts volume and snapshot data -

CloudTrail

Logs key usage for compliance -

IAM Policies

Control access to KMS keys -

Launch Templates

Enforce encryption at instance creation

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying EC2 instances with sensitive data: Enable EBS encryption by default Use customer-managed KMS keys for granular control Audit key usage with CloudTrail Apply IAM policies to restrict key access by role or service That's how AWS empowers teams to secure block storage with robust, compliant encryption using KMS as the backbone.

Question 285

******A business needs software solutions that are either hosted on the AWS platform or are linked with it. Independent software providers, as well as management and security vendors, are required to provide solutions. Which organization or team is capable of providing these solutions?

OPTIONS

- A. AWS Technical Account Managers (TAMs)
- B. AWS Partner Network (APN) Consulting Partners
- C. AWS Concierge Support
- D. AWS Partner Network (APN) Technology Partners

[!TIP]

Correct Answer:

D. AWS Partner Network (APN) Technology Partners

EXPLANATION

CONCEPTUAL FRAMING

This question targets third-party solution sourcing and platform integration, especially how AWS enables businesses to leverage certified vendors for software, security, and infrastructure tools that run on or integrate with AWS.

WHY D IS CORRECT

APN Technology Partners: Provide software solutions that are: Built on AWS Integrated with AWS services Certified and validated by AWS Include vendors in: Security (e.g., Palo Alto, Trend Micro) Monitoring (e.g., Datadog, New Relic) DevOps (e.g., HashiCorp, GitLab) Data platforms (e.g., Snowflake, MongoDB) Key benefits: Pre-integrated solutions for faster

deployment Marketplace availability for easy procurement Compliance-ready offerings Scalable licensing and support models APN Technology Partners deliver validated software and services that run on or integrate with AWS infrastructure.

SOURCES

AWS Partner Network Overview APN Technology Partners ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

TAMs Provide technical guidance — not software solutions

B.

APN Consulting Partners Offer professional services — not software products

C.

Concierge Support Handles billing and account issues — not solution delivery

REINFORCING WITH PARTNER STRATEGY

Let's map APN partner types into a solution sourcing framework: -

Partner Type

Role -

Technology Partner

Provides software and SaaS solutions -

Consulting Partner

Offers migration, architecture, and implementation services -

Marketplace Seller

Distributes solutions via AWS Marketplace -

Competency Partner

Certified in specific domains (e.g., Security, Machine Learning)

REAL-WORLD TIE-IN

If your Nairobi-based team needs a cloud-native SIEM: Browse APN Technology Partners in the AWS Marketplace Filter by security competency and regional availability Deploy via CloudFormation templates or SaaS integrations Monitor usage and billing via AWS Cost Explorer That's how AWS empowers teams to source trusted, cloud-integrated software solutions through its global APN Technology Partner ecosystem.

REFERENCES

<https://aws.amazon.com/partners/>

Question 286

**Which AWS services are available for application deployment? (Select two)

**

OPTIONS

A. AWS Elastic Beanstalk

B. AWS Config

C. AWS OpsWorks

D. AWS Application Discovery Service

E. Amazon Kinesis

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets deployment automation and configuration management, especially how AWS enables teams to launch, manage, and monitor applications using platform-native orchestration tools. 🔍 Why A and C Are Correct ✅ A. AWS Elastic Beanstalk Fully managed Platform-as-a-Service (PaaS) Supports: Web apps in Node.js, Python, Java, .NET, PHP, Go, Ruby Auto-scaling, load balancing, health monitoring Abstracts infrastructure so developers can focus on code Ideal for: Rapid prototyping Small-to-medium production apps Teams without deep ops expertise ✅ C. AWS OpsWorks Configuration management using: Chef and Puppet automation Stacks and layers for modular deployment Supports: EC2 provisioning App deployment Monitoring and lifecycle hooks Ideal for: Complex, multi-tier apps Teams with existing Chef/Puppet workflows Elastic Beanstalk and OpsWorks both streamline application deployment, but cater to different levels of control and complexity.

SOURCES

Elastic Beanstalk Overview OpsWorks Documentation ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

AWS Config Tracks resource configurations — not used for deployment

D.

Application Discovery Service Assists in migration planning — doesn't deploy applications

E.

Amazon Kinesis Handles real-time data streaming — not application deployment

REINFORCING WITH DEPLOYMENT STRATEGY

Let's map AWS deployment tools into a DevOps framework: -

Tool

Role Ideal Use Case -

Elastic Beanstalk

PaaS deployment Rapid web app launch -

OpsWorks

Config management Chef/Puppet-based automation -

CodeDeploy

CI/CD deployment Granular control over rollout -

CloudFormation

Infrastructure as code Declarative stack provisioning -

App Runner

Container deployment Serverless container apps

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a multi-tier app: Use Elastic Beanstalk for frontend deployment Use OpsWorks to manage backend services with Chef Monitor health via CloudWatch Automate rollouts with CodePipeline That's how AWS empowers teams to deploy applications with speed, control, and scalability using Elastic Beanstalk and OpsWorks.

Question 286

**Which AWS services make use of global edge locations? (Choose two)

**

OPTIONS

A. AWS Fargate

B. Amazon CloudFront

C. AWS Global Accelerator

D. AWS Wavelength

E. Amazon VPC

[!TIP]

Correct Answer:

B and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets global performance and latency optimization, especially how AWS uses edge locations to accelerate content delivery, routing, and security enforcement close to end users. 🔍 Why B and C Are Correct ✅ B. Amazon CloudFront A Content Delivery Network (CDN) Caches content at edge locations worldwide Reduces latency by serving data from the nearest location Supports: Static and dynamic content Video streaming API acceleration Integration with WAF and Shield ✅ C. AWS Global Accelerator Routes traffic through the AWS global network Starts at the nearest edge location to the user Improves: Latency Availability Failover speed Ideal for: Real-time apps Gaming Financial services CloudFront and Global Accelerator both leverage AWS edge locations to deliver fast, reliable, and secure experiences to global users.

SOURCES

Amazon CloudFront Overview AWS Global Accelerator ❌ Why the Others Are Incorrect Option
Why It's Incorrect

A.

AWS Fargate Serverless container compute — doesn't use edge locations

D.

AWS Wavelength Extends AWS to telecom networks — uses carrier edge, not AWS edge locations

E.

Amazon VPC Manages private networking — operates within regions, not edge locations

REINFORCING WITH EDGE STRATEGY

Let's map AWS edge services into a global performance framework: -

Service

Role Edge Usage -

CloudFront

CDN Caches and serves content at edge -

Global Accelerator

Traffic routing Starts at nearest edge location -

Route 53

DNS resolution Uses edge-based name servers -

AWS Shield/WAF

Security filtering Protects at edge locations

REAL-WORLD TIE-IN

If your Nairobi-based team is building a global video platform: Use CloudFront to cache and stream content Use Global Accelerator to route requests through AWS backbone Protect endpoints with WAF and Shield at edge locations Monitor latency with CloudWatch and X-Ray

That's how AWS empowers teams to deliver low-latency, high-availability experiences using edge-powered services like CloudFront and Global Accelerator.

Question 288

**Which responsibilities do customers bear while using Amazon EC2? (Select two)

**

OPTIONS

- A. Underlying hardware maintenance
- B. File-system-level encryption
- C. Guest operating system firewall configuration
- D. Hypervisor-level software patching
- E. Physical security at data center facilities

[!TIP]

Correct Answer:

B and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets the AWS shared responsibility model, especially how AWS delineates infrastructure-level responsibilities from customer-controlled configurations and data protection.

🔍 Why B and C Are Correct ☒ B. File-system-level encryption Customers are responsible for: Encrypting data at rest using tools like LUKS, BitLocker, or EBS encryption with KMS Managing encryption keys (if using customer-managed KMS keys) Ensuring compliance with data protection policies ☒ C. Guest operating system firewall configuration Customers must: Configure OS-level firewalls (e.g., iptables, ufw, Windows Firewall) Harden the instance against unauthorized access Complement AWS security groups and NACLs with internal firewall rules EC2 customers control the guest OS, including encryption, firewall settings, patching, and application security — AWS handles the underlying infrastructure.

SOURCES

Shared Responsibility Model EC2 Security Best Practices ☒ Why the Others Are Incorrect Option Why It's Incorrect

A.

Hardware maintenance AWS handles physical infrastructure and hardware lifecycle

D.

Hypervisor patching AWS maintains the virtualization layer and applies security updates

E.

Data center security AWS is responsible for physical access control and facility protection

REINFORCING WITH RESPONSIBILITY MAPPING

Let's map EC2 responsibilities into the shared model: -

Layer

AWS Responsibility Customer Responsibility -

Physical Infrastructure

Yes No -

Hypervisor & Host OS

Yes No -

Guest OS

No Yes -

Firewall (OS-level)

No Yes -

Encryption (File system)

No Yes -

Application & Data

No Yes

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying EC2 instances for a fintech app: Use EBS encryption with customer-managed KMS keys Harden the OS with firewall rules and patching Monitor access with CloudWatch and GuardDuty Delegate infrastructure security to AWS while owning data protection That's how AWS empowers teams to secure their workloads by clearly defining which layers they control within EC2 deployments.

Question 289

**A company is operating several factories where it builds products. The company needs the ability to process data, store data, and run applications with local system interdependencies that require low latency. Which AWS service should the company use to meet these requirements?

**

OPTIONS

A. AWS IoT Greengrass

B. AWS Lambda

C. AWS Outposts

D. AWS Snowball Edge

[!TIP]

Correct Answer:

C. AWS Outposts

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud deployment and edge computing, especially how AWS enables organizations to extend cloud-native services into on-premises environments for low-latency, high-throughput workloads.

WHY C IS CORRECT

AWS Outposts: Delivers AWS infrastructure and services to on-premises locations Supports: EC2, EBS, RDS, ECS, EKS, S3 (local zones) Same APIs, tools, and control plane as AWS Region Ideal for: Factory automation systems Latency-sensitive workloads Local data processing with cloud integration Key benefits: Consistent hybrid experience Local compute and storage Seamless integration with AWS Region services Compliance and data residency support AWS Outposts brings cloud-native capabilities directly into your factory floor, enabling low-latency operations with full AWS integration.

SOURCES

AWS Outposts Overview Hybrid Architecture Patterns  Why the Others Are Incorrect Option Why It's Incorrect

A.

IoT Greengrass Great for device-level logic — not full app hosting or storage

B.

Lambda Serverless compute in the cloud — not suitable for local system interdependencies

D.

Snowball Edge Designed for data migration and edge compute — not persistent app hosting

REINFORCING WITH HYBRID STRATEGY

Let's map AWS Outposts into a factory deployment model: -

Component

Role -

Outposts Rack

Local AWS infrastructure for compute/storage -

EC2 on Outposts

Run factory control apps locally -

EBS on Outposts

Store sensor and production data -

S3 Local Zones

Archive and sync data to cloud -

CloudWatch & Systems Manager

Monitor and manage workloads remotely

REAL-WORLD TIE-IN

If your Nairobi-based team is digitizing factory operations: Deploy Outposts racks in each facility
Run SCADA or MES systems locally on EC2 Store production logs in EBS volumes Sync analytics data to the cloud via S3 and Kinesis That's how AWS empowers teams to run latency-sensitive, locally integrated workloads with cloud-native tooling using Outposts.

Question 290

******A business needs to handle a huge volume of data from social network accounts using high-throughput graphical queries. Which AWS service will assist the business in developing a cloud architecture that satisfies these criteria?

OPTIONS

A. Amazon RDS

B. Amazon DynamoDB

C. Amazon Neptune

D. Amazon Redshift

[!TIP]

Correct Answer:

C. Amazon Neptune

EXPLANATION

CONCEPTUAL FRAMING

This question targets graph-based data modeling and query performance, especially how AWS enables organizations to store and traverse complex relationships at scale using purpose-built graph databases.

WHY C IS CORRECT

Amazon Neptune: Fully managed graph database service Supports: Property graph model via Apache TinkerPop RDF model via SPARQL Optimized for: Social graphs Recommendation engines Fraud detection Network topology analysis Key benefits: High-throughput traversal queries Low-latency relationship lookups ACID compliance Encryption, backup, and replication built-in Neptune is designed for applications that need to explore relationships between billions of entities with millisecond latency.

SOURCES

Amazon Neptune Overview Use Cases for Graph Databases  Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon RDS Relational — not optimized for graph traversal

B.

DynamoDB No native graph query support — better for key-value or document models

D.

Redshift Columnar data warehouse — suited for analytics, not graph queries

REINFORCING WITH GRAPH STRATEGY

Let's map Neptune into a social network architecture: -

Component

Role -

Neptune DB

Stores user relationships, likes, follows -

SPARQL/TinkerPop Queries

Traverse connections and interactions -

Lambda + API Gateway

Serve graph queries to frontend apps -

CloudWatch

Monitor query performance and usage patterns

REAL-WORLD TIE-IN

If your Nairobi-based team is building a social analytics platform: Use Neptune to model user interactions Query mutual friends, influencers, and content paths Integrate with Lambda for real-time recommendations Secure with IAM and KMS encryption That's how AWS empowers teams to build graph-powered applications with scalable, low-latency traversal using Amazon Neptune.

Question 291

**Which AWS service enables the use of the AWS Cloud to host a NoSQL database?

**

OPTIONS

A. Amazon Aurora

B. Amazon DynamoDB

C. Amazon RDS

D. Amazon Redshift

[!TIP]

Correct Answer:

B. Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question targets data modeling and scalability, especially how AWS enables organizations to host high-performance NoSQL databases with automatic scaling and millisecond latency.

WHY B IS CORRECT

Amazon DynamoDB: Fully managed NoSQL database service Supports: Key-value and document data models Single-digit millisecond latency Automatic scaling and global replication Ideal for: Real-time apps Gaming leaderboards IoT telemetry Session stores Key benefits: Serverless architecture — no provisioning required Built-in security with IAM, KMS, and encryption Event-driven integration with Lambda and Streams Global tables for multi-region replication DynamoDB is AWS's flagship NoSQL service, designed for scale, speed, and simplicity.

SOURCES

Amazon DynamoDB Overview Use Cases for DynamoDB  Why the Others Are Incorrect Option Why It's Incorrect

A.

Aurora Relational — supports MySQL/PostgreSQL, not NoSQL

C.

RDS Managed relational database — not suitable for NoSQL workloads

D.

Redshift Columnar data warehouse — optimized for analytics, not transactional NoSQL

REINFORCING WITH NOSQL STRATEGY

Let's map DynamoDB into a scalable cloud-native architecture: -

Component

Role -

DynamoDB Table

Stores key-value or document data -

Global Tables

Replicate data across regions -

Streams

Capture change events for Lambda triggers -

DAX

In-memory caching for read-heavy workloads -

IAM Policies

Secure access control

REAL-WORLD TIE-IN

If your Nairobi-based team is building a social media analytics engine: Use DynamoDB to store user interactions and metadata Trigger Lambda functions via Streams for real-time processing Scale globally with Global Tables Secure with KMS encryption and IAM roles That's how AWS empowers teams to build fast, scalable NoSQL applications using DynamoDB's fully managed infrastructure.

REFERENCES

<https://aws.amazon.com/dynamodb/>

Question 292

**Which of the following acts as an instance-level firewall to control inbound and outbound access?

**

OPTIONS

A. Network access control list

B. Security groups

C. AWS Trusted Advisor

D. Virtual private gateways

[!TIP]

Correct Answer:

B. Security groups

EXPLANATION

CONCEPTUAL FRAMING

This question targets network security and access control, especially how AWS enables organizations to define fine-grained traffic rules at the EC2 instance level using virtual firewalls.

WHY B IS CORRECT

Security Groups: Act as stateful firewalls for EC2 instances Control inbound and outbound traffic based on: Protocol Port range Source/destination IP or security group Applied per instance, not per subnet Automatically allow return traffic for allowed inbound requests Key benefits: Instance-level granularity Dynamic rule updates without rebooting Multiple groups per instance (up to 5) Audit-friendly via CloudTrail and IAM policies Security groups are the primary mechanism for controlling traffic to and from EC2 instances at the instance level.

SOURCES

Security Groups Documentation Best Practices for Security Groups ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

NACLs Operate at the subnet level — stateless and less granular than security groups

C.

Trusted Advisor Provides recommendations — doesn't enforce traffic rules

D.

Virtual Private Gateways Enable VPN connectivity — not used for firewalling EC2 traffic

REINFORCING WITH SECURITY STRATEGY

Let's map EC2 traffic control layers: -

Layer

Control Mechanism Scope -

Instance

Security Group Granular, stateful -

Subnet

NACL Broad, stateless -

VPC Edge

Virtual Gateway VPN/IP Sec routing -

Account-wide

IAM Policies Identity-based access -

Global

AWS WAF/Shield Edge-level protection

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying EC2 instances for a payment API: Use security groups to allow HTTPS traffic only from trusted IPs Block all other inbound ports Enable outbound access to S3 and RDS Monitor changes via CloudTrail and Config That's how AWS empowers teams to secure EC2 instances with precise, instance-level traffic control using security groups.

Question 292

**A business is consolidating many apps into a single AWS account. The organization wishes to keep track of the AWS Cloud fees paid by individual applications. What can the business do to ensure compliance with this requirement?

**

OPTIONS

- A. Set up invoiced billing
- B. Use AWS Artifact
- C. Set the budgets in Cost Explorer
- D. Create cost allocation tags

[!TIP]

Correct Answer:

D. Create cost allocation tags

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost attribution and resource tagging, especially how AWS enables organizations to track and allocate cloud spend across applications, teams, or environments using metadata labels.

WHY D IS CORRECT

Cost Allocation Tags: Tags are key-value pairs attached to AWS resources Used to categorize and filter costs in: Cost Explorer Billing reports Budgets and forecasts Types: User-defined tags — created and applied by the customer AWS-generated tags — automatically applied by AWS services Key benefits: Granular visibility into resource-level spend Application-level cost breakdowns Automated reporting and compliance tracking Supports chargeback and showback models Cost allocation tags are the foundation for tracking cloud spend by application, team, or environment in a consolidated AWS account.

SOURCES

AWS Cost Allocation Tags Tagging Best Practices ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Invoiced billing Changes payment method — doesn't help with cost attribution

B.

AWS Artifact Provides compliance reports — not cost tracking

C.

Budgets in Cost Explorer Useful for alerts — but doesn't categorize costs by application

REINFORCING WITH TAGGING STRATEGY

Let's map cost allocation into a financial governance model: -

Tag Key

Example Value Purpose -

Application

Swift Code Attribute spend to app -

Environment

Production Separate prod vs dev costs -

Owner

Team A Track team-level usage -

CostCenter

Finance Align with internal accounting

REAL-WORLD TIE-IN

If your Nairobi-based team is consolidating analytics and payment apps: Tag EC2, S3, Lambda, and RDS resources with Application=Analytics or Application=Payments Activate cost allocation tags in the Billing console Use Cost Explorer to filter by tag and generate monthly reports Set Budgets per tag to monitor spend thresholds That's how AWS empowers teams to track, report, and optimize cloud spend using cost allocation tags for granular visibility and accountability.

Question 294

**A batch job on an Amazon EC2 instance takes 5 hours to complete. Monthly, the quantity of data to be processed doubles, and the time required to process it is proportionate. What is the optimal cloud architecture for meeting this escalating demand?

**

OPTIONS

A. Run the application on a bigger EC2 instance size

B. Switch to an EC2 instance family that better matches batch requirements

C. Distribute the application across multiple EC2 instances and run the workload in parallel

D. Run the application on a bare metal EC2 instance

[!TIP]

Correct Answer:

C. Distribute the application across multiple EC2 instances and run the workload in parallel

EXPLANATION

CONCEPTUAL FRAMING

This question targets horizontal scalability and parallel processing, especially how AWS enables organizations to scale compute capacity elastically to meet growing data volumes and time-sensitive workloads.

WHY C IS CORRECT

Parallelization: Splits workload across multiple EC2 instances Reduces total processing time by concurrent execution Ideal for: Embarrassingly parallel tasks Batch jobs with independent data chunks MapReduce-style workflows Key benefits: Scales with data growth Minimizes latency impact Cost-efficient via Spot Instances or Auto Scaling Flexible orchestration with Step Functions, SQS, or Batch Distributing workloads across EC2 instances allows you to scale horizontally and maintain performance as data volume grows exponentially.

SOURCES

AWS Batch Overview Parallel Processing on AWS  Why the Others Are Incorrect Option Why It's Incorrect

A.

Bigger EC2 size Vertical scaling hits limits — doesn't match exponential data growth

B.

Different EC2 family May optimize compute type — but doesn't solve scaling bottleneck

D.

Bare metal instance Offers hardware access — not relevant for parallel workload distribution

REINFORCING WITH SCALABLE ARCHITECTURE

Let's map parallel batch processing into a scalable EC2 framework: -

Component

Role -

S3

Stores input/output data -

EC2 Fleet

Executes batch jobs in parallel -

Auto Scaling Group

Dynamically adjusts instance count -

AWS Batch or Step Functions

Orchestrates job execution -

CloudWatch

Monitors performance and cost metrics

REAL-WORLD TIE-IN

If your Nairobi-based team is processing monthly social media analytics: Split data into monthly shards Launch EC2 instances via AWS Batch Use Spot Instances for cost savings Monitor job completion with CloudWatch Events That's how AWS empowers teams to scale batch workloads horizontally using parallel EC2 execution and orchestration tools.

Question 295

**A company has a workload that will run continuously for 1 year. The workload cannot tolerate service interruptions. Which Amazon EC2 purchasing option will be MOST cost-effective?

**

OPTIONS

A. All Upfront Reserved Instances

B. Partial Upfront Reserved Instances

C. Dedicated Instances

D. On-Demand Instances

[!TIP]

Correct Answer:

A. All Upfront Reserved Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets long-term workload planning and cost efficiency, especially how AWS enables organizations to commit to predictable compute usage in exchange for significant discounts.

WHY A IS CORRECT

All Upfront Reserved Instances (RIs): Provide the deepest discount (up to 72%) compared to On-Demand pricing Ideal for steady-state workloads with known duration and capacity Paid entirely at the start of the term (1 or 3 years) Guarantee capacity reservation in a specific

Availability Zone Key benefits: Cost predictability No service interruptions Simplified budgeting
Optimized for compliance and audit scenarios All Upfront Reserved Instances are the most
cost-effective option for uninterrupted, long-running workloads with predictable compute needs.

SOURCES

Reserved Instances Pricing EC2 Purchasing Options ❌ Why the Others Are Incorrect Option
Why It's Incorrect

B.

Partial Upfront RIs Still discounted — but not as cost-effective as All Upfront

C.

Dedicated Instances Provide physical isolation — not optimized for cost savings

D.

On-Demand Instances Flexible — but most expensive for continuous workloads

REINFORCING WITH PURCHASING STRATEGY

Let's map EC2 purchasing options into a cost optimization framework: -

Option

Use Case Cost Efficiency -

All Upfront RI

Long-term, steady workloads 🔥 Highest -

Partial Upfront RI

Long-term with budget flexibility ⚡ High -

No Upfront RI

Long-term, cash-flow sensitive ⚖️ Moderate -

On-Demand

Short-term, unpredictable workloads 🎲 Low -

Spot Instances

Fault-tolerant, interruptible workloads  Variable -

Dedicated Hosts

Licensing or compliance needs  Specialized

REAL-WORLD TIE-IN

If your Nairobi-based team is running a 24/7 analytics engine: Choose All Upfront Reserved Instances for EC2 Lock in capacity and pricing for 1 year Monitor usage with Cost Explorer and Budgets Tag resources for cost allocation and audit tracking That's how AWS empowers teams to optimize compute costs for predictable workloads using Reserved Instances with upfront commitment.

Question 296

**How can AWS Trusted Advisor assist AWS Cloud users? (Select two)

**

OPTIONS

A. It identifies software vulnerabilities in applications running on AWS 

B. It provides a list of cost optimization recommendations based on current AWS usage 

C. It detects potential security vulnerabilities caused by permissions settings on account resources

D. It automatically corrects potential security issues caused by permissions settings on account resources

E. It provides proactive alerting whenever an Amazon EC2 instance has been compromised

[!TIP]

Correct Answer:

B and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud governance and operational excellence, especially how AWS enables organizations to optimize cost, security, performance, and reliability through automated best-practice checks. 🔍 Why B and C Are Correct ✅ B. Cost Optimization Recommendations Trusted Advisor analyzes: Idle resources (e.g., underutilized EC2, EBS, ELBs) Unattached IPs or volumes Opportunities to switch to Reserved Instances or Savings Plans Helps reduce: Wasteful spend Overprovisioning Unnecessary charges ✅ C. Security Vulnerability Detection Checks for: Overly permissive IAM policies Publicly accessible S3 buckets Unrestricted security groups Root account usage without MFA Flags misconfigurations that could lead to data exposure or privilege escalation Trusted Advisor is your cloud optimization sentinel — surfacing cost and security insights before they become problems.

SOURCES

AWS Trusted Advisor Documentation Trusted Advisor Checks ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Software vulnerabilities Requires specialized tools like Amazon Inspector — not Trusted Advisor

D.

Automatic correction Trusted Advisor is advisory only — doesn't auto-remediate

E.

EC2 compromise alerts Use GuardDuty or CloudTrail — not part of Trusted Advisor's scope

REINFORCING WITH OPTIMIZATION STRATEGY

Let's map Trusted Advisor into a proactive cloud hygiene model: -

Category

Example Checks -

Cost Optimization

Idle EC2, unused EBS, RI recommendations -

Security

Public S3 buckets, open ports, IAM misconfigurations -

Performance

EC2 limits, latency risks -

Fault Tolerance

Backup status, AZ distribution -

Service Limits

Usage thresholds for EC2, RDS, Lambda

REAL-WORLD TIE-IN

If your Nairobi-based team is managing multi-app workloads: Enable Trusted Advisor checks across all accounts Review weekly reports for cost and security flags Integrate findings into Ops workflows via AWS Support API Use Budgets and Config to enforce remediation policies That's how AWS empowers teams to maintain secure, cost-efficient, and resilient cloud environments using Trusted Advisor's automated insights.

REFERENCES

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Question 297

****Which AWS function will assist users in determining the CPU capacity of an application running on an Amazon EC2 instance?**

OPTIONS

A. Amazon CloudWatch

B. AWS Config

C. AWS CloudTrail

D. Amazon Inspector

[!TIP]

Correct Answer:

A. Amazon CloudWatch

EXPLANATION

CONCEPTUAL FRAMING

This question targets performance monitoring and resource utilization, especially how AWS enables teams to track compute metrics in real time to optimize application behavior and instance sizing.

WHY A IS CORRECT

Amazon CloudWatch: Provides real-time metrics for EC2 instances, including: CPU utilization CPU credit balance (for burstable T-series instances) Disk I/O, network throughput, memory (via agent) Supports: Custom dashboards Alarms and thresholds Log aggregation and insights

For T2/T3 burstable instances, CloudWatch tracks: CPUCreditUsage — credits consumed
CPUCreditBalance — credits remaining CPUSurplusCreditBalance — unused surplus
CPUSurplusCreditsCharged — charged surplus usage CloudWatch is the go-to tool for
monitoring EC2 performance, especially CPU metrics critical for workload tuning and scaling
decisions.

SOURCES

CloudWatch Metrics for EC2 Burstable Performance Instances  Why the Others Are Incorrect
Option Why It's Incorrect

B.

AWS Config Tracks resource configurations — not performance metrics

C.

AWS CloudTrail Logs API activity — doesn't monitor CPU usage

D.

Amazon Inspector Scans for security vulnerabilities — not resource performance

REINFORCING WITH MONITORING STRATEGY

Let's map EC2 CPU monitoring into a performance observability model: -

Metric

Purpose -

CPUUtilization

Overall CPU load -

CPUCreditBalance

Remaining burst credits -

CPUCreditUsage

Credits consumed per minute -

CPUSurplusCreditsCharged

Overuse penalties -

Custom Metrics

App-specific performance indicators

REAL-WORLD TIE-IN

If your Nairobi-based team is running analytics on T3 instances: Use CloudWatch dashboards to track CPU credit trends Set alarms for low credit balance thresholds Adjust instance type or switch to standard EC2 if consistently maxed Export metrics to CloudWatch Logs Insights for deeper analysis That's how AWS empowers teams to monitor and optimize EC2 performance using CloudWatch's rich metric and alerting capabilities.

Question 298

**A business wants to access aggregated billing data across multiple AWS accounts. What is the purpose of creating a master payer account?

**

OPTIONS

A. AWS Budgets

B. Amazon Macie

C. Amazon QuickSight →

D. AWS Organizations

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account management and consolidated billing, especially how AWS enables organizations to centralize financial oversight and governance across multiple accounts using a master-payer structure.

WHY D IS CORRECT

AWS Organizations: Enables creation of a master account (payer) and linked member accounts
Supports: Consolidated billing across all accounts Service control policies (SCPs) for governance Automated account creation via API or console Master account can: View aggregated usage and charges Set budgets and alerts Assume IAM roles in member accounts for centralized management Key benefits: Financial transparency Centralized control Scalable account structure for teams, apps, or environments Audit-friendly with role-based access and tagging AWS Organizations is the backbone for managing multiple accounts with unified billing, governance, and access control.

SOURCES

AWS Organizations Overview Consolidated Billing  Why the Others Are Incorrect Option
Why It's Incorrect

A.

AWS Budgets Sets spending thresholds — doesn't aggregate billing across accounts

B.

Amazon Macie Detects sensitive data — not related to billing

C.

Amazon QuickSight Visualizes data — doesn't manage account-level billing

REINFORCING WITH MULTI-ACCOUNT STRATEGY

Let's map AWS Organizations into a financial and governance model: -

Component

Role -

Master Account

Aggregates billing and controls policies -

Member Accounts

Isolated environments for teams/apps -

SCPs

Enforce permissions across accounts -

Tagging & Budgets

Track and control spend per account -

IAM Role Assumption

Centralized access to member accounts

REAL-WORLD TIE-IN

If your Nairobi-based team is managing analytics, dev, and prod environments: Create separate AWS accounts for each environment Link them under a master payer account via AWS Organizations Use Cost Explorer to view aggregated spend Apply SCPs to restrict risky actions in dev/test accounts That's how AWS empowers teams to scale securely and cost-effectively using Organizations for account structure and billing aggregation.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 299

**Every few years, a business invests several months in modernizing its on-premises infrastructure. The organization wishes to shorten the time required for infrastructure purchase by shifting to the AWS Cloud. What is the primary advantage of transferring this use case to the AWS Cloud?

**

OPTIONS

- A. AWS will help move the existing hardware to the AWS data centers
- B. The company will have increased agility with on-demand access to IT resources
- C. Enterprise support will be available to help with recurring application installation and setup
- D. The company will experience less downtime with Multi-AZ deployments

[!TIP]

Correct Answer:

- B. Increased agility with on-demand access to IT resources

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud agility and procurement efficiency, especially how AWS enables organizations to replace slow, capital-intensive infrastructure cycles with rapid, elastic provisioning of compute, storage, and networking resources.

WHY B IS CORRECT

Agility through On-Demand Access: AWS provides instant provisioning of infrastructure via APIs or console Eliminates: Hardware procurement delays Capacity planning bottlenecks Manual setup and configuration overhead Enables: Rapid experimentation and deployment Auto-scaling based on demand Global reach without physical expansion Key benefits: Time-to-market acceleration Pay-as-you-go pricing Elastic scaling DevOps and CI/CD enablement AWS transforms infrastructure from a capital expense into a flexible, programmable resource — unlocking agility and innovation.

SOURCES

Benefits of Cloud Computing AWS Cloud Adoption Framework ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Move existing hardware AWS doesn't migrate physical hardware — it replaces it with cloud-native services

C.

Enterprise support Helpful for guidance — but not the core advantage for agility

D.

Multi-AZ deployments Improve availability — not directly related to procurement speed

REINFORCING WITH MODERNIZATION STRATEGY

Let's map AWS agility into a modernization framework: -

Capability

AWS Benefit -

Provisioning

Launch EC2, RDS, S3 in minutes -

Scaling

Auto Scaling Groups, Lambda concurrency -

Experimentation

Sandbox environments with Cloud Formation -

Global Deployment

Launch in 30+ regions instantly -

Cost Control

Budget alerts, cost allocation tags

REAL-WORLD TIE-IN

If your Nairobi-based team is modernizing legacy analytics infrastructure: Replace physical servers with EC2 and S3 Use CloudFormation to automate environment setup Scale workloads with Auto Scaling and Lambda Monitor spend with Cost Explorer and Budgets That's how AWS empowers teams to modernize infrastructure with speed, flexibility, and global scalability — all driven by on-demand access to IT resources.

Question 300

**Which AWS service helps protect against DDoS attacks?

**

OPTIONS

A. AWS Shield

B. Amazon Inspector

C. Amazon GuardDuty

D. Amazon Detective

[!TIP]

Correct Answer:

A. AWS Shield

EXPLANATION

CONCEPTUAL FRAMING

This question targets network-layer threat mitigation, especially how AWS enables organizations to defend against volumetric and application-layer denial-of-service attacks using always-on, automated protection.

WHY A IS CORRECT

AWS Shield: Managed DDoS protection service Two tiers: Shield Standard — free, automatic protection for all AWS customers Shield Advanced — enhanced protection with cost protection, real-time metrics, and 24/7 response team Protects: CloudFront distributions Elastic Load Balancers Route 53 hosted zones Global Accelerator endpoints Key benefits: Inline mitigation — no need to engage support Real-time detection Integration with WAF and CloudWatch DDoS cost protection for Shield Advanced users AWS Shield provides scalable, automated defense against DDoS attacks — keeping applications resilient and responsive under pressure.

SOURCES

AWS Shield Overview Shield Advanced Features  Why the Others Are Incorrect Option Why It's Incorrect

B.

Amazon Inspector Scans for software vulnerabilities — not DDoS mitigation

C.

Amazon GuardDuty Detects anomalous behavior and threats — doesn't block DDoS traffic

D.

REINFORCING WITH THREAT PROTECTION STRATEGY

Let's map AWS Shield into a layered security model: -

Layer

Tool Role -

Edge

AWS Shield DDoS mitigation -

App Layer

AWS WAF Rule-based traffic filtering -

Detection

Guard Duty Threat intelligence and anomaly detection -

Investigation

Detective Root cause analysis -

Vulnerability Scanning

Inspector Software flaw detection

REAL-WORLD TIE-IN

If your Nairobi-based team is hosting a public API: Use CloudFront with Shield Standard for edge protection Add WAF rules to block suspicious patterns Monitor traffic spikes with CloudWatch Upgrade to Shield Advanced for SLA-backed response and cost protection That's how AWS empowers teams to defend against DDoS attacks with layered, automated protection using Shield and its integrated security ecosystem.

REFERENCES

<https://aws.amazon.com/shield/>

Question 301

**Which AWS offering allows customers to aggregate billing for many accounts?

**

OPTIONS

A. Amazon QuickSight

B. AWS Organizations

C. AWS Budgets

D. Amazon Forecast

[!TIP]

Correct Answer:

B. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account financial governance, especially how AWS enables organizations to centralize billing and streamline cost management across multiple linked accounts.

WHY B IS CORRECT

AWS Organizations: Allows creation of a master (payer) account and multiple linked member accounts Enables consolidated billing, where: All usage is aggregated under the master

account Volume discounts and Savings Plans apply across the organization Detailed cost breakdowns are available per account or tag Supports: Service Control Policies (SCPs) for governance Automated account creation Centralized budget and compliance enforcement AWS Organizations is the foundation for scalable, secure, and cost-efficient multi-account cloud operations.

SOURCES

Consolidated Billing in AWS Organizations AWS Organizations Best Practices  Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon QuickSight Visualizes data — doesn't manage billing aggregation

C.

AWS Budgets Sets cost thresholds — doesn't consolidate billing

D.

Amazon Forecast Predicts time-series trends — unrelated to billing or account management

REINFORCING WITH BILLING STRATEGY

Let's map AWS Organizations into a cost management framework: -

Component

Role -

Master Account

Aggregates usage and charges -

Linked Accounts

Isolated environments for teams/apps -

Consolidated Billing

Unified invoice and discount sharing -

Cost Explorer

Visualizes spend across accounts -

Budgets & Alerts

Enforce financial thresholds per account or tag

REAL-WORLD TIE-IN

If your Nairobi-based team is managing analytics, dev, and prod environments: Create separate AWS accounts for each team Link them under a master payer account via AWS Organizations Use Cost Explorer to track spend per account Apply SCPs to enforce security and budget policies That's how AWS empowers teams to scale securely and cost-effectively using Organizations for account structure and billing aggregation.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 302

**Which of the following are advantages of Amazon Web Services' cloud computing platform? (Select two)

**

OPTIONS

A. Unlimited uptime →

B. Elasticity →

C. Agility

D. Colocation

[!TIP]

Correct Answer:

B and C — Elasticity and Agility

EXPLANATION

CONCEPTUAL FRAMING

This question targets core cloud benefits, especially how AWS enables organizations to scale dynamically and innovate rapidly without traditional infrastructure constraints. 🔍 Why B and C Are Correct ✓ B. Elasticity AWS allows you to scale resources up or down based on demand Supports: Auto Scaling Groups Lambda concurrency Elastic Load Balancing Ideal for: Seasonal traffic Batch workloads Cost optimization ✓ C. Agility AWS provides on-demand access to compute, storage, and networking Enables: Rapid experimentation Continuous integration and deployment (CI/CD) Global infrastructure reach Ideal for: Startups and enterprises alike DevOps workflows Innovation cycles Elasticity and agility are the twin engines of cloud-native success — enabling scale and speed without hardware bottlenecks.

SOURCES

AWS Cloud Benefits AWS Well-Architected Framework ✗ Why the Others Are Incorrect Option Why It's Incorrect

A.

Unlimited uptime No cloud provider guarantees 100% uptime — SLAs define availability targets

D.

Colocation Refers to shared physical hosting — not a cloud-native concept

E.

Capital expenses AWS shifts spend to operational expenses — reducing upfront capital investment

REINFORCING WITH CLOUD STRATEGY

Let's map AWS advantages into a transformation framework: -

Advantage

AWS Capability -

Elasticity

Auto Scaling, Lambda, Spot Instances -

Agility

EC2, S3, Cloud Formation, global regions -

Cost Efficiency

Pay-as-you-go, Savings Plans -

Security

IAM, Shield, Guard Duty, KMS -

Innovation

AI/ML, serverless, analytics stack

REAL-WORLD TIE-IN

If your Nairobi-based team is modernizing a legacy analytics platform: Use Elasticity to scale EC2 and Lambda based on data volume Leverage Agility to deploy new features weekly via CI/CD Monitor performance with CloudWatch Optimize spend with Savings Plans and tagging That's how AWS empowers teams to build resilient, scalable, and fast-moving cloud-native applications using elasticity and agility as foundational principles.

Question 303

**An application installed in the AWS Cloud exhibits irregular consumption patterns and is responsible for non-stop workloads. Which Amazon EC2 pricing plan is the MOST cost-effective for this application?

**

OPTIONS

A. Dedicated Instances

B. Spot Instances

C. Reserved Instances →

D. On-Demand Instances

[!TIP]

Correct Answer:

D. On-Demand Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets pricing model selection based on workload behavior, especially how AWS enables organizations to balance flexibility and cost-efficiency for unpredictable or bursty compute needs.

WHY D IS CORRECT

On-Demand Instances: Ideal for irregular or unpredictable workloads No upfront commitment — pay by the second or hour Provides: Immediate provisioning Full control over instance lifecycle

No risk of interruption Key benefits: Flexibility for dynamic workloads Predictable billing without long-term lock-in Reliable performance for continuous operations On-Demand is the most cost-effective option when workloads are non-stop but usage patterns are irregular and unpredictable.

SOURCES

EC2 Pricing Models Choosing the Right EC2 Purchase Option ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Dedicated Instances Provide physical isolation — expensive and not cost-effective for irregular workloads

B.

Spot Instances Cheapest — but can be interrupted anytime, unsuitable for non-stop workloads

C.

Reserved Instances Cost-effective for predictable, steady-state usage — not for irregular patterns

REINFORCING WITH PRICING STRATEGY

Let's map EC2 pricing models to workload types: -

Pricing Model

Best For Risk Level Cost Efficiency -

On-Demand

Irregular, bursty workloads 🔒 Reliable ⚖️ Moderate -

Reserved

Predictable, long-term workloads 🔒 Reliable 💰 High -

Spot

Fault-tolerant, flexible workloads ⚠️ Interruptible 🔥 Very High -

Dedicated

REAL-WORLD TIE-IN

If your Nairobi-based team is running a real-time analytics engine with unpredictable spikes: Use On-Demand EC2 for flexibility and uptime Monitor usage with CloudWatch and Cost Explorer Consider Savings Plans if usage stabilizes over time Tag resources for cost attribution and audit tracking That's how AWS empowers teams to match pricing models to workload behavior for optimal cost and performance balance.

Question 304

**Which service's primary aim is to manage software versions?

**

OPTIONS

A. Amazon CodeStar

B. AWS Command Line Interface (AWS CLI)

C. Amazon Cognito 

D. AWS CodeCommit

[!TIP]

Correct Answer:

D. AWS CodeCommit

EXPLANATION

CONCEPTUAL FRAMING

This question targets source control and version management, especially how AWS enables teams to collaboratively manage code, track changes, and integrate with CI/CD pipelines using Git-based repositories.

WHY D IS CORRECT

AWS CodeCommit: Fully managed Git-based version control service Supports: Source code management Binary file storage Branching, merging, and tagging Integrates with: CodePipeline for CI/CD IAM for fine-grained access control CloudWatch Events for automation triggers Key benefits: Secure and scalable repository hosting No size limits on repositories Encrypted at rest and in transit Audit-friendly with commit history and access logs CodeCommit is AWS's native Git service for managing software versions, enabling secure, scalable collaboration across teams.

SOURCES

AWS CodeCommit Overview CodeCommit Best Practices  Why the Others Are Incorrect Option Why It's Incorrect

A.

CodeStar Orchestrates DevOps workflows — doesn't manage version control directly

B.

AWS CLI Command-line tool — not a version management service

C.

Cognito Manages user identities — unrelated to code or versioning

REINFORCING WITH DEVOPS STRATEGY

Let's map CodeCommit into a version control and deployment pipeline: -

Component

Role -

CodeCommit

Stores and versions source code -

CodeBuild

Compiles and tests code -

CodeDeploy

Automates deployment to EC2, Lambda, or on-prem -

CodePipeline

Orchestrates CI/CD stages -

CloudWatch Events

Triggers builds or deployments on commit

REAL-WORLD TIE-IN

If your Nairobi-based team is building a multi-service analytics platform: Use CodeCommit to manage microservice repositories Enforce branch protection rules via IAM Trigger CodePipeline builds on commit Monitor commit activity with CloudWatch and CloudTrail That's how AWS empowers teams to securely manage software versions and streamline deployment workflows using CodeCommit and its DevOps ecosystem.

Question 306

**Using AWS Config to record, audit, and evaluate changes to AWS resources to enable traceability is an example of which AWS Well-Architected Framework pillar?

**

OPTIONS

A. Security

B. Operational excellence

C. Performance efficiency

D. Cost optimization

[!TIP]

Correct Answer:

A. Security

EXPLANATION

CONCEPTUAL FRAMING

This question targets traceability and governance, especially how AWS enables organizations to monitor configuration changes and enforce security baselines through automated resource tracking.

WHY A IS CORRECT

AWS Config: Continuously records configuration changes to AWS resources Enables: Audit trails for compliance Change tracking for security investigations Conformance packs for policy enforcement Integrates with: CloudTrail for API activity Security Hub for centralized findings AWS Organizations for multi-account governance Security Pillar Focus: Traceability — who changed what, when, and how Automated alerts for non-compliant changes Evidence collection for audits and incident response AWS Config supports the Security pillar by enabling visibility, traceability, and automated compliance across your cloud environment.

SOURCES

AWS Well-Architected Framework AWS Config Documentation ❌ Why the Others Are Incorrect Pillar Why It's Incorrect Operational Excellence Focuses on operations and monitoring — not traceability or compliance Performance Efficiency Optimizes resource selection — not security or auditing Cost Optimization Manages spend — doesn't track configuration changes

REINFORCING WITH SECURITY STRATEGY

Let's map AWS Config into a security and compliance framework: -

Feature

Role -

Configuration Recorder

Tracks resource changes -

Rules & Conformance Packs

Enforce security policies -

Timeline View

Visualize change history -

Remediation Actions

Auto-correct non-compliant states -

Multi-Account Aggregator

Centralize governance across org units

REAL-WORLD TIE-IN

If your Nairobi-based team is preparing for a PCI or ISO audit: Use AWS Config to track changes to EC2, IAM, and S3 Apply conformance packs for regulatory alignment Export change logs for auditor review Integrate with Security Hub for centralized findings That's how AWS empowers teams to build secure, auditable, and compliant cloud environments using Config as a cornerstone of the Security pillar.

REFERENCES

Question 307

**Which of the following is a design concept for AWS Cloud architecture?

**

OPTIONS

A. Implement single points of failure →

B. Implement loose coupling

C. Implement monolithic design

D. Implement vertical scaling

[!TIP]

Correct Answer:

B. Implement loose coupling

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilient system design, especially how AWS encourages modular, fault-tolerant architectures that isolate failure domains and promote scalability.

WHY B IS CORRECT

Loose Coupling: Components interact via intermediaries (e.g., queues, streams, APIs) rather than direct dependencies Enables: Independent scaling Failure isolation Asynchronous processing Common tools: Amazon SQS (message queues) Amazon SNS (pub/sub notifications) Amazon EventBridge (event bus) Key benefits: Resilience — one component can fail without cascading Flexibility — swap or upgrade components independently Scalability — process spikes without bottlenecks Loose coupling is the backbone of cloud-native design — enabling agility, fault tolerance, and modular evolution.

SOURCES

AWS Well-Architected Framework Design Principles for Cloud Architecture  Why the Others Are Incorrect Option Why It's Incorrect

A.

Single points of failure Violates high availability principles — introduces fragility

C.

Monolithic design Hard to scale, update, and isolate — not cloud-native

D.

Vertical scaling Limited scalability — better suited for legacy systems

REINFORCING WITH ARCHITECTURE STRATEGY

Let's map loose coupling into a resilient AWS architecture: -

Component

Role -

SQS Queue

Buffers requests between services -

Lambda Function

Processes messages asynchronously -

SNS Topic

Broadcasts events to multiple subscribers -

Step Functions

Orchestrates loosely coupled workflows -

CloudWatch Alarms

Detects failures and triggers recovery

REAL-WORLD TIE-IN

If your Nairobi-based team is building a multi-service analytics pipeline: Use SQS to decouple ingestion from processing Trigger Lambda functions for transformation Publish results via SNS to downstream consumers Monitor with CloudWatch and auto-scale with EventBridge That's how AWS empowers teams to build resilient, scalable, and maintainable systems using loose coupling as a foundational design principle.

Question 308

**The continuous lowering in AWS Cloud price is a result of the following:

**

OPTIONS

A. Pay-as-you-go pricing

B. The AWS global infrastructure →

C. Economies of scale

D. Reserved storage pricing

[!TIP]

Correct Answer:

C. Economies of scale

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud economics and pricing dynamics, especially how AWS leverages its massive customer base and infrastructure footprint to drive down unit costs and pass savings to customers.

WHY C IS CORRECT

Economies of Scale: As AWS usage grows, infrastructure costs are spread across more customers Enables: Bulk purchasing power Operational efficiencies Lower per-unit costs AWS can: Invest in custom hardware and energy optimization Reduce pricing for services like EC2, S3, and Lambda Offer tiered pricing models and volume discounts AWS's scale is its pricing superpower — the more customers it serves, the cheaper it becomes to serve each one.

SOURCES

AWS Pricing Philosophy Economies of Scale in Cloud Computing  Why the Others Are Incorrect Option Why It's Incorrect

A.

Pay-as-you-go pricing Offers flexibility — but doesn't inherently lower prices over time

B.

Global infrastructure Enables availability and latency improvements — not directly tied to price drops

D.

Reserved storage pricing Offers discounts — but is a result of pricing strategy, not the cause of price reduction

REINFORCING WITH PRICING STRATEGY

Let's map AWS pricing levers into a cost optimization framework: -

Lever

Role -

Economies of Scale

Drives down baseline service costs -

Savings Plans & RIs

Offer discounts for predictable usage -

Spot Instances

Exploit unused capacity at deep discounts -

Tiered Pricing

Reduce cost per unit as usage grows -

Free Tier

Encourage experimentation and onboarding

REAL-WORLD TIE-IN

If your Nairobi-based team is scaling analytics workloads: Benefit from lower EC2 and S3 pricing as AWS scales globally Use Savings Plans for predictable compute Monitor spend with Cost Explorer and Budgets Tag resources for cost attribution and optimization That's how AWS empowers teams to scale affordably by passing on the benefits of global infrastructure and customer volume through economies of scale.

Question 309

**Which phase discusses agility as an advantage of AWS Cloud-based development?

**

OPTIONS

- A. Pay only when computing resources are consumed
- B. Eliminate guessing about infrastructure capacity →
- C. Support innovation by reducing time to provision IT resources
- D. Deploy applications globally in minutes

[!TIP]

Correct Answer:

C. Support innovation by reducing time to provision IT resources

EXPLANATION

CONCEPTUAL FRAMING

This question targets agility in cloud-native development, especially how AWS enables teams to accelerate innovation cycles by removing infrastructure bottlenecks and provisioning delays.

WHY C IS CORRECT

Agility in AWS Cloud: Developers can launch EC2, RDS, Lambda, and S3 in minutes No need for: Hardware procurement Manual setup Capacity planning Enables: Rapid prototyping Continuous integration and deployment (CI/CD) Fail-fast experimentation Key benefits: Time-to-market acceleration Empowered development teams Global scalability without friction Agility in AWS means developers move from idea to execution in hours — not months.

SOURCES

AWS Cloud Benefits AWS Well-Architected Framework ❌ Why the Others Are Incorrect Option
Why It's Incorrect

A.

Pay-as-you-go pricing A cost model — not directly tied to agility

B.

Capacity guessing elimination Improves scalability — but agility is about speed and responsiveness

D.

Global deployment A benefit of AWS infrastructure — not the core of agility

REINFORCING WITH DEVELOPMENT STRATEGY

Let's map agility into a cloud-native development lifecycle: -

Phase

AWS Capability -

Provisioning

EC2, Lambda, Cloud Formation -

Deployment

Code Pipeline, Code Deploy -

Monitoring

Cloud Watch, X-Ray -

Iteration

CI/CD, blue/green deployments -

Scaling

Auto Scaling, Elastic Load Balancing

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics feature: Use CloudFormation to spin up environments instantly Deploy via CodePipeline with rollback support Monitor with CloudWatch dashboards Iterate weekly with agile sprints and CI/CD That's how AWS empowers teams to innovate faster and respond to change with agility baked into every phase of cloud-native development.

Question 309

**Which phase discusses agility as an advantage of AWS Cloud-based development?

**

OPTIONS

- A. Pay only when computing resources are consumed
- B. Eliminate guessing about infrastructure capacity →
- C. Support innovation by reducing time to provision IT resources
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[!TIP]

Correct Answer:

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EXPLANATION

CONCEPTUAL FRAMING

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REINFORCING WITH DEVELOPMENT STRATEGY

Let's map agility into a cloud-native development lifecycle: -

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EC2, Lambda, Cloud Formation -

Deployment

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Monitoring

Cloud Watch, X-Ray -

Iteration

CI/CD, blue/green deployments -

Scaling

Auto Scaling, Elastic Load Balancing

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics feature: Use CloudFormation to spin up environments instantly Deploy via CodePipeline with rollback support Monitor with CloudWatch dashboards Iterate weekly with agile sprints and CI/CD That's how AWS empowers teams to innovate faster and respond to change with agility baked into every phase of cloud-native development.

Question 311

**A business wants to deliver managed Windows virtual desktops and secure network access to remote workers. Which AWS services does the business have access to in order to achieve these requirements? (Select two)

**

OPTIONS

A. Amazon Connect →

B. Amazon AppStream 2.0 →

C. Amazon WorkSpaces

D. AWS Site-to-Site VPN

E. Amazon ECS

[!TIP]

Correct Answer:

B and C — AppStream 2.0 and WorkSpaces

EXPLANATION

CONCEPTUAL FRAMING

This question targets remote workforce enablement, especially how AWS provides fully managed desktop and application streaming services to support secure, scalable access for distributed teams. 🔍 Why B and C Are Correct ✅ B. Amazon AppStream 2.0 Delivers application streaming via browser or client Ideal for: Single-app delivery Graphics-intensive workloads BYOD environments Supports: Windows desktop apps Secure access via IAM and SAML Auto-scaling fleets ✅ C. Amazon WorkSpaces Provides full virtual desktops (Windows, Linux, Ubuntu) Ideal for: Persistent desktop environments Enterprise workforce Secure remote access Supports: Directory integration Encryption at rest and in transit Usage-based billing AppStream 2.0 and WorkSpaces are AWS's twin pillars for remote productivity — one streams apps, the other delivers desktops.

SOURCES

Amazon AppStream 2.0 Overview Amazon WorkSpaces Overview ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon Connect Cloud contact center — not for desktop or app delivery

D.

AWS Site-to-Site VPN Enables secure network tunnels — doesn't deliver desktops or apps

E.

Amazon ECS Container orchestration — unrelated to end-user desktop access

REINFORCING WITH REMOTE ACCESS STRATEGY

Let's map AWS services into a remote workforce enablement model: -

Need

AWS Service Role -

App access

App Stream 2.0 Stream desktop apps securely -

Desktop access

Work Spaces Provision full virtual desktops -

Secure connectivity

VPN, Direct Connect Tunnel into VPC resources -

User identity

IAM, Cognito Authenticate and authorize users -

Monitoring

Cloud Watch, Cloud Trail Track usage and access patterns

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding remote analysts: Use WorkSpaces for full desktop environments Stream specialized tools via AppStream 2.0 Secure access with IAM and MFA Monitor usage with CloudWatch dashboards That's how AWS empowers teams to deliver secure, scalable desktop and application access for remote workers using WorkSpaces and AppStream 2.0.

Question 312

**Which of the following is an AWS shared responsibility?

**

OPTIONS

A. Identity and access management (IAM)

B. Server-side encryption (SSE)

C. Firewall configuration →

D. Maintaining physical hardware

[!TIP]

Correct Answer:

D. Maintaining physical hardware

EXPLANATION

CONCEPTUAL FRAMING

This question targets the AWS Shared Responsibility Model, which defines the boundary between what AWS secures and what the customer must manage. It's foundational to understanding cloud security and compliance.

WHY D IS CORRECT

AWS's Responsibility: Physical security of data centers
Hardware maintenance and replacement
Network infrastructure and power redundancy
Environmental controls and access restrictions
These responsibilities fall under: Security "of" the cloud — AWS ensures the cloud

infrastructure is secure and operational AWS handles the heavy lifting of physical infrastructure so customers can focus on securing their workloads “in” the cloud.

SOURCES

AWS Shared Responsibility Model AWS Security Documentation ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

IAM Customer configures IAM roles, policies, and permissions — part of security in the cloud

B.

SSE Customer chooses encryption settings and manages keys — AWS provides tools, but config is customer-managed

C.

Firewall configuration Customers define Security Groups and NACLs — AWS provides the framework, not the rules

REINFORCING WITH RESPONSIBILITY BREAKDOWN

Let's map the shared responsibility model into a clear framework: -

Responsibility

AWS Customer -

Physical infrastructure

-

Hypervisor and hardware

-

IAM policies

-

Data encryption

-

OS and patching (EC2)

-

Network configuration

REAL-WORLD TIE-IN

If your Nairobi-based team is preparing for a compliance audit: Document AWS's role in physical and infrastructure security Focus your controls on IAM, encryption, and firewall rules Use AWS Artifact to download SOC and ISO reports Monitor changes with AWS Config and CloudTrail That's how AWS empowers teams to build secure, compliant systems by clearly delineating responsibilities between infrastructure and workload management.

Question 313

**Which storage service can be utilized to host static webpages at a minimal cost?

**

OPTIONS

A. Amazon Glacier

B. Amazon DynamoDB

C. Amazon Elastic File System (Amazon EFS) 

D. Amazon Simple Storage Service (Amazon S3)

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets low-cost web hosting, especially how AWS enables teams to serve static content directly from object storage without provisioning servers or managing infrastructure.

WHY D IS CORRECT

Amazon S3 Static Website Hosting: Supports: HTML, CSS, JS, images, and client-side scripts
Custom error and index documents Public access configuration for web delivery Ideal for:
Marketing pages Documentation sites Single-page applications (SPAs) Key benefits: No server management Highly durable and available Pay only for storage and data transfer Amazon S3 turns object storage into a scalable, serverless web host — perfect for static sites with minimal cost.

SOURCES

Amazon S3 Static Website Hosting S3 Pricing  Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon Glacier Designed for archival storage — not suitable for web hosting

B.

Amazon DynamoDB No file hosting — used for NoSQL data storage

C.

Amazon EFS File system for EC2 — more expensive and complex than S3 for static hosting

REINFORCING WITH HOSTING STRATEGY

Let's map AWS services into a static vs dynamic hosting model: -

Hosting Need

AWS Service Role -

Static site

S3 Host HTML/CSS/JS with public access -

Dynamic site

EC2, Lambda, API Gateway Serve backend logic -

Global delivery

Cloud Front CDN for caching and performance -

Domain routing

Route 53 DNS and custom domain setup -

SSL/TLS

ACM Free certificates for HTTPS

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a product landing page: Upload HTML/CSS/JS to S3 bucket Enable static website hosting Use CloudFront for global caching Secure with ACM and Route 53 for HTTPS and custom domain That's how AWS empowers teams to host static websites with minimal cost and global scalability using S3 and its integrated ecosystem.

REFERENCES

<https://aws.amazon.com/s3/>

Question 315

****Which AWS services are globally specified rather than regionally defined? (Select two)**

OPTIONS



A. Amazon Route 53

B. Amazon EC2

C. Amazon S3

D. Amazon CloudFront

E. Amazon DynamoDB

[!TIP]

Correct Answer:

A and D — Route 53 and CloudFront

EXPLANATION

CONCEPTUAL FRAMING

This question targets service scope and deployment boundaries, especially how AWS distinguishes between global services (accessible across all regions) and regional services (confined to a specific AWS region). 🔍 Why A and D Are Correct ✓ A. Amazon Route 53 Global DNS and domain management service Hosted zones and domain records are not tied to a specific region Supports: Latency-based routing Geo DNS Health checks across regions ✓ D. Amazon CloudFront Global Content Delivery Network (CDN) Distributes content via edge locations worldwide Origin can be regional (e.g., S3 or EC2), but CloudFront itself is globally orchestrated Route 53 and CloudFront operate across AWS's global infrastructure — enabling worldwide reach and latency optimization.

SOURCES

AWS Global and Regional Services Route 53 Overview CloudFront Overview  Why the Others Are Incorrect Option Why It's Incorrect

B.

Amazon EC2 Regional — instances run in specific regions and AZs

C.

Amazon S3 Buckets are region-specific — even though S3 is globally accessible

E.

Amazon DynamoDB Regional — tables are created per region (Global Tables are a separate feature)

REINFORCING WITH DEPLOYMENT STRATEGY

Let's map AWS services by scope: -

Scope

Services -

Global

Route 53, Cloud Front, IAM, WAF, AWS Organizations -

Regional

EC2, S3, Dynamo DB, RDS, Lambda -

AZ-specific

EC2 instances, EBS volumes, EN Is

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a global-facing app: Use CloudFront to cache content near users worldwide Manage DNS with Route 53 for latency-based routing Host origin content in S3 (regional) Monitor performance with CloudWatch and X-Ray That's how AWS empowers

teams to build globally distributed applications using services like Route 53 and CloudFront that transcend regional boundaries.

Question 315

**A company's online program crashes when one component fails due to tight interdependencies. Which AWS Cloud design concept is most appropriate for resolving this issue?

**

OPTIONS

A. Elasticity

B. Parallel EC2 instances →

C. Decoupling components

D. Doubling EC2 compute resources

[!TIP]

Correct Answer:

C. Decoupling components

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilient system architecture, especially how AWS encourages teams to design loosely coupled services that isolate failure domains and maintain availability under

stress.

WHY C IS CORRECT

Decoupling: Breaks tight dependencies between components Enables: Independent failure recovery Asynchronous communication Scalable and modular design Common AWS tools: Amazon SQS — message queues Amazon SNS — pub/sub notifications Amazon EventBridge — event bus Step Functions — orchestrated workflows Key benefits: Resilience — one component can fail without crashing the system Flexibility — swap or upgrade services independently Scalability — handle spikes without bottlenecks Decoupling is the antidote to cascading failures — it transforms fragile systems into fault-tolerant ecosystems.

SOURCES

AWS Well-Architected Framework Decoupled Architecture Patterns ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Elasticity Helps scale — but doesn't address interdependency failures

B.

Parallel EC2 instances Improves performance — not fault isolation

D.

Doubling EC2 resources Increases capacity — but doesn't solve architectural fragility

REINFORCING WITH RESILIENCE STRATEGY

Let's map decoupling into a fault-tolerant AWS architecture: -

Component

Role -

Producer

Sends events or messages -

Queue (SQS)

Buffers and stores messages -

Consumer

Processes messages independently -

Dead-letter Queue

Captures failed messages -

Monitoring

Cloud Watch tracks health and throughput

REAL-WORLD TIE-IN

If your Nairobi-based team is building a multi-service analytics platform: Use SQS to decouple ingestion from processing Trigger Lambda or ECS tasks for transformation Monitor queue depth and failures with CloudWatch and DLQs Orchestrate workflows with Step Functions That's how AWS empowers teams to design resilient, modular systems that survive component failures through strategic decoupling.

Question 317

******A user must locate, categorize, and safeguard sensitive data stored in Amazon S3 automatically. Which AWS service satisfies these criteria?

OPTIONS

A. Amazon Inspector ➡

B. Amazon Macie

C. Amazon GuardDuty

[!TIP]

Correct Answer:

B. Amazon Macie

EXPLANATION

CONCEPTUAL FRAMING

This question targets automated data classification and privacy protection, especially how AWS enables organizations to discover, monitor, and secure sensitive information across their S3 data estate.

WHY B IS CORRECT

Amazon Macie: Uses machine learning and pattern matching to identify sensitive data
Automatically: Inventories S3 buckets Classifies data (e.g., PII, credentials, financial info)
Monitors access and policy changes Generates: Security findings for public access, unencrypted data, or misconfigurations Alerts for remediation workflows Key benefits: Continuous monitoring Automated classification Compliance support (e.g., GDPR, HIPAA)
Macie is your data privacy sentinel — scanning S3 for sensitive content and surfacing risks before they become breaches.

SOURCES

Amazon Macie Overview Macie Documentation  Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon Inspector Scans for software vulnerabilities — not data classification

C.

Amazon GuardDuty Detects threats and anomalies — doesn't inspect S3 content

D.

AWS Secrets Manager Manages secrets and credentials — not used for scanning S3 data

REINFORCING WITH DATA PROTECTION STRATEGY

Let's map Macie into a sensitive data governance framework: -

Phase

AWS Service Role -

Discovery

Macie Scan and classify sensitive data -

Monitoring

Cloud Watch, Macie Detect policy changes and access anomalies -

Remediation

Lambda, Security Hub Automate fixes and alerts -

Encryption

KMS, S3 SSE Protect data at rest -

Access Control

IAM, SC Ps Restrict who can access what

REAL-WORLD TIE-IN

If your Nairobi-based team is storing customer records in S3: Use Macie to scan for PII and financial data Monitor for public access or unencrypted buckets Trigger Security Hub findings for compliance workflows Automate bucket policy fixes with Lambda That's how AWS empowers teams to locate, classify, and protect sensitive data in S3 using Macie's intelligent scanning and alerting capabilities.

Question 318

**A business hosts an application on an Amazon EC2 instance that needs access to other AWS resources like S3 and DynamoDB. What is the MOST OPTIMAL way to delegate permissions?

**

OPTIONS



- A. Create an IAM role with the required permissions. Attach the role to the EC2 instance.
- B. Create an IAM user and use its access key and secret access key in the application.
- C. Create an IAM user and use its access key and secret access key to create a CLI profile in the EC2 instance.
- D. Create an IAM role with the required permissions. Attach the role to the administrative IAM user.

[!TIP]

Correct Answer:

A. Attach an IAM role to the EC2 instance

EXPLANATION

CONCEPTUAL FRAMING

This question targets secure and scalable identity delegation, especially how AWS enables EC2 instances to access other services without hardcoding credentials or managing secrets manually.

WHY A IS CORRECT

IAM Roles for EC2: Provide temporary security credentials via the instance metadata service
Automatically rotate credentials Enable least privilege access with fine-grained policies Avoid:
Hardcoded secrets Manual credential rotation Risk of credential leakage Key benefits: Security
— no static keys Scalability — roles can be reused across instances Auditability — integrated
with CloudTrail and IAM Access Analyzer IAM roles are the gold standard for secure, automated
permission delegation in EC2 environments.

SOURCES

IAM Roles for EC2 Best Practices for IAM ❌ Why the Others Are Incorrect Option Why It's
Incorrect

B.

IAM user with access keys Hardcoded credentials — insecure and hard to rotate

C.

IAM user with CLI profile Still relies on static keys — not optimal for automation

D.

Role attached to IAM user Doesn't help EC2 — roles must be attached to the instance itself

REINFORCING WITH ACCESS STRATEGY

Let's map EC2 access delegation into a secure workflow: -

Component

Role -

IAM Role

Defines permissions for EC2 -

Instance Profile

Binds role to EC2 instance -

Metadata Service

Supplies temporary credentials -

CloudTrail

Logs access and usage -

Access Analyzer

Detects overly permissive policies

REAL-WORLD TIE-IN

If your Nairobi-based team is building a data processing app on EC2: Create an IAM role with access to S3 and DynamoDB Attach it via an instance profile Use SDKs or CLI to access services without keys Monitor access with CloudTrail and IAM Access Analyzer That's how AWS empowers teams to delegate permissions securely and scalably using IAM roles for EC2.

Question 319

**Which of the following are support categories for AWS Trusted Advisor? (Select two)

**

OPTIONS

A. Operational excellence →

B. Cost optimization →

C. Security

D. Well-Architected Framework

E. Rightsizing

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud governance and optimization, especially how AWS Trusted Advisor helps organizations reduce costs, improve security posture, and enhance performance through automated checks and recommendations. 🔍 Why B and C Are Correct ✅ B. Cost Optimization Identifies: Idle resources (e.g., underutilized EC2, RDS) Unassociated Elastic IPs Overprovisioned services Suggests: Rightsizing Reserved instance purchases Deleting unused assets ✅ C. Security Flags: Publicly accessible S3 buckets Unrestricted security groups Root account usage Helps: Enforce least privilege Reduce attack surface Align with compliance standards Trusted Advisor is your cloud optimization co-pilot — scanning for cost leaks and security gaps across your AWS estate.

SOURCES

AWS Trusted Advisor Overview Trusted Advisor Checks ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Operational excellence Part of the Well-Architected Framework — not a Trusted Advisor category

D.

Well-Architected Framework Separate tool — not a Trusted Advisor support category

E.

Rightsizing A recommendation type, not a support category — falls under cost optimization

REINFORCING WITH OPTIMIZATION STRATEGY

Let's map Trusted Advisor into a governance framework: -

Category

Role -

Cost Optimization

Reduce spend via rightsizing and cleanup -

Security

Harden configurations and access controls -

Fault Tolerance

Improve availability and recovery -

Performance

Identify bottlenecks and misconfigurations -

Service Limits

Prevent quota breaches and outages

REAL-WORLD TIE-IN

If your Nairobi-based team is auditing cloud spend and security: Use Trusted Advisor to flag idle EC2 and unused EBS volumes Review security group rules for open ports Monitor service limits to avoid deployment failures Export findings to Security Hub or Cost Explorer for deeper analysis That's how AWS empowers teams to optimize cloud usage and secure workloads using Trusted Advisor's actionable insights across cost and security domains.

REFERENCES

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Question 320

**Which disaster recovery option is the LEAST expensive?

OPTIONS

A. Warm standby

B. Multisite →

C. Backup and restore

D. Pilot light

[!TIP]

Correct Answer:

C. Backup and restore

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost-efficient disaster recovery planning, especially how AWS enables organizations to balance recovery time objectives (RTO) and recovery point objectives (RPO) against budget constraints.

WHY C IS CORRECT

Backup and Restore: Involves periodic backups of data and configurations. No infrastructure is running until recovery is needed. **Recovery process:** Restore data from S3, Glacier, or backup vaults. **Reprovision resources** (EC2, RDS, etc.). **Ideal for:** Non-critical workloads. **Cost-sensitive environments.** **Longer RTO tolerance.** **Key benefits:** Lowest ongoing cost. Simple to implement. Scalable storage options (e.g., S3 Standard, S3 Glacier Deep Archive). Backup and restore is the most budget-friendly DR strategy — pay only for storage until disaster strikes.

SOURCES

AWS Disaster Recovery Strategies AWS Backup Overview  Why the Others Are More Expensive Option Why It's More Costly

A.

Warm standby Requires partially running infrastructure — incurs compute and licensing costs

B.

Multisite Fully active in multiple regions — highest cost, lowest RTO

D.

Pilot light Keeps core services running — cheaper than warm standby, but costlier than pure backups

Question 321

**Which type of AWS storage is ephemeral and is deleted when an Amazon EC2 instance is stopped or terminated?

**

OPTIONS

A. Amazon Elastic Block Store (Amazon EBS) 

B. Amazon EC2 instance store

C. Amazon Elastic File System (Amazon EFS)

D. Amazon S3

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets storage lifecycle and durability, especially how AWS distinguishes between persistent volumes and ephemeral storage tied to compute lifecycles.

WHY B IS CORRECT

Amazon EC2 Instance Store: Provides temporary block-level storage physically attached to the host Data is lost when the instance is stopped, terminated, or fails Ideal for: Scratch space Buffering and caching Temporary logs Key characteristics: High IOPS — good for performance-sensitive workloads No durability guarantees — not suitable for critical data Not backed by EBS — tied to the instance lifecycle Instance store is fast but fleeting — perfect for ephemeral workloads, dangerous for persistent data.

SOURCES

Amazon EC2 Instance Store Storage Options in AWS  Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon EBS Persistent block storage — survives stop/start and can be detached

C.

Amazon EFS Network file system — durable and shared across instances

D.

Amazon S3 Object storage — highly durable and independent of EC2 lifecycle

REINFORCING WITH STORAGE STRATEGY

Let's map AWS storage types by durability and use case: -

Storage Type

Durability Use Case -

Instance Store

Ephemeral Scratch space, temp logs -

EBS

Persistent Boot volumes, databases -

EFS

Persistent Shared file systems -

S3

Persistent Static assets, backups

REAL-WORLD TIE-IN

If your Nairobi-based team is running high-speed data transformations: Use instance store for temporary staging Persist results to S3 or EBS Monitor lifecycle with CloudWatch and EC2 status checks That's how AWS empowers teams to balance performance and durability by choosing the right storage type for each workload phase.

REFERENCES

<https://aws.amazon.com/ec2/>

Question 320

**A company needs fully managed, highly reliable, and scalable file storage that is accessible over the Server Message Block (SMB) protocol. Which AWS service will meet these requirements?

**

OPTIONS

A. Amazon S3

B. Amazon EFS →

C. Amazon FSx for Windows File Server

D. Amazon EBS

[!TIP]

Correct Answer:

C. Amazon FSx for Windows File Server

EXPLANATION

CONCEPTUAL FRAMING

This question targets protocol compatibility and enterprise-grade file storage, especially how AWS enables organizations to integrate Windows-native applications with fully managed SMB-accessible file systems.

WHY C IS CORRECT

Amazon FSx for Windows File Server: Built on Windows Server Supports SMB protocol for file sharing Integrates with: Active Directory Windows ACLs DFS namespaces Offers: High availability Automatic backups Multi-AZ deployment options Key benefits: Native Windows compatibility Secure access control Scalable performance tiers FSx for Windows File Server is AWS's answer to enterprise-grade SMB file sharing — fully managed, secure, and scalable.

SOURCES

Amazon FSx for Windows File Server FSx Documentation ❌ Why the Others Are Incorrect
Option Why It's Incorrect

A.

Amazon S3 Object storage — not compatible with SMB

B.

Amazon EFS NFS-based — designed for Linux workloads

D.

Amazon EBS Block storage — not a shared file system, no SMB support

REINFORCING WITH FILE STORAGE STRATEGY

Let's map AWS file storage options by protocol and use case: -

Service

Protocol Use Case -

FSx for Windows

SMB Windows apps, AD integration -

EFS

NFS Linux apps, shared file systems -

S3

REST/HTTPS Static assets, backups -

EBS

Block Boot volumes, databases

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating legacy Windows apps: Use FSx for Windows File Server for SMB-compatible storage Join to AWS Managed AD for access control Enable multi-AZ deployment for high availability Monitor usage with CloudWatch and AWS Backup That's

how AWS empowers teams to deliver enterprise-grade Windows file storage with native SMB support using FSx for Windows File Server.

REFERENCES

<https://aws.amazon.com/fsx/>

Question 323

**Which of the following is a component of the AWS Global Infrastructure?

**

OPTIONS

A. Amazon Alexa →

B. AWS Regions

C. Amazon Lightsail

D. AWS Organizations

[!TIP]

Correct Answer:

B. AWS Regions

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud infrastructure topology, especially how AWS organizes its global footprint to deliver high availability, fault tolerance, and geographic reach.

WHY B IS CORRECT

AWS Regions: Geographically isolated areas where AWS clusters data centers Each region contains multiple Availability Zones (AZs) Enables: Data residency compliance Latency optimization Disaster recovery strategies Examples: US East (N. Virginia) Europe (Frankfurt) Africa (Cape Town) Key benefits: Global scalability Redundancy and fault isolation Custom deployment strategies AWS Regions are the backbone of global cloud architecture — enabling localized control with global reach.

SOURCES

AWS Global Infrastructure Regions and Availability Zones  Why the Others Are Incorrect
Option Why It's Incorrect

A.

Amazon Alexa Consumer voice assistant — not part of AWS infrastructure

C.

Amazon Lightsail Simplified cloud service — runs within regions, not a global infra component

D.

AWS Organizations Management tool for multi-account governance — not infrastructure itself

REINFORCING WITH INFRASTRUCTURE STRATEGY

Let's map AWS infrastructure layers: -

Layer

Component Role -

Global

Regions Geographic distribution -

Regional

Availability Zones Fault isolation and redundancy -

Local

Edge Locations CDN and latency optimization -

Management

Organizations, Control Tower Governance and account structure

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a multi-region app: Choose regions based on latency and compliance Architect for multi-AZ failover Use CloudFront for edge caching Monitor health with Route 53 and CloudWatch That's how AWS empowers teams to build globally resilient applications using Regions as the foundational infrastructure component.

Question 324

**What is the purpose of having an internet gateway within a VPC?

**

OPTIONS

- A. Create a VPN connection to the VPC ➡
- B. Allow communication between the VPC and the internet
- C. Impose bandwidth constraints on internet traffic
- D. Load balance traffic from the internet across EC2 instances

[!TIP]

Correct Answer:

B. Allow communication between the VPC and the internet

EXPLANATION

CONCEPTUAL FRAMING

This question targets network connectivity and routing, especially how AWS enables VPCs to interface with the public internet through a scalable, fault-tolerant gateway.

WHY B IS CORRECT

Internet Gateway (IGW): A horizontally scaled, redundant, and highly available VPC component
Enables: Outbound internet access for public subnets Inbound access to EC2 instances with public IPs Routing via route tables and security groups Required for: Web servers API endpoints Public-facing applications The internet gateway is your VPC's bridge to the outside world — enabling secure, scalable internet access.

SOURCES

VPC Internet Gateway Documentation AWS Networking Best Practices **✗** Why the Others Are Incorrect Option Why It's Incorrect

A.

VPN connection Requires a virtual private gateway, not an internet gateway

C.

Bandwidth constraints IGW doesn't throttle traffic — use network ACLs or QoS tools

D.

Load balancing Use Elastic Load Balancer (ELB) — not IGW

REINFORCING WITH VPC CONNECTIVITY STRATEGY

Let's map AWS gateway components by function: -

Gateway Type

Purpose Use Case -

Internet Gateway

Public internet access Web servers, public AP Is -

NAT Gateway

Internet access for private subnets Backend services, patching -

Virtual Private Gateway

VPN connectivity Hybrid cloud, on-prem integration -

Transit Gateway

Multi-VPC routing Enterprise-scale architectures

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a public-facing analytics dashboard: Place EC2 instances in a public subnet Attach an internet gateway to the VPC Configure route tables to direct traffic to the IGW Secure access with security groups and NACLs That's how AWS empowers teams to enable secure, scalable internet connectivity for cloud resources using the internet gateway.

Question 325

**Which AWS service allows users to download security and compliance reports about the AWS infrastructure on demand?

**

OPTIONS

A. Amazon GuardDuty

B. AWS Security Hub 

C. AWS Artifact

D. AWS Shield

[!TIP]

Correct Answer:

C. AWS Artifact

EXPLANATION

CONCEPTUAL FRAMING

This question targets compliance visibility and audit support, especially how AWS enables organizations to access third-party attestations, certifications, and internal security reports directly from a centralized portal.

WHY C IS CORRECT

AWS Artifact: Central repository for: Security and compliance reports Audit artifacts (e.g., SOC 2, ISO 27001, PCI DSS) Agreements (e.g., Business Associate Addendum for HIPAA) Provides: On-demand access Downloadable PDFs Region-specific documentation Supports: Internal audits Regulatory compliance Third-party risk assessments AWS Artifact is your compliance command center — delivering verified reports to satisfy auditors, regulators, and security teams.

SOURCES

AWS Artifact Overview Artifact Documentation  Why the Others Are Incorrect Option Why It's Incorrect

A.

GuardDuty Threat detection — doesn't provide compliance reports

B.

Security Hub Aggregates security findings — not a source of audit documentation

D.

AWS Shield DDoS protection — unrelated to compliance reporting

REINFORCING WITH AUDIT STRATEGY

Let's map AWS services into an audit-readiness framework: -

Need

AWS Service Role -

Compliance reports

Artifact Download attestations and agreements -

Security findings

Security Hub Aggregate and prioritize alerts -

Threat detection

Guard Duty Monitor for malicious activity -

Access logs

Cloud Trail Track API calls and user actions -

Configuration drift

AWS Config Detect and remediate changes

REAL-WORLD TIE-IN

If your Nairobi-based team is preparing for a SOC 2 audit: Use AWS Artifact to download the latest SOC 2 Type II report Validate IAM policies and CloudTrail logs for access control Monitor GuardDuty and Security Hub for threat posture Document findings in your audit binder or compliance dashboard That's how AWS empowers teams to streamline audit preparation and compliance tracking using Artifact as the authoritative source for security documentation.

REFERENCES

<https://aws.amazon.com/artifact/>

Question 326

**A company is planning an infrastructure deployment to the AWS Cloud. Before the deployment, they want a cost estimate for running the infrastructure. Which AWS service or feature can provide this information?

**

OPTIONS

A. Cost Explorer

B. AWS Trusted Advisor

C. AWS Cost and Usage Report →

D. AWS Pricing Calculator

[!TIP]

Correct Answer:

D. AWS Pricing Calculator

EXPLANATION

CONCEPTUAL FRAMING

This question targets pre-deployment financial planning, especially how AWS enables teams to model and forecast cloud costs before launching any resources.

WHY D IS CORRECT

AWS Pricing Calculator: Interactive tool for estimating monthly costs across AWS services
Supports: Custom configurations (e.g., EC2 instance types, storage tiers, data transfer) Region-specific pricing Detailed breakdowns by service and usage Ideal for: Budget planning Proposal preparation Cost comparison across architectures The Pricing Calculator is your sandbox for financial modeling — helping you forecast AWS spend before a single resource is deployed.

SOURCES

AWS Pricing Calculator Cost Estimation Guide ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Cost Explorer Analyzes past usage and spend — not for pre-deployment estimates

B.

Trusted Advisor Offers cost-saving recommendations — but only after resources are deployed

C.

Cost and Usage Report Provides detailed billing data — not predictive or interactive

REINFORCING WITH BUDGETING STRATEGY

Let's map AWS cost tools by lifecycle phase: -

Phase

Tool Role -

Planning

Pricing Calculator Estimate future costs -

Deployment

Trusted Advisor Optimize spend and performance -

Monitoring

Cost Explorer Track and analyze usage -

Reporting

Cost and Usage Report Export detailed billing data

REAL-WORLD TIE-IN

If your Nairobi-based team is designing a multi-tier web app: Use Pricing Calculator to model EC2, RDS, S3, and CloudFront costs Compare on-demand vs reserved instances Export estimates for stakeholder review or budget approval That's how AWS empowers teams to forecast cloud costs with precision using the Pricing Calculator before committing to infrastructure spend.

Question 327

**How can Amazon EC2 Auto Scaling groups contribute to a web application's high availability?

**

OPTIONS

A. Add instances across multiple AWS Regions ➡

B. Add or replace instances across multiple Availability Zones

C. Serve static content closer to users

D. Distribute traffic across web servers

[!TIP]

Correct Answer:

B. Add or replace instances across multiple Availability Zones

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience and fault isolation, especially how AWS enables applications to maintain uptime and recover from failures by distributing compute across multiple Availability Zones (AZs).

WHY B IS CORRECT

Auto Scaling Groups (ASGs): Automatically launch or terminate EC2 instances based on:
Health checks Scaling policies AZ availability Distribute instances across multiple AZs for:
Redundancy High availability Resilient failover Key benefits: Self-healing — replaces unhealthy instances Load distribution — balances across AZs Cost efficiency — scales based on demand
Auto Scaling groups are the heartbeat of resilient cloud apps — dynamically balancing load and healing failures across zones.

SOURCES

Amazon EC2 Auto Scaling High Availability Best Practices  Why the Others Are Incorrect
Option Why It's Incorrect

A.

Multi-region scaling ASGs operate within a single region — use Route 53 for global routing

C.

Static content delivery Use CloudFront, not Auto Scaling

D.

Traffic distribution Use Elastic Load Balancer (ELB) — not ASG directly

REINFORCING WITH HIGH AVAILABILITY STRATEGY

Let's map EC2 Auto Scaling into a resilient architecture: -

Component

Role -

Auto Scaling Group

Launches and replaces EC2 instances -

Availability Zones

Isolate failures and distribute load -

Elastic Load Balancer

Routes traffic to healthy instances -

CloudWatch Alarms

Trigger scaling actions -

Launch Templates

Define instance configuration

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a web app: Use Auto Scaling Groups across at least two AZs Attach an Application Load Balancer for traffic routing Monitor instance health with EC2 status checks Trigger scaling via CloudWatch metrics That's how AWS empowers teams to build highly available applications that self-heal and scale across zones using EC2 Auto Scaling.

REFERENCES

<https://aws.amazon.com/ec2/>

Question 328

**A business has a stateless application workload that can withstand brief periods of outage and runs massively parallel calculations. Which Amazon EC2 pricing model should the business choose to save costs?

**

OPTIONS

A. On-Demand Instances →

B. Spot Instances

C. Reserved Instances

D. Dedicated Instances

[!TIP]

Correct Answer:

B. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization for fault-tolerant workloads, especially how AWS enables teams to leverage unused compute capacity at steep discounts for stateless, interrupt-tolerant applications.

WHY B IS CORRECT

Spot Instances: Use spare EC2 capacity at up to 90% discount compared to On-Demand
Ideal for: Stateless apps Batch processing Big data analytics CI/CD pipelines
Can be interrupted with

two minutes' notice — so workloads must be resilient Key benefits: Massive cost savings Scalable parallel compute Flexible fleet management with EC2 Auto Scaling and Spot Fleet Spot Instances are the budget-friendly powerhouse for parallel, fault-tolerant workloads that don't mind interruptions.

SOURCES

Amazon EC2 Spot Instances Spot Instance Best Practices ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

On-Demand Instances Reliable but expensive — not optimal for cost savings

C.

Reserved Instances Good for predictable workloads — not flexible or interruption-tolerant

D.

Dedicated Instances Isolated hardware — costly and unnecessary for stateless workloads

REINFORCING WITH PRICING STRATEGY

Let's map EC2 pricing models by workload type: -

Model

Best For Cost Efficiency -

Spot

Fault-tolerant, parallel jobs 💰💰💰 -

On-Demand

Short-term, unpredictable workloads 💰 -

Reserved

Steady-state, long-term workloads 💰💰 -

Dedicated

Compliance-bound, isolated workloads 💰💰💰💰

REAL-WORLD TIE-IN

If your Nairobi-based team is running a genomics pipeline: Use Spot Instances for parallel compute tasks Monitor interruptions with CloudWatch Events Use Auto Scaling with mixed instance policies Persist results to S3 or DynamoDB for durability That's how AWS empowers teams to run large-scale, cost-effective workloads using Spot Instances without sacrificing scalability.

REFERENCES

<https://aws.amazon.com/ec2/>

Question 329

**Which of the following is a method for enhancing AWS security? (Select two)

**

OPTIONS

A. Using AWS Artifact

B. Granting the broadest permissions to all IAM roles

C. Running application code with AWS Cloud9 →

D. Enabling multi-factor authentication (MFA) with Amazon Cognito →

E. Using AWS Trusted Advisor security checks

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets identity hardening and configuration hygiene, especially how AWS enables teams to enforce secure access and proactively detect misconfigurations across cloud environments. 🔍 Why D and E Are Correct ✅ D. Multi-Factor Authentication (MFA) with Amazon Cognito Adds a second layer of identity verification Supports: TOTP apps (e.g., Authy, Google Authenticator) SMS-based MFA Custom challenge flows Protects: User pools Federated identities Sensitive operations ✅ E. AWS Trusted Advisor Security Checks Scans for: Open security groups Unencrypted S3 buckets Root account usage IAM policy risks Provides: Actionable recommendations Compliance alignment Continuous posture monitoring MFA and Trusted Advisor form a powerful duo — one locks down identity, the other audits your cloud perimeter.

SOURCES

Amazon Cognito MFA Trusted Advisor Security Checks ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

AWS Artifact Provides compliance reports — not a security control

B.

Broad IAM permissions Violates least privilege principle — increases risk

C.

AWS Cloud9 Development environment — not a security enhancement tool

REINFORCING WITH SECURITY STRATEGY

Let's map AWS security enhancements by category: -

Category

Method Tool -

Identity

MFA, least privilege Cognito, IAM -

Configuration

Misconfig detection Trusted Advisor, Config -

Threat Detection

Anomaly alerts Guard Duty, Security Hub -

Encryption

Data protection KMS, S3 SSE -

Auditability

Access logs Cloud Trail, Access Analyzer

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding new users: Use Amazon Cognito with MFA for secure sign-in Run Trusted Advisor checks weekly to catch misconfigurations Enforce IAM policies with least privilege Monitor access with CloudTrail and Access Analyzer That's how AWS empowers teams to harden identity and continuously audit security posture using Cognito MFA and Trusted Advisor. **?** Question Which AWS service or tool helps to centrally manage billing and allow controlled access to resources across AWS accounts?

A.

AWS Identity and Access Management (IAM) ➡ B. AWS Organizations C. Cost Explorer D. AWS Budgets

CONCEPTUAL FRAMING

This question targets multi-account governance and financial control, especially how AWS enables enterprises to centralize billing, enforce policies, and delegate access across a fleet of accounts.

WHY B IS CORRECT

AWS Organizations: Enables: Consolidated billing across accounts Service control policies (SCPs) for permission boundaries Account automation with Organizational Units (OUs) Supports: Single payment method for all linked accounts Centralized security and compliance enforcement Delegated administration for specific services Key benefits: Financial clarity — one bill, multiple accounts Security hygiene — enforce least privilege across accounts Scalability — manage hundreds of accounts with policy inheritance AWS Organizations is your control tower for cloud governance — blending financial oversight with security and access control.

SOURCES

AWS Organizations Overview Consolidated Billing Documentation  Why the Others Are Incorrect Option Why It's Incorrect

A.

IAM Manages identities — doesn't handle billing or multi-account structure

C.

Cost Explorer Visualizes spend — doesn't manage accounts or access

D.

AWS Budgets Sets cost thresholds — doesn't centralize billing or permissions

REINFORCING WITH GOVERNANCE STRATEGY

Let's map AWS account management tools by function: -

Tool

Role -

Organizations

Multi-account structure, billing, SC Ps -

IAM

Identity and access control within accounts -

Cost Explorer

Spend analysis and forecasting -

Budgets

Alerts and thresholds for cost control -

Control Tower

Automated account provisioning and guardrails

REAL-WORLD TIE-IN

If your Nairobi-based team is managing separate accounts for dev, staging, and prod: Use AWS Organizations to link accounts under one billing umbrella Apply SCPs to restrict risky actions Delegate IAM roles for cross-account access Monitor spend with Cost Explorer and Budgets That's how AWS empowers teams to scale securely and cost-effectively across multiple accounts using Organizations as the governance backbone.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 381

**A previously used EC2 instance is no longer visible in the AWS Console. Which AWS service should be used to discover what activity led to its inaccessibility?

**

OPTIONS

A. Amazon CloudWatch Logs

B. AWS Security Hub

C. Amazon Inspector →

[!TIP]

Correct Answer:

D. AWS CloudTrail

EXPLANATION

CONCEPTUAL FRAMING

This question targets account-level visibility and forensic auditing, especially how AWS enables teams to track API activity and resource changes across services for security, compliance, and troubleshooting.

WHY D IS CORRECT

AWS CloudTrail: Records every API call and console action
Tracks: Who performed the action
What resource was affected
When and from where it occurred
Ideal for: Investigating deletions or modifications
Auditing access and usage patterns
Triggering alerts on suspicious activity
Supports: Event history search
CloudTrail Insights for anomaly detection
Integration with CloudWatch and EventBridge
CloudTrail is your audit trail — every click, every call, every change, logged and searchable.

SOURCES

AWS CloudTrail Overview
CloudTrail Event Reference  Why the Others Are Incorrect
Service Why It's Incorrect
CloudWatch Logs Monitors logs from services — doesn't track API-level resource changes
Security Hub Aggregates security findings — doesn't log operational activity
Inspector Scans for vulnerabilities — doesn't track instance lifecycle events

REINFORCING WITH AUDIT STRATEGY

Let's map AWS services by forensic role: -

Service

Role Best For -

CloudTrail

API activity logging Resource changes, deletions, access tracking -

Config

Resource state history Compliance and drift detection -

CloudWatch Logs

Service logs Application and system diagnostics -

Security Hub

Security posture Aggregated findings from Guard Duty, Inspector, etc. -

GuardDuty

Threat detection Anomaly and intrusion alerts

REAL-WORLD TIE-IN

If your Nairobi-based team suspects an EC2 instance was terminated: Use CloudTrail Event History to search for `TerminateInstances` or `DeleteTags` Filter by user identity and timestamp Enable CloudTrail Insights to detect unusual API patterns Integrate with AWS Config to correlate resource state changes That's how AWS empowers teams to investigate and respond to infrastructure changes using CloudTrail's detailed activity logs and forensic capabilities.

REFERENCES

<https://aws.amazon.com/cloudtrail/>

Question 382

******A Cloud Practitioner must retain data for seven years to meet regulatory standards. Which AWS service meets this need for the least amount of money?

OPTIONS

A. Amazon S3

B. AWS Snowball

C. Amazon Redshift 

D. Amazon S3 Glacier

[!TIP]

Correct Answer:

D. Amazon S3 Glacier

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost-efficient archival storage, especially how AWS enables teams to preserve data for compliance and disaster recovery with minimal cost and operational overhead.

WHY D IS CORRECT

Amazon S3 Glacier / Glacier Deep Archive: Designed for long-term, infrequently accessed data
Ideal for: Regulatory retention (e.g., 7–10 years) Digital preservation Tape replacement
Features: Lowest-cost storage class in S3 Durable and secure Retrieval options (standard,

expedited, bulk) Glacier is your compliance vault — cold, cost-efficient, and built for the long haul.

SOURCES

Amazon S3 Glacier Overview Glacier Deep Archive Use Cases ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

S3 Standard More expensive — designed for frequent access

B.

Snowball Physical device — not suitable for long-term cloud retention

C.

Redshift Data warehouse — optimized for analytics, not archival storage

REINFORCING WITH STORAGE STRATEGY

Let's map AWS storage classes by retention and cost: -

Storage Class

Use Case Cost Access Frequency -

S3 Standard

Active data High Frequent -

S3 IA

Infrequent access Medium Monthly -

S3 Glacier

Archival Low Yearly -

Glacier Deep Archive

Regulatory retention Lowest Rarely

REAL-WORLD TIE-IN

If your Nairobi-based team is archiving financial records: Store files in S3 Glacier Deep Archive
Apply lifecycle policies to transition from S3 Standard after 30 days Use Object Lock for WORM compliance Monitor with AWS Config and CloudTrail That's how AWS empowers teams to retain data securely and affordably for regulatory compliance using Glacier's archival storage classes.

REFERENCES

<https://aws.amazon.com/s3/>

Question 383

**Why is an AWS Well-Architected review a critical part of the cloud design process?

**

OPTIONS

- A. Mandatory before a workload can run on AWS →
- B. Helps identify design gaps and evaluate decisions
- C. Audit mechanism for service level agreements
- D. Eliminates need for ongoing auditing and compliance tests

[!TIP]

Correct Answer:

- B. Helps identify design gaps and evaluate decisions

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud architecture validation and continuous improvement, especially how AWS empowers teams to build resilient, secure, performant, and cost-effective systems by benchmarking against best practices.

WHY B IS CORRECT

AWS Well-Architected Review: Structured assessment across five pillars: Operational Excellence Security Reliability Performance Efficiency Cost Optimization Enables: Gap analysis Risk identification Design validation Outcome: Actionable insights Prioritized improvements Documentation of architectural decisions The Well-Architected Review is your architectural compass — guiding design decisions toward AWS best practices.

SOURCES

AWS Well-Architected Framework Well-Architected Tool  Why the Others Are Incorrect Option Why It's Incorrect

A.

Mandatory Not required — but strongly recommended

C.

SLA Audit Mechanism Not tied to SLAs — focuses on architecture quality

D.

Eliminates Auditing Complements audits — doesn't replace them

REINFORCING WITH REVIEW STRATEGY

Let's map the five pillars to key review questions: -

Pillar

Focus Sample Questions -

Operational Excellence

Run and monitor systems How do you anticipate and respond to failure? -

Security

Protect data and systems How do you manage identity and access? -

Reliability

Recover from disruptions How do you design for fault tolerance? -

Performance Efficiency

Use resources efficiently How do you select scalable architectures? -

Cost Optimization

Avoid unnecessary costs How do you monitor and control spend?

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics platform: Run a Well-Architected Review using the AWS tool Identify gaps in security posture and cost controls Document decisions for audit and onboarding clarity Track improvements with milestone-based remediation plans That's how AWS empowers teams to build better cloud systems by continuously evaluating architecture against proven design principles.

Question 384

**A company uses EC2 Auto Scaling and an Application Load Balancer to recover unhealthy applications. Which AWS Well-Architected Framework pillar does this action support?

**

OPTIONS

A. Security

B. Performance Efficiency

C. Operational Excellence →

D. Reliability

[!TIP]

Correct Answer:

D. Reliability

EXPLANATION

CONCEPTUAL FRAMING

This question targets system resilience and fault tolerance, especially how AWS enables teams to design architectures that automatically recover from failures and maintain availability under stress.

WHY D IS CORRECT

Reliability Pillar focuses on: Recovering from failures automatically
Scaling to meet demand
Monitoring and responding to conditions
EC2 Auto Scaling: Detects unhealthy instances
Launches replacements automatically
Application Load Balancer: Routes traffic only to healthy targets
Supports multi-AZ failover
Reliability is your uptime insurance — design for failure, recover fast, and keep serving.

SOURCES

AWS Well-Architected Reliability Pillar Auto Scaling Best Practices ❌ Why the Others Are Incorrect
Pillar Why It's Incorrect
Security Focuses on access control, encryption, and threat detection
Performance Efficiency Optimizes resource selection and scaling — not fault recovery
Operational Excellence Covers monitoring and operations — not infrastructure resilience

REINFORCING WITH RELIABILITY STRATEGY

Let's map key design principles under the Reliability pillar: -

Principle

Description -

Automatically recover from failure

Use health checks, scaling, and failover -

Test recovery procedures

Simulate failures and validate responses -

Scale horizontally

Add resources to meet demand -

Manage change

Use automation and version control for infrastructure updates

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a customer-facing app: Use Auto Scaling to maintain instance health Deploy ALB across multiple AZs for high availability Monitor with CloudWatch alarms and ELB health checks Simulate failure with AWS Fault Injection Simulator That's how AWS empowers teams to build resilient systems that recover automatically and maintain availability using the Reliability pillar's best practices.

Question 385

**What does AWS's obligation under the shared responsibility paradigm entail?

**

OPTIONS

- A. Updating network ACLs to block traffic to vulnerable ports
- B. Patching operating systems on EC2 instances →
- C. Updating firmware on underlying EC2 hosts
- D. Updating security group rules to block traffic

[!TIP]

Correct Answer:

- C. Updating firmware on underlying EC2 hosts

EXPLANATION

CONCEPTUAL FRAMING

This question targets infrastructure-level responsibilities, especially how AWS delineates security and maintenance boundaries between the cloud provider and the customer.

WHY C IS CORRECT

AWS's responsibility includes: Physical security of data centers Hardware maintenance (e.g., firmware updates, disk replacements) Hypervisor patching and host OS updates EC2 customers manage: Guest OS patching Application security Network configurations (SGs, ACLs) AWS handles the invisible scaffolding — the host firmware, hypervisor, and physical infrastructure beneath your EC2 instances.

SOURCES

AWS Shared Responsibility Model Security Best Practices ❌ Why the Others Are Incorrect
Option Responsibility Why It's Incorrect

A.

Network ACLs Customer You configure these at the VPC level

B.

EC2 OS patching Customer You control the guest OS inside your instance

D.

Security groups Customer You define inbound/outbound rules for your resources

REINFORCING WITH RESPONSIBILITY MATRIX

Let's map key responsibilities: -

Layer

AWS Responsibility Customer Responsibility -

Physical Infrastructure

-

Hypervisor & Host Firmware

-

Guest OS (EC2)

-

Security Groups & ACLs

-

Data Encryption (at rest/in transit)

(tools) (configuration) -

Application Code

REAL-WORLD TIE-IN

If your Nairobi-based team is running EC2 workloads: Trust AWS to patch host firmware and maintain physical infrastructure Use Systems Manager Patch Manager to automate guest OS updates Harden network access with Security Groups and ACLs Monitor changes with AWS Config and CloudTrail That's how AWS empowers teams to focus on workload security while AWS handles the foundational infrastructure under the shared responsibility model.

Question 386

**AWS CloudFormation is intended to assist the user in the following ways:

**

OPTIONS



A. Model and provision resources

B. Update application code

C. Set up data lakes

D. Create reports for billing

[!TIP]

Correct Answer:

A. Model and provision resources

EXPLANATION

CONCEPTUAL FRAMING

This question targets infrastructure automation and declarative provisioning, especially how AWS enables teams to define cloud resources using templates for repeatable, secure, and scalable deployments.

WHY A IS CORRECT

AWS CloudFormation: Uses YAML or JSON templates to define infrastructure Automates provisioning of: EC2 instances VPCs IAM roles S3 buckets Third-party resources via the AWS CloudFormation Registry Supports: Change sets for safe updates StackSets for multi-account/region deployment Drift detection for compliance CloudFormation is your blueprint engine — codify infrastructure, version it, and deploy it with confidence.

SOURCES

AWS CloudFormation Overview Template Anatomy  Why the Others Are Incorrect Option Why It's Incorrect

B.

Update application code Handled by CI/CD tools like CodeDeploy or CodePipeline

C.

Set up data lakes Requires services like Lake Formation and Glue — not CloudFormation alone

D.

Create billing reports Managed by Cost Explorer, CUR, and Budgets — not CloudFormation

REINFORCING WITH IAC STRATEGY

Let's map CloudFormation capabilities: -

Feature

Role -

Templates

Define infrastructure declaratively -

Stacks

Deploy and manage resource groups -

StackSets

Deploy across accounts and regions -

Change Sets

Preview updates before applying -

Drift Detection

Identify config changes outside Cloud Formation

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a multi-tier web app: Define VPC, EC2, RDS, and IAM roles in a CloudFormation template Use StackSets to deploy across dev, staging, and prod accounts Track changes with Change Sets and Drift Detection Version templates in Git for auditability That's how AWS empowers teams to codify infrastructure for repeatable, secure, and scalable deployments using CloudFormation.

REFERENCES

<https://aws.amazon.com/cloudformation/>

Question 387

****Which AWS Cloud benefit is shown by an architecture's ability to withstand failures with minimal downtime?**

OPTIONS

A. Agility

B. Elasticity

C. Scalability →

D. High availability

[!TIP]

Correct Answer:

D. High availability

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience and fault-tolerant design, especially how AWS enables teams to build systems that remain operational even when components fail, ensuring minimal disruption to users.

WHY D IS CORRECT

High Availability (HA): Refers to systems designed to remain accessible and functional during failures
Achieved through: Multi-AZ deployments Load balancing Auto Scaling Health checks and failover
Ensures: Minimal downtime Consistent user experience Business continuity
High availability is your uptime armor — built to endure, recover, and serve without skipping a beat.

SOURCES

AWS Well-Architected Reliability Pillar High Availability Concepts ❌ Why the Others Are Incorrect
Benefit Why It's Incorrect
Agility Speeds up innovation — not focused on fault tolerance
Elasticity Scales resources up/down — doesn't guarantee uptime during failure
Scalability Handles growth — not necessarily resilient to failure

REINFORCING WITH HA STRATEGY

Let's map AWS services that support high availability: -

Service

HA Feature -

EC2 + Auto Scaling

Replace unhealthy instances -

Application Load Balancer

Route traffic to healthy targets -

RDS Multi-AZ

Automatic failover for databases -

Route 53

DNS failover and latency routing -

S3

Regional durability and availability

REAL-WORLD TIE-IN

If your Nairobi-based team is running a customer-facing app: Deploy EC2 instances across multiple AZs Use ALB for traffic distribution and health checks Enable Auto Scaling for recovery and load management Monitor uptime with CloudWatch and Route 53 health checks That's how AWS empowers teams to design resilient systems that withstand failures and maintain availability using high availability principles.

Question 387

**Which AWS Cloud benefit is shown by an architecture's ability to withstand failures with minimal downtime?

**

OPTIONS

A. Agility

B. Elasticity

C. Scalability →

D. High availability

[!TIP]

Correct Answer:

D. High availability

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience and fault-tolerant design, especially how AWS enables teams to build systems that remain operational even when components fail, ensuring minimal disruption to users.

WHY D IS CORRECT

High Availability (HA): Refers to systems designed to remain accessible and functional during failures Achieved through: Multi-AZ deployments Load balancing Auto Scaling Health checks and failover Ensures: Minimal downtime Consistent user experience Business continuity High availability is your uptime armor — built to endure, recover, and serve without skipping a beat.

SOURCES

AWS Well-Architected Reliability Pillar High Availability Concepts ❌ Why the Others Are Incorrect Benefit Why It's Incorrect Agility Speeds up innovation — not focused on fault tolerance Elasticity Scales resources up/down — doesn't guarantee uptime during failure Scalability Handles growth — not necessarily resilient to failure

REINFORCING WITH HA STRATEGY

Let's map AWS services that support high availability: -

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RDS Multi-AZ

Automatic failover for databases -

Route 53

DNS failover and latency routing -

S3

Regional durability and availability

REAL-WORLD TIE-IN

If your Nairobi-based team is running a customer-facing app: Deploy EC2 instances across multiple AZs Use ALB for traffic distribution and health checks Enable Auto Scaling for recovery and load management Monitor uptime with CloudWatch and Route 53 health checks That's how AWS empowers teams to design resilient systems that withstand failures and maintain availability using high availability principles.

Question 389

**A company needs to generate reports that break down cloud costs by product, tags, and time intervals. Which AWS tool should be used?

**

OPTIONS

A. Reserved Instance utilization and coverage reports

B. Savings Plans utilization reports

C. AWS Budgets reports →

D. AWS Cost and Usage Reports

[!TIP]

Correct Answer:

D. AWS Cost and Usage Reports (CUR)

EXPLANATION

CONCEPTUAL FRAMING

This question targets granular cost visibility and financial analysis, especially how AWS enables teams to track, categorize, and analyze cloud spend across dimensions like service, tags, and time.

WHY D IS CORRECT

AWS Cost and Usage Reports (CUR): Provides line-item billing data across: Service type Usage type Linked accounts Custom tags Supports: Hourly, daily, monthly aggregation Export to Amazon S3 Integration with Athena, QuickSight, Redshift Ideal for: Chargeback models Cost

allocation by team/project Forecasting and trend analysis CUR is your financial microscope — zoom into every byte, every tag, every hour of spend.

SOURCES

AWS Cost and Usage Reports Analyzing CUR with Athena **✗** Why the Others Are Incorrect
Tool Why It's Incorrect Reserved Instance Reports Focus on RI usage — not full cost
breakdown Savings Plans Reports Track SP utilization — not all services or tags AWS Budgets
Set thresholds and alerts — not detailed reporting

REINFORCING WITH COST ANALYSIS STRATEGY

Let's map AWS cost tools by role: -

Tool

Role Best For -

CUR

Detailed billing data Full cost breakdown by tag/time/service -

Budgets

Threshold alerts Cost control and forecasting -

Cost Explorer

Visual trends Monthly spend and usage patterns -

Savings Plans Reports

Commitment tracking SP utilization and coverage -

RI Reports

Reservation tracking RI usage and savings

REAL-WORLD TIE-IN

If your Nairobi-based team is managing multi-project AWS spend: Enable CUR with tag-based cost allocation Query data using Athena or Redshift Visualize trends in QuickSight Set Budgets for each team or product line That's how AWS empowers teams to analyze, allocate, and optimize cloud spend with precision using the Cost and Usage Report.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-budgets/>

Question 390

**A business wants to grant a worker access to Amazon RDS using only the AWS CLI and SDKs. Which combination of actions satisfies this need while adhering to least privilege? (Select two)

**

OPTIONS

- A. Create IAM user with console access only →
- B. Create IAM user with programmatic access only
- C. Create IAM role with console access only
- D. Attach administrator access policy to IAM user →
- E. Attach IAM policy with Amazon RDS access to IAM user

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets access scoping and interface restriction, especially how AWS enables teams to grant precise permissions for specific services and interfaces while minimizing exposure. 🔍 Why B and E Are Correct

B.

Programmatic access only: Ensures the user can interact via CLI and SDKs only Prevents access to the AWS Management Console, reducing surface area

E.

IAM policy with Amazon RDS access: Grants least privilege by scoping permissions to only what's needed Example: `rds:DescribeDBInstances`, `rds:Connect`, etc. Least privilege is your security scalpel — cut precisely, expose minimally, and empower safely.

SOURCES

IAM Best Practices Programmatic Access Setup ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Console access only Violates CLI/SDK-only requirement

C.

IAM role with console access Roles are for temporary access — not direct user assignment

D.

Administrator access policy Violates least privilege — grants full access to all services

REINFORCING WITH IAM STRATEGY

Let's map IAM access types: -

Access Type

Interface Use Case -

Programmatic

CLI, SDK, API Automation, scripts, integrations -

Console

Web UI Manual operations, dashboards -

Roles

Temporary access Cross-account, EC2, Lambda, federated users

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding a developer for RDS scripting: Create an IAM user with programmatic access only Attach a custom policy scoped to RDS actions Rotate access keys regularly Monitor usage with CloudTrail and IAM Access Analyzer That's how AWS empowers teams to grant precise, secure access for CLI/SDK workflows while upholding the principle of least privilege.

Question 391

**Which of the following guidelines constitutes a well-architected design philosophy for cloud application development?

**

OPTIONS

A. Keep static data closer to compute resources

B. Provision resources for peak capacity →

C. Design for automated recovery from failure

D. Use tightly coupled components

[!TIP]

Correct Answer:

C. Design for automated recovery from failure

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience and fault-tolerant design, especially how AWS encourages teams to build systems that detect, respond to, and recover from failures automatically — without human intervention.

WHY C IS CORRECT

Automated recovery is a core principle of the Reliability and Operational Excellence pillars in the AWS Well-Architected Framework. Key practices include: Monitoring KPIs that reflect business impact Triggering automation when thresholds are breached Self-healing infrastructure using services like Auto Scaling, ALB, Lambda, and Step Functions Chaos engineering to validate recovery paths Automation isn't just convenience — it's continuity. Recovery should be a reflex, not a reaction.

SOURCES

AWS Well-Architected Framework Operational Excellence Pillar **✗** Why the Others Are Incorrect Option Why It's Incorrect

A.

Keep static data close to compute Helpful for latency — but not a design philosophy

B.

Provision for peak capacity Leads to overprovisioning — violates elasticity and cost optimization

D.

Tightly coupled components Reduces flexibility and fault isolation — violates modular design principles

REINFORCING WITH DESIGN PRINCIPLES

Let's map key philosophies from the Well-Architected Framework: -

Principle

Purpose -

Design for failure

Build resilient systems that recover automatically -

Perform operations as code

Automate and version control infrastructure -

Make frequent, small, reversible changes

Reduce risk and improve agility -

Use loosely coupled components

Improve fault isolation and scalability -

Monitor and learn from events

Drive continuous improvement

REAL-WORLD TIE-IN

If your Nairobi-based team is building a payment API: Use CloudWatch alarms to monitor latency and error rates Trigger Auto Scaling or Lambda fallback on failure Log and trace events with X-Ray and CloudTrail Run chaos experiments to validate recovery paths That's how AWS empowers teams to build resilient, self-healing systems by designing for automated recovery from failure.

Question 392

**A company has a serverless app with API Gateway, Lambda, and DynamoDB. Which AWS service can trace user requests across these components?

**

OPTIONS

A. AWS CloudTrail

B. Amazon CloudWatch

C. Amazon Inspector →

D. AWS X-Ray

[!TIP]

Correct Answer:

D. AWS X-Ray

EXPLANATION

CONCEPTUAL FRAMING

This question targets distributed tracing and observability, especially how AWS enables teams to visualize and analyze request flow across microservices and serverless components.

WHY D IS CORRECT

AWS X-Ray: Provides end-to-end tracing of requests Visualizes latency, errors, and bottlenecks across: API Gateway Lambda functions DynamoDB, S3, SNS, and more Features: Service

maps Trace filtering and annotations Segment and subsegment analysis X-Ray is your application MRI — trace every request, diagnose every delay, and optimize every path.

SOURCES

AWS X-Ray Overview X-Ray with Serverless  Why the Others Are Incorrect Service Why It's Incorrect CloudTrail Tracks API calls — not request flow or latency CloudWatch Monitors metrics and logs — doesn't trace request paths Inspector Scans for vulnerabilities — not observability or tracing

REINFORCING WITH OBSERVABILITY STRATEGY

Let's map AWS observability tools by role: -

Tool

Role Best For -

X-Ray

Distributed tracing Request flow, latency, bottlenecks -

CloudWatch Logs

Log aggregation Debugging and error tracking -

CloudWatch Metrics

Performance metrics CPU, memory, invocation counts -

CloudTrail

API activity logging Security and audit trails -

Inspector

Vulnerability scanning Security posture of compute resources

REAL-WORLD TIE-IN

If your Nairobi-based team is debugging a slow checkout flow: Enable X-Ray tracing on API Gateway and Lambda Use annotations to tag user sessions Visualize latency spikes in service maps Correlate with CloudWatch logs for deeper insights That's how AWS empowers teams to trace, diagnose, and optimize serverless applications using X-Ray's distributed tracing capabilities.

Question 393

**A company needs to set up a petabyte-scale data warehouse in the AWS Cloud. Which AWS service will meet this requirement?

**

OPTIONS

A. Amazon DynamoDB

B. Amazon RDS 

C. Amazon Redshift

D. Amazon ElastiCache

[!TIP]

Correct Answer:

C. Amazon Redshift

EXPLANATION

CONCEPTUAL FRAMING

This question targets big data analytics and scalable warehousing, especially how AWS enables teams to store, query, and analyze massive datasets efficiently using columnar storage and parallel processing.

WHY C IS CORRECT

Amazon Redshift: Purpose-built for petabyte-scale data warehousing Uses Massively Parallel Processing (MPP) and columnar storage Integrates with: BI tools (e.g., Tableau, QuickSight) ETL pipelines (e.g., Glue, Data Pipeline) Lakehouse architecture via Redshift Spectrum Features: Automatic scaling Materialized views Federated queries across S3 and RDS Redshift is your analytics engine — optimized for speed, scale, and deep insights across vast datasets.

SOURCES

Amazon Redshift Overview Redshift Architecture Guide  Why the Others Are Incorrect Service Why It's Incorrect DynamoDB No SQL joins or analytics — designed for key-value and document workloads RDS OLTP-focused — not optimized for analytical queries or petabyte-scale warehousing ElastiCache In-memory cache — not a storage or analytics solution

REINFORCING WITH ANALYTICS STRATEGY

Let's map AWS analytics services by role: -

Service

Role Best For -

Redshift

Data warehouse Complex queries on large datasets -

Athena

Serverless SQL on S3 Ad hoc queries on lake data -

Glue

ETL orchestration Data prep and transformation -

QuickSight

BI visualization Dashboards and reporting -

Lake Formation

Data lake governance Secure, cataloged lake access

REAL-WORLD TIE-IN

If your Nairobi-based team is analyzing customer behavior across millions of transactions: Load data into Redshift using COPY from S3 Use Spectrum to query external S3 data without loading Visualize trends in QuickSight Optimize performance with sort keys and distribution styles That's how AWS empowers teams to build scalable, performant data warehouses for deep analytics using Amazon Redshift.

REFERENCES

<https://aws.amazon.com/redshift/>

Question 394

**What are the immediate advantages of AWS Cloud computing? (Select two)

**

OPTIONS

A. Increased IT staff →

B. Capital expenses are replaced with variable expenses

C. User control of infrastructure →

D. Increased agility

E. AWS holds responsibility for security in the cloud

[!TIP]

Correct Answer:

B and D

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud financial and operational benefits, especially how AWS enables teams to shift from rigid infrastructure models to flexible, scalable, and cost-efficient cloud-native operations. 🔍 Why B and D Are Correct

B.

Capital expenses → Variable expenses: Traditional IT requires upfront investment in hardware AWS shifts this to pay-as-you-go pricing You pay for what you use — no idle capacity or sunk costs

D.

Increased agility: Provision resources in minutes, not weeks Experiment, iterate, and deploy faster Supports innovation and rapid prototyping Cloud agility is your innovation accelerator — and variable pricing is your CFO's best friend.

SOURCES

AWS Cloud Value Proposition AWS Pricing Overview ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Increased IT staff Cloud often reduces staffing needs through automation

C.

User control of infrastructure AWS abstracts infrastructure — users control configuration, not hardware

E.

AWS holds responsibility for security in the cloud Misstates the shared responsibility model — AWS secures the cloud, customers secure their workloads

REINFORCING WITH CLOUD ECONOMICS STRATEGY

Let's map AWS benefits by category: -

Benefit

Description -

Cost savings

Replace Cap Ex with Op Ex, optimize spend -

Agility

Rapid provisioning, faster innovation -

Elasticity

Scale up/down based on demand -

Global reach

Deploy in multiple regions instantly -

Security

Built-in compliance and protection tools

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new analytics platform: Use on-demand EC2 and Lambda to avoid upfront costs Iterate quickly with CloudFormation templates and CI/CD pipelines Monitor spend with AWS Budgets and CUR Scale automatically with Auto Scaling and serverless architectures That's how AWS empowers teams to move fast, spend smart, and innovate freely with cloud-native economics and agility.

Question 395

**What are the economical advantages of using AWS? (Choose two)

☒ Selected Options → A. Reduced Total Cost of Ownership (TCO) → C. Reduced Operational Expenditure (Opex)

**

[!TIP]

Correct Answer:

A and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud financial optimization, especially how AWS enables teams to minimize upfront investment and ongoing operational costs while maximizing flexibility and scalability. 🔍 Why A and C Are Correct

A.

Reduced TCO: AWS eliminates costs tied to: Hardware procurement Data center maintenance Power, cooling, and physical security Enables: Pay-as-you-go pricing Rightsizing and auto-scaling Operational efficiency through automation

C.

Reduced Opex: AWS services are fully managed, reducing: Staffing needs Manual maintenance Software patching and upgrades You shift from fixed monthly costs to usage-based billing AWS transforms infrastructure from a capital burden into a flexible, cost-efficient utility.

SOURCES

AWS Cloud Economics Center TCO Calculator  Why the Others Are Incorrect Option Why It's Incorrect

B.

Increased CapEx AWS reduces CapEx — it's a core benefit

D.

Deferred payment plans Not a standard AWS offering — billing is usage-based

E.

Business credit lines May exist via partners, but not an inherent AWS advantage

REINFORCING WITH COST STRATEGY

Let's map AWS cost benefits: -

Benefit

Description -

Lower TCO

No hardware, facilities, or long-term contracts -

Reduced Opex

Managed services reduce operational overhead -

Elastic pricing

Scale costs with usage — no overprovisioning -

Cost visibility

CUR, Budgets, and Cost Explorer for tracking -

Optimization tools

Trusted Advisor, Compute Optimizer, Savings Plans

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating from on-prem: Use the TCO Calculator to model savings Replace CapEx with on-demand EC2 and serverless Automate operations with Lambda and Systems Manager Monitor spend with Budgets and CUR That's how AWS empowers teams to reduce both capital and operational costs while gaining agility and scalability in the cloud.

Question 396

**Which AWS service is always provided at no charge?

**

OPTIONS

A. Amazon S3 

B. AWS Identity and Access Management (IAM)

C. Elastic Load Balancers

D. AWS WAF

[!TIP]

Correct Answer:

B. AWS Identity and Access Management (IAM)

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost transparency and foundational security, especially how AWS enables teams to manage access and permissions across services without incurring additional charges.

WHY B IS CORRECT

IAM (Identity and Access Management): Always free — no cost for: Users, roles, policies, and groups Federated access and permissions boundaries Core to: Least privilege enforcement Multi-account governance Audit and compliance readiness IAM is your security gatekeeper — always on, always free, and always essential.

SOURCES

AWS IAM Pricing IAM Best Practices **X** Why the Others Are Incorrect Service Why It's Not Free Amazon S3 Charges for storage, requests, and data transfer Elastic Load Balancers Billed per hour and per GB processed AWS WAF Charged per rule and request volume

REINFORCING WITH COST-FREE AWS SERVICES

Here are other AWS services that are always free: -

Service

Free Features -

IAM

Identity and access management -

Organizations

Multi-account governance -

Trusted Advisor (Basic)

Core security checks -

Billing Console

Cost tracking and forecasting -

Service Quotas

REAL-WORLD TIE-IN

If your Nairobi-based team is building a multi-account architecture: Use IAM to define roles and policies Apply least privilege across services Monitor access with CloudTrail and Access Analyzer Manage accounts with AWS Organizations — also free That's how AWS empowers teams to secure and govern cloud environments without adding cost through IAM's free, foundational capabilities.

REFERENCES

<https://aws.amazon.com/iam/>

<https://aws.amazon.com/s3/>

<https://aws.amazon.com/waf/>

Question 397

**Which acts exemplify excellent practices for AWS IAM use? (Select two)

☒ Selected Options ☒ A. Configure a strong password policy ☒ D. Rotate access keys on a regular basis

**

[!TIP]

Correct Answer:

A and D

EXPLANATION

CONCEPTUAL FRAMING

This question targets identity hygiene and credential management, especially how AWS enables teams to enforce secure access controls and minimize exposure through proactive credential rotation and policy enforcement. 🔍 Why A and D Are Correct

A.

Strong password policy: Enforces: Minimum length Complexity (uppercase, lowercase, symbols) Expiration and reuse prevention Reduces risk of brute-force and credential stuffing attacks

D.

Rotate access keys regularly: Prevents long-term exposure from leaked or compromised keys Supports: Automated rotation via IAM and Secrets Manager Audit trails via CloudTrail IAM hygiene is like brushing your teeth — strong passwords and regular key rotation keep your cloud clean.

SOURCES

IAM Best Practices Credential Reports and Access Analyzer ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

Share credentials Violates identity isolation — each user should have unique credentials

C.

Use access keys for console login Not possible — console login requires username/password or federated SSO

E.

Avoid IAM roles IAM roles are essential for secure delegation and temporary access

REINFORCING WITH IAM HYGIENE STRATEGY

Let's map key IAM best practices: -

Practice

Purpose -

Strong password policy

Prevent weak credentials -

Access key rotation

Minimize exposure risk -

Use IAM roles

Delegate access securely -

Enable MFA

Add second layer of protection -

Use groups and policies

Simplify permission management -

Monitor with CloudTrail

Audit access and changes

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding new developers: Enforce a strong password policy via IAM settings Require MFA for console access Assign permissions via groups and managed policies Rotate access keys every 90 days and monitor with CloudTrail That's how AWS empowers teams to maintain secure, scalable identity management through IAM best practices.

Question 398

**A business wants to establish reusable templates for deploying multiple AWS resources. Which AWS offering or functionality should they use?

**

OPTIONS

A. AWS Marketplace

B. Amazon Machine Image (AMI) ➡

C. AWS CloudFormation

D. AWS OpsWorks

[!TIP]

Correct Answer:

C. AWS CloudFormation

EXPLANATION

CONCEPTUAL FRAMING

This question targets infrastructure automation and repeatability, especially how AWS enables teams to codify resource configurations into reusable templates for consistent, scalable deployments.

WHY C IS CORRECT

AWS CloudFormation: Uses YAML or JSON templates to define infrastructure Automates provisioning of: EC2, RDS, S3, IAM, VPC, Lambda, and more Third-party resources via the CloudFormation Registry Supports: Stack creation and updates Cross-account and cross-region deployment via StackSets Drift detection and change sets CloudFormation is your infrastructure blueprint — versioned, repeatable, and ready to deploy at scale.

SOURCES

AWS CloudFormation Overview Template Anatomy ❌ Why the Others Are Incorrect Option
Why It's Incorrect AWS Marketplace Offers third-party solutions — not a template engine
Amazon Machine Image (AMI) Captures OS and app state — not full infrastructure AWS
OpsWorks Chef/Puppet-based — less commonly used and more complex than CloudFormation

REINFORCING WITH TEMPLATE STRATEGY

Let's map CloudFormation capabilities: -

Feature

Role -

Templates

Define infrastructure declaratively -

Stacks

Deploy and manage resource groups -

StackSets

Deploy across accounts and regions -

Change Sets

Preview updates before applying -

Drift Detection

Identify config changes outside Cloud Formation

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a multi-tier web app: Define VPC, EC2, RDS, and IAM roles in a CloudFormation template Use StackSets to deploy across dev, staging, and prod accounts Track changes with Change Sets and Drift Detection Version templates in Git for auditability That's how AWS empowers teams to codify infrastructure for repeatable, secure, and scalable deployments using CloudFormation.

REFERENCES

Question 399

**How can a business use AWS to lower its Total Cost of Ownership (TCO)?

**

OPTIONS



- A. By minimizing large capital expenditures
- B. No responsibility for third-party license costs
- C. No operational expenditures
- D. AWS manages applications

[!TIP]

Correct Answer:

- A. By minimizing large capital expenditures

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud economics and financial agility, especially how AWS enables teams to shift from upfront infrastructure investments to flexible, usage-based spending models.

WHY A IS CORRECT

Capital expenditures (CapEx) include: Buying servers, networking gear, and storage Building and maintaining data centers Long-term contracts and depreciation AWS replaces CapEx with OpEx: Pay only for what you use Scale up/down based on demand Avoid idle capacity and overprovisioning AWS turns infrastructure into a utility — no upfront costs, just on-demand power.

SOURCES

AWS Cloud Economics Center TCO Calculator  Why the Others Are Incorrect Option Why It's Incorrect

B.

No third-party license costs AWS may include license costs — not always free

C.

No operational expenditures You still pay for usage — OpEx is reduced, not eliminated

D.

AWS manages applications Only true for managed services — not all workloads

REINFORCING WITH TCO STRATEGY

Let's map AWS cost levers: -

Lever

Impact -

CapEx → OpEx

No upfront hardware or facilities -

Elastic pricing

Pay-as-you-go, scale with demand -

Managed services

Reduce staffing and maintenance costs -

Automation

Lower operational overhead -

Optimization tools

Identify and eliminate waste

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating from on-prem: Use EC2, Lambda, and S3 to replace physical servers Avoid CapEx with on-demand and reserved pricing models Automate operations with Systems Manager and CloudWatch Monitor spend with Budgets and CUR That's how AWS empowers teams to reduce TCO by eliminating capital expenditures and embracing scalable, efficient cloud-native operations.

Question 400

**A business's website is hosted on Amazon EC2. How can they ensure global accessibility and low latency for visitors?

**

OPTIONS

A. Amazon Route 53 →

B. Amazon CloudFront

C. Elastic Load Balancing

D. AWS Lambda

[!TIP]

Correct Answer:

B. Amazon CloudFront

EXPLANATION

CONCEPTUAL FRAMING

This question targets content delivery and latency reduction, especially how AWS enables teams to serve web content quickly and securely to users across the globe using edge locations.

WHY B IS CORRECT

Amazon CloudFront: A Content Delivery Network (CDN) with global edge locations Caches content close to users for low-latency access Supports: HTTPS encryption Geo-restriction and signed URLs Origin failover and DDoS protection via AWS Shield Integrates seamlessly with: EC2, S3, API Gateway, and Lambda@Edge CloudFront is your global megaphone — fast, secure, and everywhere your users are.

SOURCES

Amazon CloudFront Overview CloudFront Use Cases **✗** Why the Others Are Incorrect Service Why It's Incorrect Route 53 DNS routing — helps direct traffic, but doesn't cache or reduce latency Elastic Load Balancing Distributes traffic across EC2 — not optimized for global delivery AWS Lambda Serverless compute — not related to content distribution or latency

REINFORCING WITH GLOBAL DELIVERY STRATEGY

Let's map AWS services for global performance: -

Service

Role -

CloudFront

CDN for fast, global content delivery -

Route 53

DNS routing and health checks -

Global Accelerator

Optimized routing over AWS backbone -

Lambda@Edge

Custom logic at edge locations -

S3 + CloudFront

Static website hosting with global reach

REAL-WORLD TIE-IN

If your Nairobi-based team is serving a media-rich website: Host assets on S3 or EC2 Distribute via CloudFront for low-latency access worldwide Use Lambda@Edge for custom headers or redirects Monitor performance with CloudWatch and CloudFront metrics That's how AWS empowers teams to deliver fast, secure, and globally optimized web experiences using CloudFront.

REFERENCES

<https://aws.amazon.com/elasticloadbalancing/>

<https://aws.amazon.com/lambda/>

Question 401

******A business uses Amazon EC2 for steady-state workloads and wants to save money. Which EC2 pricing model should they choose?

OPTIONS



A. Reserved Instances

B. On-Demand Instances

C. Spot Instances

D. Dedicated Hosts

[!TIP]

Correct Answer:

A. Reserved Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost predictability and long-term savings, especially how AWS enables teams to commit to consistent usage patterns in exchange for significant discounts.

WHY A IS CORRECT

Reserved Instances (RIs): Ideal for steady-state workloads with predictable usage Offer up to 72% savings compared to On-Demand pricing Flexible options: Standard RIs for maximum savings Convertible RIs for instance type flexibility Zonal RIs for capacity reservation Commitment terms: 1-year or 3-year durations No upfront, partial upfront, or all upfront payment

models Reserved Instances are your financial scalpel — precision savings for predictable workloads.

SOURCES

EC2 Pricing Models Reserved Instance Overview  Why the Others Are Incorrect Model Why It's Incorrect On-Demand Flexible but expensive — best for short-term or unpredictable workloads Spot Instances Cheapest — but not reliable for steady-state workloads due to interruption risk Dedicated Hosts Used for licensing compliance — not cost-effective for general workloads

REINFORCING WITH EC2 PRICING STRATEGY

Let's map EC2 pricing models by use case: -

Model

Best For Savings Potential -

Reserved Instances

Steady-state workloads High (up to 72%) -

Savings Plans

Flexible compute usage High (similar to RIs) -

Spot Instances

Fault-tolerant, flexible workloads Very high (up to 90%) -

On-Demand

Short-term, unpredictable workloads Low -

Dedicated Hosts

BYOL and compliance Variable

REAL-WORLD TIE-IN

If your Nairobi-based team is running a consistent analytics engine: Use Reserved Instances for EC2 and RDS Choose partial upfront for balanced cash flow Monitor usage with Cost Explorer and CUR Combine with Savings Plans for broader flexibility That's how AWS empowers teams to optimize cloud spend for predictable workloads using Reserved Instances.

Question 402

**Which actions should a user take if they detect a hacked AWS account? (Select at least two)

☒ Selected Options → B. Rotate and delete all AWS access keys → E. Contact AWS Support

**

[!TIP]

Correct Answer:

B and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets incident response and credential hygiene, especially how AWS enables teams to contain breaches, restore control, and coordinate with AWS security teams during account compromise events. 🔍 Why B and E Are Correct

B.

Rotate and delete access keys: Prevents further unauthorized access Ensures compromised credentials are invalidated Should be followed by: Audit of IAM roles and policies CloudTrail review for suspicious activity

E.

Contact AWS Support: Initiates Security Incident Response AWS can: Help lock down the account Provide forensic guidance Assist with root account recovery In a breach, speed is

survival — rotate keys, call AWS, and lock the doors.

SOURCES

AWS Compromised Account Response Guide AWS Support Center ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Remove MFA tokens Weakens security — MFA should be enforced, not removed

C.

Move resources to another Region Doesn't contain the breach — may spread exposure

D.

Delete CloudTrail resources Destroys forensic evidence — violates audit best practices

REINFORCING WITH INCIDENT RESPONSE STRATEGY

Let's map AWS breach response steps: -

Step

Action -

1. Lock down access

Rotate keys, change passwords, enforce MFA -

2. Contact AWS Support

Initiate incident response -

3. Review logs

Use Cloud Trail and Config to trace activity -

4. Audit IAM

Check for privilege escalation or rogue policies -

5. Restore and monitor

Rebuild securely and enable alerts

REAL-WORLD TIE-IN

If your Nairobi-based team suspects a breach: Immediately rotate all IAM credentials Enable MFA on all accounts Contact AWS Support for escalation Use CloudTrail and GuardDuty to investigate Apply least privilege policies and review trust relationships That's how AWS empowers teams to respond swiftly and securely to account compromise events using best practices and support escalation.

Question 402

**Which actions should a user take if they detect a hacked AWS account? (Select at least two)

☒ Selected Options  B. Rotate and delete all AWS access keys  E. Contact AWS Support

**

[!TIP]

Correct Answer:

B and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets incident response and credential hygiene, especially how AWS enables teams to contain breaches, restore control, and coordinate with AWS security teams during account compromise events. 🔍 Why B and E Are Correct

B.

Rotate and delete access keys: Prevents further unauthorized access Ensures compromised credentials are invalidated Should be followed by: Audit of IAM roles and policies CloudTrail

review for suspicious activity

E.

Contact AWS Support: Initiates Security Incident Response AWS can: Help lock down the account Provide forensic guidance Assist with root account recovery In a breach, speed is survival — rotate keys, call AWS, and lock the doors.

SOURCES

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If your Nairobi-based team suspects a breach: Immediately rotate all IAM credentials Enable MFA on all accounts Contact AWS Support for escalation Use CloudTrail and GuardDuty to investigate Apply least privilege policies and review trust relationships That's how AWS empowers teams to respond swiftly and securely to account compromise events using best practices and support escalation.

Question 404

**How can AWS most effectively cut costs for a growing startup?

**

OPTIONS



A. It provides on-demand resources for peak usage

B. It automates developer environments

C. It automates CRM

D. It implements a fixed monthly computing budget

[!TIP]

Correct Answer:

A. On-demand resources for peak usage

EXPLANATION

CONCEPTUAL FRAMING

This question targets scalable cost efficiency, especially how AWS enables startups to avoid overprovisioning and pay only for what they use — precisely when they need it.

WHY A IS CORRECT

On-demand resources: Scale instantly to meet traffic spikes Avoid idle infrastructure during low usage Ideal for: Early-stage startups Variable workloads Rapid experimentation Supports: Elastic Load Balancing for traffic distribution Auto Scaling for dynamic provisioning S3 + CloudFront for caching and offloading compute On-demand is your startup's financial parachute — deploy when needed, pay only when used.

SOURCES

AWS Startup Guide AWS Cost Optimization **✗** Why the Others Are Incorrect Option Why It's Incorrect

B.

Automating dev environments Helpful for productivity — not a direct cost-cutting mechanism

C.

Automating CRM Not an AWS-native function — typically handled by third-party SaaS

D.

Fixed monthly budget AWS is usage-based — fixed budgeting is external, not a built-in feature

REINFORCING WITH STARTUP COST STRATEGY

Let's map AWS cost levers for startups: -

Strategy

Benefit -

On-demand compute

Pay only when needed -

Auto Scaling

Match capacity to demand -

CloudFront caching

Reduce backend load and EC2 cost -

IAM governance

Prevent cost leaks via access control -

Cost Explorer + Budgets

Track and forecast spend

REAL-WORLD TIE-IN

If your Nairobi-based startup is launching a new app: Use EC2 or Lambda with Auto Scaling
Cache assets with CloudFront to reduce compute load Monitor spend with Budgets and CUR
Enforce IAM policies to restrict costly actions That's how AWS empowers startups to scale
smartly, spend efficiently, and grow without financial drag using on-demand infrastructure.

Question 405

**Which AWS service manages objects by storing them, providing real-time access, and managing versions and lifecycles?

**

OPTIONS

A. Amazon Glacier

B. AWS Storage Gateway →

C. Amazon S3

D. Amazon EBS

[!TIP]

Correct Answer:

C. Amazon S3 (Simple Storage Service)

EXPLANATION

CONCEPTUAL FRAMING

This question targets object storage and data lifecycle governance, especially how AWS enables teams to store, retrieve, version, and transition data efficiently across storage tiers.

WHY C IS CORRECT

Amazon S3: Stores objects (files, metadata) in buckets Provides real-time access via HTTP(S)
Supports: Versioning to preserve object history Lifecycle policies to transition objects to: S3
Glacier (archival) S3 Intelligent-Tiering (cost optimization) Replication, encryption, and access control
S3 is your data vault — scalable, durable, and smart enough to manage itself over time.

SOURCES

Amazon S3 Overview S3 Lifecycle Configuration ❌ Why the Others Are Incorrect Service Why
It's Incorrect Amazon Glacier Archival storage — not real-time access AWS Storage Gateway
Hybrid storage bridge — not native object management Amazon EBS Block storage for EC2 —
not object-based or lifecycle-aware

REINFORCING WITH STORAGE STRATEGY

Let's map AWS storage services by use case: -

Service

Type Best For -

S3

Object General-purpose storage, versioning, lifecycle -

Glacier

Archival Long-term, infrequent access -

EBS

Block EC2 root volumes and databases -

EFS

File Shared file systems across EC2 -

Storage Gateway

Hybrid On-prem to cloud integration

REAL-WORLD TIE-IN

If your Nairobi-based team is storing user uploads: Use S3 with versioning to track changes
Apply lifecycle rules to move old files to Glacier Enable event notifications for Lambda triggers
Secure access with IAM policies and bucket encryption That's how AWS empowers teams to store, manage, and optimize object data with Amazon S3's lifecycle-aware architecture.

REFERENCES

<https://aws.amazon.com/s3/>

<https://aws.amazon.com/glacier/>

<https://aws.amazon.com/ebs/>

Question 406

**Which AWS service can securely store and manage source code versions?

**

OPTIONS

A. AWS CodeBuild 

B. AWS CodeCommit

C. AWS CodePipeline

D. AWS CodeStar

[!TIP]

Correct Answer:

B. AWS CodeCommit

EXPLANATION

CONCEPTUAL FRAMING

This question targets version control and secure code storage, especially how AWS enables teams to collaborate on source code with Git-based workflows inside a fully managed, encrypted environment.

WHY B IS CORRECT

AWS CodeCommit: Fully managed Git-based repository service Supports: Versioning and branching Access control via IAM Encryption at rest and in transit Integrates with: CodePipeline

for CI/CD CloudWatch Events for triggers CodeBuild for automated testing CodeCommit is your secure Git vault — scalable, integrated, and built for cloud-native development.

SOURCES

AWS CodeCommit Overview CodeCommit Security Features  Why the Others Are Incorrect Service Role Why It's Incorrect CodeBuild Build automation Doesn't store source code — pulls from repos CodePipeline CI/CD orchestration Manages workflows — not code storage CodeStar DevOps dashboard Aggregates tools — doesn't manage code directly

REINFORCING WITH DEVOPS STRATEGY

Let's map AWS developer tools by function: -

Tool

Function -

CodeCommit

Git repository and version control -

CodeBuild

Compile, test, and package code -

CodePipeline

Automate CI/CD workflows -

CodeDeploy

Deploy applications to EC2, Lambda, etc. -

CodeStar

Unified Dev Ops interface

REAL-WORLD TIE-IN

If your Nairobi-based team is building a microservices platform: Store each service in CodeCommit repos Trigger builds with CodeBuild on commit Orchestrate deployment with CodePipeline Enforce access with IAM policies and encryption That's how AWS empowers teams to securely manage source code and automate delivery pipelines using CodeCommit and its DevOps ecosystem.

Question 407

**Which AWS tool can be used to track planned infrastructure changes?

**

OPTIONS

A. AWS Personal Health Dashboard

B. AWS Trusted Advisor

C. Billing Dashboard →

D. AWS Config

[!TIP]

Correct Answer:

D. AWS Config

EXPLANATION

CONCEPTUAL FRAMING

This question targets configuration auditing and change tracking, especially how AWS enables teams to monitor, evaluate, and record changes to resource configurations over time — with compliance and automation hooks.

WHY D IS CORRECT

AWS Config: Tracks resource configuration history Detects planned and unplanned changes
Supports: Config Rules for compliance checks Change timelines and snapshots Drift detection
for CloudFormation stacks Integrates with: CloudTrail for API-level activity SNS and Lambda for
automated remediation AWS Config is your infrastructure historian — every change, every
timestamp, every compliance check.

SOURCES

AWS Config Overview Config Rules and Conformance Packs ❌ Why the Others Are Incorrect
Tool Why It's Incorrect Personal Health Dashboard Tracks AWS service health — not resource
changes Trusted Advisor Offers best practice checks — not change tracking Billing Dashboard
Monitors spend — not infrastructure configuration

REINFORCING WITH CHANGE MANAGEMENT STRATEGY

Let's map AWS tools by governance role: -

Tool

Role Best For -

AWS Config

Change tracking Resource history, compliance, drift detection -

CloudTrail

API logging Who did what, when -

Trusted Advisor

Optimization checks Cost, security, fault tolerance -

Service Catalog

Controlled provisioning Approved templates and products -

Control Tower

Multi-account governance Landing zones and guardrails

REAL-WORLD TIE-IN

If your Nairobi-based team is managing a regulated environment: Use AWS Config to track EC2, IAM, and S3 changes Apply Config Rules to enforce encryption and tagging policies Trigger Lambda remediation for non-compliant changes Export change logs to Athena or QuickSight for audit reports That's how AWS empowers teams to maintain visibility, control, and compliance across infrastructure changes using AWS Config.

REFERENCES

<https://aws.amazon.com/config/>

Question 408

**A company needs to design a disaster recovery plan that spans multiple geographic areas. Which AWS action meets this requirement?

**

OPTIONS

A. Configure multiple AWS accounts

B. Use multiple Availability Zones in one Region →

C. Configure architecture across multiple AWS Regions

D. Use many edge locations

[!TIP]

Correct Answer:

C. Configure architecture across multiple AWS Regions

EXPLANATION

CONCEPTUAL FRAMING

This question targets geo-resilience and fault isolation, especially how AWS enables teams to design architectures that withstand regional outages and ensure business continuity across continents.

WHY C IS CORRECT

Multi-Region architecture: Protects against Region-wide failures (e.g., natural disasters, major outages) Enables: Active-active or active-passive failover Data replication across Regions (e.g., S3 Cross-Region Replication, DynamoDB Global Tables) Global DNS routing via Route 53 Supports: RTO/RPO optimization Compliance with geographic data residency requirements Multi-Region is your resilience parachute — when one Region falters, another keeps you flying.

SOURCES

AWS Disaster Recovery Strategies Multi-Region Architecture Patterns  Why the Others Are Incorrect Option Why It's Incorrect

A.

Multiple accounts Useful for isolation — not geographic redundancy

B.

Multiple AZs in one Region Protects against AZ failure — not Region-wide outages

D.

Edge locations Used for CDN (CloudFront) — not infrastructure hosting or disaster recovery

REINFORCING WITH DR STRATEGY

Let's map AWS disaster recovery tiers: -

Tier

Scope Recovery Time Objective (RTO) -

Backup & Restore

Single Region Hours -

Pilot Light

Minimal standby in second Region Minutes to hours -

Warm Standby

Scaled-down replica in second Region Seconds to minutes -

Multi-Site Active/Active

Full deployment in multiple Regions Near-zero

REAL-WORLD TIE-IN

If your Nairobi-based team is building a critical fintech platform: Deploy core services in two AWS Regions (e.g., N. Virginia and Ireland) Use Route 53 failover routing for DNS-based recovery Replicate data with S3 CRR and DynamoDB Global Tables Automate failover with CloudFormation StackSets and Lambda That's how AWS empowers teams to design resilient, globally distributed architectures that survive regional failures and meet disaster recovery objectives.

Question 409

**Which are AWS-recommended security practices for managing an account's root user using IAM? (Choose two)

☒ Selected Options ☒ A. Set up multi-factor authentication (MFA) for the root user ☒ C.

Delete the root user access keys

**

[!TIP]

Correct Answer:

A and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets root account hardening, especially how AWS enables teams to minimize risk by securing the most privileged identity in the account with layered protections and strict usage boundaries. 🔍 Why A and C Are Correct

A.

Set up MFA for the root user: Adds a second layer of protection Prevents unauthorized access even if password is compromised Required for sensitive actions like: Changing support plans Closing the account Managing billing

C.

Delete root access keys: Root access keys are high-risk and cannot be protected by MFA If leaked (e.g., in a GitHub repo), they grant full control Best practice: delete or deactivate immediately The root user is your master key — lock it down, use it sparingly, and never leave it lying around.

SOURCES

AWS Root User Best Practices Security Checklist ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

Remove IAM policies from root Root user doesn't use IAM policies — it has full access by default

D.

Use root for daily tasks Violates least privilege — use IAM users or roles instead

E.

Assign read-only policy to root IAM policies don't apply to root — access is inherent

REINFORCING WITH ROOT GOVERNANCE STRATEGY

Let's map AWS root user best practices: -

Practice

Purpose -

Enable MFA

Prevent unauthorized access -

Delete access keys

Eliminate unprotected credentials -

Use IAM roles/users

Enforce least privilege -

Monitor with CloudTrail

Track root activity -

Restrict root usage

Only for account-level tasks

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding a new AWS account: Immediately enable MFA on the root user Delete any root access keys Create IAM users and roles for daily operations Monitor root activity with CloudTrail alerts That's how AWS empowers teams to secure the root user and enforce least privilege through IAM best practices.

Question 410

**Which AWS Support package is the least costly and provides 24/7 access to customer care and communities?

**

OPTIONS

A. AWS Enterprise Support

B. AWS Business Support

C. AWS Developer Support →

D. AWS Basic Support

[!TIP]

Correct Answer:

D. AWS Basic Support

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost-effective support access, especially how AWS enables teams to get foundational help and community engagement without incurring additional charges.

WHY D IS CORRECT

AWS Basic Support: Included free for all AWS accounts Provides: 24/7 access to customer service AWS documentation, whitepapers, and re:Post community Core Trusted Advisor checks

(e.g., service limits, security groups, S3 permissions) Ideal for: Startups and learners Non-critical workloads Exploratory or sandbox environments Basic Support is your always-on safety net — free, foundational, and ready to guide you.

SOURCES

AWS Support Plans Trusted Advisor Features **✗** Why the Others Are Incorrect Plan Why It's Incorrect Enterprise Support Premium tier — includes TAM, concierge, and architectural guidance Business Support Paid tier — includes technical support and full Trusted Advisor Developer Support Entry-level paid tier — includes guidance for development issues, not 24/7 customer care

REINFORCING WITH SUPPORT PLAN STRATEGY

Let's map AWS support tiers: -

Plan

Cost Key Features -

Basic

Free 24/7 customer service, re:Post, core Trusted Advisor -

Developer

Low Business hours tech support, general guidance -

Business

Medium 24/7 tech support, full Trusted Advisor, architectural guidance -

Enterprise

High TAM, concierge, proactive support, white-glove service

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a prototype: Start with Basic Support for free access to AWS docs and re:Post Use Trusted Advisor core checks to validate security and limits Upgrade to Business Support only when uptime and SLAs become critical That's how AWS empowers teams to scale support access based on workload maturity and budget using tiered support plans.

Question 411

**A firm doing online business needs to deliver new capabilities rapidly and iteratively. Which AWS Cloud function supports this need?

**

OPTIONS

A. Elasticity

B. High availability →

C. Agility

D. Reliability

[!TIP]

Correct Answer:

C. Agility

EXPLANATION

CONCEPTUAL FRAMING

This question targets deployment speed and responsiveness, especially how AWS enables teams to innovate faster, iterate quickly, and reduce time-to-market through cloud-native infrastructure and automation.

WHY C IS CORRECT

Agility in AWS means: Rapid provisioning of resources (e.g., EC2, Lambda, RDS) Quick experimentation and rollback Seamless scaling and integration with CI/CD pipelines Supports: DevOps workflows Microservices architecture Global deployment via CloudFormation and StackSets Cloud agility is your startup's superpower — deploy in minutes, pivot in seconds, and scale without friction.

SOURCES

AWS Cloud Benefits Agility in Cloud Computing ❌ Why the Others Are Incorrect Function Why It's Incorrect Elasticity Refers to scaling resources — not speed of deployment High availability Ensures uptime — not iteration speed Reliability Focuses on fault tolerance — not agility or responsiveness

REINFORCING WITH AGILITY STRATEGY

Let's map AWS features that boost agility: -

Feature

Role -

CloudFormation

Template-based infrastructure provisioning -

CodePipeline + CodeDeploy

CI/CD automation -

Lambda

Serverless execution for rapid iteration -

Elastic Beanstalk

Simplified app deployment -

Global infrastructure

Launch in multiple Regions instantly

REAL-WORLD TIE-IN

If your Nairobi-based team is iterating on a new e-commerce feature: Use Lambda and API Gateway for rapid backend deployment Automate testing and deployment with CodePipeline Launch globally with CloudFormation StackSets Monitor performance with CloudWatch and X-Ray That's how AWS empowers teams to deliver features faster, test ideas quickly, and stay ahead of the market using cloud agility.

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**

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[!TIP]

Correct Answer:

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Question 414

**To use the AWS CLI, what must a user generate?

**

OPTIONS

A. A password policy ➡

B. An access/secret key

C. A managed policy

D. An API key

[!TIP]

Correct Answer:

B. An access/secret key

EXPLANATION

CONCEPTUAL FRAMING

This question targets programmatic access and credential management, especially how AWS enables users to authenticate CLI sessions securely using IAM credentials.

WHY B IS CORRECT

Access key ID + secret access key: Generated from an IAM user or role
Used to sign CLI/API requests
Stored in: `~/.aws/credentials` file
Environment variables (`AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`)
Supports: CLI commands (e.g., `aws s3 ls`)
SDK integrations
Automation scripts and CI/CD pipelines
These keys are your CLI passport — secure them like root gold.

SOURCES

AWS CLI Configuration Guide IAM Best Practices ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Password policy Applies to console login — not CLI access

C.

Managed policy Defines permissions — doesn't authenticate CLI

D.

API key Not an AWS concept — AWS uses access/secret keys

REINFORCING WITH CLI CREDENTIAL STRATEGY

Let's map AWS CLI authentication options: -

Method

Use Case Security -

Access/Secret Key

IAM user or role (with rotation and MFA) -

Session Token

Temporary credentials (via STS) -

SSO Login

Federated access (via AWS SSO) -

Environment Variables

Automation (if rotated and scoped)

REAL-WORLD TIE-IN

If your Nairobi-based team is scripting S3 uploads: Create IAM users with scoped permissions
Generate access/secret keys Store them securely in AWS credentials file or environment variables
Rotate keys regularly and avoid hardcoding in Git That's how AWS empowers teams to securely automate cloud operations using CLI credentials and IAM best practices.

Question 415

****What's a key benefit of moving from an on-premises data center to the AWS Cloud?**

OPTIONS



- A. Compute instances can be launched and terminated as needed to optimize costs
- B. Compute costs can be viewed in the Billing console
- C. Users retain full admin access to instances
- D. Users optimize costs by permanently running peak-load instances

[!TIP]

Correct Answer:

- A. Launch and terminate compute instances as needed

EXPLANATION

CONCEPTUAL FRAMING

This question targets elastic infrastructure and cost efficiency, especially how AWS enables teams to scale compute resources dynamically based on demand — avoiding idle capacity and overprovisioning.

WHY A IS CORRECT

AWS EC2 and Lambda allow: On-demand provisioning — spin up instances in minutes Auto Scaling — match capacity to traffic Spot Instances and Savings Plans — optimize cost for flexible and predictable workloads Eliminates: Hardware procurement delays Fixed capacity planning Underutilized servers Cloud elasticity is your cost scalpel — slice away waste, deploy only what you need.

SOURCES

AWS Cloud Benefits EC2 Pricing Models **✗** Why the Others Are Incorrect Option Why It's Incorrect

B.

Billing console visibility Helpful — but not a migration benefit

C.

Full admin access True — but also applies to on-prem environments

D.

Permanent peak-load provisioning Wasteful — contradicts cloud cost optimization principles

REINFORCING WITH CLOUD MIGRATION STRATEGY

Let's map AWS benefits over on-prem: -

Benefit

AWS Advantage -

Elasticity

Scale up/down instantly -

Agility

Deploy in minutes, not weeks -

Global reach

Launch in any Region -

Security

Built-in compliance and encryption -

Cost optimization

Pay-as-you-go, no Cap Ex

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a seasonal analytics platform: Use Auto Scaling groups to handle traffic spikes Deploy Spot Instances for batch jobs Monitor usage with CloudWatch and Budgets Apply Savings Plans for predictable workloads That's how AWS empowers teams to optimize compute costs and accelerate deployment by replacing fixed infrastructure with elastic cloud resources.

Question 416

**Which AWS service monitors AWS accounts for security threats?

**

OPTIONS



A. Amazon GuardDuty

B. AWS Secrets Manager

C. Amazon Cognito

D. AWS Certificate Manager (ACM)

[!TIP]

Correct Answer:

A. Amazon GuardDuty

EXPLANATION

CONCEPTUAL FRAMING

This question targets threat detection and continuous monitoring, especially how AWS enables teams to identify suspicious activity across accounts, workloads, and data sources using machine learning and anomaly detection.

WHY A IS CORRECT

Amazon GuardDuty: A managed threat detection service Continuously monitors: VPC Flow Logs CloudTrail events DNS logs Detects: Credential compromise Reconnaissance activity Malicious API calls and data exfiltration Integrates with: Security Hub for centralized findings EventBridge and Lambda for automated remediation GuardDuty is your cloud sentinel — always watching, always learning, always alerting.

SOURCES

Amazon GuardDuty Overview GuardDuty Findings and Actions ❌ Why the Others Are Incorrect Service Role Why It's Incorrect Secrets Manager Credential storage Doesn't monitor threats Cognito Identity federation Manages users — not threat detection ACM Certificate management Handles SSL/TLS — not security monitoring

REINFORCING WITH SECURITY MONITORING STRATEGY

Let's map AWS services by security function: -

Service

Function -

GuardDuty

Threat detection and anomaly alerts -

Security Hub

Aggregated security findings -

Inspector

Vulnerability scanning -

Macie

Sensitive data discovery -

CloudTrail

API activity logging

REAL-WORLD TIE-IN

If your Nairobi-based team is running a multi-account architecture: Enable GuardDuty across all accounts via AWS Organizations Forward findings to Security Hub for centralized triage Automate responses with EventBridge and Lambda Use IAM Access Analyzer to detect risky permissions That's how AWS empowers teams to detect, investigate, and respond to threats across cloud environments using GuardDuty and its security ecosystem.

REFERENCES

<https://aws.amazon.com/guardduty/>

Question 417

**Which AWS solution enables customers to extend AWS infrastructure, services, APIs, and tools to on-premises environments?

**

OPTIONS

A. AWS Snowmobile

B. AWS Local Zones →

C. AWS Outposts

D. AWS Fargate

[!TIP]

Correct Answer:

C. AWS Outposts

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud deployment, especially how AWS enables teams to run cloud-native services within their own data centers while maintaining operational consistency with the AWS Region.

WHY C IS CORRECT

AWS Outposts: Delivers fully managed AWS infrastructure to on-premises locations Supports services like EC2, EBS, ECS, EKS, RDS, and S3 locally Uses the same: APIs and IAM policies Monitoring tools (CloudWatch, CloudTrail) Security controls and automation Ideal for: Low-latency workloads Data residency requirements Local system dependencies Outposts is your AWS Region in a rack — cloud power, local presence.

SOURCES

AWS Outposts Overview Hybrid Architecture Patterns **✗** Why the Others Are Incorrect Option
Why It's Incorrect Snowmobile Used for petabyte-scale data transfer — not hybrid infrastructure
Local Zones Extend AWS services to metro areas — not customer data centers Fargate
Serverless containers — doesn't run on-prem or extend infrastructure

REINFORCING WITH HYBRID STRATEGY

Let's map AWS hybrid solutions by use case: -

Service

Role On-Prem Integration -

Outposts

Full AWS stack -

Snowball Edge

Edge compute + storage -

Storage Gateway

Hybrid file/block/tape -

Direct Connect

Private network link -

Local Zones

Metro-based low-latency (AWS-managed)

REAL-WORLD TIE-IN

If your Nairobi-based team is running latency-sensitive workloads with local compliance: Deploy AWS Outposts in your data center Run EC2 and RDS locally with seamless AWS integration Use CloudFormation and IAM for consistent automation and access control Connect to the Region for backup, analytics, and global services That's how AWS empowers teams to extend cloud capabilities to on-prem environments using Outposts for hybrid agility and operational consistency.

Question 418

**Which AWS service allows consumers to audit API calls?

**

OPTIONS



A. AWS CloudTrail

B. AWS Trusted Advisor

C. Amazon Inspector

D. AWS X-Ray

[!TIP]

Correct Answer:

A. AWS CloudTrail

EXPLANATION

CONCEPTUAL FRAMING

This question targets auditability and operational transparency, especially how AWS enables teams to track every API call across services for security, compliance, and troubleshooting.

WHY A IS CORRECT

AWS CloudTrail: Records every API call made via: AWS Management Console AWS CLI SDKs and other AWS services Captures: Who made the call When it was made What resources were affected Stores logs in Amazon S3 for: Long-term retention Forensic analysis Compliance audits Integrates with: CloudWatch Logs for real-time alerting Athena for querying logs Security Hub for centralized findings CloudTrail is your audit ledger — every action, every actor, every timestamp.

SOURCES

AWS CloudTrail Overview CloudTrail Log Events **✗** Why the Others Are Incorrect Service Role Why It's Incorrect Trusted Advisor Best practice checks Doesn't log API activity Inspector

Vulnerability scanning Focuses on EC2 and container security — not API auditing X-Ray
Application tracing Visualizes service calls — not infrastructure-level API logs

REINFORCING WITH AUDIT STRATEGY

Let's map AWS services for security visibility: -

Service

Function -

CloudTrail

API activity logging -

Config

Resource configuration tracking -

GuardDuty

Threat detection -

Security Hub

Aggregated security findings -

IAM Access Analyzer

Permission boundary analysis

REAL-WORLD TIE-IN

If your Nairobi-based team is investigating a suspicious IAM policy change: Use CloudTrail to trace the API call Query logs with Athena for timestamp and user identity Set up CloudWatch alerts for sensitive actions (e.g., DeleteBucket, PutUserPolicy) Export logs to S3 with lifecycle policies for retention That's how AWS empowers teams to maintain full visibility and accountability across cloud operations using CloudTrail's audit trail.

REFERENCES

<https://aws.amazon.com/cloudtrail/>

Question 419

**Which AWS service enables centralized access management across multiple accounts?

**

OPTIONS

A. AWS Service Catalog

B. AWS Config

C. AWS Trusted Advisor 

D. AWS Organizations

[!TIP]

Correct Answer:

D. AWS Organizations

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account governance and policy enforcement, especially how AWS enables teams to centrally manage access, billing, and compliance across a growing cloud footprint.

WHY D IS CORRECT

AWS Organizations: Lets you create organizational units (OUs) to group accounts Apply Service Control Policies (SCPs) for permission boundaries Centralize: Billing via Consolidated

Billing Security controls Account creation and lifecycle management Integrates with: IAM for fine-grained access CloudTrail and Config for audit and compliance Control Tower for landing zone automation AWS Organizations is your cloud command center — one policy, many accounts, full control.

SOURCES

AWS Organizations Overview SCPs and Policy Management ❌ Why the Others Are Incorrect Service Role Why It's Incorrect Service Catalog Provisioning control Manages products — not account access AWS Config Resource tracking Tracks changes — doesn't manage access Trusted Advisor Optimization checks Offers guidance — not access control

REINFORCING WITH MULTI-ACCOUNT STRATEGY

Let's map AWS governance tools by function: -

Tool

Function -

Organizations

Centralized account and policy management -

SCPs

Permission boundaries across accounts -

IAM

Identity and access control -

Control Tower

Automated account setup and guardrails -

Config + CloudTrail

Audit and compliance tracking

REAL-WORLD TIE-IN

If your Nairobi-based team is scaling across departments: Use AWS Organizations to group accounts by team or function Apply SCPs to restrict risky actions (e.g., DeleteBucket, CreateUser) Enable Consolidated Billing for unified cost visibility Deploy Control Tower to automate secure account creation That's how AWS empowers teams to scale securely and manage access across environments using Organizations and its governance ecosystem.

REFERENCES

<https://aws.amazon.com/organizations/>

<https://aws.amazon.com/config/>

Question 420

**Which benefit is included with an AWS Enterprise Support plan?

**

OPTIONS

A. APN support at no cost →

B. Designated support from a Technical Account Manager (TAM)

C. On-site support from AWS engineers

D. Managed compliance as code with AWS Config

[!TIP]

Correct Answer:

B. Designated support from a TAM

EXPLANATION

CONCEPTUAL FRAMING

This question targets premium support and strategic guidance, especially how AWS enables enterprise customers to receive personalized technical oversight and proactive architectural advice through a dedicated TAM.

WHY B IS CORRECT

Technical Account Manager (TAM): Assigned to Enterprise Support customers Provides:
Proactive guidance on architecture, operations, and optimization Quarterly business reviews
Escalation management and incident coordination Acts as: Strategic advisor Single point of
contact for support and planning Bridge to AWS service teams A TAM is your cloud concierge
— guiding, optimizing, and advocating for your success.

SOURCES

AWS Enterprise Support Overview TAM Role Description ❌ Why the Others Are Incorrect
Option Why It's Incorrect

A.

APN support at no cost APN is a partner program — not a support plan feature

C.

On-site support Not included — AWS support is remote unless contracted separately

D.

Managed compliance as code AWS Config is a service — not a support plan benefit

REINFORCING WITH SUPPORT TIER STRATEGY

Let's map AWS support tiers by feature depth: -

Plan

TAM 24/7 Tech Support Architecture Guidance Cost Optimization -

Basic

-

Developer

Business hours -

Business

Limited -

Enterprise

Full

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a regulated fintech platform: Subscribe to Enterprise Support Get a TAM to guide architecture and compliance Use Well-Architected Reviews to validate design Coordinate incident response and escalation paths with AWS That's how AWS empowers teams to scale securely and strategically with personalized support through the Enterprise Support plan.

Question 421

**Which task does AWS perform automatically?

**

OPTIONS



A. Encrypt data stored in Amazon DynamoDB

B. Patch Amazon EC2 instances

C. Encrypt user network traffic

D. Create TLS certificates for user websites

[!TIP]

Correct Answer:

A. Encrypt data stored in Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question targets built-in security automation, especially how AWS enables teams to benefit from default encryption mechanisms that protect data at rest without manual configuration.

WHY A IS CORRECT

Amazon DynamoDB: Automatically encrypts all data at rest using AWS Key Management Service (KMS) No need for manual setup or key rotation Supports: Customer-managed keys (CMKs) AWS-managed keys Compliance with PCI DSS, HIPAA, and ISO standards DynamoDB encryption is always on — no switches, no scripts, just secure by design.

SOURCES

DynamoDB Encryption at Rest AWS KMS Integration  Why the Others Are Incorrect Option Why It's Incorrect

B.

Patch EC2 instances Requires manual setup via Systems Manager or automation scripts

C.

Encrypt user network traffic Depends on user configuration (e.g., TLS setup) — not automatic

D.

Create TLS certificates Requires user action via ACM — not automatic for websites

REINFORCING WITH SECURITY AUTOMATION STRATEGY

Let's map AWS services with built-in security automation: -

Service

Automatic Security Feature -

DynamoDB

Encryption at rest via KMS -

S3 (with default encryption)

Auto-encrypts new objects -

RDS

Optional encryption at rest with KMS -

CloudTrail

Automatically logs API activity -

IAM Access Analyzer

Continuously scans for risky permissions

REAL-WORLD TIE-IN

If your Nairobi-based team is building a serverless app with DynamoDB: Your data is automatically encrypted at rest You can choose AWS-managed or customer-managed KMS keys No need to write encryption logic or rotate keys manually Combine with IAM policies and VPC endpoints for secure access That's how AWS empowers teams to secure data by default and reduce operational overhead using automated encryption in DynamoDB.

Question 422

**Which AWS services provide computational capabilities? (Choose two)

☒ Selected Options ☒ A. Amazon EC2 ☒ E. AWS Lambda

**

[!TIP]

Correct Answer:

A and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets compute execution models, especially how AWS enables teams to run code and workloads using both traditional server-based and serverless architectures. 🔍 Why A and E Are Correct Amazon EC2 (Elastic Compute Cloud): Provides virtual servers (instances) Full control over OS, networking, and storage Ideal for: Long-running workloads Custom environments Legacy applications AWS Lambda: Serverless compute — runs code in response to events Automatically scales and charges only for execution time Ideal for: Microservices Event-driven architecture Lightweight automation EC2 is your customizable engine room; Lambda is your event-triggered ninja — both compute, different philosophies.

SOURCES

Amazon EC2 Overview AWS Lambda Overview ❌ Why the Others Are Incorrect Service Role Why It's Incorrect Amazon S3 Object storage Stores data — doesn't compute Amazon EBS Block storage Attaches to EC2 — doesn't compute independently Amazon Cognito Identity management Manages users — doesn't run code or workloads

REINFORCING WITH COMPUTE STRATEGY

Let's map AWS compute services by execution model: -

Service

Model Best For -

EC2

Server-based Custom environments, full control -

Lambda

Serverless Event-driven, short-lived tasks -

ECS/EKS

Container orchestration Microservices, portability -

Elastic Beanstalk

PaaS Simplified app deployment -

Fargate

Serverless containers Containerized workloads without managing servers

REAL-WORLD TIE-IN

If your Nairobi-based team is building a scalable web app: Use EC2 for backend services requiring persistent compute Use Lambda for image processing, notifications, or API triggers Combine with API Gateway, S3, and DynamoDB for a full-stack serverless architecture That's how AWS empowers teams to choose the right compute model for each workload — from full control to full abstraction.

REFERENCES

<https://aws.amazon.com/ec2/>

<https://aws.amazon.com/lambda/>

<https://aws.amazon.com/s3/>

<https://aws.amazon.com/ebs/>

Question 422

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☒ Selected Options ☒ A. Amazon EC2 ☒ E. AWS Lambda

**

[!TIP]

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EXPLANATION

CONCEPTUAL FRAMING

This question targets compute execution models, especially how AWS enables teams to run code and workloads using both traditional server-based and serverless architectures. 🔍 Why A and E Are Correct Amazon EC2 (Elastic Compute Cloud): Provides virtual servers (instances) Full control over OS, networking, and storage Ideal for: Long-running workloads Custom environments Legacy applications AWS Lambda: Serverless compute — runs code in response to events Automatically scales and charges only for execution time Ideal for: Microservices Event-driven architecture Lightweight automation EC2 is your customizable engine room; Lambda is your event-triggered ninja — both compute, different philosophies.

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Amazon EC2 Overview AWS Lambda Overview ❌ Why the Others Are Incorrect Service Role
Why It's Incorrect Amazon S3 Object storage Stores data — doesn't compute Amazon EBS
Block storage Attaches to EC2 — doesn't compute independently Amazon Cognito Identity
management Manages users — doesn't run code or workloads

REINFORCING WITH COMPUTE STRATEGY

Let's map AWS compute services by execution model: -

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Container orchestration Microservices, portability -

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PaaS Simplified app deployment -

Fargate

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REAL-WORLD TIE-IN

If your Nairobi-based team is building a scalable web app: Use EC2 for backend services requiring persistent compute Use Lambda for image processing, notifications, or API triggers Combine with API Gateway, S3, and DynamoDB for a full-stack serverless architecture That's how AWS empowers teams to choose the right compute model for each workload — from full control to full abstraction.

REFERENCES

<https://aws.amazon.com/ec2/>

<https://aws.amazon.com/lambda/>

<https://aws.amazon.com/s3/>

<https://aws.amazon.com/ebs/>

Question 423

**Which opportunities does AWS provide for clients interested in learning about cloud security in an instructor-led format? (Select two)

☒ Selected Options  B. AWS Online Tech Talks  E. AWS Classroom Training

**

[!TIP]

Correct Answer:

B and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets structured learning pathways, especially how AWS enables teams to gain cloud security expertise through interactive, expert-led sessions and technical deep dives. 🔍
Why B and E Are Correct AWS Online Tech Talks: Free, live-streamed sessions led by AWS experts Cover topics like: Security best practices Identity and access management Threat detection and compliance Include: Live Q&A Customer use cases Recorded replays for review

AWS Classroom Training: Instructor-led, immersive courses Delivered virtually or in-person
Cover: Security Engineering on AWS Architecting for Security Compliance and governance frameworks
Ideal for: Teams preparing for certification Deep technical skill development Tech Talks spark curiosity; Classroom Training builds mastery — both led by AWS experts.

SOURCES

AWS Training and Certification AWS Online Tech Talks ❌ Why the Others Are Incorrect Option
Why It's Incorrect

A.

Trusted Advisor Optimization tool — not a training resource

C.

AWS Blog Informative — but not instructor-led

D.

AWS Forums Community-driven — lacks structured instruction

REINFORCING WITH LEARNING STRATEGY

Let's map AWS learning formats by depth and delivery: -

Format

Type Best For -

Tech Talks

Live webinars Introductory and topical learning -

Classroom Training

Instructor-led Deep technical skill-building -

Digital Training

Self-paced Flexible foundational learning -

Certification Prep

Guided courses Exam readiness -

Workshops & Labs

Hands-on Practical experience

REAL-WORLD TIE-IN

If your Nairobi-based team is preparing for a security audit: Attend Tech Talks on IAM, GuardDuty, and encryption. Enroll in Classroom Training for Security Engineering on AWS. Use AWS Skill Builder to reinforce concepts. Track progress with certification paths and badges. That's how AWS empowers teams to build cloud security expertise through structured, expert-led learning formats tailored to every skill level.

Question 424

**A business wants to launch a global commercial app using EC2. What deployment offers the greatest redundancy and fault tolerance?

**

OPTIONS

A. Single Availability Zone in one Region

B. Multiple ENIs in different subnets

C. Multiple AZs in one Region →

D. Multiple AZs in two Regions

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets high availability and disaster resilience, especially how AWS enables teams to design globally distributed applications that withstand regional failures and maintain uptime across continents.

WHY D IS CORRECT

Multi-AZ, Multi-Region Deployment: Protects against: AZ-level failures (e.g., power outage, hardware fault) Region-level disruptions (e.g., natural disasters, major outages) Enables: Active-active or active-passive failover Latency-based routing via Route 53 Global load balancing with Elastic Load Balancer + Global Accelerator Ideal for: Mission-critical apps Global user bases Compliance with geographic redundancy requirements Multi-Region is your resilience blueprint — when one Region sleeps, another keeps your app alive.

SOURCES

AWS Global Infrastructure High Availability Architecture  Why the Others Are Insufficient
Option Limitation

A.

Single AZ Single point of failure — no redundancy

B.

Multiple ENIs Network-level — doesn't span AZs or Regions

C.

Multiple AZs in one Region AZ redundancy only — vulnerable to Region-wide outages

REINFORCING WITH RESILIENT ARCHITECTURE STRATEGY

Let's map EC2 deployment models by fault tolerance: -

Model

AZ Redundancy Region Redundancy Best For -

Single AZ

Dev/test -

Multi-AZ in one Region

Production workloads -

Multi-AZ in multiple Regions

Global, mission-critical apps

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a global e-commerce platform: Deploy EC2 instances in us-east-1 and eu-west-1 Use Route 53 latency-based routing Replicate data with Amazon S3 Cross-Region Replication Sync databases using Amazon Aurora Global Database That's how AWS empowers teams to build resilient, globally available applications using multi-AZ, multi-Region EC2 deployments.

Question 425

**Which tool is most suited for integrating the billing of previously distinct AWS accounts?

**

OPTIONS

A. Detailed billing report ➡

B. Consolidated billing

C. AWS Cost and Usage report

D. Cost allocation report

[!TIP]

Correct Answer:

B. Consolidated Billing

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account financial integration, especially how AWS enables organizations to streamline invoicing, optimize pricing, and centralize cost visibility across multiple accounts.

WHY B IS CORRECT

Consolidated Billing (via AWS Organizations): Combines usage from linked member accounts into a single monthly invoice Enables: Volume discounts across accounts RI and Savings Plan sharing Centralized payment and budget control Benefits: No additional cost Simplified chargeback reporting Downloadable cost and usage data Consolidated Billing is your financial unifier — one invoice, many accounts, maximum savings.

SOURCES

AWS Consolidated Billing AWS Organizations ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Detailed billing report Legacy format — replaced by Cost and Usage Report

C.

Cost and Usage Report (CUR) Provides granular data — doesn't integrate billing

D.

Cost allocation report Breaks down costs — doesn't unify invoices

REINFORCING WITH BILLING STRATEGY

Let's map AWS billing tools by function: -

Tool

Function -

Consolidated Billing

Unified invoicing across accounts -

CUR

Detailed usage and cost breakdown -

Budgets

Threshold alerts and tracking -

Cost Explorer

Visual spend analysis -

Cost Categories

Grouping and tagging for chargeback

REAL-WORLD TIE-IN

If your Nairobi-based team is managing multiple AWS accounts for dev, staging, and prod: Link accounts via AWS Organizations Enable Consolidated Billing for unified invoicing Share Savings Plans across environments Use CUR + Athena for detailed cost analysis That's how AWS empowers teams to simplify billing, optimize pricing, and maintain financial clarity across multi-account environments.

REFERENCES

Question 426

**Which AWS service or tool helps visualize, understand, and manage AWS spending and usage over time?

**

OPTIONS

A. AWS Trusted Advisor

B. Amazon CloudWatch →

C. Cost Explorer

D. AWS Budgets

[!TIP]

Correct Answer:

C. Cost Explorer

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost transparency and trend analysis, especially how AWS enables teams to track, visualize, and forecast cloud spending across services and accounts.

WHY C IS CORRECT

AWS Cost Explorer: Provides interactive dashboards to analyze: Historical spend Usage trends Forecasted costs Filters by: Service, account, region, tag, or linked account Supports: RI and Savings Plan utilization tracking Chargeback modeling Custom reports and exports Cost Explorer is your financial microscope — zoom into spend, spot trends, and steer budgets.

SOURCES

AWS Cost Explorer Overview Cost Explorer Use Cases  Why the Others Are Incorrect Tool Role Why It's Incorrect Trusted Advisor Optimization checks Flags cost risks — doesn't visualize spend CloudWatch Monitoring and alerts Tracks performance — not financial data Budgets Threshold alerts Sets limits — doesn't show historical usage or trends

REINFORCING WITH COST MANAGEMENT STRATEGY

Let's map AWS financial tools by function: -

Tool

Function -

Cost Explorer

Visualize and analyze spend over time -

Budgets

Set thresholds and receive alerts -

CUR (Cost and Usage Report)

Granular billing data for analysis -

Consolidated Billing

Unified invoicing across accounts -

Cost Categories

Group spend by team, project, or tag

REAL-WORLD TIE-IN

If your Nairobi-based team is tracking monthly S3 and EC2 costs: Use Cost Explorer to visualize usage by service and Region Filter by tags to isolate dev vs prod environments Forecast future spend based on current trends Export reports for chargeback or stakeholder reviews That's how AWS empowers teams to gain financial clarity and optimize cloud spend using Cost Explorer's visual analytics.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

Question 427

**How can deploying an application across several Availability Zones benefit you?

**

OPTIONS

- A. Lower risk from Region-wide disasters →
- B. Higher availability during AZ-level disruptions
- C. Wider geographic coverage
- D. Lower latency for users

[!TIP]

Correct Answer:

B. Higher availability because it can withstand a service disruption in one Availability Zone

EXPLANATION

CONCEPTUAL FRAMING

This question targets fault isolation and service continuity, especially how AWS enables teams to design resilient applications that survive infrastructure failures within a Region.

WHY B IS CORRECT

Availability Zones (AZs): Are physically separate data centers within a Region Connected via low-latency links Designed to fail independently Deploying across AZs: Ensures redundancy Enables failover and load balancing Minimizes impact of: Power outages Hardware failures Network disruptions Multi-AZ is your uptime insurance — when one zone falters, the others keep your app alive.

SOURCES

AWS Global Infrastructure High Availability Design  Why the Others Are Incorrect Option Why It's Incorrect

A.

Region-wide disaster protection Requires multi-Region — AZs are within one Region

C.

Wider geographic coverage AZs are close together — not designed for global reach

D.

Lower latency AZs don't reduce latency — edge services like CloudFront do

REINFORCING WITH AVAILABILITY STRATEGY

Let's map AWS deployment models by fault tolerance: -

Model

AZ Redundancy Region Redundancy Best For -

Single AZ

Dev/test -

Multi-AZ in one Region

Production workloads -

Multi-Region

Global, mission-critical apps

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a customer-facing app: Use Auto Scaling groups across multiple AZs Deploy Elastic Load Balancer to distribute traffic Store data in multi-AZ RDS or DynamoDB Monitor health with CloudWatch and Route 53 That's how AWS empowers teams to build resilient applications that maintain availability even when one AZ experiences issues.

Question 428

**Which AWS feature enables customers to control billing expense allocations?

**

OPTIONS



A. Tagging resources

B. Limiting who can create resources

C. Adding a secondary payment method

D. Running all operations on a single AWS account

[!TIP]

Correct Answer:

A. Tagging resources

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost attribution and chargeback modeling, especially how AWS enables teams to assign business context to cloud resources for granular billing and accountability.

WHY A IS CORRECT

AWS Resource Tags: Key-value pairs like: Project:Analytics Team:DevOps CostCenter:Finance
Applied to: EC2, S3, Lambda, RDS, etc. Up to 50 tags per resource Enable: Cost allocation reports Chargeback dashboards Budget tracking by team or project Tags are your billing compass — they point every dollar to its rightful owner.

SOURCES

AWS Tagging Best Practices Cost Allocation Tags **✗** Why the Others Are Incorrect Option Why It's Incorrect

B.

Limiting resource creation Controls access — doesn't allocate costs

C.

Secondary payment method Not supported — AWS uses a single payment method per account

D.

Single account operations Centralizes billing — but lacks granularity for allocations

REINFORCING WITH BILLING ALLOCATION STRATEGY

Let's map AWS cost attribution tools: -

Tool

Function -

Tags

Assign business context to resources -

Cost Categories

Group tagged resources for reporting -

Budgets

Track spend by tag or category -

CUR (Cost and Usage Report)

Export detailed usage by tag -

QuickSight

Visualize cost data by tag or team

REAL-WORLD TIE-IN

If your Nairobi-based team is running multiple projects: Tag resources with Project, Team, and Environment Activate Cost Allocation Tags in the billing console Use CUR + Athena to generate chargeback reports Visualize spend by tag in QuickSight dashboards That's how AWS empowers teams to track, allocate, and optimize cloud spend using resource tagging and cost attribution tools.

Question 429

******A business is hosting an app in one AWS Region but serving global users. Which service helps deliver low-latency access to application data?

OPTIONS



A. Amazon CloudFront

B. AWS Direct Connect

C. Amazon Route 53 global DNS

D. Amazon S3 Transfer Acceleration

[!TIP]

Correct Answer:

A. Amazon CloudFront

EXPLANATION

CONCEPTUAL FRAMING

This question targets content delivery and edge optimization, especially how AWS enables teams to serve global users with minimal latency by caching and routing content through a worldwide network of edge locations.

WHY A IS CORRECT

Amazon CloudFront: A Content Delivery Network (CDN) with hundreds of edge locations worldwide Caches content close to users for: Lower latency Reduced origin load Improved responsiveness Supports: Custom logic via Lambda@Edge TLS encryption and WAF

integration Streaming, APIs, and dynamic content delivery CloudFront is your global accelerator — it brings your app's data to the edge of the world.

SOURCES

Amazon CloudFront Overview Edge Computing with Lambda@Edge **✗** Why the Others Are Incorrect Option Why It's Incorrect

B.

Direct Connect Private link — improves bandwidth, not global latency

C.

Route 53 DNS Routes traffic — doesn't cache or accelerate content

D.

S3 Transfer Acceleration Speeds up S3 uploads — not general app delivery

REINFORCING WITH GLOBAL PERFORMANCE STRATEGY

Let's map AWS services for global delivery: -

Service

Role -

CloudFront

CDN for low-latency content delivery -

Global Accelerator

Static IP routing for TCP/UDP apps -

Route 53

DNS routing based on latency/geolocation -

Lambda@Edge

Custom logic at edge locations -

S3 Transfer Acceleration

Fast uploads to S3 via edge points

REAL-WORLD TIE-IN

If your Nairobi-based team is serving a global e-learning platform: Host the app in us-east-1 Use CloudFront to cache assets and APIs at edge locations Add Lambda@Edge for custom headers or redirects Monitor performance with CloudWatch and Real User Monitoring That's how AWS empowers teams to deliver fast, responsive applications to global users using CloudFront's edge network.

Question 430

**A business has an application with variable start and finish times. Which EC2 pricing option is most cost-effective?

**

OPTIONS

A. On-Demand Instances →

B. Spot Instances

C. Reserved Instances

D. Dedicated Hosts

[!TIP]

Correct Answer:

B. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization for flexible workloads, especially how AWS enables teams to leverage unused EC2 capacity at steep discounts for interruptible, non-critical tasks.

WHY B IS CORRECT

Spot Instances: Offer up to 90% discount compared to On-Demand pricing
Ideal for: Batch processing
Data analysis
CI/CD pipelines
Rendering and simulations
Can be: Interrupted with 2-minute warning
Replaced or re-queued automatically
Managed via: Auto Scaling groups
EC2 Fleet Spot Instance Advisor
Spot Instances are your compute bargain bin — perfect for workloads that bend with the cloud's rhythm.

SOURCES

Amazon EC2 Spot Instances Spot Instance Best Practices  Why the Others Are Less Suitable
Option Limitation

A.

On-Demand Flexible but expensive — pay full price per hour

C.

Reserved Instances Cost-effective for predictable workloads — not ideal for variable timing

D.

Dedicated Hosts Used for licensing compliance — high cost, not optimized for flexible workloads

REINFORCING WITH EC2 PRICING STRATEGY

Let's map EC2 pricing models by workload type: -

Model

Best For Cost Efficiency -

On-Demand

Unpredictable, short-term  -

Spot

Flexible, fault-tolerant    -

Reserved

Steady-state, long-term   -

Dedicated Hosts

BYOL, compliance  

REAL-WORLD TIE-IN

If your Nairobi-based team is running a nightly data crunch job: Use Spot Instances with Auto Scaling and EC2 Fleet Monitor interruptions with CloudWatch Events Re-queue failed jobs using Step Functions or Lambda Save up to 90% compared to On-Demand That's how AWS empowers teams to scale compute affordably for flexible workloads using Spot Instances.

REFERENCES

<https://aws.amazon.com/ec2/>

Question 431

**A business is developing an app to store and retrieve millions of photos and videos. Which AWS service offers the cheapest underlying storage?

**

OPTIONS

A. Amazon EC2 instance store

B. Amazon EBS →

C. Amazon S3

D. Amazon SQS

[!TIP]

Correct Answer:

C. Amazon S3

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost-effective object storage, especially how AWS enables teams to store massive volumes of unstructured data with high durability and low cost using Amazon S3.

WHY C IS CORRECT

Amazon S3 (Simple Storage Service): Designed for scalable object storage Ideal for: Photos, videos, backups, logs, documents Offers: 11 nines durability (99.999999999%) Tiered pricing: Standard Intelligent-Tiering Glacier / Glacier Deep Archive Features: Lifecycle policies for automatic tiering Event-driven triggers via S3 Events Global access via CloudFront S3 is your digital vault — cheap, durable, and built to scale with your data.

SOURCES

Amazon S3 Pricing S3 Storage Classes ❌ Why the Others Are Incorrect Option Why It's Incorrect EC2 Instance Store Ephemeral — data lost when instance stops EBS (Elastic Block Store) Persistent — but more expensive per GB than S3 SQS (Simple Queue Service) Messaging — not designed for data storage

REINFORCING WITH STORAGE STRATEGY

Let's map AWS storage services by use case and cost: -

Service

Use Case Cost Efficiency -

S3

Object storage 💰💰💰 -

EBS

Block storage for EC2 💰💰 -

EFS

Shared file system 💰 -

Glacier

Archival storage 💰💰💰💰 -

Instance Store

Temporary scratch space (non-persistent)

REAL-WORLD TIE-IN

If your Nairobi-based team is building a media-sharing app: Store user uploads in Amazon S3 Standard Use Intelligent-Tiering to auto-optimize costs Archive old content to S3 Glacier Deep Archive Serve content globally via CloudFront That's how AWS empowers teams to store massive media libraries affordably and reliably using S3's tiered object storage.

REFERENCES

<https://aws.amazon.com/s3/>

Question 432

******A business is hosting a large e-commerce app on AWS. Which services help protect against network-based threats like DDoS attacks? (Select two)

☐ Selected Options ☒ C. Amazon CloudFront ☒ D. AWS Shield

[!TIP]

Correct Answer:

C and D

EXPLANATION

CONCEPTUAL FRAMING

This question targets network-layer threat mitigation, especially how AWS enables teams to defend against volumetric attacks and malicious traffic using edge services and managed protection layers. 🔍 Why C and D Are Correct Amazon CloudFront: A global CDN that distributes content through edge locations Acts as a shielding layer between users and origin servers Integrates with: AWS WAF for custom traffic filtering Shield Standard for automatic DDoS protection Benefits: Latency reduction Traffic absorption and filtering Custom logic via Lambda@Edge AWS Shield: Managed DDoS protection service Two tiers: Shield Standard — automatic protection for all AWS customers Shield Advanced — enhanced detection, cost protection, and 24/7 response team Protects: CloudFront, Route 53, ALB, and EC2 CloudFront absorbs the blast; Shield deflects the damage — together they form AWS's DDoS defense perimeter.

SOURCES

AWS Shield Overview CloudFront Security Features ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Amazon Inspector Scans for vulnerabilities — doesn't block network threats

B.

GuardDuty Detects threats — doesn't mitigate them

E.

IAM Manages identity — not network-layer protection

REINFORCING WITH DDOS DEFENSE STRATEGY

Let's map AWS services by security layer: -

Layer

Service Role -

Edge

Cloud Front + WAF Traffic filtering and caching -

Network

AWS Shield DDoS mitigation -

Detection

Guard Duty Threat intelligence -

Identity

IAM Access control -

Vulnerability

Inspector Resource scanning

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a high-traffic e-commerce site: Serve content via CloudFront to absorb traffic and reduce latency Enable AWS Shield Standard for automatic DDoS protection Add AWS WAF for custom rules (e.g., block IPs, rate-limit requests) Monitor threats with GuardDuty and automate responses via EventBridge That's how AWS empowers teams to build resilient, secure applications that withstand network-based attacks using layered protection strategies.

REFERENCES

<https://aws.amazon.com/shield/>

Question 433

**Which AWS service or functionality enables control of application traffic between Regions?

**

OPTIONS

A. Amazon AppStream 2.0

B. Amazon VPC

C. Elastic Load Balancer →

D. Amazon Route 53

[!TIP]

Correct Answer:

D. Amazon Route 53

EXPLANATION

CONCEPTUAL FRAMING

This question targets global traffic management, especially how AWS enables teams to route user requests across Regions based on latency, geography, or custom logic using DNS-level control.

WHY D IS CORRECT

Amazon Route 53: A highly available and scalable DNS service Supports routing policies such as: Geoproximity — route based on user and resource location Latency-based — send users to the fastest Region Failover — redirect traffic during outages Weighted — distribute traffic proportionally Integrates with: CloudFront Global Accelerator Health checks for failover Route 53 is your global traffic conductor — directing users to the best Region with precision.

SOURCES

Route 53 Routing Policies Geoproximity Routing ❌ Why the Others Are Incorrect Option Why It's Incorrect AppStream 2.0 Delivers desktop apps — not traffic routing VPC Manages networking — not DNS-level traffic control Elastic Load Balancer Balances traffic within a Region — not across Regions

REINFORCING WITH ROUTING STRATEGY

Let's map AWS traffic control tools by scope: -

Tool

Scope Role -

Route 53

Global DNS-based routing across Regions -

Global Accelerator

Global Static IP routing for TCP/UDP apps -

ELB

Regional Load balancing within a Region -

CloudFront

Global CDN with edge caching and routing -

VPC Peering

Regional Network connectivity — not traffic control

REAL-WORLD TIE-IN

If your Nairobi-based team is serving users in Africa, Europe, and Asia: Use Route 53 geoproximity routing to steer traffic to the nearest Region Combine with CloudFront for edge caching Set up failover policies to reroute traffic during outages Monitor health with Route 53 health checks That's how AWS empowers teams to deliver fast, resilient applications globally using intelligent traffic routing with Route 53.

Question 434

**A company needs an isolated environment within AWS for security purposes. Which action accomplishes this?

**

OPTIONS

- A. Create a separate Availability Zone →
- B. Create a separate VPC
- C. Create a placement group
- D. Create an AWS Direct Connect connection

[!TIP]

Correct Answer:

- B. Create a separate VPC

EXPLANATION

CONCEPTUAL FRAMING

This question targets network-level isolation, especially how AWS enables teams to build logically separated environments for workloads, teams, or compliance boundaries using Virtual Private Clouds (VPCs).

WHY B IS CORRECT

Amazon VPC (Virtual Private Cloud): Creates a logically isolated network within the AWS Cloud
You control: IP address ranges Subnets (public/private) Route tables, NAT gateways, internet gateways Security groups and NACLs Enables: Workload separation Environment isolation (e.g., dev vs prod) Private-only access with no internet exposure A VPC is your fortress — custom walls, private gates, and full control over who enters.

SOURCES

Amazon VPC Overview VPC Security Best Practices ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Separate AZ AZs are part of a Region — not isolated from each other

C.

Placement Group Controls instance placement — not network isolation

D.

Direct Connect Provides private connectivity — doesn't isolate environments within AWS

REINFORCING WITH ISOLATION STRATEGY

Let's map AWS isolation tools by scope: -

Tool

Scope Role -

VPC

Logical network Full isolation and control -

Subnets

Tiered access Public vs private segmentation -

Security Groups

Instance-level Stateful traffic filtering -

NACLs

Subnet-level Stateless traffic control -

PrivateLink

Service access Secure API calls without internet

REAL-WORLD TIE-IN

If your Nairobi-based team is building a fintech app: Create separate VPCs for dev, staging, and prod Use private subnets for databases and sensitive services Connect VPCs via VPC Peering or Transit Gateway Use PrivateLink to access AWS services securely That's how AWS empowers teams to build secure, isolated environments tailored to workload and compliance needs using VPCs.

REFERENCES

<https://aws.amazon.com/vpc/>

Question 435

**What expenses should be addressed when comparing the total cost of ownership (TCO) of on-premises infrastructure vs cloud architecture? (Select two)

☒ Selected Options ☐ B. Cost of purchasing and installing server hardware ☐ C. Cost of administering infrastructure (OS, patches, backups, recovery)

**

[!TIP]

Correct Answer:

B and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets comprehensive cost modeling, especially how AWS enables teams to evaluate both capital and operational expenses when migrating from on-premises to cloud infrastructure. 🔍 Why B and C Are Correct

B.

Server Hardware Costs: Includes: Capital expenditure (CapEx) for servers, racks, cooling, power Installation and maintenance labor Cloud replaces this with pay-as-you-go compute and storage

C.

Infrastructure Administration Costs: Includes: OS and software installation Security patching and updates Backup and disaster recovery Monitoring and troubleshooting Cloud shifts this to managed services (e.g., RDS, Lambda, S3) and automation tools (e.g., Systems Manager) TCO isn't just about hardware — it's the hidden labor, risk, and downtime you eliminate with cloud.

SOURCES

AWS TCO Calculator Cloud Economics Center ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

Credit card fees Business-specific — not infrastructure-related

D.

Penetration testing costs Optional and external — not core to TCO

E.

Advertising costs Irrelevant to infrastructure — marketing expense

REINFORCING WITH TCO STRATEGY

Let's map key TCO components for on-prem vs cloud: -

Category

On-Premises Cloud -

CapEx

Hardware, facilities None -

OpEx

Admin labor, power, cooling Usage-based billing -

Security

Manual patching, audits Built-in controls and automation -

Scalability

Slow, costly Instant, elastic -

Resilience

Manual failover Multi-AZ, auto-healing

REAL-WORLD TIE-IN

If your Nairobi-based team is evaluating cloud migration: Use the AWS TCO Calculator to compare server, labor, and downtime costs Factor in automation savings from managed services Include resilience and scalability benefits in your ROI model Present findings to stakeholders with QuickSight dashboards That's how AWS empowers teams to make informed decisions by comparing full-spectrum costs across infrastructure models.

Question 436

****Which AWS service provides a managed Hadoop framework for processing massive data volumes across EC2 instances?**

OPTIONS



A. Amazon EMR

B. Amazon EC2

C. AWS Elastic Beanstalk

D. Amazon Redshift

[!TIP]

Correct Answer:

A. Amazon EMR (Elastic MapReduce)

EXPLANATION

CONCEPTUAL FRAMING

This question targets scalable big data processing, especially how AWS enables teams to run distributed analytics workloads using open-source frameworks like Hadoop, Spark, Hive, and Presto — without managing infrastructure manually.

WHY A IS CORRECT

Amazon EMR: Managed cluster platform for: Hadoop, Spark, Hive, HBase, Presto, Flink
Automatically provisions: EC2 instances Storage via S3 or HDFS Networking and security configurations Supports: Auto-scaling Spot instance integration Per-second billing Ideal for: Log analysis ETL pipelines Machine learning preprocessing EMR is your data refinery — it turns raw logs into insights with elastic, cost-effective compute.

SOURCES

Amazon EMR Overview EMR Architecture ❌ Why the Others Are Incorrect Option Why It's Incorrect EC2 Raw compute — no Hadoop orchestration or data pipeline management Elastic Beanstalk App deployment — not designed for big data frameworks Redshift Data warehouse — optimized for SQL analytics, not Hadoop-style processing

REINFORCING WITH BIG DATA STRATEGY

Let's map AWS services by data processing model: -

Service

Model Best For -

EMR

Distributed compute Hadoop/Spark workloads -

Glue

Serverless ETL Data cataloging and transformation -

Redshift

Columnar SQL BI and analytics -

Athena

Serverless SQL Querying S3 data -

Kinesis

Streaming Real-time ingestion and processing

REAL-WORLD TIE-IN

If your Nairobi-based team is analyzing clickstream data: Use Amazon EMR with Spark to process logs Store raw data in Amazon S3 Use Spot Instances for cost savings Output results to Redshift or QuickSight for visualization That's how AWS empowers teams to process massive datasets quickly and affordably using EMR's managed Hadoop ecosystem.

Question 437

**What is the best place to find AWS compliance reports and security documentation?

**

OPTIONS



A. AWS Artifact

B. AWS Marketplace

C. Amazon Inspector

D. AWS Support

[!TIP]

Correct Answer:

A. AWS Artifact

EXPLANATION

CONCEPTUAL FRAMING

This question targets compliance transparency and audit access, especially how AWS enables teams to retrieve official documentation for regulatory, legal, and security assurance purposes.

WHY A IS CORRECT

AWS Artifact: Centralized portal for: Security and compliance reports Audit artifacts (e.g., SOC 2, ISO 27001, PCI DSS) Online agreements (e.g., Business Associate Addendum for HIPAA) On-demand access to: Third-party attestations AWS control implementation details Ideal for: Auditors, compliance officers, and legal teams Risk assessments and vendor due diligence AWS Artifact is your compliance library — curated, official, and always audit-ready.

SOURCES

AWS Artifact Overview Artifact Documentation Access ❌ Why the Others Are Incorrect Option Why It's Incorrect Marketplace Sells third-party products — not compliance documents Inspector Scans for vulnerabilities — doesn't provide compliance reports Support Helps with issues — doesn't host audit documentation

REINFORCING WITH COMPLIANCE STRATEGY

Let's map AWS compliance tools by function: -

Tool

Function -

Artifact

Access official compliance reports -

Security Hub

Centralize findings from multiple services -

Inspector

Scan for vulnerabilities in EC2 and containers -

Audit Manager

Automate evidence collection for audits -

IAM Access Analyzer

Detect risky permissions and access paths

REAL-WORLD TIE-IN

If your Nairobi-based team is preparing for a PCI DSS audit: Download SOC 2 and PCI reports from AWS Artifact Use Audit Manager to automate evidence collection Validate infrastructure with Inspector and Security Hub Document IAM boundaries with Access Analyzer That's how AWS empowers teams to navigate compliance confidently using Artifact and its ecosystem of security and audit tools.

REFERENCES

<https://aws.amazon.com/artifact/>

Question 438

**A business wants to be alerted when AWS expenses or usage exceed certain limits. Which AWS service meets this need?

**

OPTIONS



A. AWS Budgets

B. Cost Explorer

C. AWS CloudTrail

[!TIP]

Correct Answer:

A. AWS Budgets

EXPLANATION

CONCEPTUAL FRAMING

This question targets financial control and proactive alerting, especially how AWS enables teams to monitor spend, forecast usage, and trigger notifications when thresholds are breached.

WHY A IS CORRECT

AWS Budgets: Lets you set: Cost budgets (actual or forecasted) Usage budgets (e.g., EC2 hours, S3 GB) RI/Savings Plan utilization thresholds Alerts via: Email Amazon SNS Ideal for: Preventing budget overruns Tracking team or project spend Automating cost governance workflows AWS Budgets is your financial watchdog — always watching, always warning, before costs spiral.

SOURCES

AWS Budgets Overview Budget Alerts Setup  Why the Others Are Incorrect Option Why It's Incorrect Cost Explorer Visualizes spend — doesn't trigger alerts CloudTrail Logs API activity — not cost-related Macie Detects sensitive data — unrelated to billing

REINFORCING WITH BUDGET STRATEGY

Let's map AWS cost control tools by function: -

Tool

Function -

Budgets

Set thresholds and receive alerts -

Cost Explorer

Analyze spend and forecast trends -

CUR (Cost and Usage Report)

Export detailed billing data -

Cost Categories

Group spend by tag or team -

QuickSight

Visualize budget performance and anomalies

REAL-WORLD TIE-IN

If your Nairobi-based team is managing multiple projects: Set monthly budgets per team using tags Trigger SNS alerts when usage exceeds thresholds Monitor RI utilization to avoid waste Integrate alerts with Slack or Jira for workflow automation That's how AWS empowers teams to stay financially disciplined and responsive using AWS Budgets' alerting and tracking capabilities.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-budgets/>

Question 438

****Which AWS service is a highly available and scalable DNS web service?**

OPTIONS

A. Amazon VPC

B. Amazon CloudFront →

C. Amazon Route 53

D. Amazon Connect

[!TIP]

Correct Answer:

C. Amazon Route 53

EXPLANATION

CONCEPTUAL FRAMING

This question targets domain name resolution and traffic routing, especially how AWS enables teams to connect users to applications with high availability, low latency, and intelligent routing policies.

WHY C IS CORRECT

Amazon Route 53: A highly available and scalable DNS service Supports: Domain registration
DNS record management Traffic routing policies: Latency-based Geoproximity Weighted
Failover Integrates with: CloudFront Global Accelerator Health checks for failover automation
Route 53 is your traffic maestro — orchestrating global requests with precision and resilience.

SOURCES

Amazon Route 53 Overview Routing Policy Documentation ❌ Why the Others Are Incorrect
Option Why It's Incorrect Amazon VPC Manages private networking — not DNS resolution
CloudFront CDN for content delivery — not DNS service Amazon Connect Contact center
platform — unrelated to DNS or routing

REINFORCING WITH DNS STRATEGY

Let's map AWS services by traffic and DNS roles: -

Service

Role -

Route 53

DNS resolution and traffic routing -

CloudFront

CDN for caching and delivery -

Global Accelerator

Static IP routing for TCP/UDP apps -

VPC

Private networking and subnet management

REAL-WORLD TIE-IN

If your Nairobi-based team is hosting a global app: Register your domain with Route 53 Use latency-based routing to direct users to the fastest Region Configure failover policies for high availability Combine with CloudFront for edge delivery That's how AWS empowers teams to deliver fast, reliable, and intelligently routed applications using Route 53's DNS capabilities.

REFERENCES

<https://aws.amazon.com/route53/>

<https://aws.amazon.com/vpc/>

Question 440

**A business needs to monitor changes to AWS resource configurations for compliance. Which AWS functionality fulfills this requirement?

**

OPTIONS

A. AWS Cost and Usage Report

B. AWS Organizations SCPs →

C. AWS Config Rules

D. VPC Flow Logs

[!TIP]

Correct Answer:

C. AWS Config Rules

EXPLANATION

CONCEPTUAL FRAMING

This question targets configuration drift detection and compliance auditing, especially how AWS enables teams to track resource changes, enforce governance policies, and maintain audit trails across cloud environments.

WHY C IS CORRECT

AWS Config: Continuously records configuration changes across supported AWS resources
Supports Config Rules: Managed rules (e.g., encryption, tagging, public access) Custom rules via Lambda Enables: Compliance dashboards Change history and relationships Automated remediation workflows AWS Config is your cloud historian — it remembers every change and flags every violation.

SOURCES

AWS Config Overview Config Rules Documentation  Why the Others Are Incorrect Option
Why It's Incorrect Cost and Usage Report Tracks spend — not resource configuration
Organizations SCPs Enforce permissions — don't monitor changes VPC Flow Logs Capture network traffic — not config state

REINFORCING WITH COMPLIANCE STRATEGY

Let's map AWS tools by compliance and monitoring role: -

Tool

Role -

AWS Config

Track and evaluate resource configurations -

CloudTrail

Log API activity and user actions -

Security Hub

Aggregate security findings across services -

Audit Manager

Automate evidence collection for audits -

IAM Access Analyzer

Detect risky access paths and permissions

REAL-WORLD TIE-IN

If your Nairobi-based team is building a regulated healthcare app: Use AWS Config to enforce encryption and tagging policies Trigger SNS alerts when non-compliant resources are detected Integrate with CloudTrail for full change attribution Export compliance reports for auditors via Audit Manager That's how AWS empowers teams to maintain continuous compliance and visibility using Config's rule-based monitoring and change tracking.

REFERENCES

<https://aws.amazon.com/config/>

Question 441

**A customer wants advice on potential cost reductions when migrating from on-premises to AWS. Which tool is most appropriate?

**

OPTIONS

A. AWS Budgets

B. Cost Explorer 

C. AWS Total Cost of Ownership (TCO) Calculator

D. AWS Well-Architected Tool

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets pre-migration financial modeling, especially how AWS enables teams to quantify savings by comparing on-premises infrastructure costs with cloud-based alternatives.

WHY C IS CORRECT

AWS TCO Calculator: Estimates cost savings across: Compute, storage, networking, and IT labor Facility and energy costs Inputs include: Server specs, utilization rates, licensing, and support costs Outputs: 5-year cost comparison Directional savings estimates Downloadable reports for stakeholders The TCO Calculator is your financial lens — it reveals the hidden costs of on-prem and the value of cloud.

SOURCES

AWS TCO Calculator Cloud Economics Center  Why the Others Are Incorrect Option Why It's Incorrect Budgets Tracks spend post-migration — doesn't estimate savings Cost Explorer Visualizes AWS usage — not comparative modeling Well-Architected Tool Evaluates architecture — not financial impact

REINFORCING WITH MIGRATION STRATEGY

Let's map AWS tools by migration phase: -

Phase

Tool Role -

Pre-migration

TCO Calculator Estimate cost savings -

Planning

Migration Evaluator Deep workload analysis -

Execution

AWS Application Migration Service Lift-and-shift automation -

Post-migration

Cost Explorer + Budgets Track and optimize spend -

Governance

Well-Architected Tool Ensure best practices and resilience

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a legacy ERP system: Use the TCO Calculator to compare server, storage, and admin costs Present savings to finance and leadership Plan migration with Migration Evaluator and Well-Architected Tool Track post-migration spend with Budgets and Cost Explorer That's how AWS empowers teams to make informed, cost-effective migration decisions using the TCO Calculator's financial modeling capabilities.

Question 442

**Which AWS services or functionalities enable customers to establish a network connection between two VPCs? (Select two)

☐ Selected Options ☒ C. VPC Peering ☒ E. AWS Transit Gateway

**

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets inter-VPC connectivity, especially how AWS enables teams to build scalable, secure, and manageable network architectures across accounts and Regions. 🔍

Why C and E Are Correct VPC Peering: Creates a direct connection between two VPCs

Supports: Same or different accounts Same or different Regions Limitations: No transitive routing Manual route table updates Best for: Simple, point-to-point VPC links AWS Transit

Gateway: Acts as a central hub for connecting multiple VPCs and on-prem networks Supports:

Transitive routing Scalable architecture Cross-account and cross-Region peering Ideal for:

Enterprise-scale networks Multi-VPC mesh topologies Peering is your direct handshake; Transit Gateway is your network switchboard.

SOURCES

VPC Peering Guide Transit Gateway Overview ❌ Why the Others Are Incorrect Option Why It's Incorrect

A.

VPC Endpoints Connect to AWS services — not VPCs

B.

Route 53 DNS routing — not network connectivity

D.

Direct Connect Connects on-prem to AWS — not VPC-to-VPC

REINFORCING WITH VPC CONNECTIVITY STRATEGY

Let's map AWS options for VPC-to-VPC networking: -

Method

Transitive Routing Cross-Account Cross-Region Best For -

VPC Peering

Simple links -

Transit Gateway

Scalable mesh -

PrivateLink

Service-level access -

VPN + Direct Connect

Hybrid connectivity

REAL-WORLD TIE-IN

If your Nairobi-based team is managing separate VPCs for dev, staging, and prod: Use Transit Gateway to centralize routing and simplify management Use VPC Peering for lightweight, direct links between isolated workloads Apply resource sharing via AWS RAM for cross-account access Monitor traffic with VPC Flow Logs and CloudWatch That's how AWS empowers teams to build secure, scalable, and maintainable multi-VPC architectures using peering and Transit Gateway.

Question 443

**Which is a recommended method for setting IAM user policies?

**

OPTIONS

A. Start with a large set of permissions and remove the ones not required

B. Use only Amazon managed policies 

C. Start with a minimum set of permissions and grant more as needed

D. Attach policies directly to each user individually

[!TIP]

Correct Answer:

C. Start with a minimum set of permissions and grant additional as necessary

EXPLANATION

CONCEPTUAL FRAMING

This question targets principle of least privilege, especially how AWS enables teams to minimize attack surface, reduce blast radius, and maintain audit-friendly access control by granting only what's needed — and nothing more.

WHY C IS CORRECT

Least Privilege Principle: Start with zero or minimal permissions Add only what's required for the task Benefits: Limits accidental exposure Reduces insider risk Improves audit clarity Best practices: Use groups and roles Prefer managed policies for common tasks Use inline policies sparingly and only when scoped tightly Least privilege is your IAM scalpel — precise, safe, and surgical in access control.

SOURCES

IAM Best Practices Least Privilege Guidance ❌ Why the Others Are Risky Option Risk

A.

Start with large permissions Overexposure — hard to audit and prune safely

B.

Use only managed policies May be too broad — lacks fine-grained control

D.

Attach policies to users directly Hard to manage — violates scalable IAM design

REINFORCING WITH IAM STRATEGY

Let's map IAM policy hygiene by practice: -

Practice

Description Risk Level -

Least privilege

Grant only what's needed Secure -

Group-based access

Assign policies to groups Scalable -

Role-based access

Use roles for services and cross-account Auditable -

Inline policies

Use sparingly for tight scope ⚠ Risky if overused -

User-attached policies

Avoid — hard to manage Fragile

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding new engineers: Assign users to IAM groups with scoped policies Use AWS managed policies as starting templates Add custom policies only when needed Audit permissions quarterly with Access Analyzer and IAM Access Advisor That's how AWS empowers teams to build secure, scalable, and audit-friendly IAM environments using least privilege and policy hygiene best practices.

Question 444

**How can consolidated billing benefit a business with many AWS accounts?

**

OPTIONS



A. Aggregates usage across accounts to reach volume discounts sooner

B. Offers an extra 5% discount on All Upfront RIs

C. Simplifies invoice processing

D. Enables AWS resellers to bill customers

[!TIP]

Correct Answer:

A. Aggregates usage across accounts to reach volume discounts sooner

EXPLANATION

CONCEPTUAL FRAMING

This question targets multi-account cost optimization, especially how AWS enables organizations to pool usage across accounts to unlock pricing tiers, share Reserved Instance benefits, and streamline financial governance.

WHY A IS CORRECT

Consolidated Billing via AWS Organizations: Combines usage from all linked accounts into a single bill Benefits: Volume discounts on services like S3, EC2, and data transfer RI and

Savings Plan sharing across accounts Centralized cost tracking and forecasting Ideal for: Enterprises with multiple teams or business units Startups managing dev/staging/prod accounts separately Consolidated billing is your financial amplifier — it turns fragmented usage into unified savings.

SOURCES

AWS Consolidated Billing AWS Organizations Overview  Why the Others Are Incorrect Option Why It's Incorrect

B.

5% RI discount No such fixed discount — RI pricing depends on term and payment option

C.

Simplified invoice True, but not the greatest benefit

D.

Reseller billing Applies to AWS Partner Network — not standard consolidated billing

REINFORCING WITH BILLING STRATEGY

Let's map AWS billing tools by function: -

Tool

Function -

Organizations

Manage accounts and enable consolidated billing -

Budgets

Set thresholds and receive alerts -

Cost Explorer

Visualize usage and forecast spend -

CUR (Cost and Usage Report)

Export detailed billing data -

Savings Plans

Flexible pricing for compute usage

REAL-WORLD TIE-IN

If your Nairobi-based team has separate accounts for dev, prod, and analytics: Link them under AWS Organizations Use consolidated billing to pool usage and unlock volume discounts Share Reserved Instances across accounts Monitor spend with Cost Explorer and Budgets That's how AWS empowers teams to maximize cost efficiency and simplify financial management using consolidated billing.

REFERENCES

<https://aws.amazon.com/organizations/>

Question 445

**Which is an AWS best practice for managing the AWS account root user?

**

OPTIONS

- A. Keep the root user password with the security team →
- B. Enable multi-factor authentication (MFA) for the root user
- C. Create an access key for the root user
- D. Keep the root user password consistent for compliance

[!TIP]

Correct Answer:

B. Enable MFA for the root user

EXPLANATION

CONCEPTUAL FRAMING

This question targets identity hardening and privileged access control, especially how AWS enables teams to protect the most powerful account credentials with layered authentication and minimal exposure.

WHY B IS CORRECT

Multi-Factor Authentication (MFA): Adds a second layer of protection beyond username and password Can use: Virtual MFA apps (e.g., Authy, Google Authenticator) Hardware tokens (e.g., YubiKey) Critical for: Root user — has full access to all resources Break-glass scenarios — e.g., account recovery, billing access MFA is your root user's shield — even if the password leaks, the fortress stays locked.

SOURCES

AWS Root User Best Practices Enabling MFA for Root **✗** Why the Others Are Risky Option Risk

A.

Share password with team Violates least privilege — root should be rarely used

C.

Create access key for root Strongly discouraged — increases exposure risk

D.

Keep password consistent Compliance ≠ security — rotation and MFA are better practices

REINFORCING WITH IDENTITY STRATEGY

Let's map AWS identity best practices for root user: -

Practice

Description -

Enable MFA

Strongest protection against unauthorized access -

Avoid root usage

Use IAM roles and users for daily tasks -

Delete root access keys

Prevent programmatic access via root -

Monitor with CloudTrail

Track root activity for audit and alerts -

Secure password storage

Use vaults like AWS Secrets Manager or external tools

REAL-WORLD TIE-IN

If your Nairobi-based team is setting up a new AWS Organization: Enable MFA for the root user immediately Create admin IAM roles for daily operations Delete any root access keys Use CloudTrail and GuardDuty to monitor root activity That's how AWS empowers teams to protect the most privileged credentials with layered security and strict usage boundaries.

Question 446

**What is the effect of AWS's pay-as-you-go pricing model?

**

OPTIONS



- A. Reduces capital expenditures
- B. Requires upfront payment
- C. Applies only to EC2, S3, and RDS
- D. Reduces operational expenditures

[!TIP]

Correct Answer:

- A. Reduces capital expenditures

EXPLANATION

CONCEPTUAL FRAMING

This question targets financial transformation through cloud adoption, especially how AWS enables teams to shift from large upfront investments to flexible, usage-based spending aligned with business demand.

WHY A IS CORRECT

Pay-as-you-go model: Eliminates need for: Data center hardware purchases Facility setup and maintenance Long-term licensing commitments Converts CapEx → OpEx: Pay only for what you use Scale up/down based on demand Avoid idle capacity and sunk costs Ideal for: Startups avoiding upfront risk Enterprises seeking agility and cost control AWS turns infrastructure into a utility — no upfront costs, just metered innovation.

SOURCES

AWS Pricing Overview Cloud Economics Center ❌ Why the Others Are Incorrect Option Why It's Incorrect

B.

Requires upfront payment False — most services are billed per use

C.

Applies only to EC2, S3, RDS False — applies across AWS (Lambda, DynamoDB, etc.)

D.

Reduces OpEx Not inherently — OpEx may increase with poor usage hygiene

REINFORCING WITH COST STRATEGY

Let's map AWS pricing models by flexibility and commitment: -

Model

Description Cap Ex Impact -

On-Demand

Pay per hour/second Eliminates Cap Ex -

Spot Instances

Use spare capacity at discount -

Savings Plans

Commit to usage for discount -

Reserved Instances

Prepay for capacity -

Free Tier

Limited usage at no cost

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new SaaS platform: Use On-Demand EC2 and Lambda to avoid upfront costs Monitor usage with AWS Budgets and Cost Explorer Scale infrastructure only when traffic grows Present CapEx savings to stakeholders using the TCO Calculator That's how AWS empowers teams to innovate without financial friction by replacing capital investment with elastic, usage-based pricing.

Question 447

**Which AWS feature relates to a customer's flexibility to scale applications up and down with changing demand?

**

OPTIONS



A. Elasticity

B. Agility

C. Security

D. Scalability

[!TIP]

Correct Answer:

A. Elasticity

EXPLANATION

CONCEPTUAL FRAMING

This question targets dynamic resource allocation, especially how AWS enables teams to automatically adjust infrastructure capacity to match real-time demand — without manual intervention or performance degradation.

WHY A IS CORRECT

Elasticity: Refers to the ability to automatically scale resources up or down
Driven by: Auto Scaling Groups, Lambda concurrency, Fargate task scaling
Benefits: Cost efficiency — pay only for what you use
Performance assurance — meet demand spikes without overprovisioning
Operational simplicity — no manual resizing
Elasticity is your cloud reflex — it expands and contracts with demand, keeping costs and performance in sync.

SOURCES

AWS Auto Scaling Elasticity vs Scalability  Why the Others Are Misleading Option Why It's Incorrect

B.

Agility Refers to speed of innovation — not resource resizing

C.

Security Protects data and access — unrelated to scaling

D.

Scalability Refers to capacity potential, not dynamic adjustment — scalability is static, elasticity is reactive  Elasticity vs Scalability Cheat Sheet Concept Definition Example Elasticity Auto-adjust resources based on demand EC2 Auto Scaling adds instances during traffic spike Scalability Ability to grow system capacity RDS scales vertically to larger instance type

REAL-WORLD TIE-IN

If your Nairobi-based team is running a flash sale: Use Auto Scaling to elastically add EC2 instances Use Lambda for serverless elasticity Monitor with CloudWatch alarms Set scaling

policies based on CPU or request count That's how AWS empowers teams to handle unpredictable demand with elastic infrastructure that grows and shrinks automatically.

REFERENCES

<https://aws.amazon.com/autoscaling/>

Question 448

**A company wants to improve security and audit posture by limiting EC2 inbound access. What should they use instead of opening SSH ports and managing keys?

**

OPTIONS

A. EC2 key pairs →

B. AWS Systems Manager Session Manager

C. IAM

D. Network ACLs

[!TIP]

Correct Answer:

B. AWS Systems Manager Session Manager

EXPLANATION

CONCEPTUAL FRAMING

This question targets secure remote access, especially how AWS enables teams to manage EC2 instances without exposing them to the internet or juggling SSH keys and bastion hosts.

WHY B IS CORRECT

Session Manager (part of AWS Systems Manager): Provides browser-based or CLI shell access to EC2 instances Requires: SSM Agent installed IAM permissions No open inbound ports Benefits: No need for SSH or bastion hosts Fully auditable via CloudTrail and SSM logs Works with private subnets and VPC endpoints Session Manager is your stealth console — secure, silent, and fully logged.

SOURCES

Session Manager Overview Best Practices for EC2 Access  Why the Others Are Risky or Insufficient Option Why It's Incorrect EC2 Key Pairs Still requires open SSH ports — not ideal for hardened environments IAM Manages identity — doesn't provide shell access Network ACLs Controls traffic — doesn't enable access or auditing

REINFORCING WITH ACCESS STRATEGY

Let's map EC2 access methods by security posture: -

Method

Inbound Ports Key Management Auditability Best For -

SSH with key pairs

Required Manual  Limited Legacy setups -

Bastion host

Required Shared keys  Complex Multi-account access -

Session Manager

Not required IAM-based Cloud Trail + S3 logs Secure, modern access

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying EC2 instances in private subnets: Use Session Manager for shell access Block port 22 in security groups Enable CloudTrail logging for session activity Use IAM roles to control who can start sessions That's how AWS empowers teams to secure EC2 access with zero exposure and full auditability using Session Manager.

Question 449

**After selecting an Amazon EC2 Dedicated Host reservation, which pricing option provides the largest discount?

**

OPTIONS

A. No upfront payment

B. Hourly on-demand payment

C. Partial upfront payment →

D. All upfront payment

[!TIP]

Correct Answer:

D. All upfront payment

EXPLANATION

CONCEPTUAL FRAMING

This question targets long-term cost efficiency, especially how AWS enables teams to maximize savings by committing to capacity with upfront payment models for Reserved Instances and Dedicated Hosts.

WHY D IS CORRECT

All Upfront Payment: Pay the full cost of the reservation term (typically 1 or 3 years) at the start
Offers the largest discount compared to: On-Demand pricing Partial upfront No upfront monthly payments
Ideal for: Predictable workloads Stable environments (e.g., prod databases, compliance-bound apps) Budget-conscious teams with CapEx flexibility All upfront is your cost scalpel — it slices through long-term spend with precision and commitment.

SOURCES

Reserved Instance Pricing Dedicated Host Reservations ✗ Why the Others Are Less Cost-Effective Option Why It's Less Optimal No upfront Highest total cost — pay monthly with minimal discount Hourly on-demand Most flexible — but no discount at all Partial upfront Some savings — but less than full upfront commitment

REINFORCING WITH PRICING STRATEGY

Let's map EC2 pricing models by flexibility and savings: -

Model

Flexibility Discount Level Best For -

On-Demand

High None Unpredictable workloads -

Spot Instances

Highest Deep Fault-tolerant tasks -

Reserved Instances (All Upfront)

Low Highest Steady-state workloads -

Savings Plans (All Upfront)

REAL-WORLD TIE-IN

If your Nairobi-based team is running a compliance-bound analytics engine on EC2: Use Dedicated Hosts for licensing and isolation Choose All Upfront Reserved Host pricing for maximum savings Track utilization with Cost Explorer and Budgets Present savings to finance using TCO Calculator That's how AWS empowers teams to optimize long-term infrastructure costs with strategic payment models like All Upfront Reserved pricing.

Question 450

**Which AWS Cloud feature enables resource supply to match changing workload demands?

**

OPTIONS

A. Security

B. Reliability →

C. Elasticity

D. High availability

[!TIP]

Correct Answer:

C. Elasticity

EXPLANATION

CONCEPTUAL FRAMING

This question targets dynamic infrastructure responsiveness, especially how AWS enables teams to automatically adjust compute, storage, and networking resources based on real-time demand — without overprovisioning or manual intervention.

WHY C IS CORRECT

Elasticity: Refers to the ability to provision and de-provision resources automatically Driven by: Auto Scaling Groups Lambda concurrency Fargate task scaling Benefits: Cost efficiency — pay only for what you use Performance assurance — meet spikes without degradation Operational agility — no manual resizing or downtime Elasticity is your cloud reflex — it stretches and contracts with demand, keeping performance and cost in harmony.

SOURCES

AWS Elasticity Explained Auto Scaling Overview  Why the Others Are Misleading Option Why It's Incorrect Security Protects data and access — unrelated to resource scaling Reliability Ensures uptime and fault tolerance — not dynamic provisioning High availability Keeps services running across failures — doesn't adjust capacity  Elasticity vs Related Concepts Concept Definition Example Elasticity Auto-adjust resources based on demand EC2 Auto Scaling adds instances during traffic spike Scalability Ability to grow system capacity RDS scales vertically to larger instance type High Availability Redundant systems to avoid downtime Multi-AZ deployment for failover Reliability Consistent performance and recovery S3 durability and replication

REAL-WORLD TIE-IN

If your Nairobi-based team is hosting a seasonal e-commerce app: Use Auto Scaling to elastically add EC2 instances during peak traffic Use Lambda for serverless elasticity Monitor with CloudWatch alarms and metrics Set scaling policies based on CPU, memory, or request

count That's how AWS empowers teams to handle unpredictable demand with elastic infrastructure that grows and shrinks automatically.

Question 451

**A business is shifting production workloads to AWS. Which activities help reduce operating expenses during migration? (Select two)

☒ Selected Options ➡ A. Reduce overprovisioned instances ➡ D. Use managed services

**

[!TIP]

Correct Answer:

A and D

EXPLANATION

CONCEPTUAL FRAMING

This question targets operational cost reduction, especially how AWS enables teams to right-size infrastructure and offload undifferentiated heavy lifting to managed services — both of which reduce labor, waste, and complexity. 🔍 Why A and D Are Correct

A.

Reduce Overprovisioned Instances: On-prem environments often overprovision for peak load AWS allows: Right-sizing with Cost Explorer and Compute Optimizer Auto Scaling to match demand dynamically Spot Instances for fault-tolerant workloads Result: Lower compute costs and better utilization

D.

Use Managed Services: Offload ops to AWS-managed offerings like: RDS (databases) Lambda (serverless compute) S3 (storage) Elastic Beanstalk (app deployment) Benefits: No patching,

backups, or scaling logic Reduced admin overhead and staffing costs Together, these strategies turn infrastructure into a lean, automated machine — optimized for cost and performance.

SOURCES

AWS Compute Optimizer AWS Managed Services  Why the Others Are Less Cost-Focused Option Why It's Not Cost-Reducing

B.

Rehost third-party licenses May increase costs — licensing varies by vendor

C.

Highly available architecture Improves resilience — may increase resource usage

E.

Improve application security Critical for risk — not directly tied to cost savings

REINFORCING WITH MIGRATION STRATEGY

Let's map AWS migration activities by cost impact: -

Activity

Cost Impact Description -

Right-sizing

High Match instance size to workload -

Managed services

High Reduce ops and maintenance -

License optimization

 Variable Depends on vendor and model -

Availability architecture

Neutral/Negative Adds redundancy -

Security hardening

Indirect Reduces risk, not cost

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a legacy ERP system: Use Compute Optimizer to downsize EC2 instances Replace self-managed MySQL with Amazon RDS Use Lambda for event-driven tasks Monitor spend with AWS Budgets and Cost Explorer That's how AWS empowers teams to cut operating costs while improving agility and resilience through right-sizing and managed services.

REFERENCES

<https://aws.amazon.com/compute-optimizer/>

Question 452

**A business wants to give a single user full access to an Amazon S3 bucket. Which element in the bucket policy defines who gets access?

**

OPTIONS



A. Principal

B. Action

C. Resource

D. Statement

[!TIP]

Correct Answer:

A. Principal

EXPLANATION

CONCEPTUAL FRAMING

This question targets fine-grained access control in S3 bucket policies, especially how AWS enables teams to specify exactly who (user, role, account) is granted permissions to perform actions on resources.

WHY A IS CORRECT

Principal: Defines the identity (user, role, account, service) that the policy applies to Can be: IAM user or role ARN AWS account ID Service principal (e.g., lambda.amazonaws.com) Used in: Bucket policies IAM policies Resource-based policies The principal is your access recipient — the named entity that gets the keys to the bucket.

SOURCES

S3 Bucket Policy Elements IAM Policy Reference **✗** Why the Others Are Misleading Element Role Why It's Incorrect Action What the principal can do Doesn't define who gets access Resource What the action applies to Doesn't define who gets access Statement Container for policy logic Doesn't define who — it holds the principal, action, and resource

REINFORCING WITH POLICY ANATOMY

Here's a breakdown of a typical S3 bucket policy:

```
json { "Version": "2012-10-17", "Statement": [ { "Effect": "Allow", "Principal": { "AWS": "arn:aws:iam::123456789012:user/Leonard" }, "Action": "s3:", "Resource": "arn:aws:s3:::my-bucket" } ] } -
```

Element

Meaning -

Principal

Who gets access (Leonard) -

Action

What they can do (s3:) - *Resource* What they can access (*my-bucket/*) -

Effect

Allow or Deny

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding a data engineer: Use a bucket policy with a Principal ARN for their IAM user Grant s3:* only on the specific bucket Monitor access with CloudTrail and S3 Access Logs Rotate credentials and enforce MFA for IAM users That's how AWS empowers teams to securely delegate access using precise Principal definitions in bucket policies.

Question 453

**Which AWS solution enables clients to acquire unused EC2 capacity at reduced prices?

**

OPTIONS

A. Reserved Instances

B. On-Demand Instances

C. Dedicated Instances 

[!TIP]

Correct Answer:

D. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization for compute workloads, especially how AWS enables teams to leverage spare EC2 capacity at steep discounts — ideal for fault-tolerant, flexible applications.

WHY D IS CORRECT

Spot Instances: Use unused EC2 capacity at up to 90% discount compared to On-Demand
Ideal for: Batch jobs CI/CD pipelines Big data processing Stateless microservices Can be:
Interrupted with 2-minute warning Managed via EC2 Auto Scaling and Spot Fleet Spot
Instances are your compute bargain bin — high performance at a fraction of the cost, if you can handle volatility.

SOURCES

Spot Instances Overview Spot Fleet and Auto Scaling  Why the Others Are Less Cost-Effective for This Use Case Option Why It's Incorrect Reserved Instances Discounted — but for predictable workloads, not unused capacity On-Demand Instances Flexible — but full price Dedicated Instances Isolated — but expensive and not discounted for spare capacity

REINFORCING WITH EC2 PRICING STRATEGY

Let's map EC2 options by cost and flexibility: -

Model

Discount Level Interruption Risk Best For -

On-Demand

None None Unpredictable workloads -

Reserved Instances

Moderate-High None Steady-state workloads -

Savings Plans

Moderate-High None Flexible compute usage -

Spot Instances

Highest Yes Fault-tolerant, flexible workloads

REAL-WORLD TIE-IN

If your Nairobi-based team is running nightly data processing jobs: Use Spot Instances for compute-heavy tasks Set up Auto Scaling with interruption handling Monitor savings with Cost Explorer and Budgets Combine with Savings Plans for predictable baseline usage That's how AWS empowers teams to maximize compute efficiency by tapping into unused EC2 capacity with Spot pricing.

Question 453

**Which AWS solution enables clients to acquire unused EC2 capacity at reduced prices?

**

OPTIONS

A. Reserved Instances

B. On-Demand Instances

C. Dedicated Instances →

D. Spot Instances

[!TIP]

Correct Answer:

D. Spot Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost optimization for compute workloads, especially how AWS enables teams to leverage spare EC2 capacity at steep discounts — ideal for fault-tolerant, flexible applications.

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SOURCES

Spot Instances Overview Spot Fleet and Auto Scaling ✗ Why the Others Are Less Cost-Effective for This Use Case Option Why It's Incorrect Reserved Instances Discounted — but for predictable workloads, not unused capacity On-Demand Instances Flexible — but full price Dedicated Instances Isolated — but expensive and not discounted for spare capacity

REINFORCING WITH EC2 PRICING STRATEGY

Let's map EC2 options by cost and flexibility: -

Model

Discount Level Interruption Risk Best For -

On-Demand

None None Unpredictable workloads -

Reserved Instances

Moderate-High None Steady-state workloads -

Savings Plans

Moderate-High None Flexible compute usage -

Spot Instances

Highest Yes Fault-tolerant, flexible workloads

REAL-WORLD TIE-IN

If your Nairobi-based team is running nightly data processing jobs: Use Spot Instances for compute-heavy tasks Set up Auto Scaling with interruption handling Monitor savings with Cost Explorer and Budgets Combine with Savings Plans for predictable baseline usage That's how AWS empowers teams to maximize compute efficiency by tapping into unused EC2 capacity with Spot pricing.

Question 455

**Which procedure should a client follow when conducting penetration testing on AWS?

**

OPTIONS

- A. Use Amazon Inspector, then notify AWS →
- B. Request and wait for approval from internal security team
- C. Notify AWS, then test immediately
- D. Request and wait for approval from AWS

[!TIP]

Correct Answer:

- B. Request and wait for approval from the customer's internal security team

EXPLANATION

CONCEPTUAL FRAMING

This question targets security assessment governance, especially how AWS enables teams to conduct penetration testing responsibly within defined service boundaries — without violating terms of service or triggering abuse alerts.

WHY B IS CORRECT

AWS Penetration Testing Policy: As of recent updates, no prior AWS approval is required for penetration testing on specific services: EC2, RDS, CloudFront, Aurora, API Gateway, Lambda, Lightsail, Elastic Beanstalk, Fargate However: Internal security teams must authorize and coordinate testing Testing must comply with AWS Acceptable Use Policy Must not target other customers or shared infrastructure AWS trusts you to test — but your own security team must greenlight the mission.

SOURCES

AWS Penetration Testing Policy Acceptable Use Policy ❌ Why the Others Are Misleading Option Why It's Incorrect

A.

Use Amazon Inspector Inspector is for automated vulnerability scanning — not manual pen testing

C.

Notify AWS and test immediately Notification is no longer required — but internal approval is

D.

Request AWS approval Obsolete — AWS no longer requires approval for listed services

REINFORCING WITH TESTING STRATEGY

Let's map AWS services by pen test approval status: -

Service

Approval Needed? -

EC2, RDS, Lambda, CloudFront, Aurora, API Gateway, Fargate, Lightsail, Elastic Beanstalk

No AWS approval needed -

Other services or cross-tenant testing

Approval required -

Inspector, GuardDuty, Macie

Automated scanning — not manual pen testing

REAL-WORLD TIE-IN

If your Nairobi-based team is planning a red team exercise: Confirm scope with internal security and legal teams Limit testing to approved AWS services Avoid denial-of-service or phishing simulations Log and monitor activity with CloudTrail and GuardDuty That's how AWS empowers teams to test their defenses responsibly while maintaining compliance and protecting shared infrastructure.

Question 456

**Which AWS service is responsible for network connection in a hybrid design that incorporates AWS?

**

OPTIONS

A. Amazon VPC 

B. AWS Direct Connect

C. AWS Directory Service

D. Amazon API Gateway

[!TIP]

Correct Answer:

B. AWS Direct Connect

EXPLANATION

CONCEPTUAL FRAMING

This question targets hybrid cloud architecture, especially how AWS enables enterprises to establish secure, high-throughput, low-latency connections between on-premises infrastructure and AWS environments.

WHY B IS CORRECT

AWS Direct Connect: Provides a dedicated physical connection from your data center to AWS
Benefits: Consistent network performance Lower latency than VPN Reduced bandwidth costs
Use cases: Hybrid workloads Data-intensive applications Compliance-sensitive environments
Direct Connect is your fiber bridge to the cloud — fast, private, and built for enterprise-grade throughput.

SOURCES

AWS Direct Connect Overview Hybrid Architecture Best Practices ❌ Why the Others Are Misleading Option Why It's Incorrect Amazon VPC Defines private cloud network — doesn't connect to on-prem directly AWS Directory Service Manages identity — not network connectivity Amazon API Gateway Manages APIs — not hybrid network links

REINFORCING WITH HYBRID CONNECTIVITY STRATEGY

Let's map AWS hybrid connection options: -

Method

Type Best For -

Direct Connect

Physical High-throughput, low-latency, secure workloads -

Site-to-Site VPN

Overlay Quick setup, encrypted traffic -

Transit Gateway

Logical hub Multi-VPC and hybrid routing -

SD-WAN via Partner Gateway

Software-defined Dynamic routing and edge integration

REAL-WORLD TIE-IN

If your Nairobi-based team is integrating an on-prem analytics cluster with AWS: Use Direct Connect for consistent, high-speed data transfer Pair with Transit Gateway for routing across multiple VPCs Monitor traffic with VPC Flow Logs and CloudWatch Secure with IAM policies and NACLs That's how AWS empowers teams to build robust hybrid architectures using Direct Connect as the backbone for secure, scalable network integration.

REFERENCES

<https://aws.amazon.com/vpc/>

Question 457

**What is the name given to the several, separate sites inside an AWS Region that are linked by low-latency networks?

**

OPTIONS

A. AWS Direct Connects

B. Amazon VPCs

C. Edge locations →

D. Availability Zones

[!TIP]

Correct Answer:

D. Availability Zones

EXPLANATION

CONCEPTUAL FRAMING

This question targets regional fault isolation and high availability, especially how AWS enables teams to deploy resilient applications across multiple physically separated data centers within a Region.

WHY D IS CORRECT

Availability Zones (AZs): Each AZ is a distinct physical location within a Region Features: Independent power, cooling, and networking Low-latency links to other AZs in the same Region Redundant infrastructure for fault isolation Benefits: High availability via multi-AZ deployments Disaster recovery within Region Scalable architecture with failover support AZs are your regional building blocks — isolated for safety, interconnected for speed.

SOURCES

AWS Global Infrastructure Availability Zone Design ❌ Why the Others Are Misleading Option Why It's Incorrect AWS Direct Connects Physical link from on-prem to AWS — not internal topology Amazon VPCs Logical network boundaries — not physical sites Edge Locations CDN endpoints — used for latency reduction, not AZ-level deployment

REINFORCING WITH INFRASTRUCTURE STRATEGY

Let's map AWS infrastructure layers by role: -

Component

Role -

Region

Geographic boundary for data sovereignty and disaster recovery -

Availability Zone

Fault-isolated deployment zone within a Region -

Edge Location

Global CDN and DNS acceleration point -

Local Zone

Low-latency compute extension near users

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a multi-tier web app: Use multi-AZ EC2 and RDS deployments for resilience Place Elastic Load Balancer across AZs for failover Monitor latency and health with CloudWatch and Route 53 That's how AWS empowers teams to build fault-tolerant, high-performance applications using Availability Zones as the backbone of regional architecture.

REFERENCES

<https://aws.amazon.com/vpc/>

Question 459

**Which EC2 pricing option is best for intermittent, spiky, or unpredictable workloads?

**

OPTIONS



A. Spot Instances

B. Dedicated Hosts 

C. On-Demand Instances

D. Reserved Instances

[!TIP]

Correct Answer:

C. On-Demand Instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets flexible compute provisioning, especially how AWS enables teams to launch EC2 instances instantly without commitments — ideal for unpredictable or short-lived workloads that must not be interrupted.

WHY C IS CORRECT

On-Demand Instances: Pay per second or hour — no upfront cost or long-term commitment
Ideal for: Spiky or unpredictable workloads Development and testing environments Critical apps that must not be interrupted Benefits: Maximum flexibility Immediate provisioning No risk of termination On-Demand is your compute parachute — deploy instantly, scale freely, and land safely without commitment.

SOURCES

EC2 Pricing Models Choosing EC2 Instances  Why the Others Are Less Suitable Option
Why It's Incorrect Spot Instances Cheapest — but can be interrupted anytime (not suitable for critical workloads) Dedicated Hosts Used for licensing and compliance — expensive and not flexible Reserved Instances Best for predictable workloads — requires upfront commitment

REINFORCING WITH EC2 STRATEGY

Let's map EC2 pricing models by workload type: -

Model

Best For Interruption Risk Commitment -

On-Demand

Unpredictable, critical workloads None None -

Spot

Fault-tolerant, batch jobs High None -

Reserved

Steady-state workloads None 1–3 years -

Savings Plans

Flexible compute usage None 1–3 years -

Dedicated Hosts

Licensing-bound workloads None High cost

REAL-WORLD TIE-IN

If your Nairobi-based team is testing a new machine learning model: Use On-Demand EC2 for training and experimentation Monitor usage with Cost Explorer and Budgets Switch to Spot Instances for non-critical batch jobs Consider Savings Plans once usage stabilizes That's how AWS empowers teams to experiment and scale without financial friction using On-Demand EC2 instances.

Question 459

**Which EC2 pricing option is best for intermittent, spiky, or unpredictable workloads?

**

OPTIONS

A. Spot Instances

B. Dedicated Hosts →

C. On-Demand Instances

D. Reserved Instances

[!TIP]

Correct Answer:

C. On-Demand Instances

EXPLANATION

CONCEPTUAL FRAMING

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REINFORCING WITH EC2 STRATEGY

Let's map EC2 pricing models by workload type: -

Model

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Spot

Fault-tolerant, batch jobs High None -

Reserved

Steady-state workloads None 1–3 years -

Savings Plans

Flexible compute usage None 1–3 years -

Dedicated Hosts

Licensing-bound workloads None High cost

REAL-WORLD TIE-IN

If your Nairobi-based team is testing a new machine learning model: Use On-Demand EC2 for training and experimentation Monitor usage with Cost Explorer and Budgets Switch to Spot Instances for non-critical batch jobs Consider Savings Plans once usage stabilizes That's how AWS empowers teams to experiment and scale without financial friction using On-Demand EC2 instances.

Question 461

**Which of the following is a duty of the client under the AWS shared responsibility model?

**

OPTIONS

A. Virtualization infrastructure

B. Network infrastructure →

C. Application security

D. Physical security of hardware

[!TIP]

Correct Answer:

C. Application security

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud security accountability, especially how AWS enables teams to focus on securing their own data, applications, and configurations while AWS handles the underlying infrastructure and physical security.

WHY C IS CORRECT

Shared Responsibility Model: AWS is responsible for: Security of the cloud — hardware, software, networking, and facilities Customers are responsible for: Security in the cloud — data, identity, applications, OS, and configurations Application security includes: IAM policies and

roles Encryption and key management Patch management and secure coding AWS locks the vault — you decide what goes inside and how it's protected.

SOURCES

AWS Shared Responsibility Model Security Best Practices  Why the Others Are AWS's Responsibility Option Responsibility Why It's Incorrect

A.

Virtualization infrastructure AWS Managed by AWS hypervisor layer

B.

Network infrastructure AWS Includes routers, switches, and backbone

D.

Physical security of hardware AWS Includes data center access, surveillance, and power systems

REINFORCING WITH RESPONSIBILITY MATRIX

Layer	AWS Responsibility	Customer Responsibility
Physical	Data center security —	
Network	Backbone and routing VPC configuration	
Compute	Hypervisor and host OS Guest OS and patching	
Storage	Disk encryption and durability Data classification and access control	
Application	— Code security, secrets management	
Identity	Root account protection IAM roles, MFA, least privilege	

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a fintech app: AWS secures the data centers and hardware You secure: IAM roles and policies Application logic and API endpoints Secrets in AWS Secrets Manager Encryption using KMS That's how AWS empowers teams to focus on securing what they build while trusting AWS to secure the foundation.

Question 462

**Which AWS service can be used to monitor illegal API calls?

**

OPTIONS

A. AWS Config 

B. AWS CloudTrail

C. AWS Trusted Advisor

D. Amazon Inspector

[!TIP]

Correct Answer:

B. AWS CloudTrail

EXPLANATION

CONCEPTUAL FRAMING

This question targets security observability and forensic auditing, especially how AWS enables teams to track every API interaction across services, users, and roles — critical for detecting unauthorized access, misconfigurations, and suspicious behavior.

WHY B IS CORRECT

AWS CloudTrail: Records every API call made via: AWS Management Console AWS SDKs CLI and other tools Captures: Caller identity Timestamp Source IP Request parameters and response Benefits: Audit trail for compliance Security incident investigation Alerts for anomalous activity via CloudWatch CloudTrail is your security camera in the cloud — always watching, always logging, always ready to rewind.

SOURCES

CloudTrail Overview Detecting Unauthorized Activity  Why the Others Are Insufficient Option Why It's Incorrect AWS Config Tracks resource configuration changes — not API calls Trusted Advisor Provides best practice checks — not real-time API monitoring Amazon Inspector Scans for vulnerabilities — doesn't monitor API activity

REINFORCING WITH MONITORING STRATEGY

Let's map AWS security tools by function: -

Tool

Function -

CloudTrail

API activity logging and auditing -

Config

Resource configuration tracking -

GuardDuty

Threat detection and anomaly alerts -

Inspector

Vulnerability scanning -

Security Hub

Centralized security findings -

CloudWatch Logs

Real-time log monitoring and alerting

REAL-WORLD TIE-IN

If your Nairobi-based team is managing sensitive workloads: Enable CloudTrail across all accounts and Regions Store logs in S3 with encryption and lifecycle policies Integrate with CloudWatch for alerting on suspicious API calls Use Athena or Lake Formation for querying logs That's how AWS empowers teams to maintain full visibility and accountability over API activity using CloudTrail as the foundation of cloud auditability.

REFERENCES

<https://aws.amazon.com/config/>

<https://aws.amazon.com/inspector/>

Question 463

**An e-commerce firm expects a surge in online traffic during major shopping holidays. Which AWS service enables dynamic resource adjustment to meet changing demand?

**

OPTIONS

A. AWS CloudTrail →

B. Amazon EC2 Auto Scaling

C. Amazon Forecast

D. AWS Config

[!TIP]

Correct Answer:

B. Amazon EC2 Auto Scaling

EXPLANATION

CONCEPTUAL FRAMING

This question targets automated infrastructure elasticity, especially how AWS enables teams to scale EC2 instances up or down based on real-time or predictive demand — ensuring performance and cost-efficiency during traffic spikes.

WHY B IS CORRECT

Amazon EC2 Auto Scaling: Automatically adjusts the number of EC2 instances in a group
Supports: Dynamic scaling — based on metrics like CPU, memory, or request count
Predictive scaling — anticipates demand based on historical patterns
Benefits: Handles traffic surges gracefully
Avoids overprovisioning
Reduces manual intervention
Auto Scaling is your cloud reflex — it expands and contracts with demand, keeping uptime and cost in balance.

SOURCES

EC2 Auto Scaling Overview
Predictive Scaling ❌ Why the Others Are Misleading
Option Why It's Incorrect
CloudTrail Logs API activity — not for scaling resources
Forecast Predicts demand — doesn't scale infrastructure
Config Tracks resource configuration — not performance or scaling

REINFORCING WITH SCALING STRATEGY

Let's map EC2 scaling options by behavior: -

Scaling Type

Trigger Use Case -

Dynamic Scaling

Real-time metrics (CPU, requests) Traffic spikes -

Predictive Scaling

Historical patterns Scheduled surges -

Manual Scaling

Admin-driven Controlled environments -

Scheduled Scaling

Time-based rules Known peak hours

REAL-WORLD TIE-IN

If your Nairobi-based team is prepping for Black Friday: Use Auto Scaling Groups with predictive scaling Set CloudWatch alarms for CPU and request thresholds Pair with Elastic Load Balancer for traffic distribution Monitor with AWS Budgets and Cost Explorer That's how AWS empowers teams to scale infrastructure dynamically and cost-effectively during high-demand events using EC2 Auto Scaling.

Question 464

**Which strategy contributes to cost optimization for consumers migrating to AWS Cloud?

**

OPTIONS



- A. Paying only for what is used
- B. Purchasing hardware before it is needed
- C. Manually provisioning cloud resources
- D. Purchasing for the maximum possible load

[!TIP]

Correct Answer:

- A. Paying only for what is used

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud-native financial efficiency, especially how AWS enables teams to shift from CapEx-heavy infrastructure to agile, usage-based pricing — unlocking cost savings, scalability, and operational flexibility.

WHY A IS CORRECT

Pay-as-you-go model: AWS charges only for: Resources consumed (compute, storage, bandwidth) Duration used (per second, minute, or hour) Benefits: No upfront investment No idle resource costs Scales with demand Ideal for: Startups and agile teams Seasonal or variable workloads Rapid experimentation and prototyping AWS turns infrastructure into a utility — always available, never overbilled, and perfectly elastic.

SOURCES

AWS Pricing Overview Cost Optimization Pillar – Well-Architected Framework ❌ Why the Others Are Costly or Inefficient Option Why It's Incorrect

B.

Purchasing hardware before needed Traditional CapEx — leads to overprovisioning and sunk costs

C.

Manual provisioning Slower, error-prone — risks idle resources and human inefficiency

D.

Purchasing for max load Overprovisioning — wastes budget during low-demand periods

REINFORCING WITH OPTIMIZATION STRATEGY

Let's map AWS cost-saving techniques: -

Strategy

Benefit -

Right-sizing

Match instance size to workload -

Auto Scaling

Scale with demand — no idle cost -

Spot Instances

Up to 90% discount for flexible workloads -

Savings Plans / Reserved Instances

Long-term discounts for predictable usage -

S3 Lifecycle Policies

Move data to cheaper storage tiers automatically

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a legacy CRM system: Use On-Demand EC2 for initial testing Monitor usage with Cost Explorer and Budgets Right-size instances with Compute Optimizer Apply S3 Intelligent-Tiering for archived data That's how AWS empowers teams to stay lean, agile, and financially efficient by paying only for what they use — and scaling only when needed.

Question 465

**According to the AWS shared responsibility model, which task is the customer's responsibility?

**

OPTIONS

- A. Maintaining infrastructure for AWS Lambda
- B. Updating OS of DynamoDB instances
- C. Maintaining Amazon S3 infrastructure →
- D. Updating the guest operating system on EC2 instances

[!TIP]

Correct Answer:

D. Updating the guest operating system on Amazon EC2 instances

EXPLANATION

CONCEPTUAL FRAMING

This question targets operational accountability in cloud environments, especially how AWS enables teams to focus on securing and maintaining their own workloads while AWS handles the underlying infrastructure and services.

WHY D IS CORRECT

Amazon EC2 is an Infrastructure-as-a-Service (IaaS) offering: AWS manages: Physical servers Networking Hypervisor Customers manage: Guest OS updates and patching Application configuration Security hardening This includes: Installing updates Managing firewalls Monitoring for vulnerabilities EC2 gives you the keys to the machine — AWS secures the garage, but you maintain the engine.

SOURCES

Shared Responsibility Model EC2 Security Best Practices ❌ Why the Others Are AWS's Responsibility Option AWS Responsibility Why It's Incorrect

A.

AWS Lambda infrastructure AWS manages all infrastructure for serverless compute

B.

DynamoDB OS updates Fully managed — no customer access to underlying OS

C.

S3 infrastructure AWS handles storage durability, availability, and scaling

REINFORCING WITH RESPONSIBILITY MATRIX

Layer	AWS Responsibility	Customer Responsibility
Physical	Data center security —	
Network	Backbone and routing VPC configuration	

Compute Hypervisor and host OS Guest OS and patching

Storage Disk durability Data classification and access control

Application — Code security, secrets management

Identity Root account protection IAM roles, MFA, least privilege

REAL-WORLD TIE-IN

If your Nairobi-based team is running EC2-based analytics workloads: You must: Patch the OS regularly Configure firewalls and security groups Monitor logs and metrics AWS will: Ensure hardware reliability Maintain network connectivity Provide availability guarantees That's how AWS empowers teams to focus on workload security and configuration while trusting AWS to manage the foundational infrastructure.

Question 466

**Which design principle is achieved by following the reliability pillar of the AWS Well-Architected Framework?

**

OPTIONS

A. Vertical scaling

B. Manual failure recovery →

C. Testing recovery procedures

D. Changing infrastructure manually

[!TIP]

Correct Answer:

C. Testing recovery procedures

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience validation, especially how AWS encourages teams to proactively simulate failures and rehearse recovery to ensure systems behave predictably under stress or disruption.

WHY C IS CORRECT

Reliability Pillar of the AWS Well-Architected Framework: Focuses on: Resilience to failures Automated recovery Monitoring and alerting Key design principles: Automatically recover from failure Test recovery procedures regularly Scale horizontally to increase availability Benefits: Confidence in disaster recovery Reduced downtime Improved incident response Recovery testing is your chaos rehearsal — break it before it breaks you.

SOURCES

AWS Well-Architected Reliability Pillar Design Principles  Why the Others Are Misaligned Option Why It's Incorrect

A.

Vertical scaling Limits fault isolation — not a reliability best practice

B.

Manual failure recovery Slow and error-prone — contradicts automation principle

D.

Changing infrastructure manually Risks drift and inconsistency — violates infrastructure-as-code principle

REINFORCING WITH RELIABILITY PRACTICES

Practice Purpose

Recovery testing Validate failover and backup mechanisms

Chaos engineering Simulate failures to improve resilience

Auto Scaling Maintain availability during demand spikes

Infrastructure as Code Ensure consistent recovery environments

Monitoring and alerting Detect and respond to failures quickly

REAL-WORLD TIE-IN

If your Nairobi-based team is building a payment gateway: Use multi-AZ deployments for EC2 and RDS Simulate failover with Route 53 health checks Automate recovery with CloudFormation and Lambda Run chaos drills to test resilience That's how AWS empowers teams to build and validate reliable systems that recover gracefully and maintain availability under pressure.

Question 467

**What enables a business to deliver a low-latency experience to global users?

**

OPTIONS

A. Use a central AWS Region

B. Add a second Availability Zone

C. Enable caching in the Region →

D. Use edge locations to put content closer to users

[!TIP]

Correct Answer:

D. Using edge locations to put content closer to all users

EXPLANATION

CONCEPTUAL FRAMING

This question targets global content delivery and latency reduction, especially how AWS enables teams to serve static and dynamic content from geographically distributed edge locations — minimizing round-trip time and improving user experience.

WHY D IS CORRECT

Edge Locations (via Amazon CloudFront): Are part of AWS's Content Delivery Network (CDN) Located in hundreds of cities worldwide Cache and deliver: Static assets (images, videos, stylesheets) Dynamic content via Lambda@Edge Streaming media and APIs Benefits: Lower latency Reduced origin load Improved scalability and fault tolerance Edge locations are your global storefronts — fast, local, and always open.

SOURCES

Amazon CloudFront Overview Latency Optimization with Edge Locations **✗** Why the Others Are Misleading Option Why It's Incorrect

A.

Central Region May reduce latency for some — but not globally

B.

Second AZ Improves availability — not latency across geographies

C.

Regional caching Helps locally — doesn't solve global latency

REINFORCING WITH DELIVERY STRATEGY

Let's map AWS latency-reduction techniques: -

Technique

Scope Benefit -

CloudFront CDN

Global Fast content delivery via edge locations -

Lambda@Edge

Global Dynamic content personalization at edge -

Global Accelerator

Global Optimized routing for TCP/UDP traffic -

S3 Transfer Acceleration

Global Faster uploads to S3 via edge routing

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a video platform: Store videos in Amazon S3 Distribute via CloudFront with edge caching Use Lambda@Edge to personalize content per region Monitor latency with CloudWatch metrics and Real User Monitoring That's how AWS empowers teams to deliver lightning-fast experiences to users worldwide using edge locations as the global delivery layer.

Question 468

**A business has optimized its workload using AWS services to increase efficiency and reduce costs. Which cost management best practice does this demonstrate?

**

OPTIONS

A. Resource controls

B. Cost allocation →

C. Architecture optimization

D. Tagging enforcement

[!TIP]

Correct Answer:

C. Architecture optimization

EXPLANATION

CONCEPTUAL FRAMING

This question targets cost-aware design, especially how AWS enables teams to refactor workloads for efficiency, scalability, and financial sustainability — not just lift-and-shift migration.

WHY C IS CORRECT

Architecture Optimization: Involves continual refinement of cloud workloads to: Reduce waste
Improve performance Align with pricing models Techniques include: Right-sizing instances
Using managed services Adopting serverless and event-driven architectures Leveraging Spot
Instances and Savings Plans Benefits: Lower operational costs Improved agility and scalability

Better alignment with business goals Architecture optimization is your cost scalpel — precise, iterative, and always aligned with performance and budget.

SOURCES

AWS Well-Architected Cost Optimization Pillar AWS Cost Optimization Best Practices ❌ Why the Others Are Misaligned Option Why It's Incorrect

A.

Resource controls Helps prevent overuse — but doesn't optimize architecture

B.

Cost allocation Tracks spend — doesn't reduce it

D.

Tagging enforcement Improves visibility — not workload efficiency

REINFORCING WITH OPTIMIZATION STRATEGY

Let's map cost optimization techniques by impact: -

Technique

Category Impact -

Right-sizing

Architecture High -

Auto Scaling

Architecture High -

Spot Instances

Pricing model High -

Savings Plans

Pricing model Moderate -

Tagging and allocation

Governance ⚠ Indirect -

Budgets and alerts

Monitoring ⚠ Preventive

REAL-WORLD TIE-IN

If your Nairobi-based team is running a data analytics pipeline: Replace EC2 clusters with AWS Glue or EMR on Spot Use S3 Intelligent-Tiering for storage Apply Compute Optimizer for instance recommendations Monitor with AWS Budgets and Cost Explorer That's how AWS empowers teams to design cost-efficient, scalable systems through architecture optimization — not just usage tracking.

Question 469

**Which feature of cloud computing does AWS demonstrate by offering reduced variable prices due to large-scale purchasing?

**

OPTIONS

A. Pay-as-you-go pricing

B. High availability

C. Global reach →

D. Economies of scale

[!TIP]

Correct Answer:

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud cost efficiency at scale, especially how AWS leverages its massive infrastructure footprint to negotiate lower costs, optimize operations, and pass savings on to customers.

WHY D IS CORRECT

Economies of Scale: AWS operates millions of servers across global Regions and AZs Benefits include: Bulk purchasing discounts Operational efficiencies Shared infrastructure costs Customers gain: Lower per-unit pricing Access to enterprise-grade infrastructure without upfront investment Scalable services at reduced cost AWS turns scale into savings — the more it builds, the less you pay.

SOURCES

AWS Cloud Economics Well-Architected Cost Optimization Pillar  Why the Others Are Misaligned Option Why It's Incorrect

A.

Pay-as-you-go pricing Pricing model — not scale-driven cost reduction

B.

High availability Reliability feature — unrelated to pricing

C.

Global reach Accessibility feature — not cost optimization

REINFORCING WITH CLOUD ECONOMICS STRATEGY

Let's map AWS cost benefits by economic principle: -

Principle

AWS Feature Customer Benefit -

Economies of scale

Global infrastructure Lower prices, better performance -

Elasticity

Auto Scaling, serverless Pay only for what you use -

Operational efficiency

Managed services Reduced overhead and complexity -

Agility

Rapid provisioning Faster time-to-market

REAL-WORLD TIE-IN

If your Nairobi-based team is scaling a SaaS platform: Use managed services like RDS, Lambda, and CloudFront Benefit from AWS's bulk infrastructure pricing Monitor spend with Cost Explorer and Budgets Optimize architecture with Compute Optimizer and Trusted Advisor That's how AWS empowers teams to build globally and scale affordably by harnessing the economic power of massive infrastructure investment.

Question 470

**A customer with AWS Basic Support discovers unauthorized use of their AWS resources. What is the preferred mechanism to notify AWS?

**

OPTIONS

A. Contact AWS Concierge Support

B. Contact a Technical Account Manager →

C. Contact the AWS Abuse team

D. Contact AWS Support

[!TIP]

Correct Answer:

C. Contact the AWS Abuse team

EXPLANATION

CONCEPTUAL FRAMING

This question targets incident escalation and abuse reporting, especially how AWS enables customers to report suspected misuse, compromised credentials, or malicious activity through a dedicated abuse channel — even under Basic Support plans.

WHY C IS CORRECT

AWS Abuse Team: Handles reports of: Unauthorized access or usage Phishing, spam, or malware hosted on AWS Compromised accounts or resources Reporting methods: AWS Abuse Form Email: abuse@amazonaws.com Required details: Plaintext logs Email headers IP addresses and timestamps The Abuse team is your security hotline — direct, responsive, and built for urgent escalation.

SOURCES

Report Amazon AWS Abuse AWS Support Tiers ❌ Why the Others Are Misaligned Option Why It's Incorrect

A.

Concierge Support Only available under Enterprise Support — focused on billing and account guidance

B.

Technical Account Manager (TAM) Assigned only to Enterprise customers — not available under Basic Support

D.

AWS Support Team General support — not specialized for abuse or security incidents

REINFORCING WITH INCIDENT RESPONSE STRATEGY

Let's map AWS escalation paths by scenario: -

Scenario

Contact -

Unauthorized usage or abuse

AWS Abuse Team -

Billing or account issues

AWS Support or Concierge -

Technical guidance (Enterprise)

Technical Account Manager -

Security best practices

AWS Security Hub, Trusted Advisor

REAL-WORLD TIE-IN

If your Nairobi-based team detects EC2 instances mining crypto unexpectedly: Immediately: Revoke compromised credentials Stop affected resources Notify AWS Abuse team with logs and timestamps Then: Enable CloudTrail and GuardDuty Rotate IAM keys and passwords Review IAM policies and access logs That's how AWS empowers teams to respond swiftly and responsibly to abuse incidents, even under Basic Support, using a dedicated escalation channel.

Question 471

**Which AWS service should a business use to continuously check the compliance of AWS resource settings?

**

OPTIONS

A. AWS Organizations 

B. AWS Config

C. AWS Artifact

D. AWS Service Catalog

[!TIP]

Correct Answer:

B. AWS Config

EXPLANATION

CONCEPTUAL FRAMING

This question targets configuration governance and compliance automation, especially how AWS enables teams to track resource states, detect drift, and enforce internal standards across cloud environments.

WHY B IS CORRECT

AWS Config: Continuously monitors and records: Resource configurations Relationships between resources Changes over time Enables: Automated compliance checks Custom rules and conformance packs Audit trails for security and operations Benefits: Simplifies compliance audits Improves change visibility Supports remediation workflows AWS Config is your cloud historian and policy enforcer — always watching, always validating, always ready to report.

SOURCES

AWS Config Overview Compliance Management with AWS Config  Why the Others Are Misaligned Option Why It's Incorrect AWS Organizations Manages multi-account structure — not resource compliance AWS Artifact Provides compliance reports — doesn't monitor configurations AWS Service Catalog Governs provisioning templates — not ongoing compliance checks

REINFORCING WITH COMPLIANCE STRATEGY

Let's map AWS tools by compliance function: -

Tool

Function -

AWS Config

Continuous monitoring and compliance enforcement -

CloudTrail

API activity logging for audit trails -

Security Hub

Aggregates security findings across services -

IAM Access Analyzer

Detects unintended access paths -

AWS Audit Manager

Automates evidence collection for audits

REAL-WORLD TIE-IN

If your Nairobi-based team is managing a regulated healthcare workload: Use AWS Config rules to enforce encryption and tagging policies Deploy conformance packs for HIPAA or ISO standards Integrate with Security Hub and CloudWatch for alerting Automate remediation with Systems Manager and Lambda That's how AWS empowers teams to maintain continuous compliance and visibility across cloud resources using AWS Config as the backbone of governance.

REFERENCES

<https://aws.amazon.com/config/>

<https://aws.amazon.com/organizations/>

<https://aws.amazon.com/artifact/>

Question 472

**Which AWS Cloud feature relieves customers from estimating future infrastructure consumption?

**

OPTIONS

A. Fast multi-Region deployment

B. Cloud security ➡

C. Elasticity of the AWS Cloud

D. Lower variable costs from economies of scale

[!TIP]

Correct Answer:

C. Elasticity of the AWS Cloud

EXPLANATION

CONCEPTUAL FRAMING

This question targets dynamic resource provisioning, especially how AWS enables teams to scale infrastructure automatically in response to demand — eliminating the need for upfront capacity planning.

WHY C IS CORRECT

Elasticity: Automatically adjusts resources to match workload demand Supports: Auto Scaling Groups Serverless compute (Lambda) Managed services with built-in scaling Benefits: No need to guess future usage Avoids overprovisioning and underutilization Improves cost efficiency and performance Elasticity is your cloud reflex — expanding and contracting with precision, so you never overbuild or underperform.

SOURCES

AWS Cloud Benefits Elasticity in AWS  Why the Others Are Misaligned Option Why It's Incorrect

A.

Multi-Region deployment Improves availability — doesn't address consumption estimation

B.

Cloud security Protects data — unrelated to resource scaling

D.

Economies of scale Reduces cost — but doesn't automate provisioning

REINFORCING WITH ELASTIC ARCHITECTURE STRATEGY

Let's map AWS elasticity features by service: -

Service

Elastic Feature -

EC2 Auto Scaling

Adjusts instance count based on demand -

Lambda

Scales automatically per request -

DynamoDB

On-demand capacity mode scales with traffic -

Aurora Serverless

Scales database capacity automatically -

Elastic Load Balancer

Distributes traffic across healthy targets dynamically

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a ticketing app for a major event: Use Auto Scaling EC2 for web servers Deploy Lambda functions for backend logic Store data in DynamoDB with on-demand mode Monitor with CloudWatch and AWS Budgets That's how AWS empowers teams to stay agile and cost-effective by scaling infrastructure elastically — no forecasting required.

Question 473

****A user needs to quickly deploy a non-relational database on AWS without managing hardware or database software. Which AWS service should they use?**

OPTIONS

A. Amazon RDS ➡

B. Amazon DynamoDB

C. Amazon Aurora

D. Amazon Redshift

[!TIP]

Correct Answer:

B. Amazon DynamoDB

EXPLANATION

CONCEPTUAL FRAMING

This question targets serverless NoSQL deployment, especially how AWS enables teams to launch scalable, fully managed databases without provisioning, patching, or maintaining infrastructure.

WHY B IS CORRECT

Amazon DynamoDB: Fully managed NoSQL database Supports key-value and document data models Features: Serverless architecture — no infrastructure to manage Automatic scaling — handles millions of requests per second Built-in security — IAM, encryption, VPC endpoints Global tables — multi-Region replication In-memory caching — via DAX for microsecond latency Use cases: Gaming leaderboards IoT telemetry E-commerce carts and catalogs DynamoDB is your zero-admin, high-speed NoSQL engine — built for scale, tuned for simplicity.

SOURCES

Amazon DynamoDB Overview DynamoDB Use Cases  Why the Others Are Misaligned
Option Why It's Incorrect Amazon RDS Relational — requires engine selection and patching
Amazon Aurora Relational — optimized for MySQL/PostgreSQL, not NoSQL Amazon Redshift
Data warehouse — designed for analytics, not transactional workloads

REINFORCING WITH DATABASE STRATEGY

Let's map AWS database services by model and management level: -

Service

Model Management Best For -

DynamoDB

No SQL Fully managed High-speed, serverless apps -

RDS

Relational Managed Traditional apps needing SQL -

Aurora

Relational Semi-serverless Scalable SQL workloads -

Redshift

Columnar Managed Analytics and BI -

DocumentDB

Document (Mongo DB) Managed JSON-based apps

REAL-WORLD TIE-IN

If your Nairobi-based team is building a mobile app with real-time user profiles: Use DynamoDB for fast, scalable NoSQL storage Enable DAX for low-latency caching Secure with IAM roles and KMS encryption Monitor with CloudWatch and AWS Config That's how AWS empowers teams to launch powerful NoSQL backends instantly using DynamoDB — no hardware, no patching, just performance.

REFERENCES

<https://aws.amazon.com/dynamodb/>

<https://aws.amazon.com/rds/>

<https://aws.amazon.com/rds/aurora/>

<https://aws.amazon.com/redshift/>

Question 474

**Which pattern is suggested for creating a highly available AWS architecture?

**

OPTIONS

- A. Ensure low-latency network connectivity
- B. Run EC2 instances at peak load →
- C. Design for failure of any single component
- D. Use a monolithic application

[!TIP]

Correct Answer:

C. Ensure that the application is designed to accommodate failure of any single component

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilience engineering, especially how AWS enables teams to build systems that remain operational even when individual components fail — a cornerstone of high availability and fault tolerance.

WHY C IS CORRECT

Design for failure: Anticipates and mitigates: Instance crashes AZ outages Network partitions
Techniques include: Multi-AZ deployments Stateless services Health checks and auto healing
Retry logic and circuit breakers Benefits: Minimized downtime Improved user experience
Robust disaster recovery High availability isn't luck — it's architecture that expects failure and refuses to flinch.

SOURCES

AWS Well-Architected Reliability Pillar Fault-Tolerant Design Patterns  Why the Others Are Misaligned Option Why It's Incorrect

A.

Low-latency connectivity Improves performance — not availability

B.

Peak-load provisioning Wasteful and static — doesn't adapt to failure

D.

Monolithic architecture Single point of failure — violates fault-tolerance principles

REINFORCING WITH AVAILABILITY STRATEGY

Let's map AWS design patterns for high availability: -

Pattern

Purpose -

Multi-AZ deployments

Survive AZ failure -

Load balancing

Distribute traffic and detect unhealthy targets -

Auto Scaling

Replace failed instances automatically -

Stateless services

Enable horizontal scaling and fast recovery -

Decoupling via queues

Isolate failures and smooth traffic spikes

REAL-WORLD TIE-IN

If your Nairobi-based team is building a logistics platform: Deploy EC2 and RDS in multiple AZs
Use Elastic Load Balancer with health checks
Store state in DynamoDB or S3, not in memory
Use SNS and SQS to decouple services
Monitor with CloudWatch and AWS Config
That's how AWS empowers teams to build resilient systems that expect failure and recover instantly — keeping uptime sacred and user trust intact.

Question 475

**When a workload is running in Amazon RDS, which task is AWS responsible for?

**

OPTIONS

A. Creating database tables

B. Updating the database schema 

C. Installing the database engine

D. Dropping database records

[!TIP]

Correct Answer:

C. Installing the database engine

EXPLANATION

CONCEPTUAL FRAMING

This question targets managed service boundaries, especially how AWS enables teams to focus on data modeling and application logic while AWS handles infrastructure provisioning, patching, and engine installation.

WHY C IS CORRECT

Amazon RDS (Relational Database Service): AWS responsibilities: Provisioning and installing the database engine Managing backups, patching, and failover Monitoring and scaling infrastructure Customer responsibilities: Schema design and updates Data modeling and query optimization Access control and application integration RDS is your database valet — AWS parks the engine, you drive the schema.

SOURCES

Amazon RDS User Guide Shared Responsibility Model for RDS  Why the Others Are Customer Responsibilities Option Why It's Incorrect

A.

Creating tables You define schema and structure

B.

Updating schema You manage migrations and changes

D.

Dropping records You control data lifecycle and deletion

REINFORCING WITH RDS RESPONSIBILITY MATRIX

Layer	AWS Responsibility	Customer Responsibility
Infrastructure	Hardware, networking, storage —	
Database engine	Installation, patching, backups	Schema, queries, data
Security	VPC, encryption, IAM integration	IAM roles, access policies
Monitoring	Cloud Watch metrics	Alerts, performance tuning

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a PostgreSQL-based analytics backend: AWS will: Install and patch PostgreSQL engine Manage backups and failover across AZs You will: Design tables and indexes Write queries and stored procedures Secure access with IAM and roles That's how AWS empowers teams to accelerate database deployment while offloading infrastructure and engine management to the cloud.

REFERENCES

<https://aws.amazon.com/rds/>

Question 476

******A business is considering migrating its on-premises data center to AWS. Which aspects should a Total Cost of Ownership (TCO) study consider? (Select two)

OPTIONS

A. Amazon EC2 instance availability →

B. Power consumption of the data center →

C. Labor costs to replace old servers

D. Application developer time

E. Database engine capacity

[!TIP]

Correct Answer:

B and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud migration financial modeling, especially how AWS encourages teams to evaluate both direct and indirect costs across the full lifecycle of infrastructure ownership — from acquisition to decommissioning. 🔍 Why B and C Are Correct
Power consumption of the data center: On-premises costs include: Electricity for servers, cooling, and lighting UPS and generator maintenance Environmental controls AWS eliminates these costs via shared cloud infrastructure
Labor costs to replace old servers: Includes: Technician time for hardware swaps Procurement and installation labor Downtime and risk during transitions AWS abstracts this with managed services and elastic provisioning
TCO is your financial microscope — it reveals the hidden costs behind the hardware curtain.

SOURCES

AWS TCO Calculator Cloud Economics Center ❌ Why the Others Are Misaligned Option Why It's Incorrect

A.

EC2 instance availability Operational feature — not a cost input

D.

Developer time Indirect and variable — not core to infrastructure TCO

E.

Database engine capacity Usage metric — not a cost driver unless tied to licensing

REINFORCING WITH TCO BREAKDOWN

Let's map TCO categories for cloud migration: -

Category

Examples -

Capital expenses (CapEx)

Hardware, racks, switches, storage -

Operational expenses (OpEx)

Power, cooling, maintenance labor -

Personnel costs

IT staff, upgrades, patching -

Downtime risk

Failures, recovery, SLA penalties -

Change management

Training, documentation, process shifts

REAL-WORLD TIE-IN

If your Nairobi-based team is migrating a legacy ERP system: Include in TCO: Power and cooling costs from your current data center Labor for server upgrades and replacements Software licensing and support contracts Compare against: AWS managed services like RDS, EC2, and S3 Elastic scaling and pay-as-you-go pricing That's how AWS empowers teams to make informed migration decisions by analyzing the full financial footprint of infrastructure ownership — not just sticker price.

Question 476

**A business is considering migrating its on-premises data center to AWS. Which aspects should a Total Cost of Ownership (TCO) study consider? (Select two)

**

OPTIONS

A. Amazon EC2 instance availability →

B. Power consumption of the data center →

C. Labor costs to replace old servers

D. Application developer time

E. Database engine capacity

[!TIP]

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EXPLANATION

CONCEPTUAL FRAMING

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REINFORCING WITH TCO BREAKDOWN

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REAL-WORLD TIE-IN

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Question 478

**A development team wants to publish and manage web services that provide REST APIs. Which AWS service meets this requirement?

**

OPTIONS

A. AWS App Mesh →

B. Amazon API Gateway

C. Amazon CloudFront

D. AWS Cloud Map

[!TIP]

Correct Answer:

B. Amazon API Gateway

EXPLANATION

CONCEPTUAL FRAMING

This question targets API lifecycle management, especially how AWS enables teams to create, deploy, secure, and monitor RESTful and WebSocket APIs at scale — without managing infrastructure.

WHY B IS CORRECT

Amazon API Gateway: Fully managed service for: Creating REST, HTTP, and WebSocket APIs
Routing requests to Lambda, EC2, or other backends
Enforcing throttling, caching, and authorization
Features: Built-in support for OpenAPI Usage plans and API keys
Integration with Cognito, IAM, and Lambda
Benefits: Scalable and secure API publishing
No server management
Real-time monitoring via CloudWatch
API Gateway is your digital front door — secure, scalable, and always ready to serve.

SOURCES

Amazon API Gateway Overview Building REST APIs with API Gateway **✗** Why the Others Are Misaligned Option Why It's Incorrect AWS App Mesh Manages service-to-service

communication — not public API exposure Amazon CloudFront CDN for content delivery — not API management AWS Cloud Map Service discovery — not API publishing or routing

REINFORCING WITH API ARCHITECTURE STRATEGY

Let's map AWS services by API-related function: -

Service

Role -

API Gateway

Publish and manage AP Is -

Lambda

Serverless backend for AP Is -

IAM & Cognito

Secure API access -

CloudWatch

Monitor API performance -

WAF & Shield

Protect AP Is from threats -

Route 53

DNS routing for custom domains

REAL-WORLD TIE-IN

If your Nairobi-based team is building a mobile app backend: Use API Gateway to expose REST endpoints Integrate with Lambda for business logic Secure with Cognito user pools Monitor usage with CloudWatch dashboards That's how AWS empowers teams to build robust, scalable APIs with zero infrastructure overhead — using API Gateway as the control plane.

Question 479

**What is an example of AWS Cloud agility?

**

OPTIONS

- A. Access to multiple instance types
- B. Access to managed services
- C. Using Consolidated Billing →
- D. Decreased acquisition time for new compute resources

[!TIP]

Correct Answer:

D. Decreased acquisition time for new compute resources

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud agility, especially how AWS enables teams to provision infrastructure in minutes — accelerating development, experimentation, and deployment cycles.

WHY D IS CORRECT

Agility in cloud computing means: Rapid provisioning of resources Quick iteration and deployment Fast response to changing business needs AWS enables: Launching EC2 instances in seconds to minutes Scaling up or down on demand Deploying global infrastructure without hardware delays Cloud agility is your launchpad — fast, flexible, and always ready to pivot.

SOURCES

AWS Cloud Benefits AWS Well-Architected Framework – Operational Excellence ❌ Why the Others Are Misaligned Option Why It's Incorrect

A.

Multiple instance types Offers flexibility — not speed of provisioning

B.

Managed services Reduces ops burden — not directly tied to agility

C.

Consolidated Billing Financial convenience — unrelated to infrastructure agility

REINFORCING WITH AGILITY STRATEGY

Let's map AWS features that enhance agility: -

Feature

Benefit -

EC2 Auto Scaling

Responds to demand spikes instantly -

Elastic Beanstalk

Rapid app deployment with minimal config -

CloudFormation

Infrastructure as code for fast, repeatable setups -

Lambda

Serverless execution with zero provisioning time -

Global infrastructure

Launch in any Region without physical setup

REAL-WORLD TIE-IN

If your Nairobi-based team is launching a new microservice: Use CloudFormation to deploy infrastructure in minutes Run compute with Lambda or EC2 based on workload Monitor agility metrics with CloudWatch dashboards That's how AWS empowers teams to move fast, iterate quickly, and deploy globally — all without waiting for hardware or manual provisioning.

Question 480

**What are the advantages of using loose coupling as a design paradigm for cloud architectures?

**

OPTIONS

- A. Facilitates low-latency request handling
- B. Allows dependent workflows →
- C. Prevents cascading failures between components
- D. Focuses on physical data center operations

[!TIP]

Correct Answer:

- C. Prevents cascading failures between different components

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilient system design, especially how AWS encourages teams to build modular, fault-tolerant architectures where components operate independently and recover gracefully from failure.

WHY C IS CORRECT

Loose Coupling: Components interact via well-defined interfaces (e.g., APIs, queues) Each service: Operates independently Can fail or change without breaking others Is easier to scale, replace, or debug Benefits: Fault isolation Improved maintainability Greater agility and resilience Loose coupling is your architectural shock absorber — it absorbs failure without spreading the damage.

SOURCES

AWS Well-Architected Framework – Reliability Pillar Design Principles for Microservices 
Why the Others Are Misaligned Option Why It's Incorrect

A.

Low-latency handling Depends on network and compute — not coupling

B.

Dependent workflows Increases coupling — opposite of the goal

D.

Physical data center ops Irrelevant in cloud-native design

REINFORCING WITH LOOSE COUPLING PATTERNS

Let's map AWS services that promote loose coupling: -

Pattern

AWS Service Benefit -

Event-driven

SNS, Event Bridge Async communication, decoupled triggers -

Queue-based

SQS Buffering and fault isolation -

API-based

API Gateway + Lambda Modular service boundaries -

Storage decoupling

S3 Durable, independent data layer -

Service discovery

Cloud Map Dynamic binding without tight dependencies

REAL-WORLD TIE-IN

If your Nairobi-based team is building a ride-hailing platform: Use API Gateway + Lambda for modular service endpoints Decouple booking and payment with SQS queues Trigger notifications via SNS Store trip logs in S3, independent of compute That's how AWS empowers teams to build robust, scalable systems that isolate failure and evolve independently — using loose coupling as the architectural backbone.

Question 481

**A business hosts a website on AWS behind an Application Load Balancer. They want to protect it from SQL injection and cross-site scripting (XSS). Which AWS service should they use?

**

OPTIONS

A. Amazon GuardDuty ➡

B. AWS WAF

C. AWS Trusted Advisor

[!TIP]

Correct Answer:

B. AWS WAF (Web Application Firewall)

EXPLANATION

CONCEPTUAL FRAMING

This question targets application-layer threat protection, especially how AWS enables teams to defend against common web exploits like SQL injection and XSS using rule-based inspection and mitigation engines.

WHY B IS CORRECT

AWS WAF: Protects against: SQL injection Cross-site scripting (XSS) HTTP flood and bot attacks Features: Custom and managed rule groups Integration with ALB, CloudFront, and API Gateway Real-time metrics and logging Benefits: Blocks malicious requests before reaching the app Inspects headers, query strings, and body Scales automatically with traffic AWS WAF is your HTTP bodyguard — inspecting every request, blocking every exploit.

SOURCES

AWS WAF Overview Managed Rule Groups **✗** Why the Others Are Misaligned Option Why It's Incorrect Amazon GuardDuty Detects threats — doesn't block web exploits directly AWS Trusted Advisor Offers best practice checks — not real-time protection Amazon Inspector Scans EC2 and container vulnerabilities — not HTTP traffic

REINFORCING WITH SECURITY STRATEGY

Let's map AWS services by security layer: -

Layer

Threat Type AWS Service -

Network

DDoS, spoofing AWS Shield, VPC ACLs -

Application

SQLi, XSS AWS WAF -

Identity

Unauthorized access IAM, Cognito -

Host/Container

Vulnerabilities Amazon Inspector -

Monitoring

Anomalies Guard Duty, Cloud Trail

REAL-WORLD TIE-IN

If your Nairobi-based team is running an e-commerce site: Deploy AWS WAF in front of your ALB. Enable managed rule groups for OWASP protections. Log blocked requests to CloudWatch or S3. Combine with Shield Standard for DDoS resilience. That's how AWS empowers teams to secure web applications against injection and scripting attacks — using WAF as the first line of defense.

REFERENCES

<https://aws.amazon.com/waf/>

<https://aws.amazon.com/guardduty/>

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

<https://aws.amazon.com/inspector/>

Question 481

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<https://aws.amazon.com/waf/>

<https://aws.amazon.com/guardduty/>

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

<https://aws.amazon.com/inspector/>

Question 483

**A company's new app is gaining popularity. They want to set up a phone number to handle increasing customer service calls. Which AWS service should they use?

**

OPTIONS



A. Amazon Connect

B. Amazon CloudFront

C. Amazon DirectConnect

[!TIP]

Correct Answer:

A. Amazon Connect

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud-native contact center deployment, especially how AWS enables teams to launch scalable, pay-as-you-go voice support systems with minimal setup and full integration capabilities.

WHY A IS CORRECT

Amazon Connect: Cloud-based omnichannel contact center Features: Claim or port phone numbers Route calls using contact flows Integrate with CRM and Lambda for automation Real-time and historical analytics Benefits: No hardware required Scales with demand Pay only for usage Global reach and high availability Amazon Connect is your voice gateway — fast to deploy, easy to scale, and built for customer satisfaction.

SOURCES

Amazon Connect Overview Amazon Connect Pricing  Why the Others Are Misaligned Option Why It's Incorrect CloudFront CDN for web content — not voice or contact center DirectConnect Dedicated network link — not customer service Trusted Advisor Optimization tool — doesn't handle communication or telephony

REINFORCING WITH CONTACT CENTER STRATEGY

Let's map Amazon Connect capabilities: -

Feature

Purpose -

Phone number management

Claim or port numbers for inbound/outbound calls -

Contact flows

Visual routing logic for customer journeys -

Lambda integration

Trigger backend logic during calls -

Real-time metrics

Monitor agent performance and call volume -

Chat and voice support

Unified customer experience across channels

REAL-WORLD TIE-IN

If your Nairobi-based team is scaling support for a fintech app: Claim a Kenyan phone number via Amazon Connect Build contact flows for billing, tech support, and onboarding Integrate with Lambda and DynamoDB for customer lookup Monitor call volume with CloudWatch dashboards That's how AWS empowers teams to launch modern, scalable contact centers in minutes — with full control and deep integration capabilities.

REFERENCES

<https://aws.amazon.com/connect/>

Question 484

******A company wants to evaluate streaming user data and respond in real time to consumer inquiries. Which AWS service satisfies this requirement?

OPTIONS

A. Amazon QuickSight

B. Amazon Redshift →

C. Amazon Kinesis Data Analytics

D. AWS Data Pipeline

[!TIP]

Correct Answer:

C. Amazon Kinesis Data Analytics

EXPLANATION

CONCEPTUAL FRAMING

This question targets real-time data stream processing, especially how AWS enables teams to ingest, analyze, and act on high-velocity data using serverless analytics engines built for scale and speed.

WHY C IS CORRECT

Amazon Kinesis Data Analytics: Processes data from: Kinesis Data Streams Kinesis Firehose
Apache Kafka Features: SQL-based stream queries Real-time dashboards and alerts
Integration with Lambda, S3, Redshift, and more Benefits: Low-latency insights No
infrastructure to manage Scales automatically with data volume Kinesis Data Analytics is your
stream-side brain — analyzing in motion, reacting in milliseconds.

SOURCES

Kinesis Data Analytics Overview Real-Time Use Cases **✗** Why the Others Are Misaligned
Option Why It's Incorrect Amazon QuickSight BI tool — visualizes static or batch data Amazon
Redshift Data warehouse — optimized for batch analytics AWS Data Pipeline ETL orchestration
— not built for real-time streaming

REINFORCING WITH STREAMING ARCHITECTURE STRATEGY

Let's map AWS services for real-time data workflows: -

Layer

AWS Service Role -

Ingestion

Kinesis Data Streams / Firehose Capture and buffer data -

Processing

Kinesis Data Analytics Analyze and transform in real time -

Action

Lambda / S3 / Redshift Trigger workflows or store results -

Monitoring

Cloud Watch / X-Ray Track performance and anomalies

REAL-WORLD TIE-IN

If your Nairobi-based team is monitoring user behavior on a mobile app: Stream events via Kinesis Data Streams Analyze click patterns with Kinesis Data Analytics (SQL) Trigger alerts via Lambda Store session summaries in S3 or Redshift That's how AWS empowers teams to turn live data into live decisions — using Kinesis as the backbone of real-time intelligence.

REFERENCES

Question 485

**Which of the following are examples of frequent IT duties that AWS can perform to free up internal resources? (Select two)

**

OPTIONS



A. Patching database software 

C. Backing up databases

B. Testing application releases

D. Creating database schema

E. Running penetration tests

[!TIP]

Correct Answer:

A and C

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud automation and managed service delegation, especially how AWS enables teams to offload repetitive, high-risk operational tasks — freeing engineers to focus on innovation and architecture. 🔍 Why A and C Are Correct

A.

Patching database software AWS automates patching via: AWS Systems Manager Patch Manager Amazon RDS automatic patching Benefits: Improved security posture Reduced manual effort Consistent compliance across environments

C.

Backing up databases AWS automates backups via: AWS Backup Amazon RDS automated snapshots Benefits: Centralized backup management Policy-driven retention and lifecycle Cross-service consistency AWS handles the heavy lifting — patching and backups become background tasks, not bottlenecks.

SOURCES

AWS Systems Manager Patch Manager AWS Backup Overview ❌ Why the Others Are Misaligned Option Why It's Incorrect

B.

Testing releases Requires custom test logic and app context — not automated by AWS

D.

Creating schema Application-specific — must be designed by developers

E.

Penetration tests Must be requested and scoped manually — not automated by AWS

REINFORCING WITH AUTOMATION STRATEGY

Let's map AWS services that automate common IT duties: -

Task

AWS Service Automation Benefit -

Patching

Systems Manager Patch Manager Scheduled, scoped, compliant updates -

Backups

AWS Backup, RDS snapshots Centralized, policy-driven protection -

Monitoring

Cloud Watch, X-Ray Real-time metrics and tracing -

Scaling

Auto Scaling, Lambda Elastic response to demand -

Security checks

Trusted Advisor, Config Rules Continuous compliance enforcement

REAL-WORLD TIE-IN

If your Nairobi-based team is managing a fleet of EC2 and RDS instances: Use Patch Manager to automate OS and DB patching Configure AWS Backup for daily snapshots and retention policies Monitor patch compliance with Systems Manager Insights That's how AWS empowers teams to shift from reactive maintenance to proactive innovation — by automating the most frequent and critical IT tasks.

REFERENCES

<https://aws.amazon.com/systems-manager/>

Question 486

**Which AWS service helps developers use loose coupling and reliable messaging between microservices?

**

OPTIONS

A. Elastic Load Balancing

B. Amazon SNS

C. Amazon CloudFront →

D. Amazon SQS

[!TIP]

Correct Answer:

D. Amazon Simple Queue Service (SQS)

EXPLANATION

CONCEPTUAL FRAMING

This question targets asynchronous communication and fault isolation, especially how AWS enables teams to decouple microservices using durable, scalable queues that buffer and relay messages reliably.

WHY D IS CORRECT

Amazon SQS: Fully managed message queuing service Supports: Standard queues for high throughput FIFO queues for ordered, exactly-once delivery Benefits: Loose coupling — producers and consumers operate independently Reliable delivery — messages persist until processed Scalable architecture — handles millions of messages per second Fault tolerance — prevents cascading failures SQS is your asynchronous glue — holding services together without holding them back.

SOURCES

Amazon SQS Overview Microservices Messaging Patterns ❌ Why the Others Are Misaligned
Option Why It's Incorrect Elastic Load Balancing Routes traffic — not designed for message
queuing Amazon SNS Pub/Sub — good for fan-out, but not reliable queuing Amazon
CloudFront CDN — accelerates content delivery, not service messaging

REINFORCING WITH MICROSERVICES MESSAGING STRATEGY

Let's map AWS services by messaging pattern: -

Pattern

AWS Service Use Case -

Queue-based

SQS Decoupled processing, retry logic -

Pub/Sub

SNS Broadcast to multiple consumers -

Event-driven

Event Bridge Rule-based routing and orchestration -

Streaming

Kinesis Real-time analytics and ingestion

REAL-WORLD TIE-IN

If your Nairobi-based team is building a ride-hailing backend: Use SQS to queue ride requests
Process them with Lambda or EC2 consumers Retry failed jobs without affecting upstream
services Monitor queue depth with CloudWatch metrics That's how AWS empowers teams to
build resilient, scalable microservices using SQS as the backbone of asynchronous
communication.

REFERENCES

<https://aws.amazon.com/sqs/>

<https://aws.amazon.com/elasticloadbalancing/>

<https://aws.amazon.com/sns/>

Question 487

**How should a web application be deployed in the AWS Cloud to guarantee high availability?

**

OPTIONS



- A. Deploy multiple instances in multiple Availability Zones
- B. Deploy multiple instances in a single Availability Zone
- C. Use a compute-optimized EC2 instance in one AZ
- D. Deploy one EC2 instance in an Auto Scaling group

[!TIP]

Correct Answer:

- A. Multiple instances across multiple Availability Zones

EXPLANATION

CONCEPTUAL FRAMING

This question targets resilient cloud architecture, especially how AWS enables teams to design fault-tolerant systems that survive infrastructure failures and maintain uptime across regions and zones.

WHY A IS CORRECT

Availability Zones (AZs): Physically isolated data centers within a Region Each AZ has independent power, cooling, and networking High Availability Strategy: Deploying across AZs ensures: Redundancy — if one AZ fails, others continue serving traffic Load balancing — distributes traffic across healthy instances Auto Scaling — adjusts capacity based on demand Benefits: Zero single point of failure Improved fault tolerance and uptime Seamless updates and patching High availability isn't a feature — it's a deployment discipline that spans zones, not just instances.

SOURCES

AWS Well-Architected Reliability Pillar Multi-AZ Deployment Best Practices  Why the Others Are Misaligned Option Why It's Incorrect

B.

Single AZ deployment Vulnerable to AZ failure — no redundancy

C.

Compute-optimized EC2 Performance-focused — not availability-focused

D.

One EC2 in Auto Scaling group No AZ diversity — still a single point of failure

REINFORCING WITH HA ARCHITECTURE PATTERNS

Let's map AWS components for high availability: -

Component

Role -

Elastic Load Balancer

Routes traffic across AZs -

Auto Scaling Group

Maintains healthy instance count -

Multi-AZ RDS

Ensures database failover -

Route 53

DNS failover across Regions -

CloudWatch Alarms

Detects and responds to failures

REAL-WORLD TIE-IN

If your Nairobi-based team is deploying a customer portal: Launch EC2 instances in at least two AZs Use Elastic Load Balancer to distribute traffic Enable Auto Scaling for demand spikes Store session data in DynamoDB or ElastiCache for resilience That's how AWS empowers teams to build web applications that stay online — even when parts of the infrastructure go offline.

Question 488

**A company needs to implement identity management for a fleet of mobile apps running in AWS. Which AWS service should they use?

**

OPTIONS



A. Amazon Cognito

B. AWS Security Hub

C. AWS Shield

D. AWS WAF

[!TIP]

Correct Answer:

A. Amazon Cognito

EXPLANATION

CONCEPTUAL FRAMING

This question targets user authentication and access control, especially how AWS enables teams to securely manage identities for mobile and web apps without building custom auth flows or storing credentials manually.

WHY A IS CORRECT

Amazon Cognito: Provides: User sign-up and sign-in flows Federated identity support (Google, Facebook, SAML, etc.) Access token issuance for API authorization Features: User pools — manage user directories and authentication Identity pools — grant temporary AWS credentials MFA, password policies, and device tracking Benefits: Scalable and secure identity management Built-in integration with mobile SDKs No need to manage passwords or auth logic manually Cognito is your identity concierge — handling sign-ups, logins, and access control with zero friction.

SOURCES

Amazon Cognito Overview Cognito for Mobile Apps  Why the Others Are Misaligned Option Why It's Incorrect AWS Security Hub Aggregates security findings — not identity management

AWS Shield Protects against DDoS — unrelated to user authentication
AWS WAF Blocks malicious HTTP requests — not designed for identity flows

REINFORCING WITH MOBILE IDENTITY STRATEGY

Let's map Cognito's role in mobile app security: -

Feature

Purpose -

User pools

Manage user accounts and authentication -

Identity pools

Grant AWS credentials for resource access -

Federation

Support social and enterprise identity providers -

Tokens

Secure API access via JWT -

MFA and password policies

Strengthen account security

REAL-WORLD TIE-IN

If your Nairobi-based team is building a mobile banking app: Use Cognito user pools for sign-up/sign-in Enable MFA and password strength policies Use identity pools to access S3 or DynamoDB securely Integrate with Facebook or Google login for convenience That's how AWS empowers teams to secure mobile apps with scalable, standards-based identity management — using Cognito as the authentication backbone.

REFERENCES

<https://aws.amazon.com/shield/>

Question 489

**Which AWS service or feature enables a business to track, monitor, and control its AWS expenses and consumption over time?

**

OPTIONS



A. AWS Budgets 

B. AWS Cost Explorer

C. AWS Organizations

D. Consolidated billing

[!TIP]

Correct Answer:

B. AWS Cost Explorer

EXPLANATION

CONCEPTUAL FRAMING

This question targets cloud financial transparency, especially how AWS enables teams to visualize usage trends, identify cost drivers, and optimize spending with interactive dashboards

and filtering tools.

WHY B IS CORRECT

AWS Cost Explorer: Provides: Historical usage and cost data Forecasting and trend analysis
Filtering by service, account, tag, or region Features: Graphical dashboards Daily and monthly granularity Exportable reports Benefits: Improved budget planning Cost optimization insights
Visibility across accounts and services Cost Explorer is your financial radar — scanning usage patterns and guiding spend decisions with precision.

SOURCES

AWS Cost Explorer Overview Managing AWS Costs **✗** Why the Others Are Misaligned Option
Why It's Incorrect AWS Budgets Sets thresholds and alerts — doesn't visualize trends AWS
Organizations Manages multi-account structure — not cost tracking Consolidated billing
Aggregates charges — no analytics or monitoring tools

REINFORCING WITH COST MANAGEMENT STRATEGY

Let's map AWS tools by financial function: -

Tool

Role -

Cost Explorer

Visualize and analyze usage and spend -

Budgets

Set limits and receive alerts -

Savings Plans & RI Reports

Optimize long-term pricing -

Trusted Advisor

Recommend cost-saving actions -

Billing Console

REAL-WORLD TIE-IN

If your Nairobi-based team is managing multiple AWS accounts: Use Cost Explorer to track EC2, S3, and Lambda usage Filter by tags to isolate dev vs prod environments Forecast next month's spend based on current trends Pair with Budgets to trigger alerts when thresholds are breached That's how AWS empowers teams to gain full visibility into cloud spend — turning cost management into a strategic advantage.

REFERENCES

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

<https://aws.amazon.com/aws-cost-management/aws-budgets/>

<https://aws.amazon.com/organizations/>

Question 490

**A large corporation hires a developer who needs AWS credentials. Which security best practices should be followed? (Select two)

**

OPTIONS



A. Grant access only to needed resources

B. Share root user credentials

C. Add to administrator group →

D. Prevent password changes →

E. Enforce minimum password length

[!TIP]

Correct Answer:

A and E

EXPLANATION

CONCEPTUAL FRAMING

This question targets identity and access management (IAM) hygiene, especially how AWS enables teams to enforce least privilege, secure credential policies, and avoid risky practices like root access sharing. 🔍 Why A and E Are Correct

A.

Grant access only to needed resources Principle of least privilege: Assign only the permissions required for the job Use IAM roles and policies to scope access Reduces blast radius in case of compromise

E.

Enforce minimum password length Part of a strong password policy: AWS IAM allows setting: Minimum length Complexity (uppercase, symbols, etc.) Rotation and reuse rules Helps prevent brute-force and weak password attacks IAM hygiene is like brushing your teeth — skip it, and the decay spreads fast.

SOURCES

IAM Best Practices Password Policy Configuration ❌ Why the Others Are Misaligned Option Why It's Incorrect

B.

Share root credentials  Major security risk — root access should never be shared

C.

Add to admin group Violates least privilege — only grant admin if absolutely necessary

D.

Prevent password changes Reduces security — users should be able to rotate passwords

REINFORCING WITH IAM SECURITY STRATEGY

Let's map IAM best practices for onboarding developers: -

Practice

Benefit -

Least privilege access

Limits exposure and risk -

Strong password policy

Prevents weak credentials -

MFA enforcement

Adds second layer of protection -

No root access

Preserves account integrity -

Role-based access

Simplifies permission management

REAL-WORLD TIE-IN

If your Nairobi-based team is onboarding a new backend developer: Create an IAM user or role with scoped access to Lambda and S3 Enforce a password policy with 12+ characters and symbols Enable MFA for console access Monitor activity with CloudTrail and IAM Access

Analyzer That's how AWS empowers teams to onboard securely, minimize risk, and maintain audit-ready identity hygiene.

Question 493

**

**

[!TIP]

Correct Answer:

D. Reserved Instances

EXPLANATION

CONCEPTUAL FRAMING

 Why D Is Correct Reserved Instances Cost Savings: Up to 75% cheaper than On-Demand when committing to 1–3 years. Predictability: Ideal for workloads with consistent usage patterns. Capacity Assurance: Guarantees availability in a chosen Availability Zone. Flexibility: Convertible RIs allow changes in instance families, OS, or tenancy. Reserved Instances are essentially a billing discount mechanism applied to matching On-Demand usage, ensuring both savings and reliability.  Why the Others Are Misaligned Option Why It's Incorrect

A.

Dedicated Hosts  Overkill unless you need compliance or BYOL licensing; higher cost, less efficient.

B.

On-Demand Instances Flexible but expensive long-term; suited for short-term or unpredictable workloads.

C.

Spot Instances Cheapest option but unreliable; can be interrupted anytime, unsuitable for steady workloads.

REINFORCING WITH EC2 PRICING STRATEGY

Strategy Best Use Case Cost Profile

Reserved Instances Steady, long-term workloads High savings, guaranteed capacity

On-Demand Short-term, unpredictable demand Moderate cost, high flexibility

Spot Fault-tolerant, flexible jobs Very low cost, low reliability

Dedicated Hosts Compliance/licensing needs High cost, specialized use

REAL-WORLD TIE-IN

Lock in 3-year Reserved Instances for maximum savings. Use Convertible RIs if you anticipate changing instance families as workloads evolve. Monitor usage with AWS Cost Explorer and Trusted Advisor to validate savings. That's how AWS empowers enterprises to balance cost efficiency with operational reliability in production workloads.

Question 494

**

**

[!TIP]

Correct Answer:

A. Standard Reserved Instance

EXPLANATION

CONCEPTUAL FRAMING

 Why A Is Correct – Standard Reserved Instance Maximum Discount: Offers the highest savings—up to 75% compared to On-Demand pricing when committed for 1 or 3 years. Predictable Usage: Perfect for workloads like database servers that run non-stop and have stable resource needs. No Need for Flexibility: Since the instance type and configuration are unlikely to change, the flexibility of Convertible RIs isn't necessary. Capacity Reservation: Ensures availability in the chosen Availability Zone. Standard Reserved Instances are ideal when you know your compute needs in advance and want to optimize cost without requiring configuration changes. ❌ Why the Others Are Misaligned Option Why It's Incorrect

B.

Convertible Reserved Instance  More flexible but slightly more expensive; better suited for evolving workloads.

C.

On-Demand Instance  Most expensive over time; good for short-term or unpredictable usage, not long-term stability.

D.

Spot Instance  Cheapest but unreliable; can be interrupted anytime, making it unsuitable for critical workloads like databases.

REINFORCING WITH EC2 PRICING STRATEGY

Strategy	Best Use Case	Cost Profile
Standard Reserved Instances	Predictable, long-term workloads	Highest savings, guaranteed capacity

Convertible RIs Long-term but evolving workloads Good savings, flexible configuration

On-Demand Short-term, bursty workloads High cost, high flexibility

Spot Fault-tolerant, flexible workloads Lowest cost, lowest reliability

REAL-WORLD TIE-IN

Choose a 1-year Standard Reserved Instance for the database server. Use AWS Cost Explorer to validate savings and monitor usage. Rely on Trusted Advisor to ensure you're maximizing cost efficiency. This approach ensures your critical workload remains available while minimizing your cloud spend.

Question 496

**A company has multiple applications and is now building a new multi-tier application. The company will host the new application on Amazon EC2 instances. The company wants the network routing and traffic between the various applications to follow the security principle of least privilege.

Which AWS service or feature should the company use to enforce this principle?

**

OPTIONS



A. Security groups

B. AWS Shield

C. AWS Global Accelerator

D. AWS Direct Connect gateway

[!TIP]

Correct Answer:

A

EXPLANATION

(Only one correct answer in the original question)

REINFORCING WITH VPC NETWORK SECURITY STRATEGY

Security Groups Scope: Instance/ENI level **Granularity:** Port + Protocol + Source SG/CIDR **Use Case Example:** Web tier → App tier only on 3000, DB tier only 3306

Network ACLs (NACLs) Scope: Subnet level **Granularity:** Stateless, numbered rules **Use Case Example:** Additional stateless hardening or deny-specific IPs

VPC Flow Logs Scope: Monitoring **Granularity:** All traffic (accept/reject) **Use Case Example:** Auditing and detecting anomalous flows

AWS Network Firewall Scope: VPC-to-VPC or Internet gateway **Granularity:** Stateful IDS/IPS + deep packet **Use Case Example:** Advanced threat protection across many VPCs

Prefix Lists / Managed Scope: Reusable CIDR groups **Granularity:** — **Use Case Example:** Avoid hardcoding IP ranges in SGs For most multi-tier applications, **Security Groups alone are sufficient and are the primary mechanism** to achieve least-privilege network segmentation.

Question 497

******A business requires 24/7 phone, email, and chat support, with a response time of less than one hour in the event of a service outage to a production system.

Which AWS Support plan best matches these needs for the least amount of money?

OPTIONS

A. Basic

B. Developer



C. Business

D. Enterprise

[!TIP]

Correct Answer:

C (Business)

Question 498

******A customer is considering migrating an application burden to the Amazon Web Services (AWS) Cloud.

Which control becomes AWS's responsibility after the migration?

OPTIONS

A. Patching the guest operating system



B. Maintaining physical and environmental controls

C. Protecting communications and maintaining zone security

D. Patching specific applications

[!TIP]

Correct Answer:

B (Maintaining physical and environmental controls)

Question 499

**What is the function of a VPC's internet gateway?

**

OPTIONS

A. To create a VPN connection to the VPC



B. To allow communication between the VPC and the Internet

C. To impose bandwidth constraints on internet traffic

D. To load balance traffic from the Internet across Amazon EC2 instances

[!TIP]

Correct Answer:

B

Question 500

**Which AWS IAM feature enables developers to use the AWS CLI to access AWS services?

**

OPTIONS

A. API keys



B. Access keys

C. User names/Passwords

D. SSH keys

[!TIP]

Correct Answer:

B (Access keys)

Question 501

******A company's web application requires AWS credentials and authorizations to use an AWS service.

Which IAM entity should the company use as best practice?

OPTIONS



A. IAM role

B. IAM user

C. IAM group

D. IAM multi-factor authentication (MFA)

[!TIP]

Correct Answer:

A (IAM role)

Question 502

******Which AWS service or feature gives a company the ability to control incoming traffic and outgoing traffic for Amazon EC2 instances?

**

OPTIONS



A. Security groups

B. Amazon Route 53

C. AWS Direct Connect

D. Amazon VPC

[!TIP]

Correct Answer:

A (Security groups)

Question 503

**On AWS, a corporation is constructing a new archiving system capable of storing terabytes of data. The firm will NOT often retrieve the data.

Which Amazon S3 storage type will result in the LOWEST SYSTEM COST?

**

OPTIONS

A. S3 Standard-Infrequent Access (S3 Standard-IA)



B. S3 Glacier

C. S3 Intelligent-Tiering

D. S3 One Zone-Infrequent Access (S3 One Zone-IA)

[!TIP]

Correct Answer:

B (S3 Glacier)

Question 504

**Which AWS Support package includes access to architectural and operational assessments, as well as 24/7 email, online chat, and phone support from Senior Cloud Support Engineers?

**

OPTIONS

A. Basic

B. Business

C. Developer



D. Enterprise

[!TIP]

Correct Answer:

D (Enterprise)

Question 505

**Which of the following describes the root user of an AWS account?

**

OPTIONS

A. The root user is the only user that can be configured with multi-factor authentication (MFA).

B. The root user is the only user that can access the AWS Management Console.



C. The root user is the first sign-in identity that is available when an AWS account is created.

D. The root user has a password that cannot be changed.

[!TIP]

Correct Answer:

C

Question 506

**The AWS IAM recommended practice for providing the fewest possible privileges is as follows:

**

OPTIONS

A. apply an IAM policy to an IAM group and limit the size of the group.

B. require multi-factor authentication (MFA) for all IAM users.

C. require each IAM user who has different permissions to have multiple passwords.



D. apply an IAM policy only to IAM users who require it.

[!TIP]

Correct Answer:

D

Question 507

To prevent fraudulent compute activity, a user needs a simple method to detect whether any Amazon EC2 instances have

unlimited access to their ports**.

Which Amazon Web Services (AWS) offering will meet this requirement?

**

OPTIONS

A. VPC Flow Logs

B. AWS WAF

C. AWS CloudTrail



D. AWS Trusted Advisor

[!TIP]

Correct Answer:

D (AWS Trusted Advisor)

Question 508

**Which AWS service is a content delivery network that safely and quickly distributes data, video, and apps to consumers worldwide?

**

OPTIONS

A. AWS CloudFormation

B. AWS Direct Connect



C. Amazon CloudFront

D. Amazon Pinpoint

[!TIP]

Correct Answer:

C (Amazon CloudFront)

Question 510

******A company is migrating its public website to AWS. The company wants to host the domain name for the website on AWS.

Which AWS service should the company use to meet this requirement?

OPTIONS

A. AWS Lambda



B. Amazon Route 53

C. Amazon CloudFront

D. AWS Direct Connect

[!TIP]

Correct Answer:

B (Amazon Route 53)

Question 511

**A company needs to evaluate its AWS environment and provide best practice recommendations in five categories: cost, performance, service limits, fault tolerance, and security.

Which AWS service can the company use to meet these requirements?

**

OPTIONS

A. AWS Shield

B. AWS WAF



C. AWS Trusted Advisor

D. AWS Service Catalog

[!TIP]

Correct Answer:

C (AWS Trusted Advisor)

Question 512

**Which AWS feature is exemplified by on-demand technology services that allow businesses to substitute variable expenditures for upfront fixed expenses?

**

OPTIONS

A. High availability

B. Economies of scale



C. Pay-as-you-go pricing

D. Global reach

[!TIP]

Correct Answer:

C (Pay-as-you-go pricing)

Question 513

A business is relocating and need an

encrypted connection to AWS**.

Which AWS service will assist you in fulfilling this requirement?

**

OPTIONS



A. AWS VPN

B. Amazon Route 53

C. Amazon API Gateway

D. Amazon Connect

[!TIP]

Correct Answer:

A (AWS VPN)

Question 514

**What technology permits compute capacity to alter in response to changing load conditions?

**

OPTIONS

A. Load balancing

B. Automatic failover

C. Round robin



D. Auto Scaling

[!TIP]

Correct Answer:

D (Auto Scaling)

Question 515

****Which AWS service provides the capability to view end-to-end performance metrics and troubleshoot distributed applications?**

OPTIONS

A. AWS Cloud9

B. AWS CodeStar

C. AWS Cloud Map



D. AWS X-Ray

[!TIP]

Correct Answer:

D (AWS X-Ray)

Question 516

A major corporation has a workload that

demands on-premises hardware

.

The organization want to continue using the

same management and control plane services** as it does on AWS.

Which Amazon Web Services (AWS) offering should the business employ to achieve these requirements?

**

OPTIONS

A. AWS Device Farm

B. AWS Fargate



C. AWS Outposts

D. AWS Ground Station

[!TIP]

Correct Answer:

C (AWS Outposts)

Question 517

**A business operates an e-commerce application that is hosted in Europe. To reduce latency for international customers accessing the website, the firm would want to cache frequently viewed static information closer to the consumers.

Which Amazon Web Services (AWS) offering will meet these requirements?

**

OPTIONS

A. Amazon ElastiCache



B. Amazon CloudFront

C. Amazon Elastic File System (Amazon EFS)

D. Amazon Elastic Block Store (Amazon EBS)

[!TIP]

Correct Answer:

B (Amazon CloudFront)

Question 518

Which of the following is

not** a recommended approach for IAM user management? (Select two.)

**

OPTIONS

A. Require IAM users to change their passwords after a specified period of time

B. Prevent IAM users from reusing previous passwords

C. Recommend that the same password be used on AWS and other sites



D. Require IAM users to store their passwords in raw text



E. Disable multi-factor authentication (MFA) for IAM users

[!TIP]

Correct Answer:

D and E

Question 519

**Which Amazon Web Services feature assists in identifying harmful or illegal activity in AWS accounts and workloads?

**

OPTIONS

A. Amazon Rekognition

B. AWS Trusted Advisor



C. Amazon GuardDuty

D. Amazon CloudWatch

[!TIP]

Correct Answer:

C (Amazon GuardDuty)

Question 520

Which AWS service or functionality provides

technical support

to users who subscribe to the

AWS Basic Support plan**?

**

OPTIONS

A. AWS senior support engineers

B. AWS technical account manager (TAM)



C. AWS Trusted Advisor

D. AWS Discussion Forums

[!TIP]

Correct Answer:

C (AWS Trusted Advisor)

Question 521

****Which AWS service provides threat detection by monitoring for malicious activities and unauthorized actions to protect AWS accounts, workloads, and data that is stored in Amazon S3?**

OPTIONS

A. AWS Shield

B. AWS Firewall Manager



C. Amazon GuardDuty

D. Amazon Inspector

[!TIP]

Correct Answer:

C (Amazon GuardDuty)

Question 522

**Which AWS service enables you to get AWS security and compliance information on-demand?

**

OPTIONS

A. AWS CloudTrail



B. AWS Artifact

C. AWS Health

D. Amazon CloudWatch

[!TIP]

Correct Answer:

B (AWS Artifact)

Question 523

Which Amazon Web Services (AWS) service makes use of edge locations**?

**

OPTIONS

A. Amazon Aurora



B. AWS Global Accelerator

C. Amazon Connect

D. AWS Outposts

[!TIP]

Correct Answer:

B (AWS Global Accelerator)

Question 524

**Which of the following AWS capabilities allows a user to deploy an Amazon Elastic Compute Cloud (Amazon EC2) instance that has already been configured?

**

OPTIONS

A. Amazon Elastic Block Store (Amazon EBS)



B. Amazon Machine Image

C. Amazon EC2 Systems Manager

D. Amazon AppStream 2.0

[!TIP]

Correct Answer:

B (Amazon Machine Image)

Question 525

Which AWS service can a company use to

store and manage Docker images**?

**

OPTIONS

A. Amazon DynamoDB

B. Amazon Kinesis Data Streams



C. Amazon Elastic Container Registry (Amazon ECR)

D. Amazon Elastic File System (Amazon EFS)

[!TIP]

Correct Answer:

C (Amazon ECR)

Question 526

A business wishes to send its traffic

directly and confidentially** to a virtual private cloud (VPC) rather than through the public Internet.

Which mode of connection enables this capability?

**

OPTIONS

A. AWS VPN



B. AWS Direct Connect

C. VPC NAT gateway

D. VPC Internet gateway

[!TIP]

Correct Answer:

Question 528

A company needs an

automated security assessment report

that will identify

unintended network access to Amazon EC2 instances. The report also must identify

operating system vulnerabilities

on those instances.

Which AWS service or feature should the company use to meet this requirement?

**

OPTIONS

A. AWS Trusted Advisor

B. Security groups



C. Amazon Macie

D. Amazon Inspector

[!TIP]

Correct Answer:

D (Amazon Inspector)

Question 529

Which AWS Support plan is the

LEAST EXPENSIVE

that provides for a

one-hour goal response time** for support cases?

**

OPTIONS

A. Enterprise



B. Business

C. Developer

D. Basic

[!TIP]

Correct Answer:

B (Business)

Question 530

Which AWS hybrid storage offering allows users to

effortlessly integrate on-premises applications with AWS Cloud storage**?

**

OPTIONS

A. AWS Backup

B. Amazon Connect

C. AWS Direct Connect



D. AWS Storage Gateway

[!TIP]

Correct Answer:

D (AWS Storage Gateway)

Question 531

**A user needs to prepare a report that summarizes the status of AWS account's major security checks.

The report must contain the following:

Permissions on Amazon S3 buckets are now inactive.

Whether or not multi-factor authentication is enabled for the root user of the AWS account.

If any security groups are set to enable unlimited access, this will be shown.

Where can I get all of this information in
one place

?

**

OPTIONS

A. Amazon QuickSight dashboard

B. AWS CloudTrail trails



C. AWS Trusted Advisor report

D. IAM credential report

[!TIP]

Correct Answer:

C (AWS Trusted Advisor)

Question 532

A pharmaceutical company's infrastructure is managed in a single AWS Region.
The organization want to link

hundreds of VPCs

across

many AWS accounts

.

Which AWS service or feature should the business use to

streamline administration and save operating costs**?

**

OPTIONS

A. VPC endpoints

B. AWS Direct Connect



C. AWS Transit Gateway

D. VPC peering

[!TIP]

Correct Answer:

C (AWS Transit Gateway)

Question 533

A global company is building a simple time-tracking mobile app. The app needs to operate

globally

and must store collected data in a

database

. **Data must be**

accessible from the AWS Region that is closest to the user

.

What should the company do to meet these data storage requirements with the

LEAST amount of operational overhead**?

**

OPTIONS

A. Use Amazon EC2 in multiple Regions to host separate databases

B. Use Amazon RDS cross-Region replication



C. Use Amazon DynamoDB global tables

D. Use AWS Database Migration Service (AWS DMS)

[!TIP]

Correct Answer:

C (Amazon DynamoDB global tables)

Question 534

One advantage of Amazon Elastic Compute Cloud (Amazon EC2)

on-demand pricing** is the following:

**

OPTIONS

A. the ability to bid for a lower hourly cost.

B. paying a daily rate regardless of time used.



C. paying only for time used.

D. pre-paying for instances and paying a lower hourly rate.

[!TIP]

Correct Answer:

C (paying only for time used)

Question 535

Which task is a

customer's responsibility**, according to the AWS Shared Responsibility Model?

**

OPTIONS



- A. Management of the guest operating systems
- B. Maintenance of the configuration of infrastructure devices
- C. Management of the host operating systems and virtualization
- D. Maintenance of the software that powers Availability Zones

[!TIP]

Correct Answer:

A (Management of the guest operating systems)

Question 536

Which AWS service enables you to

swiftly conduct one-time queries** on Amazon S3 data?

**

OPTIONS

- A. Amazon EMR
- B. Amazon DynamoDB
- C. Amazon Redshift
-  D. Amazon Athena

[!TIP]

Correct Answer:

D (Amazon Athena)

Question 537

A company needs to deliver new website features

quickly

in an

iterative manner

to

minimize the time to market**.

Which AWS Cloud concept does this requirement represent?

**

OPTIONS

A. Reliability

B. Elasticity



C. Agility

D. High availability

[!TIP]

Correct Answer:

C (Agility)

Question 538

Which VPC component adds an

additional layer of protection
to the

subnet**?

**

OPTIONS

A. Security groups



B. Network ACLs

C. NAT gateways

D. Route tables

[!TIP]

Correct Answer:

B (Network ACLs)

Question 539

**A company wants to increase its ability to recover its infrastructure in the case of a natural disaster.

Which pillar of the AWS Well-Architected Framework does this ability represent?

**

OPTIONS

A. Cost optimization

B. Performance efficiency



C. Reliability

D. Security

[!TIP]

Correct Answer:

C (Reliability)

Question 540

Which AWS Support package is the

LEAST EXPENSIVE

that includes a

dedicated AWS Technical Account Manager (TAM)**?

**

OPTIONS

A. AWS Developer Support



B. AWS Enterprise Support

C. AWS Basic Support

D. AWS Business Support

[!TIP]

Correct Answer:

B (AWS Enterprise Support)

Question 541

Which tool is suitable for

monitoring Amazon Web Services service limits**?

**

OPTIONS

A. AWS Total Cost of Ownership (TCO) Calculator



B. AWS Trusted Advisor

C. AWS Personal Health Dashboard

D. AWS Cost and Usage report

[!TIP]

Correct Answer:

B (AWS Trusted Advisor)

Question 542

Which AWS service or functionality involves the implementation of an

internet service provider (ISP)

and a

colocation facility**?

**

OPTIONS

A. AWS VPN

B. Amazon Connect



C. AWS Direct Connect

D. Internet gateway

[!TIP]

Correct Answer:

C (AWS Direct Connect)

Question 543

Which AWS service tracks

API calls and user activity**?

**

OPTIONS

A. AWS Organizations

B. AWS Config

C. Amazon CloudWatch



D. AWS CloudTrail

[!TIP]

Correct Answer:

D (AWS CloudTrail)

Question 544

Which cloud architectural design concepts are advised for re-architecting a

huge monolithic application**? (Select two.)

**

OPTIONS

A. Use manual monitoring.

B. Use fixed servers.



C. Implement loose coupling.

D. Rely on individual components.



E. Design for scalability.

[!TIP]

Correct Answer:

C and E

Question 545

The worldwide architecture of Amazon Web Services is comprised of

Regions, Availability Zones, and what else**?

**

OPTIONS

A. VPCs



B. Data centers

C. Dark fiber network links

D. Edge locations

[!TIP]

Correct Answer:

B (Data centers)

Question 546

A business must

monitor its AWS accounts and determine when an API request is performed against its AWS resources**.

Which AWS product or service is most appropriate for meeting these requirements?

**

OPTIONS

A. Amazon Inspector

B. Amazon CloudWatch



C. AWS CloudTrail

D. AWS IAM

[!TIP]

Correct Answer:

C (AWS CloudTrail)

Question 547

Which AWS service, feature, or tool uses

machine learning to continuously monitor cost and usage
for

unusual cloud spending**?

**

OPTIONS

A. Amazon Lookout for Metrics

B. AWS Budgets

C. Amazon CloudWatch



D. AWS Cost Anomaly Detection

[!TIP]

Correct Answer:

D (AWS Cost Anomaly Detection)

Question 548

Which AWS Cloud feature will enable a multinational corporation to meet its demand for low latency to all of its customers?**

**

OPTIONS

A. Fault tolerance



B. Global reach

C. Pay-as-you-go pricing

D. High availability

[!TIP]

Correct Answer:

B (Global reach)

Question 549

A business currently operates in one AWS Region and is extending operations to a second. In the second Region, the organization is utilizing the

identical AWS CloudFormation template** as in the original Region. When the organization seeks to deploy Amazon EC2 On-Demand Instances in the second Region, it encounters errors.

What might possibly be the source of these error messages?

**

OPTIONS



A. A new EC2 key pair has not been created for the EC2 instances.

B. The requested EC2 instance types are not available in the second Region.

C. The company cannot operate in a second Region until it updates its AWS contact.

D. The company has not configured AWS Budgets to monitor the budget for the EC2 instances.

[!TIP]

Correct Answer:

A

Question 550

**Which principles are used while architecting apps for AWS Cloud reliability? (Select two.)

**

OPTIONS



A. Design for automated failure recovery

B. Use multiple Availability Zones

C. Manage changes via documented processes

D. Test for moderate demand to ensure reliability

E. Backup recovery to an on-premises environment

[!TIP]

Correct Answer:

A and B

Question 551

A company wants to migrate to AWS and use the

same security software it uses on premises

. The security software vendor offers its security software

as a service on AWS**.

Where can the company purchase the security solution?

**

OPTIONS

A. AWS Partner Solutions Finder

B. AWS Support Center

C. AWS Management Console



D. AWS Marketplace

[!TIP]

Correct Answer:

D (AWS Marketplace)

Question 552

**AWS is responsible for which of the following security-related elements of hosting an Amazon Elastic Compute Cloud (Amazon EC2) instance?

**

OPTIONS

A. Security of private keys



B. Hypervisor software updates

C. Security updates to software running on the instance

D. Policies controlling instance access

[!TIP]

Correct Answer:

B (Hypervisor software updates)

Question 553

Which of the following statements most accurately characterizes

Elastic Load Balancing**?

**

OPTIONS

A. It translates a domain name into an IP address using DNS.



B. It distributes incoming application traffic across one or more Amazon EC2 instances.

C. It collects metrics on connected Amazon EC2 instances.

D. It automatically adjusts the number of Amazon EC2 instances to support incoming traffic.

[!TIP]

Correct Answer:

B

Question 554

Which feature of the AWS Cloud enables customers to

reduce idle CPU capacity**?

**

OPTIONS

A. Agility



B. Elasticity

C. Reliability

D. Durability

[!TIP]

Correct Answer:

B (Elasticity)

Question 555

Which of the following is a

managed AWS service

that is used

specifically for extract, transform, and load (ETL)** data?

**

OPTIONS

A. Amazon Athena



B. AWS Glue

C. Amazon S3

D. AWS Snowball Edge

[!TIP]

Correct Answer:

B (AWS Glue)

Question 556

A corporation is developing a mobile application to give its clients with

shopping suggestions

. The business intends to include a

graph database** into the shopping recommendation engine.

Which Amazon Web Services database service should the business use?

**

OPTIONS

A. Amazon DynamoDB

B. Amazon Aurora



C. Amazon Neptune

D. Amazon DocumentDB (with MongoDB compatibility)

[!TIP]

Correct Answer:

C (Amazon Neptune)

Question 557

Which of the following actions are controlled with

AWS Identity and Access Management (IAM)**? (Choose two.)

**

OPTIONS



A. Control access to AWS service APIs and to other specific resources.

B. Provide intelligent threat detection and continuous monitoring.



C. Protect the AWS environment using **multi-factor authentication (MFA)**.

D. Grant users access to AWS data centers.

E. Provide firewall protection for applications from common web attacks.

[!TIP]

Correct Answer:

A and C

Question 558

A business wishes to establish a

dedicated link

between its on-premises IT infrastructure and

AWS Region resources

. Additionally, the organization wishes to

decrease network latency and congestion**.

Which Amazon Web Services (AWS) service or functionality should the business select?

**

OPTIONS

A. AWS VPN

B. AWS PrivateLink

C. Amazon Connect



D. AWS Direct Connect

[!TIP]

Correct Answer:

D (AWS Direct Connect)

Question 559

Which AWS service allows clients to

audit and monitor AWS resource changes**?

**

OPTIONS

A. AWS Trusted Advisor

B. Amazon GuardDuty

C. Amazon Inspector



D. AWS Config

[!TIP]

Correct Answer:

D (AWS Config)

Question 560

Which of the following is an

advantage** of using AWS's cloud computing platform?

**

OPTIONS

A. Permissive security removes the administrative burden.



B. Ability to focus on revenue-generating activities.

C. Control over cloud network hardware.

D. Choice of specific cloud hardware vendors.

[!TIP]

Correct Answer:

B (Ability to focus on revenue-generating activities)

Question 561

Which of the following are

shared controls

that apply to both AWS and the customer, according to the

AWS Shared Responsibility Model**? (Choose two.)

**

OPTIONS



A. Resource configuration management

B. Network data integrity



C. Employee awareness and training

D. Physical and environmental security

E. Replacement and disposal of disk drives

[!TIP]

Correct Answer:

A and C

Question 562

Which AWS service or feature gives

information about planned events that are now occurring or may occur in the near future and may impact an AWS account**?

**

OPTIONS

A. AWS Config

B. AWS Systems Manager



C. AWS Personal Health Dashboard

D. AWS Trusted Advisor

[!TIP]

Correct Answer:

C (AWS Personal Health Dashboard)

Question 563

A business has many Amazon Web Services accounts and wants to

streamline and unify its billing process**.

Which AWS service is capable of doing this?

**

OPTIONS

A. AWS Cost and Usage Reports



B. AWS Organizations

C. AWS Cost Explorer

D. AWS Budgets

[!TIP]

Correct Answer:

B (AWS Organizations)

Question 564

What does it imply when a customer uses AWS to construct a hybrid cloud architecture**?

**

OPTIONS

A. All resources run using on-premises infrastructure.

B. Some resources run on-premises and some run in a colocation center.

C. All resources run in the AWS Cloud.



D. Some resources run on-premises and some run in the AWS Cloud.

[!TIP]

Correct Answer:

D

Question 565

What is the

LEAST EXPENSIVE

AWS Support plan that contains a

full set of AWS Trusted Advisor best practice checks**?

**

OPTIONS

A. AWS Enterprise Support



B. AWS Business Support

C. AWS Developer Support

D. AWS Basic Support

[!TIP]

Correct Answer:

B (AWS Business Support)

Question 567

Which AWS service provides

domain registration, DNS routing, and service health checks**?

**

OPTIONS

A. AWS Direct Connect



B. Amazon Route 53

C. Amazon CloudFront

D. Amazon API Gateway

[!TIP]

Correct Answer:

B (Amazon Route 53)

Question 568

Which AWS service or feature can help to improve network security by

restricting requests for a web application hosted on AWS from a certain network**? (Select two.)

**

OPTIONS



A. AWS WAF

B. AWS Trusted Advisor

C. AWS Direct Connect

D. AWS Organizations



E. Network ACLs

[!TIP]

Correct Answer:

A and E

Question 569

Which components are

necessary

to configure an AWS

site-to-site VPN** connection successfully? (Select two.)

**

OPTIONS

A. Internet gateway

B. NAT gateway



C. Customer gateway

D. Transit gateway



E. Virtual private gateway

[!TIP]

Correct Answer:

C and E

Question 570

Which AWS service should be used to migrate a company's on-premises MySQL database to Amazon RDS?**

**

OPTIONS

A. AWS Direct Connect

B. AWS Server Migration Service (AWS SMS)



C. AWS Database Migration Service (AWS DMS)

D. AWS Schema Conversion Tool (AWS SCT)

[!TIP]

Correct Answer:

Question 571

**Which benefits does a company gain when the company moves from on-premises IT architecture to the AWS Cloud? (Choose two.)

**

OPTIONS



A. Reduced or eliminated tasks for hardware troubleshooting, capacity planning, and procurement

B. Elimination of the need for trained IT staff

C. Automatic security configuration of all applications that are migrated to the cloud

D. Elimination of the need for disaster recovery planning



E. Faster deployment of new features and applications

[!TIP]

Correct Answer:

A and E

Question 572

A business needs

security against increased distributed denial of service (DDoS) assaults on its website
, as well as help from

AWS professionals** professionals in the case of such an attack.

Which AWS managed service will satisfy these criteria?

**

OPTIONS



A. AWS Shield Advanced

B. AWS Firewall Manager

C. AWS WAF

D. Amazon GuardDuty

[!TIP]

Correct Answer:

A (AWS Shield Advanced)

Question 573

Which of the following is a

benefit of decoupling** an AWS Cloud architecture?

**

OPTIONS

A. Reduced latency



B. Ability to upgrade components independently

C. Decreased costs

D. Fewer components to manage

[!TIP]

Correct Answer:

B (Ability to upgrade components independently)

Question 574

A Cloud Practitioner requires a

dedicated link

between AWS resources and an

on-premises system

that is

constant and devoted**.

Which AWS service satisfies this criteria?

**

OPTIONS



A. AWS Direct Connect

B. AWS VPN

C. Amazon Connect

D. AWS Data Pipeline

[!TIP]

Correct Answer:

A (AWS Direct Connect)

Question 575

Which task is the

responsibility of the customer** according to the AWS shared responsibility model?

**

OPTIONS

A. Maintain the security of the hardware that runs Amazon EC2 instances.



B. Patch the guest operating system of Amazon EC2 instances.

C. Protect the security of the AWS global infrastructure.

D. Patch Amazon RDS software.

[!TIP]

Correct Answer:

B (Patch the guest operating system of Amazon EC2 instances)